

Filter Summary Report: CG,TIA,Automated,simple,Z1,Z3,Z4

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Contents

1 Examined $H(z)$ for CG TIA Automated simple Z1 Z3 Z4: $\frac{Z_1 Z_3 Z_4 g_m}{2Z_1 Z_3 g_m + Z_1 Z_4 g_m + 2Z_3 + Z_4}$

$$H(z) = \frac{Z_1 Z_3 Z_4 g_m}{2Z_1 Z_3 g_m + Z_1 Z_4 g_m + 2Z_3 + Z_4}$$

2 AP

3 BP

3.1 BP-1 $Z(s) = \left(R_1, \infty, R_3, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{L_4 R_1 R_3 g_m s}{2R_1 R_3 g_m + 2R_3 + s^2 (2C_4 L_4 R_1 R_3 g_m + 2C_4 L_4 R_3) + s (L_4 R_1 g_m + L_4)}$$

Parameters:

Q: $\frac{2\sqrt{C_4} R_3}{\sqrt{L_4}}$
 wo: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$
 bandwidth: $\frac{1}{2C_4 R_3}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
 Qz: None
 Wz: None

3.2 BP-2 $Z(s) = \left(R_1, \infty, R_3, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{L_4 R_1 R_3 R_4 g_m s}{2R_1 R_3 R_4 g_m + 2R_3 R_4 + s^2 (2C_4 L_4 R_1 R_3 R_4 g_m + 2C_4 L_4 R_3 R_4) + s (2L_4 R_1 R_3 g_m + L_4 R_1 R_4 g_m + 2L_4 R_3 + L_4 R_4)}$$

Parameters:

Q: $\frac{2\sqrt{C_4} R_3 R_4}{2\sqrt{L_4} R_3 + \sqrt{L_4} R_4}$
 wo: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$
 bandwidth: $\frac{2\sqrt{L_4} R_3 + \sqrt{L_4} R_4}{2C_4 \sqrt{L_4} R_3 R_4}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}$
 Qz: None
 Wz: None

3.3 BP-3 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{L_4 R_1 R_4 g_m s}{2R_1 R_4 g_m + 2R_4 + s^2 (C_3 L_4 R_1 R_4 g_m + C_3 L_4 R_4 + 2C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_4) + s (2L_4 R_1 g_m + 2L_4)}$$

Parameters:

Q: $\frac{\sqrt{2} C_3 R_4 + 2\sqrt{2} C_4 R_4}{2\sqrt{L_4} \sqrt{C_3 + 2C_4}}$
 wo: $\frac{\sqrt{2}}{\sqrt{C_3 L_4 + 2C_4 L_4}}$
 bandwidth: $\frac{2\sqrt{2} \sqrt{L_4} \sqrt{C_3 + 2C_4}}{\sqrt{C_3 L_4 + 2C_4 L_4} (\sqrt{2} C_3 R_4 + 2\sqrt{2} C_4 R_4)}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
 Qz: None
 Wz: None

3.4 BP-4 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{L_4 R_1 R_3 g_m s}{2 R_1 R_3 g_m + 2 R_3 + s^2 (C_3 L_4 R_1 R_3 g_m + C_3 L_4 R_3 + 2 C_4 L_4 R_1 R_3 g_m + 2 C_4 L_4 R_3) + s (L_4 R_1 g_m + L_4)}$$

Parameters:

Q: $\frac{\sqrt{2} C_3 R_3 + 2 \sqrt{2} C_4 R_3}{\sqrt{L_4} \sqrt{C_3 + 2 C_4}}$
wo: $\frac{\sqrt{2}}{\sqrt{C_3 L_4 + 2 C_4 L_4}}$
bandwidth: $\frac{\sqrt{2} \sqrt{L_4} \sqrt{C_3 + 2 C_4}}{\sqrt{C_3 L_4 + 2 C_4 L_4} (\sqrt{2} C_3 R_3 + 2 \sqrt{2} C_4 R_3)}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
Qz: None
Wz: None

3.5 BP-5 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{L_4 R_1 R_3 R_4 g_m s}{2 R_1 R_3 R_4 g_m + 2 R_3 R_4 + s^2 (C_3 L_4 R_1 R_3 R_4 g_m + C_3 L_4 R_3 R_4 + 2 C_4 L_4 R_1 R_3 R_4 g_m + 2 C_4 L_4 R_3 R_4) + s (2 L_4 R_1 R_3 g_m + L_4 R_1 R_4 g_m + 2 L_4 R_3 + L_4 R_4)}$$

Parameters:

Q: $\frac{\sqrt{2} C_3 R_3 R_4 + 2 \sqrt{2} C_4 R_3 R_4}{2 \sqrt{L_4} R_3 \sqrt{C_3 + 2 C_4} + \sqrt{L_4} R_4 \sqrt{C_3 + 2 C_4}}$
wo: $\frac{\sqrt{2}}{\sqrt{C_3 L_4 + 2 C_4 L_4}}$
bandwidth: $\frac{\sqrt{2} (2 \sqrt{L_4} R_3 \sqrt{C_3 + 2 C_4} + \sqrt{L_4} R_4 \sqrt{C_3 + 2 C_4})}{\sqrt{C_3 L_4 + 2 C_4 L_4} (\sqrt{2} C_3 R_3 R_4 + 2 \sqrt{2} C_4 R_3 R_4)}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}$
Qz: None
Wz: None

3.6 BP-6 $Z(s) = \left(R_1, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{L_3 R_1 R_4 g_m s}{R_1 R_4 g_m + R_4 + s^2 (C_3 L_3 R_1 R_4 g_m + C_3 L_3 R_4) + s (2 L_3 R_1 g_m + 2 L_3)}$$

Parameters:

Q: $\frac{\sqrt{C_3} R_4}{2 \sqrt{L_3}}$
wo: $\frac{1}{\sqrt{C_3} \sqrt{L_3}}$
bandwidth: $\frac{2}{C_3 R_4}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1 R_4 g_m}{2 R_1 g_m + 2}$
Qz: None
Wz: None

3.7 BP-7 $Z(s) = \left(R_1, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{L_3 R_1 R_4 g_m s}{R_1 R_4 g_m + R_4 + s^2 (C_3 L_3 R_1 R_4 g_m + C_3 L_3 R_4 + 2 C_4 L_3 R_1 R_4 g_m + 2 C_4 L_3 R_4) + s (2 L_3 R_1 g_m + 2 L_3)}$$

Parameters:

Q: $\frac{C_3 R_4 + 2 C_4 R_4}{2 \sqrt{L_3} \sqrt{C_3 + 2 C_4}}$
wo: $\frac{1}{\sqrt{C_3 L_3 + 2 C_4 L_3}}$
bandwidth: $\frac{2 \sqrt{L_3} \sqrt{C_3 + 2 C_4}}{\sqrt{C_3 L_3 + 2 C_4 L_3} (C_3 R_4 + 2 C_4 R_4)}$

K-LP: 0
K-HP: 0
K-BP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
Qz: None
Wz: None

3.8 BP-8 $Z(s) = \left(R_1, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{L_3 L_4 R_1 R_4 g_m s}{2L_3 R_1 R_4 g_m + 2L_3 R_4 + L_4 R_1 R_4 g_m + L_4 R_4 + s^2 (C_3 L_3 L_4 R_1 R_4 g_m + C_3 L_3 L_4 R_4 + 2C_4 L_3 L_4 R_1 R_4 g_m + 2C_4 L_3 L_4 R_4) + s (2L_3 L_4 R_1 g_m + 2L_3 L_4)}$$

Parameters:

Q: $\frac{C_3 R_4 \sqrt{2L_3 + L_4} + 2C_4 R_4 \sqrt{2L_3 + L_4}}{2\sqrt{L_3} \sqrt{L_4} \sqrt{C_3 + 2C_4}}$
wo: $\frac{\sqrt{2L_3 + L_4}}{\sqrt{C_3 L_3 L_4 + 2C_4 L_3 L_4}}$
bandwidth: $\frac{2\sqrt{L_3} \sqrt{L_4} \sqrt{C_3 + 2C_4} \sqrt{2L_3 + L_4}}{\sqrt{C_3 L_3 L_4 + 2C_4 L_3 L_4} (C_3 R_4 \sqrt{2L_3 + L_4} + 2C_4 R_4 \sqrt{2L_3 + L_4})}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
Qz: None
Wz: None

3.9 BP-9 $Z(s) = \left(R_1, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, R_4, \infty, \infty \right)$

$$H(s) = \frac{L_3 R_1 R_3 R_4 g_m s}{R_1 R_3 R_4 g_m + R_3 R_4 + s^2 (C_3 L_3 R_1 R_3 R_4 g_m + C_3 L_3 R_3 R_4) + s (2L_3 R_1 R_3 g_m + L_3 R_1 R_4 g_m + 2L_3 R_3 + L_3 R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_3} R_3 R_4}{2\sqrt{L_3} R_3 + \sqrt{L_3} R_4}$
wo: $\frac{1}{\sqrt{C_3} \sqrt{L_3}}$
bandwidth: $\frac{2\sqrt{L_3} R_3 + \sqrt{L_3} R_4}{C_3 \sqrt{L_3} R_3 R_4}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}$
Qz: None
Wz: None

3.10 BP-10 $Z(s) = \left(R_1, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{L_3 R_1 R_3 g_m s}{R_1 R_3 g_m + R_3 + s^2 (C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + 2C_4 L_3 R_3) + s (L_3 R_1 g_m + L_3)}$$

Parameters:

Q: $\frac{C_3 R_3 + 2C_4 R_3}{\sqrt{L_3} \sqrt{C_3 + 2C_4}}$
wo: $\frac{1}{\sqrt{C_3 L_3 + 2C_4 L_3}}$
bandwidth: $\frac{\sqrt{L_3} \sqrt{C_3 + 2C_4}}{\sqrt{C_3 L_3 + 2C_4 L_3} (C_3 R_3 + 2C_4 R_3)}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
Qz: None
Wz: None

$$\mathbf{3.11 \quad BP-11} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 R_1 R_3 R_4 g_m s}{R_1 R_3 R_4 g_m + R_3 R_4 + s^2 (C_3 L_3 R_1 R_3 R_4 g_m + C_3 L_3 R_3 R_4 + 2C_4 L_3 R_1 R_3 R_4 g_m + 2C_4 L_3 R_3 R_4) + s (2L_3 R_1 R_3 g_m + L_3 R_1 R_4 g_m + 2L_3 R_3 + L_3 R_4)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{C_3 R_3 R_4 + 2C_4 R_3 R_4}{2\sqrt{L_3 R_3} \sqrt{C_3 + 2C_4} + \sqrt{L_3 R_4} \sqrt{C_3 + 2C_4}} \\ \text{wo: } & \frac{1}{\sqrt{C_3 L_3 + 2C_4 L_3}} \\ \text{bandwidth: } & \frac{2\sqrt{L_3 R_3} \sqrt{C_3 + 2C_4} + \sqrt{L_3 R_4} \sqrt{C_3 + 2C_4}}{\sqrt{C_3 L_3 + 2C_4 L_3} (C_3 R_3 R_4 + 2C_4 R_3 R_4)} \\ \text{K-LP: } & 0 \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

$$\mathbf{3.12 \quad BP-12} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 L_4 R_1 R_3 g_m s}{2L_3 R_1 R_3 g_m + 2L_3 R_3 + L_4 R_1 R_3 g_m + L_4 R_3 + s^2 (C_3 L_3 L_4 R_1 R_3 g_m + C_3 L_3 L_4 R_3 + 2C_4 L_3 L_4 R_1 R_3 g_m + 2C_4 L_3 L_4 R_3) + s (L_3 L_4 R_1 g_m + L_3 L_4)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{C_3 R_3 \sqrt{2L_3 + L_4} + 2C_4 R_3 \sqrt{2L_3 + L_4}}{\sqrt{L_3} \sqrt{L_4} \sqrt{C_3 + 2C_4}} \\ \text{wo: } & \frac{\sqrt{2L_3 + L_4}}{\sqrt{C_3 L_3 L_4 + 2C_4 L_3 L_4}} \\ \text{bandwidth: } & \frac{\sqrt{L_3} \sqrt{L_4} \sqrt{C_3 + 2C_4} \sqrt{2L_3 + L_4}}{\sqrt{C_3 L_3 L_4 + 2C_4 L_3 L_4} (C_3 R_3 \sqrt{2L_3 + L_4} + 2C_4 R_3 \sqrt{2L_3 + L_4})} \\ \text{K-LP: } & 0 \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{R_1 R_3 g_m}{R_1 g_m + 1} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

$$\mathbf{3.13 \quad BP-13} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 L_4 R_1 R_3 R_4 g_m s}{2L_3 R_1 R_3 R_4 g_m + 2L_3 R_3 R_4 + L_4 R_1 R_3 R_4 g_m + L_4 R_3 R_4 + s^2 (C_3 L_3 L_4 R_1 R_3 R_4 g_m + C_3 L_3 L_4 R_3 R_4 + 2C_4 L_3 L_4 R_1 R_3 R_4 g_m + 2C_4 L_3 L_4 R_3 R_4) + s (2L_3 L_4 R_1 R_3 g_m + L_3 L_4 R_1 R_4 g_m + 2L_3 L_4 R_3 + L_3 L_4 R_4)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{C_3 R_3 R_4 \sqrt{2L_3 + L_4} + 2C_4 R_3 R_4 \sqrt{2L_3 + L_4}}{2\sqrt{L_3} \sqrt{L_4} R_3 \sqrt{C_3 + 2C_4} + \sqrt{L_3} \sqrt{L_4} R_4 \sqrt{C_3 + 2C_4}} \\ \text{wo: } & \frac{\sqrt{2L_3 + L_4}}{\sqrt{C_3 L_3 L_4 + 2C_4 L_3 L_4}} \\ \text{bandwidth: } & \frac{\sqrt{2L_3 + L_4} (2\sqrt{L_3} \sqrt{L_4} R_3 \sqrt{C_3 + 2C_4} + \sqrt{L_3} \sqrt{L_4} R_4 \sqrt{C_3 + 2C_4})}{\sqrt{C_3 L_3 L_4 + 2C_4 L_3 L_4} (C_3 R_3 R_4 \sqrt{2L_3 + L_4} + 2C_4 R_3 R_4 \sqrt{2L_3 + L_4})} \\ \text{K-LP: } & 0 \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

$$\mathbf{3.14 \quad BP-14} \quad Z(s) = \left(L_1 s, \quad \infty, \quad R_3, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_3 g_m s}{2C_4 L_1 R_3 g_m s^2 + s (2C_4 R_3 + L_1 g_m) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2} \sqrt{C_4} \sqrt{L_1} \sqrt{R_3} g_m \sqrt{\frac{1}{g_m}}}{2C_4 R_3 + L_1 g_m} \\ \text{wo: } & \frac{\sqrt{2} \sqrt{\frac{1}{g_m}}}{2\sqrt{C_4} \sqrt{L_1} \sqrt{R_3}} \end{aligned}$$

bandwidth: $\frac{2C_4R_3+L_1g_m}{2C_4L_1R_3g_m}$
K-LP: 0
K-HP: 0
K-BP: $\frac{L_1R_3g_m}{2C_4R_3+L_1g_m}$
Qz: None
Wz: None

3.15 BP-15 $Z(s) = \left(L_1s, \infty, R_3, \frac{R_4}{C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{L_1R_3R_4g_ms}{2C_4L_1R_3R_4g_ms^2 + 2R_3 + R_4 + s(2C_4R_3R_4 + 2L_1R_3g_m + L_1R_4g_m)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_4}\sqrt{L_1}\sqrt{R_3}\sqrt{R_4g_m}\sqrt{2R_3+R_4}\sqrt{\frac{1}{g_m}}}{2C_4R_3R_4+2L_1R_3g_m+L_1R_4g_m}$
wo: $\frac{\sqrt{2}\sqrt{2R_3+R_4}\sqrt{\frac{1}{g_m}}}{2\sqrt{C_4}\sqrt{L_1}\sqrt{R_3}\sqrt{R_4}}$
bandwidth: $\frac{2C_4R_3R_4+2L_1R_3g_m+L_1R_4g_m}{2C_4L_1R_3R_4g_m}$
K-LP: 0
K-HP: 0
K-BP: $\frac{L_1R_3R_4g_m}{2C_4R_3R_4+2L_1R_3g_m+L_1R_4g_m}$
Qz: None
Wz: None

3.16 BP-16 $Z(s) = \left(L_1s, \infty, \frac{1}{C_3s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{L_1R_4g_ms}{C_3L_1R_4g_ms^2 + s(C_3R_4 + 2L_1g_m) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_3}\sqrt{L_1}\sqrt{R_4g_m}\sqrt{\frac{1}{g_m}}}{C_3R_4+2L_1g_m}$
wo: $\frac{\sqrt{2}\sqrt{\frac{1}{g_m}}}{\sqrt{C_3}\sqrt{L_1}\sqrt{R_4}}$
bandwidth: $\frac{C_3R_4+2L_1g_m}{C_3L_1R_4g_m}$
K-LP: 0
K-HP: 0
K-BP: $\frac{L_1R_4g_m}{C_3R_4+2L_1g_m}$
Qz: None
Wz: None

3.17 BP-17 $Z(s) = \left(L_1s, \infty, \frac{1}{C_3s}, \frac{R_4}{C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{L_1R_4g_ms}{s^2(C_3L_1R_4g_m + 2C_4L_1R_4g_m) + s(C_3R_4 + 2C_4R_4 + 2L_1g_m) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{L_1}\sqrt{R_4g_m}\sqrt{C_3+2C_4}\sqrt{\frac{1}{g_m}}}{C_3R_4+2C_4R_4+2L_1g_m}$
wo: $\sqrt{2}\sqrt{\frac{1}{C_3L_1R_4g_m+2C_4L_1R_4g_m}}$
bandwidth: $\frac{(C_3R_4+2C_4R_4+2L_1g_m)\sqrt{\frac{1}{C_3L_1R_4g_m+2C_4L_1R_4g_m}}}{\sqrt{L_1}\sqrt{R_4g_m}\sqrt{C_3+2C_4}\sqrt{\frac{1}{g_m}}}$
K-LP: 0
K-HP: 0
K-BP: $\frac{L_1R_4g_m}{C_3R_4+2C_4R_4+2L_1g_m}$
Qz: None
Wz: None

3.18 BP-18 $Z(s) = \left(L_1 s, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{C_3} \sqrt{L_1} \sqrt{R_3} \sqrt{R_4} g_m \sqrt{2R_3 + R_4} \sqrt{\frac{1}{g_m}}}{C_3 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m}$
wo: $\frac{\sqrt{2R_3 + R_4} \sqrt{\frac{1}{g_m}}}{\sqrt{C_3} \sqrt{L_1} \sqrt{R_3} \sqrt{R_4}}$
bandwidth: $\frac{C_3 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m}{C_3 L_1 R_3 R_4 g_m}$
K-LP: 0
K-HP: 0
K-BP: $\frac{L_1 R_3 R_4 g_m}{C_3 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m}$
Qz: None
Wz: None

3.19 BP-19 $Z(s) = \left(L_1 s, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{L_1} \sqrt{R_3} g_m \sqrt{C_3 + 2C_4} \sqrt{\frac{1}{g_m}}}{C_3 R_3 + 2C_4 R_3 + L_1 g_m}$
wo: $\sqrt{\frac{1}{C_3 L_1 R_3 g_m + 2C_4 L_1 R_3 g_m}}$
bandwidth: $\frac{(C_3 R_3 + 2C_4 R_3 + L_1 g_m) \sqrt{\frac{1}{C_3 L_1 R_3 g_m + 2C_4 L_1 R_3 g_m}}}{\sqrt{L_1} \sqrt{R_3} g_m \sqrt{C_3 + 2C_4} \sqrt{\frac{1}{g_m}}}$
K-LP: 0
K-HP: 0
K-BP: $\frac{L_1 R_3 g_m}{C_3 R_3 + 2C_4 R_3 + L_1 g_m}$
Qz: None
Wz: None

3.20 BP-20 $Z(s) = \left(L_1 s, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{L_1} \sqrt{R_3} \sqrt{R_4} g_m \sqrt{C_3 + 2C_4} \sqrt{2R_3 + R_4} \sqrt{\frac{1}{g_m}}}{C_3 R_3 R_4 + 2C_4 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m}$
wo: $\sqrt{2R_3 + R_4} \sqrt{\frac{1}{C_3 L_1 R_3 R_4 g_m + 2C_4 L_1 R_3 R_4 g_m}}$
bandwidth: $\frac{(C_3 R_3 R_4 + 2C_4 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m) \sqrt{\frac{1}{C_3 L_1 R_3 R_4 g_m + 2C_4 L_1 R_3 R_4 g_m}}}{\sqrt{L_1} \sqrt{R_3} \sqrt{R_4} g_m \sqrt{C_3 + 2C_4} \sqrt{\frac{1}{g_m}}}$
K-LP: 0
K-HP: 0
K-BP: $\frac{L_1 R_3 R_4 g_m}{C_3 R_3 R_4 + 2C_4 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m}$
Qz: None
Wz: None

3.21 BP-21 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, R_3, R_4, \infty, \infty \right)$

Parameters:

$$H(s) = \frac{L_1 R_3 R_4 g_m s}{C_3 L_1 R_3 R_4 g_m s^2 + 2R_3 + R_4 + s(C_3 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$H(s) = \frac{L_1 R_3 g_m s}{s^2(C_3 L_1 R_3 g_m + 2C_4 L_1 R_3 g_m) + s(C_3 R_3 + 2C_4 R_3 + L_1 g_m) + 1}$$

$$H(s) = \frac{L_1 R_3 R_4 g_m s}{2R_3 + R_4 + s^2(C_3 L_1 R_3 R_4 g_m + 2C_4 L_1 R_3 R_4 g_m) + s(C_3 R_3 R_4 + 2C_4 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$H(s) = \frac{L_1 R_3 R_4 g_m s}{2R_3 + R_4 + s^2(2C_1 L_1 R_3 + C_1 L_1 R_4) + s(2L_1 R_3 g_m + L_1 R_4 g_m)}$$

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1}g_m}$
 wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
 bandwidth: $\frac{g_m}{C_1}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_3R_4}{2R_3+R_4}$
 Qz: None
 Wz: None

3.22 BP-22 $Z(s) = \left(\frac{L_1R_1s}{C_1L_1R_1s^2+L_1s+R_1}, \infty, R_3, R_4, \infty, \infty \right)$

$$H(s) = \frac{L_1R_1R_3R_4g_ms}{2R_1R_3 + R_1R_4 + s^2(2C_1L_1R_1R_3 + C_1L_1R_1R_4) + s(2L_1R_1R_3g_m + L_1R_1R_4g_m + 2L_1R_3 + L_1R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_1}R_1}{\sqrt{L_1R_1g_m+\sqrt{L_1}}}$
 wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
 bandwidth: $\frac{\sqrt{L_1R_1g_m+\sqrt{L_1}}}{C_1\sqrt{L_1}R_1}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_1R_3R_4g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}$
 Qz: None
 Wz: None

4 BS

4.1 BS-1 $Z(s) = \left(R_1, \infty, R_3, L_4s + \frac{1}{C_4s}, \infty, \infty \right)$

$$H(s) = \frac{C_4L_4R_1R_3g_ms^2 + R_1R_3g_m}{R_1g_m + s^2(C_4L_4R_1g_m + C_4L_4) + s(2C_4R_1R_3g_m + 2C_4R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{L_4}}{2\sqrt{C_4}R_3}$
 wo: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$
 bandwidth: $\frac{2R_3}{L_4}$
 K-LP: $\frac{R_1R_3g_m}{R_1g_m+1}$
 K-HP: $\frac{R_1R_3g_m}{R_1g_m+1}$
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$

4.2 BS-2 $Z(s) = \left(R_1, \infty, R_3, \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{C_4L_4R_1R_3R_4g_ms^2 + R_1R_3R_4g_m}{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^2(2C_4L_4R_1R_3g_m + C_4L_4R_1R_4g_m + 2C_4L_4R_3 + C_4L_4R_4) + s(2C_4R_1R_3R_4g_m + 2C_4R_3R_4)}$$

Parameters:

Q: $\frac{2\sqrt{L_4}R_3+\sqrt{L_4}R_4}{2\sqrt{C_4}R_3R_4}$
 wo: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$
 bandwidth: $\frac{2R_3R_4}{\sqrt{L_4}(2\sqrt{L_4}R_3+\sqrt{L_4}R_4)}$
 K-LP: $\frac{R_1R_3R_4g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}$
 K-HP: $\frac{R_1R_3R_4g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}$

K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$

4.3 BS-3 $Z(s) = \left(R_1, \infty, L_3 s + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_3 R_1 R_4 g_m s^2 + R_1 R_4 g_m}{2 R_1 g_m + s^2 (2 C_3 L_3 R_1 g_m + 2 C_3 L_3) + s (C_3 R_1 R_4 g_m + C_3 R_4) + 2}$$

Parameters:

Q: $\frac{2\sqrt{L_3}}{\sqrt{C_3} R_4}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
bandwidth: $\frac{R_4}{2L_3}$
K-LP: $\frac{R_1 R_4 g_m}{2 R_1 g_m + 2}$
K-HP: $\frac{R_1 R_4 g_m}{2 R_1 g_m + 2}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

4.4 BS-4 $Z(s) = \left(R_1, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_3 R_1 R_3 R_4 g_m s^2 + R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^2 (2 C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_1 R_4 g_m + 2 C_3 L_3 R_3 + C_3 L_3 R_4) + s (C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4)}$$

Parameters:

Q: $\frac{2\sqrt{L_3} R_3 + \sqrt{L_3} R_4}{\sqrt{C_3} R_3 R_4}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
bandwidth: $\frac{R_3 R_4}{\sqrt{L_3} (2\sqrt{L_3} R_3 + \sqrt{L_3} R_4)}$
K-LP: $\frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}$
K-HP: $\frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

4.5 BS-5 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, R_3, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 R_3 R_4 g_m s^2 + R_3 R_4 g_m}{2 R_3 g_m + R_4 g_m + s^2 (2 C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m) + s (2 C_1 R_3 + C_1 R_4)}$$

Parameters:

Q: $\frac{\sqrt{L_1} g_m}{\sqrt{C_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{1}{L_1 g_m}$
K-LP: $\frac{R_3 R_4}{2 R_3 + R_4}$
K-HP: $\frac{R_3 R_4}{2 R_3 + R_4}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$

4.6 BS-6 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, R_3, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 R_1 R_3 R_4 g_m s^2 + R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^2 (2 C_1 L_1 R_1 R_3 g_m + C_1 L_1 R_1 R_4 g_m + 2 C_1 L_1 R_3 + C_1 L_1 R_4) + s (2 C_1 R_1 R_3 + C_1 R_1 R_4)}$$

Parameters:

Q: $\frac{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}{\sqrt{C_1} R_1}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{R_1}{\sqrt{L_1} (\sqrt{L_1} R_1 g_m + \sqrt{L_1})}$
 K-LP: $\frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}$
 K-HP: $\frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}$
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$

5 GE

5.1 GE-1 $Z(s) = \left(R_1, \infty, R_3, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_4 R_1 R_3 g_m s^2 + C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{R_1 g_m + s^2 (C_4 L_4 R_1 g_m + C_4 L_4) + s (2 C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2 C_4 R_3 + C_4 R_4) + 1}$$

Parameters:

Q: $\frac{\sqrt{L_4}}{2 \sqrt{C_4} R_3 + \sqrt{C_4} R_4}$
 wo: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$
 bandwidth: $\frac{2 \sqrt{C_4} R_3 + \sqrt{C_4} R_4}{\sqrt{C_4} L_4}$
 K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
 K-HP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
 K-BP: $\frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}$
 Qz: $\frac{\sqrt{L_4}}{\sqrt{C_4} R_4}$
 Wz: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$

5.2 GE-2 $Z(s) = \left(R_1, \infty, R_3, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_4 R_1 R_3 R_4 g_m s^2 + L_4 R_1 R_3 g_m s + R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^2 (2 C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_3 + C_4 L_4 R_4) + s (L_4 R_1 g_m + L_4)}$$

Parameters:

Q: $\frac{2 \sqrt{C_4} R_3 + \sqrt{C_4} R_4}{\sqrt{L_4}}$
 wo: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$
 bandwidth: $\frac{1}{\sqrt{C_4} (2 \sqrt{C_4} R_3 + \sqrt{C_4} R_4)}$
 K-LP: $\frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}$
 K-HP: $\frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}$
 K-BP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
 Qz: $\frac{\sqrt{C_4} R_4}{\sqrt{L_4}}$
 Wz: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$

5.3 GE-3 $Z(s) = \left(R_1, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_3 R_1 R_4 g_m s^2 + C_3 R_1 R_3 R_4 g_m s + R_1 R_4 g_m}{2 R_1 g_m + s^2 (2 C_3 L_3 R_1 g_m + 2 C_3 L_3) + s (2 C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2 C_3 R_3 + C_3 R_4) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{L_3}}{2\sqrt{C_3}R_3 + \sqrt{C_3}R_4} \\ \text{wo: } & \frac{1}{\sqrt{C_3}\sqrt{L_3}} \\ \text{bandwidth: } & \frac{2\sqrt{C_3}R_3 + \sqrt{C_3}R_4}{2\sqrt{C_3}L_3} \\ \text{K-LP: } & \frac{R_1 R_4 g_m}{2 R_1 g_m + 2} \\ \text{K-HP: } & \frac{R_1 R_4 g_m}{2 R_1 g_m + 2} \\ \text{K-BP: } & \frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4} \\ \text{QZ: } & \frac{\sqrt{L_3}}{\sqrt{C_3}R_3} \\ \text{WZ: } & \frac{1}{\sqrt{C_3}\sqrt{L_3}} \end{aligned}$$

5.4 GE-4 $Z(s) = \left(R_1, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_3 R_1 R_3 R_4 g_m s^2 + L_3 R_1 R_4 g_m s + R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^2 (2 C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_1 R_4 g_m + 2 C_3 L_3 R_3 + C_3 L_3 R_4) + s (2 L_3 R_1 g_m + 2 L_3)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_3}R_3 + \sqrt{C_3}R_4}{2\sqrt{L_3}} \\ \text{wo: } & \frac{1}{\sqrt{C_3}\sqrt{L_3}} \\ \text{bandwidth: } & \frac{2}{\sqrt{C_3}(2\sqrt{C_3}R_3 + \sqrt{C_3}R_4)} \\ \text{K-LP: } & \frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4} \\ \text{K-HP: } & \frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4} \\ \text{K-BP: } & \frac{R_1 R_4 g_m}{2 R_1 g_m + 2} \\ \text{QZ: } & \frac{\sqrt{C_3}R_3}{\sqrt{L_3}} \\ \text{WZ: } & \frac{1}{\sqrt{C_3}\sqrt{L_3}} \end{aligned}$$

5.5 GE-5 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, R_3, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 R_3 R_4 g_m s^2 + C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m}{2 R_3 g_m + R_4 g_m + s^2 (2 C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m) + s (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{L_1}g_m}{\sqrt{C_1}R_1 g_m + \sqrt{C_1}} \\ \text{wo: } & \frac{1}{\sqrt{C_1}\sqrt{L_1}} \\ \text{bandwidth: } & \frac{\sqrt{C_1}R_1 g_m + \sqrt{C_1}}{\sqrt{C_1}L_1 g_m} \\ \text{K-LP: } & \frac{R_3 R_4}{2 R_3 + R_4} \\ \text{K-HP: } & \frac{R_3 R_4}{2 R_3 + R_4} \\ \text{K-BP: } & \frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4} \\ \text{QZ: } & \frac{\sqrt{L_1}}{\sqrt{C_1}R_1} \\ \text{WZ: } & \frac{1}{\sqrt{C_1}\sqrt{L_1}} \end{aligned}$$

5.6 GE-6 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 R_1 R_3 R_4 g_m s^2 + L_1 R_3 R_4 g_m s + R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^2 (2 C_1 L_1 R_1 R_3 g_m + C_1 L_1 R_1 R_4 g_m + 2 C_1 L_1 R_3 + C_1 L_1 R_4) + s (2 L_1 R_3 g_m + L_1 R_4 g_m)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_1 R_1 g_m} + \sqrt{C_1}}{\sqrt{L_1 g_m}} \\
\text{wo: } & \frac{1}{\sqrt{C_1} \sqrt{L_1}} \\
\text{bandwidth: } & \frac{g_m}{\sqrt{C_1} (\sqrt{C_1 R_1 g_m} + \sqrt{C_1})} \\
\text{K-LP: } & \frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4} \\
\text{K-HP: } & \frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4} \\
\text{K-BP: } & \frac{R_3 R_4}{2 R_3 + R_4} \\
\text{Qz: } & \frac{\sqrt{C_1 R_1}}{\sqrt{L_1}} \\
\text{Wz: } & \frac{1}{\sqrt{C_1} \sqrt{L_1}}
\end{aligned}$$

6 HP

7 LP

$$7.1 \quad \text{LP-1} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_3 g_m}{2 C_1 C_4 R_3 s^2 + g_m + s (C_1 + 2 C_4 R_3 g_m)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_3} \sqrt{g_m}}{C_1 + 2 C_4 R_3 g_m} \\
\text{wo: } & \frac{\sqrt{2} \sqrt{g_m}}{2 \sqrt{C_1} \sqrt{C_4} \sqrt{R_3}} \\
\text{bandwidth: } & \frac{C_1 + 2 C_4 R_3 g_m}{2 C_1 C_4 R_3} \\
\text{K-LP: } & R_3 \\
\text{K-HP: } & 0 \\
\text{K-BP: } & 0 \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

$$7.2 \quad \text{LP-2} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_4 g_m}{2 C_1 C_4 R_3 R_4 s^2 + 2 R_3 g_m + R_4 g_m + s (2 C_1 R_3 + C_1 R_4 + 2 C_4 R_3 R_4 g_m)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_3} \sqrt{R_4} \sqrt{g_m} \sqrt{2 R_3 + R_4}}{2 C_1 R_3 + C_1 R_4 + 2 C_4 R_3 R_4 g_m} \\
\text{wo: } & \frac{\sqrt{2} \sqrt{2 R_3 g_m + R_4 g_m}}{2 \sqrt{C_1} \sqrt{C_4} \sqrt{R_3} \sqrt{R_4}} \\
\text{bandwidth: } & \frac{\sqrt{2 R_3 g_m + R_4 g_m} (2 C_1 R_3 + C_1 R_4 + 2 C_4 R_3 R_4 g_m)}{2 C_1 C_4 R_3 R_4 \sqrt{g_m} \sqrt{2 R_3 + R_4}} \\
\text{K-LP: } & \frac{R_3 R_4}{2 R_3 + R_4} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & 0 \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

$$7.3 \quad \text{LP-3} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_4 g_m}{C_1 C_3 R_4 s^2 + 2 g_m + s (2 C_1 + C_3 R_4 g_m)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{2} \sqrt{C_1} \sqrt{C_3} \sqrt{R_4} \sqrt{g_m}}{2 C_1 + C_3 R_4 g_m} \\
\text{wo: } & \frac{\sqrt{2} \sqrt{g_m}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_4}}
\end{aligned}$$

bandwidth: $\frac{2C_1+C_3R_4g_m}{C_1C_3R_4}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.4 LP-4 $Z(s) = \left(\frac{1}{C_1s}, \infty, \frac{1}{C_3s}, \frac{R_4}{C_4R_4s+1}, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{R_4}\sqrt{g_m}\sqrt{C_3+2C_4}}{2C_1+C_3R_4g_m+2C_4R_4g_m}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1C_3R_4+2C_1C_4R_4}}$
bandwidth: $\frac{2C_1+C_3R_4g_m+2C_4R_4g_m}{\sqrt{C_1}\sqrt{R_4}\sqrt{C_3+2C_4}\sqrt{C_1C_3R_4+2C_1C_4R_4}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.5 LP-5 $Z(s) = \left(\frac{1}{C_1s}, \infty, \frac{R_3}{C_3R_3s+1}, R_4, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{R_4}\sqrt{g_m}\sqrt{2R_3+R_4}}{2C_1R_3+C_1R_4+C_3R_3R_4g_m}$
wo: $\frac{\sqrt{2R_3g_m+R_4g_m}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{R_4}}$
bandwidth: $\frac{\sqrt{2R_3g_m+R_4g_m}(2C_1R_3+C_1R_4+C_3R_3R_4g_m)}{C_1C_3R_3R_4\sqrt{g_m}\sqrt{2R_3+R_4}}$
K-LP: $\frac{R_3R_4}{2R_3+R_4}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.6 LP-6 $Z(s) = \left(\frac{1}{C_1s}, \infty, \frac{R_3}{C_3R_3s+1}, \frac{1}{C_4s}, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_3}\sqrt{g_m}\sqrt{C_3+2C_4}}{C_1+C_3R_3g_m+2C_4R_3g_m}$
wo: $\frac{\sqrt{g_m}}{\sqrt{C_1C_3R_3+2C_1C_4R_3}}$
bandwidth: $\frac{C_1+C_3R_3g_m+2C_4R_3g_m}{\sqrt{C_1}\sqrt{R_3}\sqrt{C_3+2C_4}\sqrt{C_1C_3R_3+2C_1C_4R_3}}$
K-LP: R_3
K-HP: 0
K-BP: 0
Qz: None
Wz: None

$$H(s) = \frac{R_4g_m}{2g_m + s^2 (C_1C_3R_4 + 2C_1C_4R_4) + s (2C_1 + C_3R_4g_m + 2C_4R_4g_m)}$$

$$H(s) = \frac{R_3R_4g_m}{C_1C_3R_3R_4s^2 + 2R_3g_m + R_4g_m + s (2C_1R_3 + C_1R_4 + C_3R_3R_4g_m)}$$

$$H(s) = \frac{R_3g_m}{g_m + s^2 (C_1C_3R_3 + 2C_1C_4R_3) + s (C_1 + C_3R_3g_m + 2C_4R_3g_m)}$$

7.7 LP-7 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s^2 (C_1 C_3 R_3 R_4 + 2C_1 C_4 R_3 R_4) + s (2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_3} \sqrt{R_4} \sqrt{g_m} \sqrt{C_3 + 2C_4} \sqrt{2R_3 + R_4}}{2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m}$
wo: $\frac{\sqrt{2R_3 g_m + R_4 g_m}}{\sqrt{C_1 C_3 R_3 R_4 + 2C_1 C_4 R_3 R_4}}$
bandwidth: $\frac{\sqrt{2R_3 g_m + R_4 g_m} (2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m)}{\sqrt{C_1} \sqrt{R_3} \sqrt{R_4} \sqrt{g_m} \sqrt{C_3 + 2C_4} \sqrt{2R_3 + R_4} \sqrt{C_1 C_3 R_3 R_4 + 2C_1 C_4 R_3 R_4}}$
K-LP: $\frac{R_3 R_4}{2R_3 + R_4}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.8 LP-8 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_3 g_m}{2C_1 C_4 R_1 R_3 s^2 + R_1 g_m + s (C_1 R_1 + 2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_1} \sqrt{R_3} \sqrt{R_1 g_m + 1}}{C_1 R_1 + 2C_4 R_1 R_3 g_m + 2C_4 R_3}$
wo: $\frac{\sqrt{2} \sqrt{R_1 g_m + 1}}{2\sqrt{C_1} \sqrt{C_4} \sqrt{R_1} \sqrt{R_3}}$
bandwidth: $\frac{C_1 R_1 + 2C_4 R_1 R_3 g_m + 2C_4 R_3}{2C_1 C_4 R_1 R_3}$
K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.9 LP-9 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_4 g_m}{2C_1 C_4 R_1 R_3 R_4 s^2 + 2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4}$
wo: $\frac{\sqrt{2} \sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{2\sqrt{C_1} \sqrt{C_4} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4}}$
bandwidth: $\frac{2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4}{2C_1 C_4 R_1 R_3 R_4}$
K-LP: $\frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.10 LP-10 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m}{C_1 C_3 R_1 R_4 s^2 + 2R_1 g_m + s (2C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_4} \sqrt{R_1 g_m + 1}}{2C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4}$
wo: $\frac{\sqrt{2R_1 g_m + 2}}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_4}}$

bandwidth: $\frac{\sqrt{2}\sqrt{2R_1g_m+2}(2C_1R_1+C_3R_1R_4g_m+C_3R_4)}{2C_1C_3R_1R_4\sqrt{R_1g_m+1}}$
K-LP: $\frac{R_1R_4g_m}{2R_1g_m+2}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.11 LP-11 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \infty, \frac{1}{C_3s}, \frac{R_4}{C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{R_1R_4g_m}{2R_1g_m + s^2 (C_1C_3R_1R_4 + 2C_1C_4R_1R_4) + s (2C_1R_1 + C_3R_1R_4g_m + C_3R_4 + 2C_4R_1R_4g_m + 2C_4R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{R_1}\sqrt{R_4}\sqrt{C_3+2C_4}\sqrt{R_1g_m+1}}{2C_1R_1+C_3R_1R_4g_m+C_3R_4+2C_4R_1R_4g_m+2C_4R_4}$
wo: $\frac{\sqrt{2R_1g_m+2}}{\sqrt{C_1C_3R_1R_4+2C_1C_4R_1R_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_1g_m+2}(2C_1R_1+C_3R_1R_4g_m+C_3R_4+2C_4R_1R_4g_m+2C_4R_4)}{2\sqrt{C_1}\sqrt{R_1}\sqrt{R_4}\sqrt{C_3+2C_4}\sqrt{R_1g_m+1}\sqrt{C_1C_3R_1R_4+2C_1C_4R_1R_4}}$
K-LP: $\frac{R_1R_4g_m}{2R_1g_m+2}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.12 LP-12 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \infty, \frac{R_3}{C_3R_3s+1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{R_1R_3R_4g_m}{C_1C_3R_1R_3R_4s^2 + 2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s (2C_1R_1R_3 + C_1R_1R_4 + C_3R_1R_3R_4g_m + C_3R_3R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}\sqrt{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}{2C_1R_1R_3+C_1R_1R_4+C_3R_1R_3R_4g_m+C_3R_3R_4}$
wo: $\frac{\sqrt{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_3}\sqrt{R_4}}$
bandwidth: $\frac{2C_1R_1R_3+C_1R_1R_4+C_3R_1R_3R_4g_m+C_3R_3R_4}{C_1C_3R_1R_3R_4}$
K-LP: $\frac{R_1R_3R_4g_m}{2R_1R_3g_m+R_1R_4g_m+2R_3+R_4}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.13 LP-13 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \infty, \frac{R_3}{C_3R_3s+1}, \frac{1}{C_4s}, \infty, \infty \right)$

$$H(s) = \frac{R_1R_3g_m}{R_1g_m + s^2 (C_1C_3R_1R_3 + 2C_1C_4R_1R_3) + s (C_1R_1 + C_3R_1R_3g_m + C_3R_3 + 2C_4R_1R_3g_m + 2C_4R_3) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_1}\sqrt{R_3}\sqrt{C_3+2C_4}\sqrt{R_1g_m+1}}{C_1R_1+C_3R_1R_3g_m+C_3R_3+2C_4R_1R_3g_m+2C_4R_3}$
wo: $\frac{\sqrt{R_1g_m+1}}{\sqrt{C_1C_3R_1R_3+2C_1C_4R_1R_3}}$
bandwidth: $\frac{C_1R_1+C_3R_1R_3g_m+C_3R_3+2C_4R_1R_3g_m+2C_4R_3}{\sqrt{C_1}\sqrt{R_1}\sqrt{R_3}\sqrt{C_3+2C_4}\sqrt{C_1C_3R_1R_3+2C_1C_4R_1R_3}}$
K-LP: $\frac{R_1R_3g_m}{R_1g_m+1}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.14 LP-14 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^2 (C_1 C_3 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{C_3 + 2C_4} \sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4}$
 wo: $\frac{\sqrt{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}{\sqrt{C_1 C_3 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4}}$
 bandwidth: $\frac{2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4}{\sqrt{C_1} \sqrt{R_1} \sqrt{R_3} \sqrt{R_4} \sqrt{C_3 + 2C_4} \sqrt{C_1 C_3 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4}}$
 K-LP: $\frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.15 LP-15 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{L_1 g_m}{C_3 + 2C_4 + s^2 (C_1 C_3 L_1 + 2C_1 C_4 L_1) + s (C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1 g_m}}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{g_m}{C_1}$
 K-LP: $\frac{L_1 g_m}{C_3 + 2C_4}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.16 LP-16 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{L_1 R_1 g_m}{C_3 R_1 + 2C_4 R_1 + s^2 (C_1 C_3 L_1 R_1 + 2C_1 C_4 L_1 R_1) + s (C_3 L_1 R_1 g_m + C_3 L_1 + 2C_4 L_1 R_1 g_m + 2C_4 L_1)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1 R_1 g_m + \sqrt{L_1}}}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{\sqrt{L_1 R_1 g_m + \sqrt{L_1}}}{C_1 \sqrt{L_1 R_1}}$
 K-LP: $\frac{L_1 g_m}{C_3 + 2C_4}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{R_1 g_m + s^2 (C_3 C_4 R_1 R_3 R_4 g_m + C_3 C_4 R_3 R_4) + s (C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}{C_3R_3+2C_4R_3+C_4R_4}$
 wo: $\frac{1}{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}$
 bandwidth: $\frac{C_3R_3+2C_4R_3+C_4R_4}{C_3C_4R_3R_4}$
 K-LP: $\frac{R_1R_3g_m}{R_1g_m+1}$
 K-HP: 0
 K-BP: $\frac{C_4R_1R_3R_4g_m}{C_3R_1R_3g_m+C_3R_3+2C_4R_1R_3g_m+C_4R_1R_4g_m+2C_4R_3+C_4R_4}$
 Qz: None
 Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3s}, \frac{R_4}{C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{C_3R_1R_3R_4g_ms + R_1R_4g_m}{2R_1g_m + s^2(2C_3C_4R_1R_3R_4g_m + 2C_3C_4R_3R_4) + s(2C_3R_1R_3g_m + C_3R_1R_4g_m + 2C_3R_3 + C_3R_4 + 2C_4R_1R_4g_m + 2C_4R_4) + 2}$$

Parameters:

Q: $\frac{2\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}{2C_3R_3+C_3R_4+2C_4R_4}$
 wo: $\frac{1}{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}$
 bandwidth: $\frac{2C_3R_3+C_3R_4+2C_4R_4}{2C_3C_4R_3R_4}$
 K-LP: $\frac{R_1R_4g_m}{2R_1g_m+2}$
 K-HP: 0
 K-BP: $\frac{C_3R_1R_3R_4g_m}{2C_3R_1R_3g_m+C_3R_1R_4g_m+2C_3R_3+C_3R_4+2C_4R_1R_4g_m+2C_4R_4}$
 Qz: None
 Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(L_1s, \infty, R_3, R_4 + \frac{1}{C_4s}, \infty, \infty \right)$

$$H(s) = \frac{C_4L_1R_3R_4g_ms^2 + L_1R_3g_ms}{s^2(2C_4L_1R_3g_m + C_4L_1R_4g_m) + s(2C_4R_3 + C_4R_4 + L_1g_m) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_4}\sqrt{L_1g_m}\sqrt{2R_3+R_4}\sqrt{\frac{1}{g_m}}}{2C_4R_3+C_4R_4+L_1g_m}$
 wo: $\frac{1}{\sqrt{2C_4L_1R_3g_m+C_4L_1R_4g_m}}$
 bandwidth: $\frac{(2C_4R_3+C_4R_4+L_1g_m)\sqrt{\frac{1}{2C_4L_1R_3g_m+C_4L_1R_4g_m}}}{\sqrt{C_4}\sqrt{L_1g_m}\sqrt{2R_3+R_4}\sqrt{\frac{1}{g_m}}}$
 K-LP: 0
 K-HP: $\frac{R_3R_4}{2R_3+R_4}$
 K-BP: $\frac{L_1R_3g_m}{2C_4R_3+C_4R_4+L_1g_m}$
 Qz: None
 Wz: None

8.4 X-INVALID-NUMER-4 $Z(s) = \left(L_1s, \infty, \frac{1}{C_3s}, R_4 + \frac{1}{C_4s}, \infty, \infty \right)$

$$H(s) = \frac{C_4L_1R_4g_ms + L_1g_m}{C_3C_4L_1R_4g_ms^2 + C_3 + 2C_4 + s(C_3C_4R_4 + C_3L_1g_m + 2C_4L_1g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{C_4}\sqrt{L_1}\sqrt{R_4}g_m\sqrt{C_3+2C_4}\sqrt{\frac{1}{g_m}}}{C_3C_4R_4+C_3L_1g_m+2C_4L_1g_m}$
 wo: $\frac{\sqrt{C_3+2C_4}\sqrt{\frac{1}{g_m}}}{\sqrt{C_3}\sqrt{C_4}\sqrt{L_1}\sqrt{R_4}}$
 bandwidth: $\frac{C_3C_4R_4+C_3L_1g_m+2C_4L_1g_m}{C_3C_4L_1R_4g_m}$
 K-LP: $\frac{L_1g_m}{C_3+2C_4}$
 K-HP: 0
 K-BP: $\frac{C_4L_1R_4g_m}{C_3C_4R_4+C_3L_1g_m+2C_4L_1g_m}$
 Qz: None

Wz: None

8.5 X-INVALID-NUMER-5 $Z(s) = \left(L_1 s, \infty, R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 R_3 R_4 g_m s^2 + L_1 R_4 g_m s}{s^2 (2C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m) + s (2C_3 R_3 + C_3 R_4 + 2L_1 g_m) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_3}\sqrt{L_1 g_m}\sqrt{2R_3+R_4}\sqrt{\frac{1}{g_m}}}{2C_3 R_3 + C_3 R_4 + 2L_1 g_m} \\ \text{wo: } & \sqrt{2}\sqrt{\frac{1}{2C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m}} \\ \text{bandwidth: } & \frac{(2C_3 R_3 + C_3 R_4 + 2L_1 g_m)\sqrt{\frac{1}{2C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m}}}{\sqrt{C_3}\sqrt{L_1 g_m}\sqrt{2R_3+R_4}\sqrt{\frac{1}{g_m}}} \end{aligned}$$

K-LP: 0

K-HP: $\frac{R_3 R_4}{2R_3+R_4}$

K-BP: $\frac{L_1 R_4 g_m}{2C_3 R_3 + C_3 R_4 + 2L_1 g_m}$

Qz: None

Wz: None

8.6 X-INVALID-NUMER-6 $Z(s) = \left(L_1 s, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 R_3 g_m s + L_1 g_m}{2C_3 C_4 L_1 R_3 g_m s^2 + C_3 + 2C_4 + s (2C_3 C_4 R_3 + C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_3}\sqrt{C_4}\sqrt{L_1}\sqrt{R_3 g_m}\sqrt{C_3+2C_4}\sqrt{\frac{1}{g_m}}}{2C_3 C_4 R_3 + C_3 L_1 g_m + 2C_4 L_1 g_m} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{C_3+2C_4}\sqrt{\frac{1}{g_m}}}{2\sqrt{C_3}\sqrt{C_4}\sqrt{L_1}\sqrt{R_3}} \\ \text{bandwidth: } & \frac{2C_3 C_4 R_3 + C_3 L_1 g_m + 2C_4 L_1 g_m}{2C_3 C_4 L_1 R_3 g_m} \end{aligned}$$

K-LP: $\frac{L_1 g_m}{C_3+2C_4}$

K-HP: 0

K-BP: $\frac{C_3 L_1 R_3 g_m}{2C_3 C_4 R_3 + C_3 L_1 g_m + 2C_4 L_1 g_m}$

Qz: None

Wz: None

8.7 X-INVALID-NUMER-7 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 R_3 R_4 g_m s + R_3 g_m}{g_m + s^2 (2C_1 C_4 R_3 + C_1 C_4 R_4) + s (C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1}\sqrt{C_4}\sqrt{g_m}\sqrt{2R_3+R_4}}{C_1+2C_4 R_3 g_m + C_4 R_4 g_m} \\ \text{wo: } & \frac{\sqrt{g_m}}{\sqrt{2C_1 C_4 R_3 + C_1 C_4 R_4}} \\ \text{bandwidth: } & \frac{C_1+2C_4 R_3 g_m + C_4 R_4 g_m}{\sqrt{C_1}\sqrt{C_4}\sqrt{2R_3+R_4}\sqrt{2C_1 C_4 R_3 + C_1 C_4 R_4}} \end{aligned}$$

K-LP: R_3

K-HP: 0

K-BP: $\frac{C_4 R_3 R_4 g_m}{C_1+2C_4 R_3 g_m + C_4 R_4 g_m}$

Qz: None

Wz: None

8.8 X-INVALID-NUMER-8 $Z(s) = \left(\frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 R_3 R_4 g_m s + R_4 g_m}{2g_m + s^2 (2C_1 C_3 R_3 + C_1 C_3 R_4) + s (2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_3}\sqrt{g_m}\sqrt{2R_3+R_4}}{2C_1+2C_3R_3g_m+C_3R_4g_m} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{g_m}}{\sqrt{2C_1C_3R_3+C_1C_3R_4}} \\ \text{bandwidth: } & \frac{2C_1+2C_3R_3g_m+C_3R_4g_m}{\sqrt{C_1}\sqrt{C_3}\sqrt{2R_3+R_4}\sqrt{2C_1C_3R_3+C_1C_3R_4}} \\ \text{K-LP: } & \frac{R_4}{2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_3R_3R_4g_m}{2C_1+2C_3R_3g_m+C_3R_4g_m} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.9 X-INVALID-NUMER-9 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{R_1 g_m + s^2 (2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4) + s (C_1 R_1 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{2R_3+R_4}\sqrt{R_1g_m+1}}{C_1R_1+2C_4R_1R_3g_m+C_4R_1R_4g_m+2C_4R_3+C_4R_4} \\ \text{wo: } & \frac{\sqrt{R_1g_m+1}}{\sqrt{2C_1C_4R_1R_3+C_1C_4R_1R_4}} \\ \text{bandwidth: } & \frac{C_1R_1+2C_4R_1R_3g_m+C_4R_1R_4g_m+2C_4R_3+C_4R_4}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{2R_3+R_4}\sqrt{2C_1C_4R_1R_3+C_1C_4R_1R_4}} \\ \text{K-LP: } & \frac{R_1R_3g_m}{R_1g_m+1} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_4R_1R_3R_4g_m}{C_1R_1+2C_4R_1R_3g_m+C_4R_1R_4g_m+2C_4R_3+C_4R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.10 X-INVALID-NUMER-10 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 R_1 R_3 R_4 g_m s + R_1 R_4 g_m}{2R_1 g_m + s^2 (2C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4) + s (2C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{2R_3+R_4}\sqrt{R_1g_m+1}}{2C_1R_1+2C_3R_1R_3g_m+C_3R_1R_4g_m+2C_3R_3+C_3R_4} \\ \text{wo: } & \frac{\sqrt{2R_1g_m+2}}{\sqrt{2C_1C_3R_1R_3+C_1C_3R_1R_4}} \\ \text{bandwidth: } & \frac{\sqrt{2}\sqrt{2R_1g_m+2}(2C_1R_1+2C_3R_1R_3g_m+C_3R_1R_4g_m+2C_3R_3+C_3R_4)}{2\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{2R_3+R_4}\sqrt{R_1g_m+1}\sqrt{2C_1C_3R_1R_3+C_1C_3R_1R_4}} \\ \text{K-LP: } & \frac{R_1R_4g_m}{2R_1g_m+2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_3R_1R_3R_4g_m}{2C_1R_1+2C_3R_1R_3g_m+C_3R_1R_4g_m+2C_3R_3+C_3R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.11 X-INVALID-NUMER-11 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 R_3 g_m s + R_3 g_m}{g_m + s^2 (2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3) + s (C_1 R_1 g_m + C_1 + 2C_4 R_3 g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_3g_m}\sqrt{\frac{g_m}{R_1g_m+1}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_3}\sqrt{\frac{g_m}{R_1g_m+1}}}{C_1R_1g_m+C_1+2C_4R_3g_m} \\ \text{wo: } & \sqrt{\frac{g_m}{2C_1C_4R_1R_3g_m+2C_1C_4R_3}} \end{aligned}$$

bandwidth: $\frac{\sqrt{\frac{gm}{2C_1C_4R_1R_3gm+2C_1C_4R_3}}(C_1R_1gm+C_1+2C_4R_3gm)}{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_3gm}\sqrt{\frac{gm}{R_1gm+1}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_3}\sqrt{\frac{gm}{R_1gm+1}}}$

K-LP: R_3
K-HP: 0
K-BP: $\frac{C_1R_1R_3gm}{C_1R_1gm+C_1+2C_4R_3gm}$
Qz: None
Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(R_1 + \frac{1}{C_1s}, \ \infty, \ R_3, \ \frac{R_4}{C_4R_4s+1}, \ \infty, \ \infty\right)$

$$H(s) = \frac{C_1R_1R_3R_4g_ms + R_3R_4g_m}{2R_3g_m + R_4g_m + s^2(2C_1C_4R_1R_3R_4g_m + 2C_1C_4R_3R_4) + s(2C_1R_1R_3g_m + C_1R_1R_4g_m + 2C_1R_3 + C_1R_4 + 2C_4R_3R_4g_m)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{R_4g_m}\sqrt{\frac{gm}{R_1gm+1}}\sqrt{2R_3+R_4}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{gm}{R_1gm+1}}\sqrt{2R_3+R_4}}{2C_1R_1R_3gm+C_1R_1R_4gm+2C_1R_3+C_1R_4+2C_4R_3R_4gm}$

wo: $\sqrt{\frac{2R_3gm+R_4gm}{2C_1C_4R_1R_3R_4gm+2C_1C_4R_3R_4}}$

bandwidth: $\frac{\sqrt{\frac{2R_3gm+R_4gm}{2C_1C_4R_1R_3R_4gm+2C_1C_4R_3R_4}}(2C_1R_1R_3gm+C_1R_1R_4gm+2C_1R_3+C_1R_4+2C_4R_3R_4gm)}{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_3}\sqrt{R_4gm}\sqrt{\frac{gm}{R_1gm+1}}\sqrt{2R_3+R_4}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{gm}{R_1gm+1}}\sqrt{2R_3+R_4}}$

K-LP: $\frac{R_3R_4}{2R_3+R_4}$
K-HP: 0
K-BP: $\frac{C_1R_1R_3R_4gm}{2C_1R_1R_3gm+C_1R_1R_4gm+2C_1R_3+C_1R_4+2C_4R_3R_4gm}$
Qz: None
Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(R_1 + \frac{1}{C_1s}, \ \infty, \ \frac{1}{C_3s}, \ R_4, \ \infty, \ \infty\right)$

$$H(s) = \frac{C_1R_1R_4g_ms + R_4g_m}{2g_m + s^2(C_1C_3R_1R_4g_m + C_1C_3R_4) + s(2C_1R_1g_m + 2C_1 + C_3R_4g_m)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_4gm}\sqrt{\frac{gm}{R_1gm+1}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}\sqrt{R_4}\sqrt{\frac{gm}{R_1gm+1}}}{2C_1R_1gm+2C_1+C_3R_4gm}$

wo: $\sqrt{2}\sqrt{\frac{gm}{C_1C_3R_1R_4gm+C_1C_3R_4}}$

bandwidth: $\frac{\sqrt{2}\sqrt{\frac{gm}{C_1C_3R_1R_4gm+C_1C_3R_4}}(2C_1R_1gm+2C_1+C_3R_4gm)}{\sqrt{2}\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_4gm}\sqrt{\frac{gm}{R_1gm+1}}+\sqrt{2}\sqrt{C_1}\sqrt{C_3}\sqrt{R_4}\sqrt{\frac{gm}{R_1gm+1}}}$

K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_1R_1R_4gm}{2C_1R_1gm+2C_1+C_3R_4gm}$
Qz: None
Wz: None

8.14 X-INVALID-NUMER-14 $Z(s) = \left(R_1 + \frac{1}{C_1s}, \ \infty, \ \frac{1}{C_3s}, \ \frac{R_4}{C_4R_4s+1}, \ \infty, \ \infty\right)$

$$H(s) = \frac{C_1R_1R_4g_ms + R_4g_m}{2g_m + s^2(C_1C_3R_1R_4g_m + C_1C_3R_4 + 2C_1C_4R_1R_4g_m + 2C_1C_4R_4) + s(2C_1R_1g_m + 2C_1 + C_3R_4g_m + 2C_4R_4g_m)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}C_3R_1\sqrt{R_4gm}\sqrt{\frac{gm}{C_3R_1gm+C_3+2C_4R_1gm+2C_4}}+\sqrt{2}\sqrt{C_1}C_3\sqrt{R_4}\sqrt{\frac{gm}{C_3R_1gm+C_3+2C_4R_1gm+2C_4}}+2\sqrt{2}\sqrt{C_1}C_4R_1\sqrt{R_4gm}\sqrt{\frac{gm}{C_3R_1gm+C_3+2C_4R_1gm+2C_4}}+2\sqrt{2}\sqrt{C_1}C_4\sqrt{R_4}\sqrt{\frac{gm}{C_3R_1gm+C_3+2C_4R_1gm+2C_4}}}{2C_1R_1gm+2C_1+C_3R_4gm+2C_4R_4gm}$

wo: $\sqrt{2}\sqrt{\frac{gm}{C_1C_3R_1R_4gm+C_1C_3R_4+2C_1C_4R_1R_4gm+2C_1C_4R_4}}$

bandwidth: $\frac{\sqrt{2}\sqrt{\frac{gm}{C_1C_3R_1R_4gm+C_1C_3R_4+2C_1C_4R_1R_4gm+2C_1C_4R_4}}(2C_1R_1gm+2C_1+C_3R_4gm+2C_4R_4gm)}{\sqrt{2}\sqrt{C_1}C_3R_1\sqrt{R_4gm}\sqrt{\frac{gm}{C_3R_1gm+C_3+2C_4R_1gm+2C_4}}+\sqrt{2}\sqrt{C_1}C_3\sqrt{R_4}\sqrt{\frac{gm}{C_3R_1gm+C_3+2C_4R_1gm+2C_4}}+2\sqrt{2}\sqrt{C_1}C_4R_1\sqrt{R_4gm}\sqrt{\frac{gm}{C_3R_1gm+C_3+2C_4R_1gm+2C_4}}+2\sqrt{2}\sqrt{C_1}C_4\sqrt{R_4}\sqrt{\frac{gm}{C_3R_1gm+C_3+2C_4R_1gm+2C_4}}}$

K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_1R_1R_4gm}{2C_1R_1gm+2C_1+C_3R_4gm+2C_4R_4gm}$

Qz: None
Wz: None

8.15 X-INVALID-NUMER-15 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m}{2 R_3 g_m + R_4 g_m + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4) + s (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{C_1} \sqrt{C_3} R_1 \sqrt{R_3} \sqrt{R_4} g_m \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{2 R_3 + R_4} + \sqrt{C_1} \sqrt{C_3} \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{2 R_3 + R_4}}{2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m}$$

$$\text{wo: } \sqrt{\frac{2 R_3 g_m + R_4 g_m}{C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{2 R_3 g_m + R_4 g_m}{C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4}} (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m)}{\sqrt{C_1} \sqrt{C_3} R_1 \sqrt{R_3} \sqrt{R_4} g_m \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{2 R_3 + R_4} + \sqrt{C_1} \sqrt{C_3} \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{R_1 g_m + 1}} \sqrt{2 R_3 + R_4}}$$

$$\text{K-LP: } \frac{R_3 R_4}{2 R_3 + R_4}$$

$$\text{K-HP: } 0$$

$$\text{K-BP: } \frac{C_1 R_1 R_3 R_4 g_m}{2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m}$$

Qz: None

Wz: None

8.16 X-INVALID-NUMER-16 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 R_3 g_m s + R_3 g_m}{g_m + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2 C_1 C_4 R_1 R_3 g_m + 2 C_1 C_4 R_3) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2 C_4 R_3 g_m)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{C_1} C_3 R_1 \sqrt{R_3} g_m \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} + \sqrt{C_1} C_3 \sqrt{R_3} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} + 2 \sqrt{C_1} C_4 R_1 \sqrt{R_3} g_m \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} + 2 \sqrt{C_1} C_4 \sqrt{R_3} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}}}{C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2 C_4 R_3 g_m}$$

$$\text{wo: } \sqrt{\frac{g_m}{C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2 C_1 C_4 R_1 R_3 g_m + 2 C_1 C_4 R_3}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{g_m}{C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2 C_1 C_4 R_1 R_3 g_m + 2 C_1 C_4 R_3}} (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2 C_4 R_3 g_m)}{\sqrt{C_1} C_3 R_1 \sqrt{R_3} g_m \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} + \sqrt{C_1} C_3 \sqrt{R_3} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} + 2 \sqrt{C_1} C_4 R_1 \sqrt{R_3} g_m \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} + 2 \sqrt{C_1} C_4 \sqrt{R_3} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}}}$$

$$\text{K-LP: } R_3$$

$$\text{K-HP: } 0$$

$$\text{K-BP: } \frac{C_1 R_1 R_3 g_m}{C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2 C_4 R_3 g_m}$$

Qz: None

Wz: None

8.17 X-INVALID-NUMER-17 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m}{2 R_3 g_m + R_4 g_m + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4) + s (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2 C_4 R_3 R_4 g_m)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{C_1} C_3 R_1 \sqrt{R_3} \sqrt{R_4} g_m \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4} + \sqrt{C_1} C_3 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4} + 2 \sqrt{C_1} C_4 R_1 \sqrt{R_3} \sqrt{R_4} g_m \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4} + 2 \sqrt{C_1} C_4 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4}}{2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2 C_4 R_3 R_4 g_m}$$

$$\text{wo: } \sqrt{\frac{2 R_3 g_m + R_4 g_m}{C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{2 R_3 g_m + R_4 g_m}{C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4}} (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2 C_4 R_3 R_4 g_m)}{\sqrt{C_1} C_3 R_1 \sqrt{R_3} \sqrt{R_4} g_m \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4} + \sqrt{C_1} C_3 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4} + 2 \sqrt{C_1} C_4 R_1 \sqrt{R_3} \sqrt{R_4} g_m \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4} + 2 \sqrt{C_1} C_4 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4}} \sqrt{2 R_3 + R_4}}$$

$$\text{K-LP: } \frac{R_3 R_4}{2 R_3 + R_4}$$

$$\text{K-HP: } 0$$

$$\text{K-BP: } \frac{C_1 R_1 R_3 R_4 g_m}{2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2 C_4 R_3 R_4 g_m}$$

Qz: None

Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, \infty, R_3, R_4, \infty, \infty)$

$$H(s) = \frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1, \infty, R_3, \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{R_1 R_3 g_m}{R_1 g_m + s(2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty\right)$

$$H(s) = \frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s(2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1, \infty, R_3, R_4 + \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{R_1 g_m + s(2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4) + 1}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, R_4, \infty, \infty\right)$

$$H(s) = \frac{R_1 R_4 g_m}{2R_1 g_m + s(C_3 R_1 R_4 g_m + C_3 R_4) + 2}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{R_1 g_m}{s(C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty\right)$

$$H(s) = \frac{R_1 R_4 g_m}{2R_1 g_m + s(C_3 R_1 R_4 g_m + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4) + 2}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{C_4 R_1 R_4 g_m s + R_1 g_m}{s^2(C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4) + s(C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{C_4 L_4 R_1 g_m s^2 + R_1 g_m}{s^3(C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s(C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{L_4 R_1 g_m s}{2R_1 g_m + s^2 (C_3 L_4 R_1 g_m + C_3 L_4 + 2C_4 L_4 R_1 g_m + 2C_4 L_4) + 2}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 g_m s^2 + C_4 R_1 R_4 g_m s + R_1 g_m}{s^3 (C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_4 g_m s^2 + L_4 R_1 g_m s + R_1 R_4 g_m}{2R_1 g_m + s^3 (C_3 C_4 L_4 R_1 R_4 g_m + C_3 C_4 L_4 R_4) + s^2 (C_3 L_4 R_1 g_m + C_3 L_4 + 2C_4 L_4 R_1 g_m + 2C_4 L_4) + s (C_3 R_1 R_4 g_m + C_3 R_4) + 2}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(R_1, \infty, \frac{1}{C_3 s}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_4 g_m s^2 + R_1 R_4 g_m}{2R_1 g_m + s^3 (C_3 C_4 L_4 R_1 R_4 g_m + C_3 C_4 L_4 R_4) + s^2 (2C_4 L_4 R_1 g_m + 2C_4 L_4) + s (C_3 R_1 R_4 g_m + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4) + 2}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4, \infty, \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s (C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4)}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{R_1 R_3 g_m}{R_1 g_m + s (C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s (C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_3 g_m s^2 + R_1 R_3 g_m}{R_1 g_m + s^3 (C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_4 R_3) + s^2 (C_4 L_4 R_1 g_m + C_4 L_4) + s (C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_3 g_m s^2 + C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{R_1 g_m + s^3 (C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_4 R_3) + s^2 (C_3 C_4 R_1 R_3 R_4 g_m + C_3 C_4 R_3 R_4 + C_4 L_4 R_1 g_m + C_4 L_4) + s (C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4) + 1}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_3 R_4 g_m s^2 + L_4 R_1 R_3 g_m s + R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^3 (C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_3 C_4 L_4 R_3 R_4) + s^2 (C_3 L_4 R_1 R_3 g_m + C_3 L_4 R_3 + 2C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_3 + C_4 L_4 R_4) + s (C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + L_4 R_1 g_m + L_4)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(R_1, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_3 R_4 g_m s^2 + R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^3 (C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_3 C_4 L_4 R_3 R_4) + s^2 (2C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_3 + C_4 L_4 R_4) + s (C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$$

$$H(s) = \frac{C_3 R_1 R_3 R_4 g_m s + R_1 R_4 g_m}{2R_1 g_m + s (2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + 2}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 R_1 R_3 g_m s + R_1 g_m}{s^2 (2C_3 C_4 R_1 R_3 g_m + 2C_3 C_4 R_3) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 R_1 R_3 R_4 g_m s^2 + R_1 g_m + s (C_3 R_1 R_3 g_m + C_4 R_1 R_4 g_m)}{s^2 (2C_3 C_4 R_1 R_3 g_m + C_3 C_4 R_1 R_4 g_m + 2C_3 C_4 R_3 + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_1 R_3 g_m s^3 + C_3 R_1 R_3 g_m s + C_4 L_4 R_1 g_m s^2 + R_1 g_m}{s^3 (C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (2C_3 C_4 R_1 R_3 g_m + 2C_3 C_4 R_3) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_4 R_1 R_3 g_m s^2 + L_4 R_1 g_m s}{2R_1 g_m + s^3 (2C_3 C_4 L_4 R_1 R_3 g_m + 2C_3 C_4 L_4 R_3) + s^2 (C_3 L_4 R_1 g_m + C_3 L_4 + 2C_4 L_4 R_1 g_m + 2C_4 L_4) + s (2C_3 R_1 R_3 g_m + 2C_3 R_3) + 2}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_1 R_3 g_m s^3 + R_1 g_m + s^2 (C_3 C_4 R_1 R_3 R_4 g_m + C_4 L_4 R_1 g_m) + s (C_3 R_1 R_3 g_m + C_4 R_1 R_4 g_m)}{s^3 (C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (2C_3 C_4 R_1 R_3 g_m + C_3 C_4 R_1 R_4 g_m + 2C_3 C_4 R_3 + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_4 R_1 R_3 R_4 g_m s^2 + L_4 R_1 R_4 g_m s}{2R_1 R_4 g_m + 2R_4 + s^3 (2C_3 C_4 L_4 R_1 R_3 R_4 g_m + 2C_3 C_4 L_4 R_3 R_4) + s^2 (2C_3 L_4 R_1 R_3 g_m + C_3 L_4 R_1 R_4 g_m + 2C_3 L_4 R_3 + C_3 L_4 R_4 + 2C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_4) + s (2C_3 R_1 R_3 R_4 g_m + 2C_3 R_3 R_4 + 2L_4 R_1 g_m + 2L_4)}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_1 R_3 R_4 g_m s^3 + R_1 R_4 g_m + s^2 (C_3 L_4 R_1 R_3 g_m + C_4 L_4 R_1 R_4 g_m) + s (C_3 R_1 R_3 R_4 g_m + L_4 R_1 g_m)}{2R_1 g_m + s^3 (2C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_4 R_1 R_4 g_m + 2C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4) + s^2 (C_3 L_4 R_1 g_m + C_3 L_4 + 2C_4 L_4 R_1 g_m + 2C_4 L_4) + s (2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + 2}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(R_1, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_1 R_3 R_4 g_m s^3 + C_3 R_1 R_3 R_4 g_m s + C_4 L_4 R_1 R_4 g_m s^2 + R_1 R_4 g_m}{2 R_1 g_m + s^3 (2 C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_4 R_1 R_4 g_m + 2 C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4) + s^2 (2 C_3 C_4 R_1 R_3 R_4 g_m + 2 C_3 C_4 R_3 R_4 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4) + s (2 C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2 C_3 R_3 + C_3 R_4 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_4) + 2}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(R_1, \infty, L_3 s + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 g_m s^2 + R_1 g_m}{s^3 (2 C_3 C_4 L_3 R_1 g_m + 2 C_3 C_4 L_3) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.31 \quad X-INVALID-ORDER-31} \quad Z(s) = \left(R_1, \infty, L_3 s + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_4 g_m s^2 + R_1 R_4 g_m}{2 R_1 g_m + s^3 (2 C_3 C_4 L_3 R_1 R_4 g_m + 2 C_3 C_4 L_3 R_4) + s^2 (2 C_3 L_3 R_1 g_m + 2 C_3 L_3) + s (C_3 R_1 R_4 g_m + C_3 R_4 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_4) + 2}$$

$$\mathbf{9.32 \quad X-INVALID-ORDER-32} \quad Z(s) = \left(R_1, \infty, L_3 s + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 R_1 R_4 g_m s^3 + C_3 L_3 R_1 g_m s^2 + C_4 R_1 R_4 g_m s + R_1 g_m}{s^3 (2 C_3 C_4 L_3 R_1 g_m + 2 C_3 C_4 L_3) + s^2 (C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.33 \quad X-INVALID-ORDER-33} \quad Z(s) = \left(R_1, \infty, L_3 s + \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 g_m s^4 + R_1 g_m + s^2 (C_3 L_3 R_1 g_m + C_4 L_4 R_1 g_m)}{s^3 (2 C_3 C_4 L_3 R_1 g_m + 2 C_3 C_4 L_3 + C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.34 \quad X-INVALID-ORDER-34} \quad Z(s) = \left(R_1, \infty, L_3 s + \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_1 g_m s^3 + L_4 R_1 g_m s}{2 R_1 g_m + s^4 (2 C_3 C_4 L_3 L_4 R_1 g_m + 2 C_3 C_4 L_3 L_4) + s^2 (2 C_3 L_3 R_1 g_m + 2 C_3 L_3 + C_3 L_4 R_1 g_m + C_3 L_4 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4) + 2}$$

$$\mathbf{9.35 \quad X-INVALID-ORDER-35} \quad Z(s) = \left(R_1, \infty, L_3 s + \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 g_m s^4 + C_3 C_4 L_3 R_1 R_4 g_m s^3 + C_4 R_1 R_4 g_m s + R_1 g_m + s^2 (C_3 L_3 R_1 g_m + C_4 L_4 R_1 g_m)}{s^3 (2 C_3 C_4 L_3 R_1 g_m + 2 C_3 C_4 L_3 + C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.36 \quad X-INVALID-ORDER-36} \quad Z(s) = \left(R_1, \infty, L_3 s + \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_1 R_4 g_m s^3 + L_4 R_1 R_4 g_m s}{2 R_1 R_4 g_m + 2 R_4 + s^4 (2 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2 C_3 C_4 L_3 L_4 R_4) + s^3 (2 C_3 L_3 L_4 R_1 g_m + 2 C_3 L_3 L_4) + s^2 (2 C_3 L_3 R_1 R_4 g_m + 2 C_3 L_3 R_4 + C_3 L_4 R_1 R_4 g_m + C_3 L_4 R_4 + 2 C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_4) + s (2 L_4 R_1 g_m + 2 L_4)}$$

$$\mathbf{9.37 \quad X-INVALID-ORDER-37} \quad Z(s) = \left(R_1, \infty, L_3 s + \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_4 g_m s^4 + C_3 L_3 L_4 R_1 g_m s^3 + L_4 R_1 g_m s + R_1 R_4 g_m + s^2 (C_3 L_3 R_1 R_4 g_m + C_4 L_4 R_1 R_4 g_m)}{2 R_1 g_m + s^4 (2 C_3 C_4 L_3 L_4 R_1 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (C_3 C_4 L_4 R_1 R_4 g_m + C_3 C_4 L_4 R_4) + s^2 (2 C_3 L_3 R_1 g_m + 2 C_3 L_3 + C_3 L_4 R_1 g_m + C_3 L_4 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4) + s (C_3 R_1 R_4 g_m + C_3 R_4) + 2}$$

$$\mathbf{9.38 \quad X-INVALID-ORDER-38} \quad Z(s) = \left(R_1, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_4 g_m s^4 + R_1 R_4 g_m + s^2 (C_3 L_3 R_1 R_4 g_m + C_4 L_4 R_1 R_4 g_m)}{2 R_1 g_m + s^4 (2 C_3 C_4 L_3 L_4 R_1 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (2 C_3 C_4 L_3 R_1 R_4 g_m + 2 C_3 C_4 L_3 R_4 + C_3 C_4 L_4 R_1 R_4 g_m + C_3 C_4 L_4 R_4) + s^2 (2 C_3 L_3 R_1 g_m + 2 C_3 L_3 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4) + s (C_3 R_1 R_4 g_m + C_3 R_4 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_4) + 2}$$

$$\mathbf{9.39 \quad X-INVALID-ORDER-39} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 R_1 g_m s}{R_1 g_m + s^2 (C_3 L_3 R_1 g_m + C_3 L_3 + 2 C_4 L_3 R_1 g_m + 2 C_4 L_3) + 1}$$

$$\mathbf{9.40 \quad X-INVALID-ORDER-40} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 R_1 R_4 g_m s^2 + L_3 R_1 g_m s}{R_1 g_m + s^3 (C_3 C_4 L_3 R_1 R_4 g_m + C_3 C_4 L_3 R_4) + s^2 (C_3 L_3 R_1 g_m + C_3 L_3 + 2 C_4 L_3 R_1 g_m + 2 C_4 L_3) + s (C_4 R_1 R_4 g_m + C_4 R_4) + 1}$$

$$\mathbf{9.41 \quad X-INVALID-ORDER-41} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 g_m s^3 + L_3 R_1 g_m s}{R_1 g_m + s^4 (C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4) + s^2 (C_3 L_3 R_1 g_m + C_3 L_3 + 2 C_4 L_3 R_1 g_m + 2 C_4 L_3 + C_4 L_4 R_1 g_m + C_4 L_4) + 1}$$

$$\mathbf{9.42 \quad X-INVALID-ORDER-42} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 L_4 R_1 g_m s}{2 L_3 R_1 g_m + 2 L_3 + L_4 R_1 g_m + L_4 + s^2 (C_3 L_3 L_4 R_1 g_m + C_3 L_3 L_4 + 2 C_4 L_3 L_4 R_1 g_m + 2 C_4 L_3 L_4)}$$

$$\mathbf{9.43 \quad X-INVALID-ORDER-43} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 g_m s^3 + C_4 L_3 R_1 R_4 g_m s^2 + L_3 R_1 g_m s}{R_1 g_m + s^4 (C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4) + s^3 (C_3 C_4 L_3 R_1 R_4 g_m + C_3 C_4 L_3 R_4) + s^2 (C_3 L_3 R_1 g_m + C_3 L_3 + 2 C_4 L_3 R_1 g_m + 2 C_4 L_3 + C_4 L_4 R_1 g_m + C_4 L_4) + s (C_4 R_1 R_4 g_m + C_4 R_4) + 1}$$

$$\mathbf{9.44 \quad X-INVALID-ORDER-44} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 R_4 g_m s^3 + L_3 L_4 R_1 g_m s^2 + L_3 R_1 R_4 g_m s}{R_1 R_4 g_m + R_4 + s^4 (C_3 C_4 L_3 L_4 R_1 R_4 g_m + C_3 C_4 L_3 L_4 R_4) + s^3 (C_3 L_3 L_4 R_1 g_m + C_3 L_3 L_4 + 2 C_4 L_3 L_4 R_1 g_m + 2 C_4 L_3 L_4) + s^2 (C_3 L_3 R_1 R_4 g_m + C_3 L_3 R_4 + C_4 L_4 R_1 R_4 g_m + C_4 L_4 R_4) + s (2 L_3 R_1 g_m + 2 L_3 + L_4 R_1 g_m + L_4)}$$

$$\mathbf{9.45 \quad X-INVALID-ORDER-45} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 R_4 g_m s^3 + L_3 R_1 R_4 g_m s}{R_1 R_4 g_m + R_4 + s^4 (C_3 C_4 L_3 L_4 R_1 R_4 g_m + C_3 C_4 L_3 L_4 R_4) + s^3 (2 C_4 L_3 L_4 R_1 g_m + 2 C_4 L_3 L_4) + s^2 (C_3 L_3 R_1 R_4 g_m + C_3 L_3 R_4 + 2 C_4 L_3 R_1 R_4 g_m + 2 C_4 L_3 R_4 + C_4 L_4 R_1 R_4 g_m + C_4 L_4 R_4) + s (2 L_3 R_1 g_m + 2 L_3)}$$

$$\mathbf{9.46 \quad X-INVALID-ORDER-46} \quad Z(s) = \left(R_1, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 g_m s^2 + C_3 R_1 R_3 g_m s + R_1 g_m}{s^3 (2 C_3 C_4 L_3 R_1 g_m + 2 C_3 C_4 L_3) + s^2 (2 C_3 C_4 R_1 R_3 g_m + 2 C_3 C_4 R_3) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.47 \quad X-INVALID-ORDER-47} \quad Z(s) = \left(R_1, \infty, L_3s + R_3 + \frac{1}{C_3s}, \frac{R_4}{C_4R_4s+1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3L_3R_1R_4g_ms^2 + C_3R_1R_3R_4g_ms + R_1R_4g_m}{2R_1g_m + s^3(2C_3C_4L_3R_1R_4g_m + 2C_3C_4L_3R_4) + s^2(2C_3C_4R_1R_3R_4g_m + 2C_3C_4R_3R_4 + 2C_3L_3R_1g_m + 2C_3L_3) + s(2C_3R_1R_3g_m + C_3R_1R_4g_m + 2C_3R_3 + C_3R_4 + 2C_4R_1R_4g_m + 2C_4R_4) + 2}$$

$$\mathbf{9.48 \quad X-INVALID-ORDER-48} \quad Z(s) = \left(R_1, \infty, L_3s + R_3 + \frac{1}{C_3s}, R_4 + \frac{1}{C_4s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3C_4L_3R_1R_4g_ms^3 + R_1g_m + s^2(C_3C_4R_1R_3R_4g_m + C_3L_3R_1g_m) + s(C_3R_1R_3g_m + C_4R_1R_4g_m)}{s^3(2C_3C_4L_3R_1g_m + 2C_3C_4L_3) + s^2(2C_3C_4R_1R_3g_m + C_3C_4R_1R_4g_m + 2C_3C_4R_3 + C_3C_4R_4) + s(C_3R_1g_m + C_3 + 2C_4R_1g_m + 2C_4)}$$

$$\mathbf{9.49 \quad X-INVALID-ORDER-49} \quad Z(s) = \left(R_1, \infty, L_3s + R_3 + \frac{1}{C_3s}, L_4s + \frac{1}{C_4s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3C_4L_3L_4R_1g_ms^4 + C_3C_4L_4R_1R_3g_ms^3 + C_3R_1R_3g_ms + R_1g_m + s^2(C_3L_3R_1g_m + C_4L_4R_1g_m)}{s^3(2C_3C_4L_3R_1g_m + 2C_3C_4L_3 + C_3C_4L_4R_1g_m + C_3C_4L_4) + s^2(2C_3C_4R_1R_3g_m + 2C_3C_4R_3) + s(C_3R_1g_m + C_3 + 2C_4R_1g_m + 2C_4)}$$

$$\mathbf{9.50 \quad X-INVALID-ORDER-50} \quad Z(s) = \left(R_1, \infty, L_3s + R_3 + \frac{1}{C_3s}, \frac{L_4s}{C_4L_4s^2+1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3L_3L_4R_1g_ms^3 + C_3L_4R_1R_3g_ms^2 + L_4R_1g_ms}{2R_1g_m + s^4(2C_3C_4L_3L_4R_1g_m + 2C_3C_4L_3L_4) + s^3(2C_3C_4L_4R_1R_3g_m + 2C_3C_4L_4R_3) + s^2(2C_3L_3R_1g_m + 2C_3L_3 + C_3L_4R_1g_m + C_3L_4 + 2C_4L_4R_1g_m + 2C_4L_4) + s(2C_3R_1R_3g_m + 2C_3R_3) + 2}$$

$$\mathbf{9.51 \quad X-INVALID-ORDER-51} \quad Z(s) = \left(R_1, \infty, L_3s + R_3 + \frac{1}{C_3s}, L_4s + R_4 + \frac{1}{C_4s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3C_4L_3L_4R_1g_ms^4 + R_1g_m + s^3(C_3C_4L_3R_1R_4g_m + C_3C_4L_4R_1R_3g_m) + s^2(C_3C_4R_1R_3R_4g_m + C_3L_3R_1g_m + C_4L_4R_1g_m) + s(C_3R_1R_3g_m + C_4R_1R_4g_m)}{s^3(2C_3C_4L_3R_1g_m + 2C_3C_4L_3 + C_3C_4L_4R_1g_m + C_3C_4L_4) + s^2(2C_3C_4R_1R_3g_m + C_3C_4R_1R_4g_m + 2C_3C_4R_3 + C_3C_4R_4) + s(C_3R_1g_m + C_3 + 2C_4R_1g_m + 2C_4)}$$

$$\mathbf{9.52 \quad X-INVALID-ORDER-52} \quad Z(s) = \left(R_1, \infty, L_3s + R_3 + \frac{1}{C_3s}, \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4}, \infty, \infty \right)$$

$$H(s) = \frac{C_3L_3L_4R_1R_4g_ms^3 + C_3L_4R_1R_3R_4g_ms^2 + L_4R_1R_4g_ms}{2R_1R_4g_m + 2R_4 + s^4(2C_3C_4L_3L_4R_1R_4g_m + 2C_3C_4L_3L_4R_4) + s^3(2C_3C_4L_4R_1R_3R_4g_m + 2C_3C_4L_4R_3R_4 + 2C_3L_3L_4R_1g_m + 2C_3L_3L_4) + s^2(2C_3L_3R_1R_4g_m + 2C_3L_3R_4 + 2C_3L_4R_1R_3g_m + C_3L_4R_1R_4g_m + 2C_3L_4R_3 + C_3L_4R_4 + 2C_4L_4R_1R_4g_m + 2C_4L_4R_4) + s(2C_3R_1R_3g_m + C_3R_1R_4g_m + 2C_3R_3 + C_3R_4) + 2}$$

$$\mathbf{9.53 \quad X-INVALID-ORDER-53} \quad Z(s) = \left(R_1, \infty, L_3s + R_3 + \frac{1}{C_3s}, \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3C_4L_3L_4R_1R_4g_ms^4 + R_1R_4g_m + s^3(C_3C_4L_4R_1R_3R_4g_m + C_3L_3L_4R_1g_m) + s^2(C_3L_3R_1R_4g_m + C_3L_4R_1R_3g_m + C_4L_4R_1R_4g_m) + s(C_3R_1R_3R_4g_m + L_4R_1g_m)}{2R_1g_m + s^4(2C_3C_4L_3L_4R_1g_m + 2C_3C_4L_3L_4) + s^3(2C_3C_4L_4R_1R_3g_m + C_3C_4L_4R_1R_4g_m + 2C_3C_4L_4R_3 + C_3C_4L_4R_4) + s^2(2C_3L_3R_1g_m + 2C_3L_3 + C_3L_4R_1g_m + C_3L_4 + 2C_4L_4R_1g_m + 2C_4L_4) + s(2C_3R_1R_3g_m + C_3R_1R_4g_m + 2C_3R_3 + C_3R_4) + 2}$$

$$\mathbf{9.54 \quad X-INVALID-ORDER-54} \quad Z(s) = \left(R_1, \infty, L_3s + R_3 + \frac{1}{C_3s}, \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3C_4L_3L_4R_1R_4g_ms^4 + C_3C_4L_4R_1R_3R_4g_ms^3 + C_3R_1R_3R_4g_ms + R_1R_4g_m + s^2(C_3L_3R_1R_4g_m + C_4L_4R_1R_4g_m)}{2R_1g_m + s^4(2C_3C_4L_3L_4R_1g_m + 2C_3C_4L_3L_4) + s^3(2C_3C_4L_3R_1R_4g_m + 2C_3C_4L_3R_4 + 2C_3C_4L_4R_1R_3g_m + C_3C_4L_4R_1R_4g_m + 2C_3C_4L_4R_3 + C_3C_4L_4R_4) + s^2(2C_3C_4R_1R_3R_4g_m + 2C_3C_4R_3R_4 + 2C_3L_3R_1g_m + 2C_3L_3 + 2C_4L_4R_1g_m + 2C_4L_4) + s(2C_3R_1R_3g_m + C_3R_1R_4g_m + 2C_3R_3 + C_3R_4) + 2}$$

$$\mathbf{9.55 \quad X-INVALID-ORDER-55} \quad Z(s) = \left(R_1, \infty, \frac{L_3R_3s}{C_3L_3R_3s^2+L_3s+R_3}, R_4 + \frac{1}{C_4s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4L_3R_1R_3R_4g_ms^2 + L_3R_1R_3g_ms}{R_1R_3g_m + R_3 + s^3(C_3C_4L_3R_1R_3R_4g_m + C_3C_4L_3R_3R_4) + s^2(C_3L_3R_1R_3g_m + C_3L_3R_3 + 2C_4L_3R_1R_3g_m + C_4L_3R_1R_4g_m + 2C_4L_3R_3 + C_4L_3R_4) + s(C_4R_1R_3R_4g_m + C_4R_3R_4 + L_3R_1g_m + L_3)}$$

$$\mathbf{9.56 \quad X-INVALID-ORDER-56} \quad Z(s) = \left(R_1, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 R_3 g_m s^3 + L_3 R_1 R_3 g_m s}{R_1 R_3 g_m + R_3 + s^4 (C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_3 L_4 R_3) + s^3 (C_4 L_3 L_4 R_1 g_m + C_4 L_3 L_4) + s^2 (C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + 2C_4 L_3 R_3 + C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_3) + s (L_3 R_1 g_m + L_3)}$$

$$\mathbf{9.57 \quad X-INVALID-ORDER-57} \quad Z(s) = \left(R_1, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 R_3 g_m s^3 + C_4 L_3 R_1 R_3 R_4 g_m s^2 + L_3 R_1 R_3 g_m s}{R_1 R_3 g_m + R_3 + s^4 (C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_3 L_4 R_3) + s^3 (C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_3 C_4 L_3 R_3 R_4 + C_4 L_3 L_4 R_1 g_m + C_4 L_3 L_4) + s^2 (C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + C_4 L_3 R_1 R_4 g_m + 2C_4 L_3 R_3 + C_4 L_3 R_4 + C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_3) + s (C_4 R_1 R_3 R_4 g_m + L_3)}$$

$$\mathbf{9.58 \quad X-INVALID-ORDER-58} \quad Z(s) = \left(R_1, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 R_3 R_4 g_m s^3 + L_3 L_4 R_1 R_3 g_m s^2 + L_3 R_1 R_3 R_4 g_m s}{R_1 R_3 R_4 g_m + R_3 R_4 + s^4 (C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + C_3 C_4 L_3 L_4 R_3 R_4) + s^3 (C_3 L_3 L_4 R_1 R_3 g_m + C_3 L_3 L_4 R_3 + 2C_4 L_3 L_4 R_1 R_3 g_m + C_4 L_3 L_4 R_1 R_4 g_m + 2C_4 L_3 L_4 R_3 + C_4 L_3 L_4 R_4) + s^2 (C_3 L_3 R_1 R_3 R_4 g_m + C_3 L_3 R_3 R_4 + C_4 L_4 R_1 R_3 R_4 g_m + C_4 L_4 R_3 R_4 + L_3 L_4 R_1 g_m + L_3)}$$

$$\mathbf{9.59 \quad X-INVALID-ORDER-59} \quad Z(s) = \left(R_1, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 R_3 R_4 g_m s^3 + L_3 R_1 R_3 R_4 g_m s}{R_1 R_3 R_4 g_m + R_3 R_4 + s^4 (C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + C_3 C_4 L_3 L_4 R_3 R_4) + s^3 (2C_4 L_3 L_4 R_1 R_3 g_m + C_4 L_3 L_4 R_1 R_4 g_m + 2C_4 L_3 L_4 R_3 + C_4 L_3 L_4 R_4) + s^2 (C_3 L_3 R_1 R_3 R_4 g_m + C_3 L_3 R_3 R_4 + 2C_4 L_3 R_1 R_3 R_4 g_m + 2C_4 L_3 R_3 R_4 + C_4 L_4 R_1 R_3 R_4 g_m + C_4 L_4 R_3 R_4) + s (2L_3 R_1 R_3 R_4 g_m + L_3)}$$

$$\mathbf{9.60 \quad X-INVALID-ORDER-60} \quad Z(s) = \left(R_1, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_3 g_m s^2 + L_3 R_1 g_m s + R_1 R_3 g_m}{R_1 g_m + s^3 (2C_3 C_4 L_3 R_1 R_3 g_m + 2C_3 C_4 L_3 R_3) + s^2 (C_3 L_3 R_1 g_m + C_3 L_3 + 2C_4 L_3 R_1 g_m + 2C_4 L_3) + s (2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

$$\mathbf{9.61 \quad X-INVALID-ORDER-61} \quad Z(s) = \left(R_1, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_3 R_4 g_m s^2 + L_3 R_1 R_4 g_m s + R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^3 (2C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_3 C_4 L_3 R_3 R_4) + s^2 (2C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_1 R_4 g_m + 2C_3 L_3 R_3 + C_3 L_3 R_4 + 2C_4 L_3 R_1 R_4 g_m + 2C_4 L_3 R_4) + s (2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4 + 2L_3 R_1 g_m + 2L_3)}$$

$$\mathbf{9.62 \quad X-INVALID-ORDER-62} \quad Z(s) = \left(R_1, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 R_1 R_3 R_4 g_m s^3 + R_1 R_3 g_m + s^2 (C_3 L_3 R_1 R_3 g_m + C_4 L_3 R_1 R_4 g_m) + s (C_4 R_1 R_3 R_4 g_m + L_3 R_1 g_m)}{R_1 g_m + s^3 (2C_3 C_4 L_3 R_1 R_3 g_m + C_3 C_4 L_3 R_1 R_4 g_m + 2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4) + s^2 (C_3 L_3 R_1 g_m + C_3 L_3 + 2C_4 L_3 R_1 g_m + 2C_4 L_3) + s (2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4) + 1}$$

$$\mathbf{9.63 \quad X-INVALID-ORDER-63} \quad Z(s) = \left(R_1, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_3 g_m s^4 + C_4 L_3 L_4 R_1 g_m s^3 + L_3 R_1 g_m s + R_1 R_3 g_m + s^2 (C_3 L_3 R_1 R_3 g_m + C_4 L_4 R_1 R_3 g_m)}{R_1 g_m + s^4 (C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4) + s^3 (2C_3 C_4 L_3 R_1 R_3 g_m + 2C_3 C_4 L_3 R_3) + s^2 (C_3 L_3 R_1 g_m + C_3 L_3 + 2C_4 L_3 R_1 g_m + 2C_4 L_3 + C_4 L_4 R_1 g_m + C_4 L_4) + s (2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

$$\mathbf{9.64 \quad X-INVALID-ORDER-64} \quad Z(s) = \left(R_1, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_1 R_3 g_m s^3 + L_3 L_4 R_1 g_m s^2 + L_4 R_1 R_3 g_m s}{2R_1 R_3 g_m + 2R_3 + s^4 (2C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_3 C_4 L_3 L_4 R_3) + s^3 (C_3 L_3 L_4 R_1 g_m + C_3 L_3 L_4 + 2C_4 L_3 L_4 R_1 g_m + 2C_4 L_3 L_4) + s^2 (2C_3 L_3 R_1 R_3 g_m + 2C_3 L_3 R_3 + 2C_4 L_4 R_1 R_3 g_m + 2C_4 L_4 R_3) + s (2L_3 R_1 g_m + 2L_3 + L_4 R_1 g_m + L_4)}$$

$$\mathbf{9.65 \quad X-INVALID-ORDER-65} \quad Z(s) = \left(R_1, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_3 g_m s^4 + R_1 R_3 g_m + s^3 (C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_4 L_3 L_4 R_1 g_m) + s^2 (C_3 L_3 R_1 R_3 g_m + C_4 L_3 R_1 R_4 g_m + C_4 L_4 R_1 R_3 g_m) + s (C_4 R_1 R_3 R_4 g_m + L_3 R_1 g_m)}{R_1 g_m + s^4 (C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4) + s^3 (2C_3 C_4 L_3 R_1 R_3 g_m + C_3 C_4 L_3 R_1 R_4 g_m + 2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4) + s^2 (C_3 L_3 R_1 g_m + C_3 L_3 + 2C_4 L_3 R_1 g_m + 2C_4 L_3 + C_4 L_4 R_1 g_m + C_4 L_4) + s (2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4) + 1}$$

$$\mathbf{9.66 \quad X-INVALID-ORDER-66} \quad Z(s) = \left(R_1, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_1 R_3 R_4 g_m s^3 + L_3 L_4 R_1 R_4 g_m s^2 + L_4 R_1 R_3 R_4 g_m s}{2R_1 R_3 R_4 g_m + 2R_3 R_4 + s^4 (2C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + 2C_3 C_4 L_3 L_4 R_3 R_4) + s^3 (2C_3 L_3 L_4 R_1 R_3 g_m + C_3 L_3 L_4 R_1 R_4 g_m + 2C_3 L_3 L_4 R_3 + C_3 L_3 L_4 R_4 + 2C_4 L_3 L_4 R_1 R_4 g_m + 2C_4 L_3 L_4 R_4) + s^2 (2C_3 L_3 R_1 R_3 R_4 g_m + 2C_3 L_3 R_3 R_4 + 2C_4 L_4 R_1 R_3 R_4 g_m + 2C_4 L_4 R_3 R_4 + 2L_3 R_1 g_m + L_4 R_1 g_m) + s (C_4 R_1 R_3 R_4 g_m + L_3 R_1 g_m)}$$

$$\mathbf{9.67 \quad X-INVALID-ORDER-67} \quad Z(s) = \left(R_1, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m s^4 + R_1 R_3 R_4 g_m + s^3 (C_3 L_3 L_4 R_1 R_3 g_m + C_4 L_3 L_4 R_1 R_4 g_m) + s^2 (C_3 L_3 R_1 R_3 R_4 g_m + C_4 L_4 R_1 R_3 R_4 g_m + L_3 L_4 R_1 g_m) + s (L_3 R_1 R_4 g_m + L_4 R_1 R_3 g_m)}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^4 (2C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_3 C_4 L_3 L_4 R_3 + C_3 C_4 L_3 L_4 R_4) + s^3 (C_3 L_3 L_4 R_1 g_m + C_3 L_3 L_4 + 2C_4 L_3 L_4 R_1 g_m + 2C_4 L_3 L_4) + s^2 (2C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_1 R_4 g_m + 2C_3 L_3 R_3 + C_3 L_3 R_4 + 2C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_3 + C_4 L_4 R_4) + s (C_4 R_1 R_3 R_4 g_m + L_3 R_1 g_m)}$$

$$\mathbf{9.68 \quad X-INVALID-ORDER-68} \quad Z(s) = \left(R_1, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m s^4 + C_4 L_3 L_4 R_1 R_4 g_m s^3 + L_3 R_1 R_4 g_m s + R_1 R_3 R_4 g_m + s^2 (C_3 L_3 R_1 R_3 R_4 g_m + C_4 L_4 R_1 R_3 R_4 g_m)}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^4 (2C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_3 C_4 L_3 L_4 R_3 + C_3 C_4 L_3 L_4 R_4) + s^3 (2C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_3 C_4 L_3 R_3 R_4 + 2C_4 L_3 L_4 R_1 g_m + 2C_4 L_3 L_4) + s^2 (2C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_1 R_4 g_m + 2C_3 L_3 R_3 + C_3 L_3 R_4 + 2C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_3 + C_4 L_4 R_4) + s (C_4 R_1 R_3 R_4 g_m + L_3 R_1 g_m)}$$

$$\mathbf{9.69 \quad X-INVALID-ORDER-69} \quad Z(s) = \left(R_1, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_3 g_m s^2 + R_1 R_3 g_m}{R_1 g_m + s^3 (2C_3 C_4 L_3 R_1 R_3 g_m + 2C_3 C_4 L_3 R_3) + s^2 (C_3 L_3 R_1 g_m + C_3 L_3) + s (C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

$$\mathbf{9.70 \quad X-INVALID-ORDER-70} \quad Z(s) = \left(R_1, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_3 R_4 g_m s^2 + R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^3 (2C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_3 C_4 L_3 R_3 R_4) + s^2 (2C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_1 R_4 g_m + 2C_3 L_3 R_3 + C_3 L_3 R_4) + s (C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4) + 1}$$

$$\mathbf{9.71 \quad X-INVALID-ORDER-71} \quad Z(s) = \left(R_1, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 R_1 R_3 R_4 g_m s^3 + C_3 L_3 R_1 R_3 g_m s^2 + C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{R_1 g_m + s^3 (2C_3 C_4 L_3 R_1 R_3 g_m + C_3 C_4 L_3 R_1 R_4 g_m + 2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4) + s^2 (C_3 C_4 R_1 R_3 R_4 g_m + C_3 C_4 R_3 R_4 + C_3 L_3 R_1 g_m + C_3 L_3) + s (C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4) + 1}$$

$$\mathbf{9.72 \quad X-INVALID-ORDER-72} \quad Z(s) = \left(R_1, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_3 g_m s^4 + R_1 R_3 g_m + s^2 (C_3 L_3 R_1 R_3 g_m + C_4 L_4 R_1 R_3 g_m)}{R_1 g_m + s^4 (C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4) + s^3 (2C_3 C_4 L_3 R_1 R_3 g_m + 2C_3 C_4 L_3 R_3 + C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_4 R_3) + s^2 (C_3 L_3 R_1 g_m + C_3 L_3 + C_4 L_4 R_1 g_m + C_4 L_4) + s (C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

$$\mathbf{9.73 \quad X-INVALID-ORDER-73} \quad Z(s) = \left(R_1, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_1 R_3 g_m s^3 + L_4 R_1 R_3 g_m s}{2R_1 R_3 g_m + 2R_3 + s^4 (2C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_3 C_4 L_3 L_4 R_3) + s^3 (C_3 L_3 L_4 R_1 g_m + C_3 L_3 L_4) + s^2 (2C_3 L_3 R_1 R_3 g_m + 2C_3 L_3 R_3 + C_3 L_4 R_1 R_3 g_m + C_3 L_4 R_3 + 2C_4 L_4 R_1 R_3 g_m + 2C_4 L_4 R_3) + s (L_4 R_1 g_m + L_4)}$$

$$\mathbf{9.74 \quad X-INVALID-ORDER-74} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \quad L_4s + R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3C_4L_3L_4R_1R_3g_ms^4 + C_3C_4L_3R_1R_3R_4g_ms^3 + C_4R_1R_3R_4g_ms + R_1R_3g_m + s^2(C_3L_3R_1R_3g_m + C_4L_4R_1R_3g_m)}{R_1g_m + s^4(C_3C_4L_3L_4R_1g_m + C_3C_4L_3L_4) + s^3(2C_3C_4L_3R_1R_3g_m + C_3C_4L_3R_1R_4g_m + 2C_3C_4L_3R_3 + C_3C_4L_3R_4 + C_3C_4L_4R_1R_3g_m + C_3C_4L_4R_3) + s^2(C_3C_4R_1R_3R_4g_m + C_3C_4R_3R_4 + C_3L_3R_1g_m + C_3L_3 + C_4L_4R_1g_m + C_4L_4) + s(C_3R_1R_3g_m + C_3R_3 + 2C_3R_4) + 1}$$

$$\mathbf{9.75 \quad X-INVALID-ORDER-75} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \quad \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3L_3L_4R_1R_3R_4g_ms^3 + L_4R_1R_3R_4g_ms}{2R_1R_3R_4g_m + 2R_3R_4 + s^4(2C_3C_4L_3L_4R_1R_3R_4g_m + 2C_3C_4L_3L_4R_3R_4) + s^3(2C_3L_3L_4R_1R_3g_m + C_3L_3L_4R_1R_4g_m + 2C_3L_3L_4R_3 + C_3L_3L_4R_4) + s^2(2C_3L_3R_1R_3R_4g_m + 2C_3L_3R_3R_4 + C_3L_4R_1R_3R_4g_m + C_3L_4R_3R_4 + 2C_4L_4R_1R_3R_4g_m + 2C_4L_4R_3R_4) + s(2L_1R_3g_m + L_1R_4g_m) + 1}$$

$$\mathbf{9.76 \quad X-INVALID-ORDER-76} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \quad \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3C_4L_3L_4R_1R_3R_4g_ms^4 + C_3L_3L_4R_1R_3g_ms^3 + L_4R_1R_3g_ms + R_1R_3R_4g_m + s^2(C_3L_3R_1R_3R_4g_m + C_4L_4R_1R_3R_4g_m)}{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^4(2C_3C_4L_3L_4R_1R_3g_m + C_3C_4L_3L_4R_1R_4g_m + 2C_3C_4L_3L_4R_3 + C_3C_4L_3L_4R_4) + s^3(C_3C_4L_4R_1R_3R_4g_m + C_3C_4L_4R_3R_4 + C_3L_3L_4R_1g_m + C_3L_3L_4) + s^2(2C_3L_3R_1R_3g_m + C_3L_3R_1R_4g_m + 2C_3L_3R_3 + C_3L_3R_4 + C_3L_4R_1R_3R_4g_m + C_3L_4R_3R_4) + s(2L_1R_3g_m + L_1R_4g_m) + 1}$$

$$\mathbf{9.77 \quad X-INVALID-ORDER-77} \quad Z(s) = \left(R_1, \quad \infty, \quad \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3C_4L_3L_4R_1R_3R_4g_ms^4 + R_1R_3R_4g_m + s^2(C_3L_3R_1R_3R_4g_m + C_4L_4R_1R_3R_4g_m)}{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^4(2C_3C_4L_3L_4R_1R_3g_m + C_3C_4L_3L_4R_1R_4g_m + 2C_3C_4L_3L_4R_3 + C_3C_4L_3L_4R_4) + s^3(2C_3C_4L_3R_1R_3R_4g_m + 2C_3C_4L_3R_3R_4 + C_3C_4L_4R_1R_3R_4g_m + C_3C_4L_4R_3R_4) + s^2(2C_3L_3R_1R_3g_m + C_3L_3R_1R_4g_m + 2C_3L_3R_3 + C_3L_3R_4 + C_3L_4R_1R_3R_4g_m + C_3L_4R_3R_4) + s(2L_1R_3g_m + L_1R_4g_m) + 1}$$

$$\mathbf{9.78 \quad X-INVALID-ORDER-78} \quad Z(s) = (L_1s, \quad \infty, \quad R_3, \quad R_4, \quad \infty, \quad \infty)$$

$$H(s) = \frac{L_1R_3R_4g_ms}{2R_3 + R_4 + s(2L_1R_3g_m + L_1R_4g_m)}$$

$$\mathbf{9.79 \quad X-INVALID-ORDER-79} \quad Z(s) = \left(L_1s, \quad \infty, \quad R_3, \quad L_4s + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4L_1L_4R_3g_ms^3 + L_1R_3g_ms}{C_4L_1L_4g_ms^3 + s^2(2C_4L_1R_3g_m + C_4L_4) + s(2C_4R_3 + L_1g_m) + 1}$$

$$\mathbf{9.80 \quad X-INVALID-ORDER-80} \quad Z(s) = \left(L_1s, \quad \infty, \quad R_3, \quad \frac{L_4s}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1L_4R_3g_ms^2}{2C_4L_1L_4R_3g_ms^3 + 2R_3 + s^2(2C_4L_4R_3 + L_1L_4g_m) + s(2L_1R_3g_m + L_4)}$$

$$\mathbf{9.81 \quad X-INVALID-ORDER-81} \quad Z(s) = \left(L_1s, \quad \infty, \quad R_3, \quad L_4s + R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4L_1L_4R_3g_ms^3 + C_4L_1R_3R_4g_ms^2 + L_1R_3g_ms}{C_4L_1L_4g_ms^3 + s^2(2C_4L_1R_3g_m + C_4L_1R_4g_m + C_4L_4) + s(2C_4R_3 + C_4R_4 + L_1g_m) + 1}$$

$$\mathbf{9.82 \quad X-INVALID-ORDER-82} \quad Z(s) = \left(L_1s, \quad \infty, \quad R_3, \quad \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1L_4R_3R_4g_ms^2}{2C_4L_1L_4R_3R_4g_ms^3 + 2R_3R_4 + s^2(2C_4L_4R_3R_4 + 2L_1L_4R_3g_m + L_1L_4R_4g_m) + s(2L_1R_3R_4g_m + 2L_4R_3 + L_4R_4)}$$

$$\mathbf{9.83 \quad X-INVALID-ORDER-83} \quad Z(s) = \left(L_1 s, \quad \infty, \quad R_3, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_3 R_4 g_m s^3 + L_1 L_4 R_3 g_m s^2 + L_1 R_3 R_4 g_m s}{2R_3 + R_4 + s^3 (2C_4 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (2C_4 L_4 R_3 + C_4 L_4 R_4 + L_1 L_4 g_m) + s (2L_1 R_3 g_m + L_1 R_4 g_m + L_4)}$$

$$\mathbf{9.84 \quad X-INVALID-ORDER-84} \quad Z(s) = \left(L_1 s, \quad \infty, \quad R_3, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_3 R_4 g_m s^3 + L_1 R_3 R_4 g_m s}{2R_3 + R_4 + s^3 (2C_4 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (2C_4 L_1 R_3 R_4 g_m + 2C_4 L_4 R_3 + C_4 L_4 R_4) + s (2C_4 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.85 \quad X-INVALID-ORDER-85} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 g_m}{C_3 + 2C_4 + s (C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

$$\mathbf{9.86 \quad X-INVALID-ORDER-86} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 g_m s^2 + L_1 g_m}{C_3 C_4 L_1 L_4 g_m s^3 + C_3 C_4 L_4 s^2 + C_3 + 2C_4 + s (C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

$$\mathbf{9.87 \quad X-INVALID-ORDER-87} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 g_m s^2}{2L_1 g_m s + s^3 (C_3 L_1 L_4 g_m + 2C_4 L_1 L_4 g_m) + s^2 (C_3 L_4 + 2C_4 L_4) + 2}$$

$$\mathbf{9.88 \quad X-INVALID-ORDER-88} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 g_m s^2 + C_4 L_1 R_4 g_m s + L_1 g_m}{C_3 C_4 L_1 L_4 g_m s^3 + C_3 + 2C_4 + s^2 (C_3 C_4 L_1 R_4 g_m + C_3 C_4 L_4) + s (C_3 C_4 R_4 + C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

$$\mathbf{9.89 \quad X-INVALID-ORDER-89} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_4 g_m s^2}{2R_4 + s^3 (C_3 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 R_4 g_m) + s^2 (C_3 L_4 R_4 + 2C_4 L_4 R_4 + 2L_1 L_4 g_m) + s (2L_1 R_4 g_m + 2L_4)}$$

$$\mathbf{9.90 \quad X-INVALID-ORDER-90} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_4 g_m s^3 + L_1 L_4 g_m s^2 + L_1 R_4 g_m s}{C_3 C_4 L_1 L_4 R_4 g_m s^4 + s^3 (C_3 C_4 L_4 R_4 + C_3 L_1 L_4 g_m + 2C_4 L_1 L_4 g_m) + s^2 (C_3 L_1 R_4 g_m + C_3 L_4 + 2C_4 L_4) + s (C_3 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.91 \quad X-INVALID-ORDER-91} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_4 g_m s^3 + L_1 R_4 g_m s}{C_3 C_4 L_1 L_4 R_4 g_m s^4 + s^3 (C_3 C_4 L_4 R_4 + 2C_4 L_1 L_4 g_m) + s^2 (C_3 L_1 R_4 g_m + 2C_4 L_1 R_4 g_m + 2C_4 L_4) + s (C_3 R_4 + 2C_4 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.92 \quad X-INVALID-ORDER-92} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 R_3 R_4 g_m s^2 + L_1 R_3 g_m s}{C_3 C_4 L_1 R_3 R_4 g_m s^3 + s^2 (C_3 C_4 R_3 R_4 + C_3 L_1 R_3 g_m + 2 C_4 L_1 R_3 g_m + C_4 L_1 R_4 g_m) + s (C_3 R_3 + 2 C_4 R_3 + C_4 R_4 + L_1 g_m) + 1}$$

$$\mathbf{9.93 \quad X-INVALID-ORDER-93} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_3 g_m s^3 + L_1 R_3 g_m s}{C_3 C_4 L_1 L_4 R_3 g_m s^4 + s^3 (C_3 C_4 L_4 R_3 + C_4 L_1 L_4 g_m) + s^2 (C_3 L_1 R_3 g_m + 2 C_4 L_1 R_3 g_m + C_4 L_4) + s (C_3 R_3 + 2 C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.94 \quad X-INVALID-ORDER-94} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_3 g_m s^2}{2 R_3 + s^3 (C_3 L_1 L_4 R_3 g_m + 2 C_4 L_1 L_4 R_3 g_m) + s^2 (C_3 L_4 R_3 + 2 C_4 L_4 R_3 + L_1 L_4 g_m) + s (2 L_1 R_3 g_m + L_4)}$$

$$\mathbf{9.95 \quad X-INVALID-ORDER-95} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_3 g_m s^3 + C_4 L_1 R_3 R_4 g_m s^2 + L_1 R_3 g_m s}{C_3 C_4 L_1 L_4 R_3 g_m s^4 + s^3 (C_3 C_4 L_1 R_3 R_4 g_m + C_3 C_4 L_4 R_3 + C_4 L_1 L_4 g_m) + s^2 (C_3 C_4 R_3 R_4 + C_3 L_1 R_3 g_m + 2 C_4 L_1 R_3 g_m + C_4 L_1 R_4 g_m + C_4 L_4) + s (C_3 R_3 + 2 C_4 R_3 + C_4 R_4 + L_1 g_m) + 1}$$

$$\mathbf{9.96 \quad X-INVALID-ORDER-96} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_3 R_4 g_m s^2}{2 R_3 R_4 + s^3 (C_3 L_1 L_4 R_3 R_4 g_m + 2 C_4 L_1 L_4 R_3 R_4 g_m) + s^2 (C_3 L_4 R_3 R_4 + 2 C_4 L_4 R_3 R_4 + 2 L_1 L_4 R_3 g_m + L_1 L_4 R_4 g_m) + s (2 L_1 R_3 R_4 g_m + 2 L_4 R_3 + L_4 R_4)}$$

$$\mathbf{9.97 \quad X-INVALID-ORDER-97} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_3 R_4 g_m s^3 + L_1 L_4 R_3 g_m s^2 + L_1 R_3 R_4 g_m s}{C_3 C_4 L_1 L_4 R_3 R_4 g_m s^4 + 2 R_3 + R_4 + s^3 (C_3 C_4 L_4 R_3 R_4 + C_3 L_1 L_4 R_3 g_m + 2 C_4 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (C_3 L_1 R_3 R_4 g_m + C_3 L_4 R_3 + 2 C_4 L_4 R_3 + C_4 L_4 R_4 + L_1 L_4 g_m) + s (C_3 R_3 R_4 + 2 L_1 R_3 g_m + L_1 R_4 g_m + L_4)}$$

$$\mathbf{9.98 \quad X-INVALID-ORDER-98} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_3 R_4 g_m s^3 + L_1 R_3 R_4 g_m s}{C_3 C_4 L_1 L_4 R_3 R_4 g_m s^4 + 2 R_3 + R_4 + s^3 (C_3 C_4 L_4 R_3 R_4 + 2 C_4 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (C_3 L_1 R_3 R_4 g_m + 2 C_4 L_1 R_3 R_4 g_m + 2 C_4 L_4 R_3 + C_4 L_4 R_4) + s (C_3 R_3 R_4 + 2 C_4 R_3 R_4 + 2 L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.99 \quad X-INVALID-ORDER-99} \quad Z(s) = \left(L_1 s, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 R_3 R_4 g_m s^2 + L_1 R_4 g_m s}{2 C_3 C_4 L_1 R_3 R_4 g_m s^3 + s^2 (2 C_3 C_4 R_3 R_4 + 2 C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m + 2 C_4 L_1 R_4 g_m) + s (2 C_3 R_3 + C_3 R_4 + 2 C_4 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.100 \quad X-INVALID-ORDER-100} \quad Z(s) = \left(L_1 s, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_4 R_3 g_m s^3 + C_3 L_1 R_3 g_m s + C_4 L_1 L_4 g_m s^2 + L_1 g_m}{C_3 C_4 L_1 L_4 g_m s^3 + C_3 + 2 C_4 + s^2 (2 C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_4) + s (2 C_3 C_4 R_3 + C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

$$\mathbf{9.101 \quad X-INVALID-ORDER-101} \quad Z(s) = \left(L_1 s, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_4 R_3 g_m s^3 + L_1 L_4 g_m s^2}{2C_3 C_4 L_1 L_4 R_3 g_m s^4 + s^3 (2C_3 C_4 L_4 R_3 + C_3 L_1 L_4 g_m + 2C_4 L_1 L_4 g_m) + s^2 (2C_3 L_1 R_3 g_m + C_3 L_4 + 2C_4 L_4) + s (2C_3 R_3 + 2L_1 g_m) + 2}$$

$$\mathbf{9.102 \quad X-INVALID-ORDER-102} \quad Z(s) = \left(L_1 s, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_4 R_3 g_m s^3 + L_1 g_m + s^2 (C_3 C_4 L_1 R_3 R_4 g_m + C_4 L_1 L_4 g_m) + s (C_3 L_1 R_3 g_m + C_4 L_1 R_4 g_m)}{C_3 C_4 L_1 L_4 g_m s^3 + C_3 + 2C_4 + s^2 (2C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_1 R_4 g_m + C_3 C_4 L_4) + s (2C_3 C_4 R_3 + C_3 C_4 R_4 + C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

$$\mathbf{9.103 \quad X-INVALID-ORDER-103} \quad Z(s) = \left(L_1 s, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_4 R_3 R_4 g_m s^3 + L_1 L_4 R_4 g_m s^2}{2C_3 C_4 L_1 L_4 R_3 R_4 g_m s^4 + 2R_4 + s^3 (2C_3 C_4 L_4 R_3 R_4 + 2C_3 L_1 L_4 R_3 g_m + C_3 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 R_4 g_m) + s^2 (2C_3 L_1 R_3 R_4 g_m + 2C_3 L_4 R_3 + C_3 L_4 R_4 + 2C_4 L_4 R_4 + 2L_1 L_4 g_m) + s (2C_3 R_3 R_4 + 2L_1 R_4 g_m + 2L_4)}$$

$$\mathbf{9.104 \quad X-INVALID-ORDER-104} \quad Z(s) = \left(L_1 s, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_4 R_3 R_4 g_m s^4 + L_1 R_4 g_m s + s^3 (C_3 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (C_3 L_1 R_3 R_4 g_m + L_1 L_4 g_m)}{s^4 (2C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_1 L_4 R_4 g_m) + s^3 (2C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4 + C_3 L_1 L_4 g_m + 2C_4 L_1 L_4 g_m) + s^2 (2C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m + C_3 L_4 + 2C_4 L_4) + s (2C_3 R_3 + C_3 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.105 \quad X-INVALID-ORDER-105} \quad Z(s) = \left(L_1 s, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_4 R_3 R_4 g_m s^4 + C_3 L_1 R_3 R_4 g_m s^2 + C_4 L_1 L_4 R_4 g_m s^3 + L_1 R_4 g_m s}{s^4 (2C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_1 L_4 R_4 g_m) + s^3 (2C_3 C_4 L_1 R_3 R_4 g_m + 2C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4 + 2C_4 L_1 L_4 g_m) + s^2 (2C_3 C_4 R_3 R_4 + 2C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m + 2C_4 L_1 R_4 g_m + 2C_4 L_4) + s (2C_3 R_3 + C_3 R_4 + 2C_4 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.106 \quad X-INVALID-ORDER-106} \quad Z(s) = \left(L_1 s, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_4 g_m s^3 + L_1 R_4 g_m s}{2C_3 L_1 L_3 g_m s^3 + s^2 (C_3 L_1 R_4 g_m + 2C_3 L_3) + s (C_3 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.107 \quad X-INVALID-ORDER-107} \quad Z(s) = \left(L_1 s, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 g_m s^2 + L_1 g_m}{2C_3 C_4 L_1 L_3 g_m s^3 + 2C_3 C_4 L_3 s^2 + C_3 + 2C_4 + s (C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

$$\mathbf{9.108 \quad X-INVALID-ORDER-108} \quad Z(s) = \left(L_1 s, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_4 g_m s^3 + L_1 R_4 g_m s}{2C_3 C_4 L_1 L_3 R_4 g_m s^4 + s^3 (2C_3 C_4 L_3 R_4 + 2C_3 L_1 L_3 g_m) + s^2 (C_3 L_1 R_4 g_m + 2C_3 L_3 + 2C_4 L_1 R_4 g_m) + s (C_3 R_4 + 2C_4 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.109 \quad X-INVALID-ORDER-109} \quad Z(s) = \left(L_1 s, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 R_4 g_m s^3 + C_3 L_1 L_3 g_m s^2 + C_4 L_1 R_4 g_m s + L_1 g_m}{2C_3 C_4 L_1 L_3 g_m s^3 + C_3 + 2C_4 + s^2 (C_3 C_4 L_1 R_4 g_m + 2C_3 C_4 L_3) + s (C_3 C_4 R_4 + C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

9.110 X-INVALID-ORDER-110 $Z(s) = \left(L_1 s, \infty, L_3 s + \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 g_m s^4 + L_1 g_m + s^2 (C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m)}{C_3 + 2C_4 + s^3 (2C_3 C_4 L_1 L_3 g_m + C_3 C_4 L_1 L_4 g_m) + s^2 (2C_3 C_4 L_3 + C_3 C_4 L_4) + s (C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

9.111 X-INVALID-ORDER-111 $Z(s) = \left(L_1 s, \infty, L_3 s + \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 L_4 g_m s^4 + L_1 L_4 g_m s^2}{2C_3 C_4 L_1 L_3 L_4 g_m s^5 + 2C_3 C_4 L_3 L_4 s^4 + 2L_1 g_m s + s^3 (2C_3 L_1 L_3 g_m + C_3 L_1 L_4 g_m + 2C_4 L_1 L_4 g_m) + s^2 (2C_3 L_3 + C_3 L_4 + 2C_4 L_4) + 2}$$

9.112 X-INVALID-ORDER-112 $Z(s) = \left(L_1 s, \infty, L_3 s + \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 g_m s^4 + C_3 C_4 L_1 L_3 R_4 g_m s^3 + C_4 L_1 R_4 g_m s + L_1 g_m + s^2 (C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m)}{C_3 + 2C_4 + s^3 (2C_3 C_4 L_1 L_3 g_m + C_3 C_4 L_1 L_4 g_m) + s^2 (C_3 C_4 L_1 R_4 g_m + 2C_3 C_4 L_3 + C_3 C_4 L_4) + s (C_3 C_4 R_4 + C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

9.113 X-INVALID-ORDER-113 $Z(s) = \left(L_1 s, \infty, L_3 s + \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_4 g_m s^4 + L_1 L_4 R_4 g_m s^2}{2C_3 C_4 L_1 L_3 L_4 R_4 g_m s^5 + 2R_4 + s^4 (2C_3 C_4 L_3 L_4 R_4 + 2C_3 L_1 L_3 L_4 g_m) + s^3 (2C_3 L_1 L_3 R_4 g_m + C_3 L_1 L_4 R_4 g_m + 2C_3 L_3 L_4 + 2C_4 L_1 L_4 R_4 g_m) + s^2 (2C_3 L_3 R_4 + C_3 L_4 R_4 + 2C_4 L_4 R_4 + 2L_1 L_4 g_m) + s (2L_1 R_4 g_m + 2L_4)}$$

9.114 X-INVALID-ORDER-114 $Z(s) = \left(L_1 s, \infty, L_3 s + \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_4 g_m s^5 + C_3 L_1 L_3 L_4 g_m s^4 + L_1 L_4 g_m s^2 + L_1 R_4 g_m s + s^3 (C_3 L_1 L_3 R_4 g_m + C_4 L_1 L_4 R_4 g_m)}{2C_3 C_4 L_1 L_3 L_4 g_m s^5 + s^4 (C_3 C_4 L_1 L_4 R_4 g_m + 2C_3 C_4 L_3 L_4) + s^3 (C_3 C_4 L_4 R_4 + 2C_3 L_1 L_3 g_m + C_3 L_1 L_4 g_m + 2C_4 L_1 L_4 g_m) + s^2 (C_3 L_1 R_4 g_m + 2C_3 L_3 + C_3 L_4 + 2C_4 L_4) + s (C_3 R_4 + 2L_1 g_m) + 2}$$

9.115 X-INVALID-ORDER-115 $Z(s) = \left(L_1 s, \infty, L_3 s + \frac{1}{C_3 s}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_4 g_m s^5 + L_1 R_4 g_m s + s^3 (C_3 L_1 L_3 R_4 g_m + C_4 L_1 L_4 R_4 g_m)}{2C_3 C_4 L_1 L_3 L_4 g_m s^5 + s^4 (2C_3 C_4 L_1 L_3 R_4 g_m + C_3 C_4 L_1 L_4 R_4 g_m + 2C_3 C_4 L_3 L_4) + s^3 (2C_3 C_4 L_3 R_4 + C_3 C_4 L_4 R_4 + 2C_3 L_1 L_3 g_m + 2C_4 L_1 L_4 g_m) + s^2 (C_3 L_1 R_4 g_m + 2C_3 L_3 + 2C_4 L_1 R_4 g_m + 2C_4 L_4) + s (C_3 R_4 + 2C_4 R_4 + 2L_1 g_m) + 2}$$

9.116 X-INVALID-ORDER-116 $Z(s) = \left(L_1 s, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{L_1 L_3 R_4 g_m s^2}{C_3 L_1 L_3 R_4 g_m s^3 + R_4 + s^2 (C_3 L_3 R_4 + 2L_1 L_3 g_m) + s (L_1 R_4 g_m + 2L_3)}$$

9.117 X-INVALID-ORDER-117 $Z(s) = \left(L_1 s, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{L_1 L_3 g_m s^2}{L_1 g_m s + s^3 (C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s^2 (C_3 L_3 + 2C_4 L_3) + 1}$$

9.118 X-INVALID-ORDER-118 $Z(s) = \left(L_1 s, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{L_1 L_3 R_4 g_m s^2}{R_4 + s^3 (C_3 L_1 L_3 R_4 g_m + 2C_4 L_1 L_3 R_4 g_m) + s^2 (C_3 L_3 R_4 + 2C_4 L_3 R_4 + 2L_1 L_3 g_m) + s (L_1 R_4 g_m + 2L_3)}$$

$$\mathbf{9.119 \quad X-INVALID-ORDER-119} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 R_4 g_m s^3 + L_1 L_3 g_m s^2}{C_3 C_4 L_1 L_3 R_4 g_m s^4 + s^3 (C_3 C_4 L_3 R_4 + C_3 L_1 L_3 g_m + 2 C_4 L_1 L_3 g_m) + s^2 (C_3 L_3 + C_4 L_1 R_4 g_m + 2 C_4 L_3) + s (C_4 R_4 + L_1 g_m) + 1}$$

$$\mathbf{9.120 \quad X-INVALID-ORDER-120} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 g_m s^4 + L_1 L_3 g_m s^2}{C_3 C_4 L_1 L_3 L_4 g_m s^5 + C_3 C_4 L_3 L_4 s^4 + L_1 g_m s + s^3 (C_3 L_1 L_3 g_m + 2 C_4 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s^2 (C_3 L_3 + 2 C_4 L_3 + C_4 L_4) + 1}$$

$$\mathbf{9.121 \quad X-INVALID-ORDER-121} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 L_4 g_m s^2}{2 L_3 + L_4 + s^3 (C_3 L_1 L_3 L_4 g_m + 2 C_4 L_1 L_3 L_4 g_m) + s^2 (C_3 L_3 L_4 + 2 C_4 L_3 L_4) + s (2 L_1 L_3 g_m + L_1 L_4 g_m)}$$

$$\mathbf{9.122 \quad X-INVALID-ORDER-122} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 g_m s^4 + C_4 L_1 L_3 R_4 g_m s^3 + L_1 L_3 g_m s^2}{C_3 C_4 L_1 L_3 L_4 g_m s^5 + s^4 (C_3 C_4 L_1 L_3 R_4 g_m + C_3 C_4 L_3 L_4) + s^3 (C_3 C_4 L_3 R_4 + C_3 L_1 L_3 g_m + 2 C_4 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s^2 (C_3 L_3 + C_4 L_1 R_4 g_m + 2 C_4 L_3 + C_4 L_4) + s (C_4 R_4 + L_1 g_m) + 1}$$

$$\mathbf{9.123 \quad X-INVALID-ORDER-123} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 L_4 R_4 g_m s^2}{2 L_3 R_4 + L_4 R_4 + s^3 (C_3 L_1 L_3 L_4 R_4 g_m + 2 C_4 L_1 L_3 L_4 R_4 g_m) + s^2 (C_3 L_3 L_4 R_4 + 2 C_4 L_3 L_4 R_4 + 2 L_1 L_3 L_4 g_m) + s (2 L_1 L_3 R_4 g_m + L_1 L_4 R_4 g_m + 2 L_3 L_4)}$$

$$\mathbf{9.124 \quad X-INVALID-ORDER-124} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_4 g_m s^4 + L_1 L_3 L_4 g_m s^3 + L_1 L_3 R_4 g_m s^2}{C_3 C_4 L_1 L_3 L_4 R_4 g_m s^5 + R_4 + s^4 (C_3 C_4 L_3 L_4 R_4 + C_3 L_1 L_3 L_4 g_m + 2 C_4 L_1 L_3 L_4 g_m) + s^3 (C_3 L_1 L_3 R_4 g_m + C_3 L_3 L_4 + C_4 L_1 L_4 R_4 g_m + 2 C_4 L_3 L_4) + s^2 (C_3 L_3 R_4 + C_4 L_4 R_4 + 2 L_1 L_3 g_m + L_1 L_4 g_m) + s (L_1 R_4 g_m + 2 L_3 + L_4)}$$

$$\mathbf{9.125 \quad X-INVALID-ORDER-125} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_4 g_m s^4 + L_1 L_3 R_4 g_m s^2}{C_3 C_4 L_1 L_3 L_4 R_4 g_m s^5 + R_4 + s^4 (C_3 C_4 L_3 L_4 R_4 + 2 C_4 L_1 L_3 L_4 g_m) + s^3 (C_3 L_1 L_3 R_4 g_m + 2 C_4 L_1 L_3 R_4 g_m + C_4 L_1 L_4 R_4 g_m + 2 C_4 L_3 L_4) + s^2 (C_3 L_3 R_4 + 2 C_4 L_3 R_4 + C_4 L_4 R_4 + 2 L_1 L_3 g_m) + s (L_1 R_4 g_m + 2 L_3)}$$

$$\mathbf{9.126 \quad X-INVALID-ORDER-126} \quad Z(s) = \left(L_1 s, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_4 g_m s^3 + C_3 L_1 R_3 R_4 g_m s^2 + L_1 R_4 g_m s}{2 C_3 L_1 L_3 g_m s^3 + s^2 (2 C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m + 2 C_3 L_3) + s (2 C_3 R_3 + C_3 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.127 \quad X-INVALID-ORDER-127} \quad Z(s) = \left(L_1 s, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 g_m s^2 + C_3 L_1 R_3 g_m s + L_1 g_m}{2 C_3 C_4 L_1 L_3 g_m s^3 + C_3 + 2 C_4 + s^2 (2 C_3 C_4 L_1 R_3 g_m + 2 C_3 C_4 L_3) + s (2 C_3 C_4 R_3 + C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

$$\mathbf{9.128 \quad X-INVALID-ORDER-128} \quad Z(s) = \left(L_1 s, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_4 g_m s^3 + C_3 L_1 R_3 R_4 g_m s^2 + L_1 R_4 g_m s}{2 C_3 C_4 L_1 L_3 R_4 g_m s^4 + s^3 (2 C_3 C_4 L_1 R_3 R_4 g_m + 2 C_3 C_4 L_3 R_4 + 2 C_3 L_1 L_3 g_m) + s^2 (2 C_3 C_4 R_3 R_4 + 2 C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m + 2 C_3 L_3 + 2 C_4 L_1 R_4 g_m) + s (2 C_3 R_3 + C_3 R_4 + 2 C_4 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.129 \quad X-INVALID-ORDER-129} \quad Z(s) = \left(L_1 s, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 R_4 g_m s^3 + L_1 g_m + s^2 (C_3 C_4 L_1 R_3 R_4 g_m + C_3 L_1 L_3 g_m) + s (C_3 L_1 R_3 g_m + C_4 L_1 R_4 g_m)}{2 C_3 C_4 L_1 L_3 g_m s^3 + C_3 + 2 C_4 + s^2 (2 C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_1 R_4 g_m + 2 C_3 C_4 L_3) + s (2 C_3 C_4 R_3 + C_3 C_4 R_4 + C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

$$\mathbf{9.130 \quad X-INVALID-ORDER-130} \quad Z(s) = \left(L_1 s, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 g_m s^4 + C_3 C_4 L_1 L_4 R_3 g_m s^3 + C_3 L_1 R_3 g_m s + L_1 g_m + s^2 (C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m)}{C_3 + 2 C_4 + s^3 (2 C_3 C_4 L_1 L_3 g_m + C_3 C_4 L_1 L_4 g_m) + s^2 (2 C_3 C_4 L_1 R_3 g_m + 2 C_3 C_4 L_3 + C_3 C_4 L_4) + s (2 C_3 C_4 R_3 + C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

$$\mathbf{9.131 \quad X-INVALID-ORDER-131} \quad Z(s) = \left(L_1 s, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 g_m s^4 + C_3 L_1 L_4 R_3 g_m s^3 + L_1 L_4 g_m s^2}{2 C_3 C_4 L_1 L_3 L_4 g_m s^5 + s^4 (2 C_3 C_4 L_1 L_4 R_3 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (2 C_3 C_4 L_4 R_3 + 2 C_3 L_1 L_3 g_m + C_3 L_1 L_4 g_m + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_3 L_1 R_3 g_m + 2 C_3 L_3 + C_3 L_4 + 2 C_4 L_4) + s (2 C_3 R_3 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.132 \quad X-INVALID-ORDER-132} \quad Z(s) = \left(L_1 s, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 g_m s^4 + L_1 g_m + s^3 (C_3 C_4 L_1 L_3 R_4 g_m + C_3 C_4 L_1 L_4 R_3 g_m) + s^2 (C_3 C_4 L_1 R_3 R_4 g_m + C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s (C_3 L_1 R_3 g_m + C_4 L_1 R_4 g_m)}{C_3 + 2 C_4 + s^3 (2 C_3 C_4 L_1 L_3 g_m + C_3 C_4 L_1 L_4 g_m) + s^2 (2 C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_1 R_4 g_m + 2 C_3 C_4 L_3 + C_3 C_4 L_4) + s (2 C_3 C_4 R_3 + C_3 C_4 R_4 + C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

$$\mathbf{9.133 \quad X-INVALID-ORDER-133} \quad Z(s) = \left(L_1 s, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_4 g_m s^4 + C_3 L_1 L_4 R_3 R_4 g_m s^3 + L_1 L_4 R_4 g_m s^2}{2 C_3 C_4 L_1 L_3 L_4 R_4 g_m s^5 + 2 R_4 + s^4 (2 C_3 C_4 L_1 L_4 R_3 R_4 g_m + 2 C_3 C_4 L_3 L_4 R_4 + 2 C_3 L_1 L_3 L_4 g_m) + s^3 (2 C_3 C_4 L_4 R_3 R_4 + 2 C_3 L_1 L_3 R_4 g_m + 2 C_3 L_1 L_4 R_3 g_m + C_3 L_1 L_4 R_4 g_m + 2 C_3 L_3 L_4 + 2 C_4 L_1 L_4 R_4 g_m) + s^2 (2 C_3 L_1 R_3 R_4 g_m + 2 C_3 L_3 R_4 + 2 C_3 L_4 R_3 + C_3 L_4 R_4 + 2 C_4 L_4)}$$

$$\mathbf{9.134 \quad X-INVALID-ORDER-134} \quad Z(s) = \left(L_1 s, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_4 g_m s^5 + L_1 R_4 g_m s + s^4 (C_3 C_4 L_1 L_4 R_3 R_4 g_m + C_3 L_1 L_3 L_4 g_m) + s^3 (C_3 L_1 L_3 R_4 g_m + C_3 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (C_3 L_1 R_3 R_4 g_m + L_1 L_4 g_m)}{2 C_3 C_4 L_1 L_3 L_4 g_m s^5 + s^4 (2 C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_1 L_4 R_4 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (2 C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4 + 2 C_3 L_1 L_3 g_m + C_3 L_1 L_4 g_m + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m + 2 C_3 L_3 + C_3 L_4 + 2 C_4 L_4) + s (2 C_3 R_3 + C_3 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.135 \quad X-INVALID-ORDER-135} \quad Z(s) = \left(L_1 s, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_4 g_m s^5 + C_3 C_4 L_1 L_4 R_3 R_4 g_m s^4 + C_3 L_1 R_3 R_4 g_m s^2 + L_1 R_4 g_m s + s^3 (C_3 L_1 L_3 R_4 g_m + C_4 L_1 L_4 R_4 g_m)}{2 C_3 C_4 L_1 L_3 L_4 g_m s^5 + s^4 (2 C_3 C_4 L_1 L_3 R_4 g_m + 2 C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_1 L_4 R_4 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (2 C_3 C_4 L_1 R_3 R_4 g_m + 2 C_3 C_4 L_3 R_4 + 2 C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4 + 2 C_3 L_1 L_3 g_m + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_3 C_4 R_3 R_4 + 2 C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m + 2 C_3 L_3 + 2 C_4 L_4)}$$

$$\mathbf{9.136 \quad X-INVALID-ORDER-136} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 R_3 R_4 g_m s^2}{C_3 L_1 L_3 R_3 R_4 g_m s^3 + R_3 R_4 + s^2 (C_3 L_3 R_3 R_4 + 2 L_1 L_3 R_3 g_m + L_1 L_3 R_4 g_m) + s (L_1 R_3 R_4 g_m + 2 L_3 R_3 + L_3 R_4)}$$

$$\mathbf{9.137 \quad X-INVALID-ORDER-137} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 R_3 g_m s^2}{R_3 + s^3 (C_3 L_1 L_3 R_3 g_m + 2 C_4 L_1 L_3 R_3 g_m) + s^2 (C_3 L_3 R_3 + 2 C_4 L_3 R_3 + L_1 L_3 g_m) + s (L_1 R_3 g_m + L_3)}$$

$$\mathbf{9.138 \quad X-INVALID-ORDER-138} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 R_3 R_4 g_m s^2}{R_3 R_4 + s^3 (C_3 L_1 L_3 R_3 R_4 g_m + 2 C_4 L_1 L_3 R_3 R_4 g_m) + s^2 (C_3 L_3 R_3 R_4 + 2 C_4 L_3 R_3 R_4 + 2 L_1 L_3 R_3 g_m + L_1 L_3 R_4 g_m) + s (L_1 R_3 R_4 g_m + 2 L_3 R_3 + L_3 R_4)}$$

$$\mathbf{9.139 \quad X-INVALID-ORDER-139} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 R_3 R_4 g_m s^3 + L_1 L_3 R_3 g_m s^2}{C_3 C_4 L_1 L_3 R_3 R_4 g_m s^4 + R_3 + s^3 (C_3 C_4 L_3 R_3 R_4 + C_3 L_1 L_3 R_3 g_m + 2 C_4 L_1 L_3 R_3 g_m + C_4 L_1 L_3 R_4 g_m) + s^2 (C_3 L_3 R_3 + C_4 L_1 R_3 R_4 g_m + 2 C_4 L_3 R_3 + C_4 L_3 R_4 + L_1 L_3 g_m) + s (C_4 R_3 R_4 + L_1 R_3 g_m + L_3)}$$

$$\mathbf{9.140 \quad X-INVALID-ORDER-140} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_3 g_m s^4 + L_1 L_3 R_3 g_m s^2}{C_3 C_4 L_1 L_3 L_4 R_3 g_m s^5 + R_3 + s^4 (C_3 C_4 L_3 L_4 R_3 + C_4 L_1 L_3 L_4 g_m) + s^3 (C_3 L_1 L_3 R_3 g_m + 2 C_4 L_1 L_3 R_3 g_m + C_4 L_1 L_4 R_3 g_m + C_4 L_3 L_4) + s^2 (C_3 L_3 R_3 + 2 C_4 L_3 R_3 + C_4 L_4 R_3 + L_1 L_3 g_m) + s (L_1 R_3 g_m + L_3)}$$

$$\mathbf{9.141 \quad X-INVALID-ORDER-141} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 L_4 R_3 g_m s^2}{2 L_3 R_3 + L_4 R_3 + s^3 (C_3 L_1 L_3 L_4 R_3 g_m + 2 C_4 L_1 L_3 L_4 R_3 g_m) + s^2 (C_3 L_3 L_4 R_3 + 2 C_4 L_3 L_4 R_3 + L_1 L_3 L_4 g_m) + s (2 L_1 L_3 R_3 g_m + L_1 L_4 R_3 g_m + L_3 L_4)}$$

$$\mathbf{9.142 \quad X-INVALID-ORDER-142} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_3 g_m s^4 + C_4 L_1 L_3 R_3 R_4 g_m s^3 + L_1 L_3 R_3 g_m s^2}{C_3 C_4 L_1 L_3 L_4 R_3 g_m s^5 + R_3 + s^4 (C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_3 C_4 L_3 L_4 R_3 + C_4 L_1 L_3 L_4 g_m) + s^3 (C_3 C_4 L_3 R_3 R_4 + C_3 L_1 L_3 R_3 g_m + 2 C_4 L_1 L_3 R_3 g_m + C_4 L_1 L_3 R_4 g_m + C_4 L_1 L_4 R_3 g_m + C_4 L_3 L_4) + s^2 (C_3 L_3 R_3 + C_4 L_1 R_3 R_4 g_m + 2 C_4 L_3 R_3 + C_4 L_3 R_4 + C_4 L_4 R_3 + L_1 L_3 g_m)}$$

$$\mathbf{9.143 \quad X-INVALID-ORDER-143} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 L_4 R_3 R_4 g_m s^2}{2 L_3 R_3 R_4 + L_4 R_3 R_4 + s^3 (C_3 L_1 L_3 L_4 R_3 R_4 g_m + 2 C_4 L_1 L_3 L_4 R_3 R_4 g_m) + s^2 (C_3 L_3 L_4 R_3 R_4 + 2 C_4 L_3 L_4 R_3 R_4 + 2 L_1 L_3 L_4 R_3 g_m + L_1 L_3 L_4 R_4 g_m) + s (2 L_1 L_3 R_3 R_4 g_m + L_1 L_4 R_3 R_4 g_m + 2 L_3 L_4 R_3 + L_3 L_4 R_4)}$$

$$\mathbf{9.144 \quad X-INVALID-ORDER-144} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_3 R_4 g_m s^4 + L_1 L_3 L_4 R_3 g_m s^3 + L_1 L_3 R_3 R_4 g_m s^2}{C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^5 + R_3 R_4 + s^4 (C_3 C_4 L_3 L_4 R_3 R_4 + C_3 L_1 L_3 L_4 R_3 g_m + 2 C_4 L_1 L_3 L_4 R_3 g_m + C_4 L_1 L_3 L_4 R_4 g_m) + s^3 (C_3 L_1 L_3 R_3 R_4 g_m + C_3 L_3 L_4 R_3 + C_4 L_1 L_4 R_3 R_4 g_m + 2 C_4 L_3 L_4 R_3 + C_4 L_3 L_4 R_4 + L_1 L_3 L_4 g_m) + s^2 (C_3 L_3 R_3 R_4 + C_4 L_4 R_3 R_4 + 2 L_1 L_3 R_3 g_m + L_1 L_3 R_4 g_m)}$$

$$\mathbf{9.145 \quad X-INVALID-ORDER-145} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_3 R_4 g_m s^4 + L_1 L_3 R_3 R_4 g_m s^2}{C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^5 + R_3 R_4 + s^4 (C_3 C_4 L_3 L_4 R_3 R_4 + 2 C_4 L_1 L_3 L_4 R_3 g_m + C_4 L_1 L_3 L_4 R_4 g_m) + s^3 (C_3 L_1 L_3 R_3 R_4 g_m + 2 C_4 L_1 L_3 R_3 R_4 g_m + C_4 L_1 L_4 R_3 R_4 g_m + 2 C_4 L_3 L_4 R_3 + C_4 L_3 L_4 R_4) + s^2 (C_3 L_3 R_3 R_4 + 2 C_4 L_3 R_3 R_4 + C_4 L_4 R_3 R_4 + 2 L_1 L_3 R_3 g_m + L_1 L_3 R_4 g_m)}$$

$$\mathbf{9.146 \quad X-INVALID-ORDER-146} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_3 R_4 g_m s^3 + L_1 L_3 R_4 g_m s^2 + L_1 R_3 R_4 g_m s}{2R_3 + R_4 + s^3 (2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 g_m) + s^2 (2C_3 L_3 R_3 + C_3 L_3 R_4 + 2L_1 L_3 g_m) + s (2L_1 R_3 g_m + L_1 R_4 g_m + 2L_3)}$$

$$\mathbf{9.147 \quad X-INVALID-ORDER-147} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_3 g_m s^3 + L_1 L_3 g_m s^2 + L_1 R_3 g_m s}{2C_3 C_4 L_1 L_3 R_3 g_m s^4 + s^3 (2C_3 C_4 L_3 R_3 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s^2 (C_3 L_3 + 2C_4 L_1 R_3 g_m + 2C_4 L_3) + s (2C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.148 \quad X-INVALID-ORDER-148} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_3 R_4 g_m s^3 + L_1 L_3 R_4 g_m s^2 + L_1 R_3 R_4 g_m s}{2C_3 C_4 L_1 L_3 R_3 R_4 g_m s^4 + 2R_3 + R_4 + s^3 (2C_3 C_4 L_3 R_3 R_4 + 2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 g_m + 2C_4 L_1 L_3 R_4 g_m) + s^2 (2C_3 L_3 R_3 + C_3 L_3 R_4 + 2C_4 L_1 R_3 g_m + 2C_4 L_3 R_4 + 2L_1 L_3 g_m) + s (2C_4 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m + 2L_3)}$$

$$\mathbf{9.149 \quad X-INVALID-ORDER-149} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 R_3 R_4 g_m s^4 + L_1 R_3 g_m s + s^3 (C_3 L_1 L_3 R_3 g_m + C_4 L_1 L_3 R_4 g_m) + s^2 (C_4 L_1 R_3 R_4 g_m + L_1 L_3 g_m)}{s^4 (2C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s^2 (C_3 L_3 + 2C_4 L_1 R_3 g_m + C_4 L_1 R_4 g_m + 2C_4 L_3) + s (2C_4 R_3 + C_4 R_4 + L_1 g_m) + 1}$$

$$\mathbf{9.150 \quad X-INVALID-ORDER-150} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 g_m s^5 + C_4 L_1 L_3 L_4 g_m s^4 + L_1 L_3 g_m s^2 + L_1 R_3 g_m s + s^3 (C_3 L_1 L_3 R_3 g_m + C_4 L_1 L_4 R_3 g_m)}{C_3 C_4 L_1 L_3 L_4 g_m s^5 + s^4 (2C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_3 L_4) + s^3 (2C_3 C_4 L_3 R_3 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s^2 (C_3 L_3 + 2C_4 L_1 R_3 g_m + 2C_4 L_3 + C_4 L_4) + s (2C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.151 \quad X-INVALID-ORDER-151} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_3 g_m s^4 + L_1 L_3 L_4 g_m s^3 + L_1 L_4 R_3 g_m s^2}{2C_3 C_4 L_1 L_3 L_4 R_3 g_m s^5 + 2R_3 + s^4 (2C_3 C_4 L_3 L_4 R_3 + C_3 L_1 L_3 L_4 g_m + 2C_4 L_1 L_3 L_4 g_m) + s^3 (2C_3 L_1 L_3 R_3 g_m + C_3 L_3 L_4 + 2C_4 L_1 L_4 R_3 g_m + 2C_4 L_3 L_4) + s^2 (2C_3 L_3 R_3 + 2C_4 L_4 R_3 + 2L_1 L_3 g_m + L_1 L_4 g_m) + s (2L_1 R_3 g_m + 2L_3 + L_4)}$$

$$\mathbf{9.152 \quad X-INVALID-ORDER-152} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 g_m s^5 + L_1 R_3 g_m s + s^4 (C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_4 L_1 L_3 L_4 g_m) + s^3 (C_3 L_1 L_3 R_3 g_m + C_4 L_1 L_3 R_4 g_m + C_4 L_1 L_4 R_3 g_m) + s^2 (C_4 L_1 R_3 R_4 g_m + L_1 L_3 g_m)}{C_3 C_4 L_1 L_3 L_4 g_m s^5 + s^4 (2C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_3 R_4 g_m + C_3 C_4 L_3 L_4) + s^3 (2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s^2 (C_3 L_3 + 2C_4 L_1 R_3 g_m + C_4 L_1 R_4 g_m + 2C_4 L_3 + C_4 L_4) + s (2C_4 R_3 + C_4 R_4 + L_1 g_m) + 1}$$

$$\mathbf{9.153 \quad X-INVALID-ORDER-153} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_3 R_4 g_m s^4 + L_1 L_3 L_4 R_4 g_m s^3 + L_1 L_4 R_3 R_4 g_m s^2}{2C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^5 + 2R_3 R_4 + s^4 (2C_3 C_4 L_3 L_4 R_3 R_4 + 2C_3 L_1 L_3 L_4 R_3 g_m + C_3 L_1 L_3 L_4 R_4 g_m + 2C_4 L_1 L_3 L_4 R_4 g_m) + s^3 (2C_3 L_1 L_3 R_3 R_4 g_m + 2C_3 L_3 L_4 R_3 + C_3 L_3 L_4 R_4 + 2C_4 L_1 L_4 R_3 R_4 g_m + 2C_4 L_3 L_4 R_4 + 2L_1 L_3 L_4 g_m) + s^2 (2C_3 L_3 R_3 R_4 + 2C_4 L_4 R_3 R_4 + 2L_1 L_3 R_3 R_4 + L_1 L_4 R_3 R_4) + s (2L_1 R_3 R_4 + 2L_3 + L_4)}$$

$$\mathbf{9.154 \quad X-INVALID-ORDER-154} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^5 + L_1 R_3 R_4 g_m s + s^4 (C_3 L_1 L_3 L_4 R_3 g_m + C_4 L_1 L_3 L_4 R_4 g_m) + s^3 (C_3 L_1 L_3 R_3 R_4 g_m + C_4 L_1 L_4 R_3 R_4 g_m + L_1 L_3 L_4 g_m) + s^2 (L_1 L_3 R_4 g_m + L_1 L_4 R_3 g_m)}{2R_3 + R_4 + s^5 (2C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2C_3 C_4 L_3 L_4 R_3 + C_3 C_4 L_3 L_4 R_4 + C_3 L_1 L_3 L_4 g_m + 2C_4 L_1 L_3 L_4 g_m) + s^3 (2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 g_m + C_3 L_3 L_4 + 2C_4 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m + 2C_4 L_3 L_4) + s^2 (2C_3 L_3 R_3 + C_3 L_3 R_4 + 2C_4 L_3 R_3 + 2C_4 L_3 R_4) + s (2L_1 R_3 + L_1 R_4 + L_1 g_m)}$$

$$\mathbf{9.155 \quad X-INVALID-ORDER-155} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^5 + C_4 L_1 L_3 L_4 R_4 g_m s^4 + L_1 L_3 R_4 g_m s^2 + L_1 R_3 R_4 g_m s + s^3 (C_3 L_1 L_3 R_3 R_4 g_m + C_4 L_1 L_4 R_3 R_4 g_m)}{2R_3 + R_4 + s^5 (2C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2C_3 C_4 L_1 L_3 R_3 R_4 g_m + 2C_3 C_4 L_3 L_4 R_3 + C_3 C_4 L_3 L_4 R_4 + 2C_4 L_1 L_3 L_4 g_m) + s^3 (2C_3 C_4 L_3 R_3 R_4 + 2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 g_m + 2C_4 L_1 L_3 R_4 g_m + 2C_4 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m + 2C_4 L_3 L_4) + s^2 (2C_3 L_3 R_3 + C_3 L_3 R_4 + 2C_4 L_3 R_3 + 2C_4 L_3 R_4) + s (2L_1 R_3 + L_1 R_4 + L_1 g_m)}$$

$$\mathbf{9.156 \quad X-INVALID-ORDER-156} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_3 R_4 g_m s^3 + L_1 R_3 R_4 g_m s}{2 R_3 + R_4 + s^3 (2 C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 g_m) + s^2 (C_3 L_1 R_3 R_4 g_m + 2 C_3 L_3 R_3 + C_3 L_3 R_4) + s (C_3 R_3 R_4 + 2 L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.157 \quad X-INVALID-ORDER-157} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_3 g_m s^3 + L_1 R_3 g_m s}{2 C_3 C_4 L_1 L_3 R_3 g_m s^4 + s^3 (2 C_3 C_4 L_3 R_3 + C_3 L_1 L_3 g_m) + s^2 (C_3 L_1 R_3 g_m + C_3 L_3 + 2 C_4 L_1 R_3 g_m) + s (C_3 R_3 + 2 C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.158 \quad X-INVALID-ORDER-158} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_3 R_4 g_m s^3 + L_1 R_3 R_4 g_m s}{2 C_3 C_4 L_1 L_3 R_3 R_4 g_m s^4 + 2 R_3 + R_4 + s^3 (2 C_3 C_4 L_3 R_3 R_4 + 2 C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 g_m) + s^2 (C_3 L_1 R_3 R_4 g_m + 2 C_3 L_3 R_3 + C_3 L_3 R_4 + 2 C_4 L_1 R_3 R_4 g_m) + s (C_3 R_3 R_4 + 2 C_4 R_3 R_4 + 2 L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.159 \quad X-INVALID-ORDER-159} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 R_3 R_4 g_m s^4 + C_3 L_1 L_3 R_3 g_m s^3 + C_4 L_1 R_3 R_4 g_m s^2 + L_1 R_3 g_m s}{s^4 (2 C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (C_3 C_4 L_1 R_3 R_4 g_m + 2 C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4 + C_3 L_1 L_3 g_m) + s^2 (C_3 C_4 R_3 R_4 + C_3 L_1 R_3 g_m + C_3 L_3 + 2 C_4 L_1 R_3 g_m + C_4 L_1 R_4 g_m) + s (C_3 R_3 + 2 C_4 R_3 + C_4 R_4 + L_1 g_m) + 1}$$

$$\mathbf{9.160 \quad X-INVALID-ORDER-160} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 g_m s^5 + L_1 R_3 g_m s + s^3 (C_3 L_1 L_3 R_3 g_m + C_4 L_1 L_4 R_3 g_m)}{C_3 C_4 L_1 L_3 L_4 g_m s^5 + s^4 (2 C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_3 L_4) + s^3 (2 C_3 C_4 L_3 R_3 + C_3 C_4 L_4 R_3 + C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s^2 (C_3 L_1 R_3 g_m + C_3 L_3 + 2 C_4 L_1 R_3 g_m + C_4 L_4) + s (C_3 R_3 + 2 C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.161 \quad X-INVALID-ORDER-161} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_3 g_m s^4 + L_1 L_4 R_3 g_m s^2}{2 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^5 + 2 R_3 + s^4 (2 C_3 C_4 L_3 L_4 R_3 + C_3 L_1 L_3 L_4 g_m) + s^3 (2 C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_4 R_3 g_m + C_3 L_3 L_4 + 2 C_4 L_1 L_4 R_3 g_m) + s^2 (2 C_3 L_3 R_3 + C_3 L_4 R_3 + 2 C_4 L_4 R_3 + L_1 L_4 g_m) + s (2 L_1 R_3 g_m + L_4)}$$

$$\mathbf{9.162 \quad X-INVALID-ORDER-162} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 g_m s^5 + C_3 C_4 L_1 L_3 R_3 R_4 g_m s^4 + C_4 L_1 R_3 R_4 g_m s^2 + L_1 R_3 g_m s + s^3 (C_3 L_1 L_3 R_3 g_m + C_4 L_1 L_4 R_3 g_m)}{C_3 C_4 L_1 L_3 L_4 g_m s^5 + s^4 (2 C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_3 R_4 g_m + C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_3 L_4) + s^3 (C_3 C_4 L_1 R_3 R_4 g_m + 2 C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4 + C_3 C_4 L_4 R_3 + C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s^2 (C_3 C_4 R_3 R_4 + C_3 L_1 R_3 g_m + C_3 L_3 + 2 C_4 L_1 R_3 g_m + C_4 L_1 R_4 g_m + C_4 L_4) + s (C_3 R_3 + 2 C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.163 \quad X-INVALID-ORDER-163} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_3 R_4 g_m s^4 + L_1 L_4 R_3 R_4 g_m s^2}{2 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^5 + 2 R_3 R_4 + s^4 (2 C_3 C_4 L_3 L_4 R_3 R_4 + 2 C_3 L_1 L_3 L_4 R_3 g_m + C_3 L_1 L_3 L_4 R_4 g_m) + s^3 (2 C_3 L_1 L_3 R_3 R_4 g_m + C_3 L_1 L_4 R_3 R_4 g_m + 2 C_3 L_3 L_4 R_3 + C_3 L_3 L_4 R_4 + 2 C_4 L_1 L_4 R_3 R_4 g_m) + s^2 (2 C_3 L_3 R_3 R_4 + C_3 L_4 R_3 R_4 + 2 C_4 L_4 R_3 R_4 + 2 L_1 L_4 R_3 g_m + L_1 L_4 R_4 g_m) + s (C_3 R_3 R_4 + L_1 R_3 g_m + L_1 R_4 g_m) + 1}$$

$$\mathbf{9.164 \quad X-INVALID-ORDER-164} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^5 + C_3 L_1 L_3 L_4 R_3 g_m s^4 + L_1 L_4 R_3 g_m s^2 + L_1 R_3 R_4 g_m s + s^3 (C_3 L_1 L_3 R_3 R_4 g_m + C_4 L_1 L_4 R_3 R_4 g_m)}{2 R_3 + R_4 + s^5 (2 C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (C_3 C_4 L_1 L_4 R_3 R_4 g_m + 2 C_3 C_4 L_3 L_4 R_3 + C_3 C_4 L_3 L_4 R_4 + C_3 L_1 L_3 L_4 g_m) + s^3 (C_3 C_4 L_4 R_3 R_4 + 2 C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 g_m + C_3 L_1 L_4 R_3 g_m + C_3 L_3 L_4 + 2 C_4 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (C_3 R_3 R_4 + L_1 R_3 g_m + L_1 R_4 g_m) + s (C_3 R_3 + 2 C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.165 \quad X-INVALID-ORDER-165} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^5 + L_1 R_3 R_4 g_m s + s^3 (C_3 L_1 L_3 R_3 R_4 g_m + C_4 L_1 L_4 R_3 R_4 g_m)}{2 R_3 + R_4 + s^5 (2 C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2 C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_3 C_4 L_1 L_4 R_3 R_4 g_m + 2 C_3 C_4 L_3 L_4 R_3 + C_3 C_4 L_3 L_4 R_4) + s^3 (2 C_3 C_4 L_3 R_3 R_4 + C_3 C_4 L_4 R_3 R_4 + 2 C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 g_m + 2 C_4 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (C_3 L_1 L_3 R_3 + C_3 L_1 L_3 R_4 + C_4 L_1 L_4 R_3 + C_4 L_1 L_4 R_4) + s (C_3 R_3 + C_3 R_4 + C_4 R_3 + C_4 R_4) + C_3 + C_4}$$

$$\mathbf{9.166 \quad X-INVALID-ORDER-166} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_4 g_m}{2 R_3 g_m + R_4 g_m + s (2 C_1 R_3 + C_1 R_4)}$$

$$\mathbf{9.167 \quad X-INVALID-ORDER-167} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_3 g_m s^2 + R_3 g_m}{C_1 C_4 L_4 s^3 + g_m + s^2 (2 C_1 C_4 R_3 + C_4 L_4 g_m) + s (C_1 + 2 C_4 R_3 g_m)}$$

$$\mathbf{9.168 \quad X-INVALID-ORDER-168} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_4 R_3 g_m s}{2 C_1 C_4 L_4 R_3 s^3 + 2 R_3 g_m + s^2 (C_1 L_4 + 2 C_4 L_4 R_3 g_m) + s (2 C_1 R_3 + L_4 g_m)}$$

$$\mathbf{9.169 \quad X-INVALID-ORDER-169} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_3 g_m s^2 + C_4 R_3 R_4 g_m s + R_3 g_m}{C_1 C_4 L_4 s^3 + g_m + s^2 (2 C_1 C_4 R_3 + C_1 C_4 R_4 + C_4 L_4 g_m) + s (C_1 + 2 C_4 R_3 g_m + C_4 R_4 g_m)}$$

$$\mathbf{9.170 \quad X-INVALID-ORDER-170} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_4 R_3 R_4 g_m s}{2 C_1 C_4 L_4 R_3 R_4 s^3 + 2 R_3 R_4 g_m + s^2 (2 C_1 L_4 R_3 + C_1 L_4 R_4 + 2 C_4 L_4 R_3 R_4 g_m) + s (2 C_1 R_3 R_4 + 2 L_4 R_3 g_m + L_4 R_4 g_m)}$$

$$\mathbf{9.171 \quad X-INVALID-ORDER-171} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_3 R_4 g_m s^2 + L_4 R_3 g_m s + R_3 R_4 g_m}{2 R_3 g_m + R_4 g_m + s^3 (2 C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4) + s^2 (C_1 L_4 + 2 C_4 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (2 C_1 R_3 + C_1 R_4 + L_4 g_m)}$$

$$\mathbf{9.172 \quad X-INVALID-ORDER-172} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_3 R_4 g_m s^2 + R_3 R_4 g_m}{2 R_3 g_m + R_4 g_m + s^3 (2 C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4) + s^2 (2 C_1 C_4 R_3 R_4 + 2 C_4 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (2 C_1 R_3 + C_1 R_4 + 2 C_4 R_3 R_4 g_m)}$$

$$\mathbf{9.173 \quad X-INVALID-ORDER-173} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{g_m}{s^2 (C_1 C_3 + 2 C_1 C_4) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.174 \quad X-INVALID-ORDER-174} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 R_4 g_m s + g_m}{C_1 C_3 C_4 R_4 s^3 + s^2 (C_1 C_3 + 2 C_1 C_4 + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

9.175 X-INVALID-ORDER-175 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_4 g_m s^2 + g_m}{C_1 C_3 C_4 L_4 s^4 + C_3 C_4 L_4 g_m s^3 + s^2 (C_1 C_3 + 2C_1 C_4) + s (C_3 g_m + 2C_4 g_m)}$$

9.176 X-INVALID-ORDER-176 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{L_4 g_m s}{2C_1 s + 2g_m + s^3 (C_1 C_3 L_4 + 2C_1 C_4 L_4) + s^2 (C_3 L_4 g_m + 2C_4 L_4 g_m)}$$

9.177 X-INVALID-ORDER-177 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_4 g_m s^2 + C_4 R_4 g_m s + g_m}{C_1 C_3 C_4 L_4 s^4 + s^3 (C_1 C_3 C_4 R_4 + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

9.178 X-INVALID-ORDER-178 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{L_4 R_4 g_m s}{2R_4 g_m + s^3 (C_1 C_3 L_4 R_4 + 2C_1 C_4 L_4 R_4) + s^2 (2C_1 L_4 + C_3 L_4 R_4 g_m + 2C_4 L_4 R_4 g_m) + s (2C_1 R_4 + 2L_4 g_m)}$$

9.179 X-INVALID-ORDER-179 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_4 R_4 g_m s^2 + L_4 g_m s + R_4 g_m}{C_1 C_3 C_4 L_4 R_4 s^4 + 2g_m + s^3 (C_1 C_3 L_4 + 2C_1 C_4 L_4 + C_3 C_4 L_4 R_4 g_m) + s^2 (C_1 C_3 R_4 + C_3 L_4 g_m + 2C_4 L_4 g_m) + s (2C_1 + C_3 R_4 g_m)}$$

9.180 X-INVALID-ORDER-180 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_4 R_4 g_m s^2 + R_4 g_m}{C_1 C_3 C_4 L_4 R_4 s^4 + 2g_m + s^3 (2C_1 C_4 L_4 + C_3 C_4 L_4 R_4 g_m) + s^2 (C_1 C_3 R_4 + 2C_1 C_4 R_4 + 2C_4 L_4 g_m) + s (2C_1 + C_3 R_4 g_m + 2C_4 R_4 g_m)}$$

9.181 X-INVALID-ORDER-181 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 R_3 R_4 g_m s + R_3 g_m}{C_1 C_3 C_4 R_3 R_4 s^3 + g_m + s^2 (C_1 C_3 R_3 + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_3 C_4 R_3 R_4 g_m) + s (C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

9.182 X-INVALID-ORDER-182 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_4 R_3 g_m s^2 + R_3 g_m}{C_1 C_3 C_4 L_4 R_3 s^4 + g_m + s^3 (C_1 C_4 L_4 + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_3 + 2C_1 C_4 R_3 + C_4 L_4 g_m) + s (C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m)}$$

9.183 X-INVALID-ORDER-183 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{L_4 R_3 g_m s}{2R_3 g_m + s^3 (C_1 C_3 L_4 R_3 + 2C_1 C_4 L_4 R_3) + s^2 (C_1 L_4 + C_3 L_4 R_3 g_m + 2C_4 L_4 R_3 g_m) + s (2C_1 R_3 + L_4 g_m)}$$

9.184 X-INVALID-ORDER-184 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_4 R_3 g_m s^2 + C_4 R_3 R_4 g_m s + R_3 g_m}{C_1 C_3 C_4 L_4 R_3 s^4 + g_m + s^3 (C_1 C_3 C_4 R_3 R_4 + C_1 C_4 L_4 + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_3 + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_3 C_4 R_3 R_4 g_m + C_4 L_4 g_m) + s (C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

$$\mathbf{9.185 \quad X-INVALID-ORDER-185} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_4 R_3 R_4 g_m s}{2 R_3 R_4 g_m + s^3 (C_1 C_3 L_4 R_3 R_4 + 2 C_1 C_4 L_4 R_3 R_4) + s^2 (2 C_1 L_4 R_3 + C_1 L_4 R_4 + C_3 L_4 R_3 R_4 g_m + 2 C_4 L_4 R_3 R_4 g_m) + s (2 C_1 R_3 R_4 + 2 L_4 R_3 g_m + L_4 R_4 g_m)}$$

$$\mathbf{9.186 \quad X-INVALID-ORDER-186} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_3 R_4 g_m s^2 + L_4 R_3 g_m s + R_3 R_4 g_m}{C_1 C_3 C_4 L_4 R_3 R_4 s^4 + 2 R_3 g_m + R_4 g_m + s^3 (C_1 C_3 L_4 R_3 + 2 C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (C_1 C_3 R_3 R_4 + C_1 L_4 + C_3 L_4 R_3 g_m + 2 C_4 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + L_4 g_m)}$$

$$\mathbf{9.187 \quad X-INVALID-ORDER-187} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_3 R_4 g_m s^2 + R_3 R_4 g_m}{C_1 C_3 C_4 L_4 R_3 R_4 s^4 + 2 R_3 g_m + R_4 g_m + s^3 (2 C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (C_1 C_3 R_3 R_4 + 2 C_1 C_4 R_3 R_4 + 2 C_4 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2 C_4 R_3 R_4 g_m)}$$

$$\mathbf{9.188 \quad X-INVALID-ORDER-188} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 R_3 g_m s + g_m}{2 C_1 C_3 C_4 R_3 s^3 + s^2 (C_1 C_3 + 2 C_1 C_4 + 2 C_3 C_4 R_3 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.189 \quad X-INVALID-ORDER-189} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 R_3 R_4 g_m s + R_4 g_m}{2 C_1 C_3 C_4 R_3 R_4 s^3 + 2 g_m + s^2 (2 C_1 C_3 R_3 + C_1 C_3 R_4 + 2 C_1 C_4 R_4 + 2 C_3 C_4 R_3 R_4 g_m) + s (2 C_1 + 2 C_3 R_3 g_m + C_3 R_4 g_m + 2 C_4 R_4 g_m)}$$

$$\mathbf{9.190 \quad X-INVALID-ORDER-190} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 R_3 R_4 g_m s^2 + g_m + s (C_3 R_3 g_m + C_4 R_4 g_m)}{s^3 (2 C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4) + s^2 (C_1 C_3 + 2 C_1 C_4 + 2 C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.191 \quad X-INVALID-ORDER-191} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_3 g_m s^3 + C_3 R_3 g_m s + C_4 L_4 g_m s^2 + g_m}{C_1 C_3 C_4 L_4 s^4 + s^3 (2 C_1 C_3 C_4 R_3 + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 + 2 C_1 C_4 + 2 C_3 C_4 R_3 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.192 \quad X-INVALID-ORDER-192} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_4 R_3 g_m s^2 + L_4 g_m s}{2 C_1 C_3 C_4 L_4 R_3 s^4 + 2 g_m + s^3 (C_1 C_3 L_4 + 2 C_1 C_4 L_4 + 2 C_3 C_4 L_4 R_3 g_m) + s^2 (2 C_1 C_3 R_3 + C_3 L_4 g_m + 2 C_4 L_4 g_m) + s (2 C_1 + 2 C_3 R_3 g_m)}$$

$$\mathbf{9.193 \quad X-INVALID-ORDER-193} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_3 g_m s^3 + g_m + s^2 (C_3 C_4 R_3 R_4 g_m + C_4 L_4 g_m) + s (C_3 R_3 g_m + C_4 R_4 g_m)}{C_1 C_3 C_4 L_4 s^4 + s^3 (2 C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 + 2 C_1 C_4 + 2 C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.194 \quad X-INVALID-ORDER-194} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_4 R_3 R_4 g_m s^2 + L_4 R_4 g_m s}{2 C_1 C_3 C_4 L_4 R_3 R_4 s^4 + 2 R_4 g_m + s^3 (2 C_1 C_3 L_4 R_3 + C_1 C_3 L_4 R_4 + 2 C_1 C_4 L_4 R_4 + 2 C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (2 C_1 C_3 R_3 R_4 + 2 C_1 L_4 + 2 C_3 L_4 R_3 g_m + C_3 L_4 R_4 g_m + 2 C_4 L_4 R_4 g_m) + s (2 C_1 R_4 + 2 C_3 R_3 R_4 g_m + 2 L_4 g_m)}$$

$$\mathbf{9.195 \quad X-INVALID-ORDER-195} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_3 R_4 g_m s^3 + R_4 g_m + s^2 (C_3 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (C_3 R_3 R_4 g_m + L_4 g_m)}{2g_m + s^4 (2C_1 C_3 C_4 L_4 R_3 + C_1 C_3 C_4 L_4 R_4) + s^3 (C_1 C_3 L_4 + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m + C_3 C_4 L_4 R_4 g_m) + s^2 (2C_1 C_3 R_3 + C_1 C_3 R_4 + C_3 L_4 g_m + 2C_4 L_4 g_m) + s (2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m)}$$

$$\mathbf{9.196 \quad X-INVALID-ORDER-196} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_3 R_4 g_m s^3 + C_3 R_3 R_4 g_m s + C_4 L_4 R_4 g_m s^2 + R_4 g_m}{2g_m + s^4 (2C_1 C_3 C_4 L_4 R_3 + C_1 C_3 C_4 L_4 R_4) + s^3 (2C_1 C_3 C_4 R_3 R_4 + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m + C_3 C_4 L_4 R_4 g_m) + s^2 (2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 C_4 R_4 + 2C_3 C_4 R_3 R_4 g_m + 2C_4 L_4 g_m) + s (2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m + 2C_4 R_4 g_m)}$$

$$\mathbf{9.197 \quad X-INVALID-ORDER-197} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_4 g_m s^2 + R_4 g_m}{2C_1 C_3 L_3 s^3 + 2g_m + s^2 (C_1 C_3 R_4 + 2C_3 L_3 g_m) + s (2C_1 + C_3 R_4 g_m)}$$

$$\mathbf{9.198 \quad X-INVALID-ORDER-198} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 g_m s^2 + g_m}{2C_1 C_3 C_4 L_3 s^4 + 2C_3 C_4 L_3 g_m s^3 + s^2 (C_1 C_3 + 2C_1 C_4) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.199 \quad X-INVALID-ORDER-199} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_4 g_m s^2 + R_4 g_m}{2C_1 C_3 C_4 L_3 R_4 s^4 + 2g_m + s^3 (2C_1 C_3 L_3 + 2C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 C_3 R_4 + 2C_1 C_4 R_4 + 2C_3 L_3 g_m) + s (2C_1 + C_3 R_4 g_m + 2C_4 R_4 g_m)}$$

$$\mathbf{9.200 \quad X-INVALID-ORDER-200} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 R_4 g_m s^3 + C_3 L_3 g_m s^2 + C_4 R_4 g_m s + g_m}{2C_1 C_3 C_4 L_3 s^4 + s^3 (C_1 C_3 C_4 R_4 + 2C_3 C_4 L_3 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.201 \quad X-INVALID-ORDER-201} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 g_m s^4 + g_m + s^2 (C_3 L_3 g_m + C_4 L_4 g_m)}{s^4 (2C_1 C_3 C_4 L_3 + C_1 C_3 C_4 L_4) + s^3 (2C_3 C_4 L_3 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 + 2C_1 C_4) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.202 \quad X-INVALID-ORDER-202} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 g_m s^3 + L_4 g_m s}{2C_1 C_3 C_4 L_3 L_4 s^5 + 2C_1 s + 2C_3 C_4 L_3 L_4 g_m s^4 + 2g_m + s^3 (2C_1 C_3 L_3 + C_1 C_3 L_4 + 2C_1 C_4 L_4) + s^2 (2C_3 L_3 g_m + C_3 L_4 g_m + 2C_4 L_4 g_m)}$$

$$\mathbf{9.203 \quad X-INVALID-ORDER-203} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 g_m s^4 + C_3 C_4 L_3 R_4 g_m s^3 + C_4 R_4 g_m s + g_m + s^2 (C_3 L_3 g_m + C_4 L_4 g_m)}{s^4 (2C_1 C_3 C_4 L_3 + C_1 C_3 C_4 L_4) + s^3 (C_1 C_3 C_4 R_4 + 2C_3 C_4 L_3 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.204 \quad X-INVALID-ORDER-204} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_4 g_m s^3 + L_4 R_4 g_m s}{2C_1 C_3 C_4 L_3 L_4 R_4 s^5 + 2R_4 g_m + s^4 (2C_1 C_3 L_3 L_4 + 2C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (2C_1 C_3 L_3 R_4 + C_1 C_3 L_4 R_4 + 2C_1 C_4 L_4 R_4 + 2C_3 L_3 L_4 g_m) + s^2 (2C_1 L_4 + 2C_3 L_3 R_4 g_m + C_3 L_4 R_4 g_m + 2C_4 L_4 R_4 g_m) + s (2C_1 R_4 + 2L_4 g_m)}$$

$$\mathbf{9.205 \quad X-INVALID-ORDER-205} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_4 g_m s^4 + C_3 L_3 L_4 g_m s^3 + L_4 g_m s + R_4 g_m + s^2 (C_3 L_3 R_4 g_m + C_4 L_4 R_4 g_m)}{2C_1 C_3 C_4 L_3 L_4 s^5 + 2g_m + s^4 (C_1 C_3 C_4 L_4 R_4 + 2C_3 C_4 L_3 L_4 g_m) + s^3 (2C_1 C_3 L_3 + C_1 C_3 L_4 + 2C_1 C_4 L_4 + C_3 C_4 L_4 R_4 g_m) + s^2 (C_1 C_3 R_4 + 2C_3 L_3 g_m + C_3 L_4 g_m + 2C_4 L_4 g_m) + s (2C_1 + C_3 R_4 g_m)}$$

$$\mathbf{9.206 \quad X-INVALID-ORDER-206} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_4 g_m s^4 + R_4 g_m + s^2 (C_3 L_3 R_4 g_m + C_4 L_4 R_4 g_m)}{2C_1 C_3 C_4 L_3 L_4 s^5 + 2g_m + s^4 (2C_1 C_3 C_4 L_3 R_4 + C_1 C_3 C_4 L_4 R_4 + 2C_3 C_4 L_3 L_4 g_m) + s^3 (2C_1 C_3 L_3 + 2C_1 C_4 L_4 + 2C_3 C_4 L_3 R_4 g_m + C_3 C_4 L_4 R_4 g_m) + s^2 (C_1 C_3 R_4 + 2C_1 C_4 R_4 + 2C_3 L_3 g_m + 2C_4 L_4 g_m) + s (2C_1 + C_3 R_4 g_m + 2C_4 R_4 g_m)}$$

$$\mathbf{9.207 \quad X-INVALID-ORDER-207} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 R_4 g_m s}{C_1 C_3 L_3 R_4 s^3 + R_4 g_m + s^2 (2C_1 L_3 + C_3 L_3 R_4 g_m) + s (C_1 R_4 + 2L_3 g_m)}$$

$$\mathbf{9.208 \quad X-INVALID-ORDER-208} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 g_m s}{C_1 s + g_m + s^3 (C_1 C_3 L_3 + 2C_1 C_4 L_3) + s^2 (C_3 L_3 g_m + 2C_4 L_3 g_m)}$$

$$\mathbf{9.209 \quad X-INVALID-ORDER-209} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 R_4 g_m s}{R_4 g_m + s^3 (C_1 C_3 L_3 R_4 + 2C_1 C_4 L_3 R_4) + s^2 (2C_1 L_3 + C_3 L_3 R_4 g_m + 2C_4 L_3 R_4 g_m) + s (C_1 R_4 + 2L_3 g_m)}$$

$$\mathbf{9.210 \quad X-INVALID-ORDER-210} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 R_4 g_m s^2 + L_3 g_m s}{C_1 C_3 C_4 L_3 R_4 s^4 + g_m + s^3 (C_1 C_3 L_3 + 2C_1 C_4 L_3 + C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 C_4 R_4 + C_3 L_3 g_m + 2C_4 L_3 g_m) + s (C_1 + C_4 R_4 g_m)}$$

$$\mathbf{9.211 \quad X-INVALID-ORDER-211} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 g_m s^3 + L_3 g_m s}{C_1 C_3 C_4 L_3 L_4 s^5 + C_1 s + C_3 C_4 L_3 L_4 g_m s^4 + g_m + s^3 (C_1 C_3 L_3 + 2C_1 C_4 L_3 + C_1 C_4 L_4) + s^2 (C_3 L_3 g_m + 2C_4 L_3 g_m + C_4 L_4 g_m)}$$

$$\mathbf{9.212 \quad X-INVALID-ORDER-212} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 L_4 g_m s}{2L_3 g_m + L_4 g_m + s^3 (C_1 C_3 L_3 L_4 + 2C_1 C_4 L_3 L_4) + s^2 (C_3 L_3 L_4 g_m + 2C_4 L_3 L_4 g_m) + s (2C_1 L_3 + C_1 L_4)}$$

$$\mathbf{9.213 \quad X-INVALID-ORDER-213} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 g_m s^3 + C_4 L_3 R_4 g_m s^2 + L_3 g_m s}{C_1 C_3 C_4 L_3 L_4 s^5 + g_m + s^4 (C_1 C_3 C_4 L_3 R_4 + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 + 2C_1 C_4 L_3 + C_1 C_4 L_4 + C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 C_4 R_4 + C_3 L_3 g_m + 2C_4 L_3 g_m + C_4 L_4 g_m) + s (C_1 + C_4 R_4 g_m)}$$

$$\mathbf{9.214 \quad X-INVALID-ORDER-214} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 L_4 R_4 g_m s}{2L_3 R_4 g_m + L_4 R_4 g_m + s^3 (C_1 C_3 L_3 L_4 R_4 + 2C_1 C_4 L_3 L_4 R_4) + s^2 (2C_1 L_3 L_4 + C_3 L_3 L_4 R_4 g_m + 2C_4 L_3 L_4 R_4 g_m) + s (2C_1 L_3 R_4 + C_1 L_4 R_4 + 2L_3 L_4 g_m)}$$

$$\mathbf{9.215 \quad X-INVALID-ORDER-215} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_4 g_m s^3 + L_3 L_4 g_m s^2 + L_3 R_4 g_m s}{C_1 C_3 C_4 L_3 L_4 R_4 s^5 + R_4 g_m + s^4 (C_1 C_3 L_3 L_4 + 2C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (C_1 C_3 L_3 R_4 + C_1 C_4 L_4 R_4 + C_3 L_3 L_4 g_m + 2C_4 L_3 L_4 g_m) + s^2 (2C_1 L_3 + C_1 L_4 + C_3 L_3 R_4 g_m + C_4 L_4 R_4 g_m) + s (C_1 R_4 + 2L_3 g_m + L_4 g_m)}$$

$$\mathbf{9.216 \quad X-INVALID-ORDER-216} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_4 g_m s^3 + L_3 R_4 g_m s}{C_1 C_3 C_4 L_3 L_4 R_4 s^5 + R_4 g_m + s^4 (2C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (C_1 C_3 L_3 R_4 + 2C_1 C_4 L_3 R_4 + C_1 C_4 L_4 R_4 + 2C_4 L_3 L_4 g_m) + s^2 (2C_1 L_3 + C_3 L_3 R_4 g_m + 2C_4 L_3 R_4 g_m + C_4 L_4 R_4 g_m) + s (C_1 R_4 + 2L_3 g_m)}$$

$$\mathbf{9.217 \quad X-INVALID-ORDER-217} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_4 g_m s^2 + C_3 R_3 R_4 g_m s + R_4 g_m}{2C_1 C_3 L_3 s^3 + 2g_m + s^2 (2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_3 L_3 g_m) + s (2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m)}$$

$$\mathbf{9.218 \quad X-INVALID-ORDER-218} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 g_m s^2 + C_3 R_3 g_m s + g_m}{2C_1 C_3 C_4 L_3 s^4 + s^3 (2C_1 C_3 C_4 R_3 + 2C_3 C_4 L_3 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.219 \quad X-INVALID-ORDER-219} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_4 g_m s^2 + C_3 R_3 R_4 g_m s + R_4 g_m}{2C_1 C_3 C_4 L_3 R_4 s^4 + 2g_m + s^3 (2C_1 C_3 C_4 R_3 R_4 + 2C_1 C_3 L_3 + 2C_3 C_4 L_3 R_4 g_m) + s^2 (2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 C_4 R_4 + 2C_3 C_4 R_3 R_4 g_m + 2C_3 L_3 g_m) + s (2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m + 2C_4 R_4 g_m)}$$

$$\mathbf{9.220 \quad X-INVALID-ORDER-220} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 R_4 g_m s^3 + g_m + s^2 (C_3 C_4 R_3 R_4 g_m + C_3 L_3 g_m) + s (C_3 R_3 g_m + C_4 R_4 g_m)}{2C_1 C_3 C_4 L_3 s^4 + s^3 (2C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + 2C_3 C_4 L_3 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.221 \quad X-INVALID-ORDER-221} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 g_m s^4 + C_3 C_4 L_4 R_3 g_m s^3 + C_3 R_3 g_m s + g_m + s^2 (C_3 L_3 g_m + C_4 L_4 g_m)}{s^4 (2C_1 C_3 C_4 L_3 + C_1 C_3 C_4 L_4) + s^3 (2C_1 C_3 C_4 R_3 + 2C_3 C_4 L_3 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.222 \quad X-INVALID-ORDER-222} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 g_m s^3 + C_3 L_4 R_3 g_m s^2 + L_4 g_m s}{2C_1 C_3 C_4 L_3 L_4 s^5 + 2g_m + s^4 (2C_1 C_3 C_4 L_4 R_3 + 2C_3 C_4 L_3 L_4 g_m) + s^3 (2C_1 C_3 L_3 + C_1 C_3 L_4 + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m) + s^2 (2C_1 C_3 R_3 + 2C_3 L_3 g_m + C_3 L_4 g_m + 2C_4 L_4 g_m) + s (2C_1 + 2C_3 R_3 g_m)}$$

$$\mathbf{9.223 \quad X-INVALID-ORDER-223} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 g_m s^4 + g_m + s^3 (C_3 C_4 L_3 R_4 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_3 C_4 R_3 R_4 g_m + C_3 L_3 g_m + C_4 L_4 g_m) + s (C_3 R_3 g_m + C_4 R_4 g_m)}{s^4 (2C_1 C_3 C_4 L_3 + C_1 C_3 C_4 L_4) + s^3 (2C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + 2C_3 C_4 L_3 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.224 \quad X-INVALID-ORDER-224} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_4 g_m s^3 + C_3 L_4 R_3 R_4 g_m s^2 + L_4 R_4 g_m s}{2 C_1 C_3 C_4 L_3 L_4 R_4 s^5 + 2 R_4 g_m + s^4 (2 C_1 C_3 C_4 L_4 R_3 R_4 + 2 C_1 C_3 L_3 L_4 + 2 C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (2 C_1 C_3 L_3 R_4 + 2 C_1 C_3 L_4 R_3 + C_1 C_3 L_4 R_4 + 2 C_1 C_4 L_4 R_4 + 2 C_3 C_4 L_4 R_3 R_4 g_m + 2 C_3 L_3 L_4 g_m) + s^2 (2 C_1 C_3 R_3 R_4 + 2 C_1 L_4 + 2 C_3 L_3 R_4 g_m + 2 C_3 L_4 R_3 g_m + C_3 L_4 R_4 g_m) + s (C_1 R_3 R_4 + L_4 g_m)}$$

$$\mathbf{9.225 \quad X-INVALID-ORDER-225} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_4 g_m s^4 + R_4 g_m + s^3 (C_3 C_4 L_4 R_3 R_4 g_m + C_3 L_3 L_4 g_m) + s^2 (C_3 L_3 R_4 g_m + C_3 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (C_3 R_3 R_4 g_m + L_4 g_m)}{2 C_1 C_3 C_4 L_3 L_4 s^5 + 2 g_m + s^4 (2 C_1 C_3 C_4 L_4 R_3 + C_1 C_3 C_4 L_4 R_4 + 2 C_3 C_4 L_3 L_4 g_m) + s^3 (2 C_1 C_3 L_3 + C_1 C_3 L_4 + 2 C_1 C_4 L_4 + 2 C_3 C_4 L_4 R_3 g_m + C_3 C_4 L_4 R_4 g_m) + s^2 (2 C_1 C_3 R_3 + C_1 C_3 R_4 + 2 C_3 L_3 g_m + C_3 L_4 g_m + 2 C_4 L_4 g_m) + s (2 C_1 + 2 C_3 R_3 g_m + C_3 R_4 g_m)}$$

$$\mathbf{9.226 \quad X-INVALID-ORDER-226} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_4 g_m s^4 + C_3 C_4 L_4 R_3 R_4 g_m s^3 + C_3 R_3 R_4 g_m s + R_4 g_m + s^2 (C_3 L_3 R_4 g_m + C_4 L_4 R_4 g_m)}{2 C_1 C_3 C_4 L_3 L_4 s^5 + 2 g_m + s^4 (2 C_1 C_3 C_4 L_3 R_4 + 2 C_1 C_3 C_4 L_4 R_3 + C_1 C_3 C_4 L_4 R_4 + 2 C_3 C_4 L_3 L_4 g_m) + s^3 (2 C_1 C_3 C_4 R_3 R_4 + 2 C_1 C_3 L_3 + 2 C_1 C_4 L_4 + 2 C_3 C_4 L_3 R_4 g_m + 2 C_3 C_4 L_4 R_3 g_m + C_3 C_4 L_4 R_4 g_m) + s^2 (2 C_1 C_3 R_3 + C_1 C_3 R_4 + 2 C_1 C_4 R_4 + 2 C_3 C_4 R_3 R_4 g_m + 2 C_3 L_3 R_4 g_m + C_4 L_4 R_4 g_m) + s (2 C_1 + 2 C_3 R_3 g_m + C_3 R_4 g_m)}$$

$$\mathbf{9.227 \quad X-INVALID-ORDER-227} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 R_3 R_4 g_m s}{C_1 C_3 L_3 R_3 R_4 s^3 + R_3 R_4 g_m + s^2 (2 C_1 L_3 R_3 + C_1 L_3 R_4 + C_3 L_3 R_3 R_4 g_m) + s (C_1 R_3 R_4 + 2 L_3 R_3 g_m + L_3 R_4 g_m)}$$

$$\mathbf{9.228 \quad X-INVALID-ORDER-228} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 R_3 g_m s}{R_3 g_m + s^3 (C_1 C_3 L_3 R_3 + 2 C_1 C_4 L_3 R_3) + s^2 (C_1 L_3 + C_3 L_3 R_3 g_m + 2 C_4 L_3 R_3 g_m) + s (C_1 R_3 + L_3 g_m)}$$

$$\mathbf{9.229 \quad X-INVALID-ORDER-229} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 R_3 R_4 g_m s}{R_3 R_4 g_m + s^3 (C_1 C_3 L_3 R_3 R_4 + 2 C_1 C_4 L_3 R_3 R_4) + s^2 (2 C_1 L_3 R_3 + C_1 L_3 R_4 + C_3 L_3 R_3 R_4 g_m + 2 C_4 L_3 R_3 R_4 g_m) + s (C_1 R_3 R_4 + 2 L_3 R_3 g_m + L_3 R_4 g_m)}$$

$$\mathbf{9.230 \quad X-INVALID-ORDER-230} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 R_3 R_4 g_m s^2 + L_3 R_3 g_m s}{C_1 C_3 C_4 L_3 R_3 R_4 s^4 + R_3 g_m + s^3 (C_1 C_3 L_3 R_3 + 2 C_1 C_4 L_3 R_3 + C_1 C_4 L_3 R_4 + C_3 C_4 L_3 R_3 R_4 g_m) + s^2 (C_1 C_4 R_3 R_4 + C_1 L_3 + C_3 L_3 R_3 g_m + 2 C_4 L_3 R_3 g_m + C_4 L_3 R_4 g_m) + s (C_1 R_3 + C_4 R_3 R_4 g_m + L_3 g_m)}$$

$$\mathbf{9.231 \quad X-INVALID-ORDER-231} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_3 g_m s^3 + L_3 R_3 g_m s}{C_1 C_3 C_4 L_3 L_4 R_3 s^5 + R_3 g_m + s^4 (C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (C_1 C_3 L_3 R_3 + 2 C_1 C_4 L_3 R_3 + C_1 C_4 L_4 R_3 + C_4 L_3 L_4 g_m) + s^2 (C_1 L_3 + C_3 L_3 R_3 g_m + 2 C_4 L_3 R_3 g_m + C_4 L_4 R_3 g_m) + s (C_1 R_3 + L_3 g_m)}$$

$$\mathbf{9.232 \quad X-INVALID-ORDER-232} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 L_4 R_3 g_m s}{2 L_3 R_3 g_m + L_4 R_3 g_m + s^3 (C_1 C_3 L_3 L_4 R_3 + 2 C_1 C_4 L_3 L_4 R_3) + s^2 (C_1 L_3 L_4 + C_3 L_3 L_4 R_3 g_m + 2 C_4 L_3 L_4 R_3 g_m) + s (2 C_1 L_3 R_3 + C_1 L_4 R_3 + L_3 L_4 g_m)}$$

$$\mathbf{9.233 \quad X-INVALID-ORDER-233} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_3 g_m s^3 + C_4 L_3 R_3 R_4 g_m s^2 + L_3 R_3 g_m s}{C_1 C_3 C_4 L_3 L_4 R_3 s^5 + R_3 g_m + s^4 (C_1 C_3 C_4 L_3 R_3 R_4 + C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (C_1 C_3 L_3 R_3 + 2 C_1 C_4 L_3 R_3 + C_1 C_4 L_3 R_4 + C_1 C_4 L_4 R_3 + C_3 C_4 L_3 R_3 R_4 g_m + C_4 L_3 L_4 g_m) + s^2 (C_1 C_4 R_3 R_4 + C_1 L_3 + C_3 L_3 R_3 g_m + 2 C_4 L_3 R_3 g_m + C_4 L_3 R_4 g_m + C_4 L_4 R_3 g_m) + s (C_1 R_3 + L_3 g_m)}$$

$$\mathbf{9.234 \quad X-INVALID-ORDER-234} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 L_4 R_3 R_4 g_m s}{2 L_3 R_3 R_4 g_m + L_4 R_3 R_4 g_m + s^3 (C_1 C_3 L_3 L_4 R_3 R_4 + 2 C_1 C_4 L_3 L_4 R_3 R_4) + s^2 (2 C_1 L_3 L_4 R_3 + C_1 L_3 L_4 R_4 + C_3 L_3 L_4 R_3 R_4 g_m + 2 C_4 L_3 L_4 R_3 R_4 g_m) + s (2 C_1 L_3 R_3 R_4 + C_1 L_4 R_3 R_4 + 2 L_3 L_4 R_3 g_m + L_3 L_4 R_4 g_m)}$$

$$\mathbf{9.235 \quad X-INVALID-ORDER-235} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_3 R_4 g_m s^3 + L_3 L_4 R_3 g_m s^2 + L_3 R_3 R_4 g_m s}{C_1 C_3 C_4 L_3 L_4 R_3 R_4 s^5 + R_3 R_4 g_m + s^4 (C_1 C_3 L_3 L_4 R_3 + 2 C_1 C_4 L_3 L_4 R_3 + C_1 C_4 L_3 L_4 R_4 + C_3 C_4 L_3 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_3 R_4 + C_1 C_4 L_4 R_3 R_4 + C_1 L_3 L_4 + C_3 L_3 L_4 R_3 g_m + 2 C_4 L_3 L_4 R_3 g_m + C_4 L_3 L_4 R_4 g_m) + s^2 (2 C_1 L_3 R_3 + C_1 L_3 R_4 + C_1 L_4 R_3 + C_3 L_3 R_3 R_4 g_m)}$$

$$\mathbf{9.236 \quad X-INVALID-ORDER-236} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_3 R_4 g_m s^3 + L_3 R_3 R_4 g_m s}{C_1 C_3 C_4 L_3 L_4 R_3 R_4 s^5 + R_3 R_4 g_m + s^4 (2 C_1 C_4 L_3 L_4 R_3 + C_1 C_4 L_3 L_4 R_4 + C_3 C_4 L_3 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_3 R_4 + 2 C_1 C_4 L_3 R_3 R_4 + C_1 C_4 L_4 R_3 R_4 + 2 C_4 L_3 L_4 R_3 g_m + C_4 L_3 L_4 R_4 g_m) + s^2 (2 C_1 L_3 R_3 + C_1 L_3 R_4 + C_3 L_3 R_3 R_4 g_m + 2 C_4 L_3 R_3 R_4 g_m + C_4 L_4 R_3 R_4 g_m)}$$

$$\mathbf{9.237 \quad X-INVALID-ORDER-237} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 R_4 g_m s^2 + L_3 R_4 g_m s + R_3 R_4 g_m}{2 R_3 g_m + R_4 g_m + s^3 (2 C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4) + s^2 (2 C_1 L_3 + 2 C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m) + s (2 C_1 R_3 + C_1 R_4 + 2 L_3 g_m)}$$

$$\mathbf{9.238 \quad X-INVALID-ORDER-238} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 g_m s^2 + L_3 g_m s + R_3 g_m}{2 C_1 C_3 C_4 L_3 R_3 s^4 + g_m + s^3 (C_1 C_3 L_3 + 2 C_1 C_4 L_3 + 2 C_3 C_4 L_3 R_3 g_m) + s^2 (2 C_1 C_4 R_3 + C_3 L_3 g_m + 2 C_4 L_3 g_m) + s (C_1 + 2 C_4 R_3 g_m)}$$

$$\mathbf{9.239 \quad X-INVALID-ORDER-239} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 R_4 g_m s^2 + L_3 R_4 g_m s + R_3 R_4 g_m}{2 C_1 C_3 C_4 L_3 R_3 R_4 s^4 + 2 R_3 g_m + R_4 g_m + s^3 (2 C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4 + 2 C_1 C_4 L_3 R_4 + 2 C_3 C_4 L_3 R_3 R_4 g_m) + s^2 (2 C_1 C_4 R_3 R_4 + 2 C_1 L_3 + 2 C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m + 2 C_4 L_3 R_4 g_m) + s (2 C_1 R_3 + C_1 R_4 + 2 C_4 R_3 R_4 g_m + 2 L_3 g_m)}$$

$$\mathbf{9.240 \quad X-INVALID-ORDER-240} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 R_3 R_4 g_m s^3 + R_3 g_m + s^2 (C_3 L_3 R_3 g_m + C_4 L_3 R_4 g_m) + s (C_4 R_3 R_4 g_m + L_3 g_m)}{g_m + s^4 (2 C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4) + s^3 (C_1 C_3 L_3 + 2 C_1 C_4 L_3 + 2 C_3 C_4 L_3 R_3 g_m + C_3 C_4 L_3 R_4 g_m) + s^2 (2 C_1 C_4 R_3 + C_1 C_4 R_4 + C_3 L_3 g_m + 2 C_4 L_3 g_m) + s (C_1 + 2 C_4 R_3 g_m + C_4 R_4 g_m)}$$

$$\mathbf{9.241 \quad X-INVALID-ORDER-241} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_3 g_m s^4 + C_4 L_3 L_4 g_m s^3 + L_3 g_m s + R_3 g_m + s^2 (C_3 L_3 R_3 g_m + C_4 L_4 R_3 g_m)}{C_1 C_3 C_4 L_3 L_4 s^5 + g_m + s^4 (2 C_1 C_3 C_4 L_3 R_3 + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 + 2 C_1 C_4 L_3 + C_1 C_4 L_4 + 2 C_3 C_4 L_3 R_3 g_m) + s^2 (2 C_1 C_4 R_3 + C_3 L_3 g_m + 2 C_4 L_3 g_m + C_4 L_4 g_m) + s (C_1 + 2 C_4 R_3 g_m)}$$

$$\mathbf{9.242 \quad X-INVALID-ORDER-242} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_3 g_m s^3 + L_3 L_4 g_m s^2 + L_4 R_3 g_m s}{2 C_1 C_3 C_4 L_3 L_4 R_3 s^5 + 2 R_3 g_m + s^4 (C_1 C_3 L_3 L_4 + 2 C_1 C_4 L_3 L_4 + 2 C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (2 C_1 C_3 L_3 R_3 + 2 C_1 C_4 L_4 R_3 + C_3 L_3 L_4 g_m + 2 C_4 L_3 L_4 g_m) + s^2 (2 C_1 L_3 + C_1 L_4 + 2 C_3 L_3 R_3 g_m + 2 C_4 L_4 R_3 g_m) + s (2 C_1 R_3 + 2 L_3 g_m + L_4 g_m)}$$

$$\mathbf{9.243 \quad X-INVALID-ORDER-243} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_3 g_m s^4 + R_3 g_m + s^3 (C_3 C_4 L_3 R_3 R_4 g_m + C_4 L_3 L_4 g_m) + s^2 (C_3 L_3 R_3 g_m + C_4 L_3 R_4 g_m + C_4 L_4 R_3 g_m) + s (C_4 R_3 R_4 g_m + L_3 g_m)}{C_1 C_3 C_4 L_3 L_4 s^5 + g_m + s^4 (2 C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 + 2 C_1 C_4 L_3 + C_1 C_4 L_4 + 2 C_3 C_4 L_3 R_3 g_m + C_3 C_4 L_3 R_4 g_m) + s^2 (2 C_1 C_4 R_3 + C_1 C_4 R_4 + C_3 L_3 g_m + 2 C_4 L_3 g_m + C_4 L_4 g_m) + s (C_1 + 2 C_4 R_3 g_m + C_4 R_4 g_m)}$$

$$\mathbf{9.244 \quad X-INVALID-ORDER-244} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_3 R_4 g_m s^3 + L_3 L_4 R_4 g_m s^2 + L_4 R_3 R_4 g_m s}{2C_1 C_3 C_4 L_3 L_4 R_3 R_4 s^5 + 2R_3 R_4 g_m + s^4 (2C_1 C_3 L_3 L_4 R_3 + C_1 C_3 L_3 L_4 R_4 + 2C_1 C_4 L_3 L_4 R_4 + 2C_3 C_4 L_3 L_4 R_3 R_4 g_m) + s^3 (2C_1 C_3 L_3 R_3 R_4 + 2C_1 C_4 L_4 R_3 R_4 + 2C_1 L_3 L_4 + 2C_3 L_3 L_4 R_3 g_m + C_3 L_3 L_4 R_4 g_m + 2C_4 L_3 L_4 R_4 g_m) + s^2 (2C_1 L_3 R_4 + 2C_1 L_4 R_3 + C_1 L_4 R_4 + 2C_3 L_3 R_3 R_4 g_m + C_3 L_3 R_4 R_3 g_m + C_4 L_3 R_3 R_4 g_m) + s (L_3 R_4 g_m + L_4 R_3 g_m)}$$

$$\mathbf{9.245 \quad X-INVALID-ORDER-245} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_3 R_4 g_m s^4 + R_3 R_4 g_m + s^3 (C_3 L_3 L_4 R_3 g_m + C_4 L_3 L_4 R_4 g_m) + s^2 (C_3 L_3 R_3 R_4 g_m + C_4 L_4 R_3 R_4 g_m + L_3 L_4 g_m) + s (L_3 R_4 g_m + L_4 R_3 g_m)}{2R_3 g_m + R_4 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (C_1 C_3 L_3 L_4 + 2C_1 C_4 L_3 L_4 + 2C_3 C_4 L_3 L_4 R_3 g_m + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (2C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4 + 2C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + C_3 L_3 L_4 g_m + 2C_4 L_3 L_4 g_m) + s^2 (2C_1 L_3 + C_1 L_4 + 2C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m + C_4 L_3 R_3 g_m) + s (L_3 R_4 g_m + L_4 R_3 g_m)}$$

$$\mathbf{9.246 \quad X-INVALID-ORDER-246} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_3 R_4 g_m s^4 + C_4 L_3 L_4 R_4 g_m s^3 + L_3 R_4 g_m s + R_3 R_4 g_m + s^2 (C_3 L_3 R_3 R_4 g_m + C_4 L_4 R_3 R_4 g_m)}{2R_3 g_m + R_4 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (2C_1 C_3 C_4 L_3 R_3 R_4 + 2C_1 C_4 L_3 L_4 + 2C_3 C_4 L_3 L_4 R_3 g_m + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (2C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4 + 2C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + 2C_3 C_4 L_3 R_3 R_4 g_m + 2C_4 L_3 L_4 g_m) + s^2 (2C_1 L_3 + C_1 L_4 + 2C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m + C_4 L_3 R_3 g_m) + s (L_3 R_4 g_m + L_4 R_3 g_m)}$$

$$\mathbf{9.247 \quad X-INVALID-ORDER-247} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 R_4 g_m s^2 + R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s^3 (2C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4) + s^2 (C_1 C_3 R_3 R_4 + 2C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m) + s (2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m)}$$

$$\mathbf{9.248 \quad X-INVALID-ORDER-248} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 g_m s^2 + R_3 g_m}{2C_1 C_3 C_4 L_3 R_3 s^4 + g_m + s^3 (C_1 C_3 L_3 + 2C_3 C_4 L_3 R_3 g_m) + s^2 (C_1 C_3 R_3 + 2C_1 C_4 R_3 + C_3 L_3 g_m) + s (C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m)}$$

$$\mathbf{9.249 \quad X-INVALID-ORDER-249} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 R_4 g_m s^2 + R_3 R_4 g_m}{2C_1 C_3 C_4 L_3 R_3 R_4 s^4 + 2R_3 g_m + R_4 g_m + s^3 (2C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4 + 2C_3 C_4 L_3 R_3 R_4 g_m) + s^2 (C_1 C_3 R_3 R_4 + 2C_1 C_4 R_3 R_4 + 2C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m) + s (2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m)}$$

$$\mathbf{9.250 \quad X-INVALID-ORDER-250} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 R_3 R_4 g_m s^3 + C_3 L_3 R_3 g_m s^2 + C_4 R_3 R_4 g_m s + R_3 g_m}{g_m + s^4 (2C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4) + s^3 (C_1 C_3 C_4 R_3 R_4 + C_1 C_3 L_3 + 2C_3 C_4 L_3 R_3 g_m + C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 C_3 R_3 + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_3 C_4 R_3 R_4 g_m + C_3 L_3 g_m) + s (C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

$$\mathbf{9.251 \quad X-INVALID-ORDER-251} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_3 g_m s^4 + R_3 g_m + s^2 (C_3 L_3 R_3 g_m + C_4 L_4 R_3 g_m)}{C_1 C_3 C_4 L_3 L_4 s^5 + g_m + s^4 (2C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_4 R_3 + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 + C_1 C_4 L_4 + 2C_3 C_4 L_3 R_3 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_3 + 2C_1 C_4 R_3 + C_3 L_3 g_m + C_4 L_4 g_m) + s (C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m)}$$

$$\mathbf{9.252 \quad X-INVALID-ORDER-252} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_3 g_m s^3 + L_4 R_3 g_m s}{2C_1 C_3 C_4 L_3 L_4 R_3 s^5 + 2R_3 g_m + s^4 (C_1 C_3 L_3 L_4 + 2C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (2C_1 C_3 L_3 R_3 + C_1 C_3 L_4 R_3 + 2C_1 C_4 L_4 R_3 + C_3 L_3 L_4 g_m) + s^2 (C_1 L_4 + 2C_3 L_3 R_3 g_m + C_3 L_4 R_3 g_m + 2C_4 L_4 R_3 g_m) + s (2C_1 R_3 + L_4 g_m)}$$

$$\mathbf{9.253 \quad X-INVALID-ORDER-253} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_3 g_m s^4 + C_3 C_4 L_3 R_3 R_4 g_m s^3 + C_4 R_3 R_4 g_m s + R_3 g_m + s^2 (C_3 L_3 R_3 g_m + C_4 L_4 R_3 g_m)}{C_1 C_3 C_4 L_3 L_4 s^5 + g_m + s^4 (2C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_1 C_3 C_4 L_4 R_3 + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 C_4 R_3 R_4 + C_1 C_3 L_3 + C_1 C_4 L_4 + 2C_3 C_4 L_3 R_3 g_m + C_3 C_4 L_3 R_4 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_3 + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_3 C_4 R_3 R_4 g_m + C_3 L_3 g_m + C_4 L_4 g_m)}$$

$$\mathbf{9.254 \quad X-INVALID-ORDER-254} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_3 R_4 g_m s^3 + L_4 R_3 R_4 g_m s}{2C_1 C_3 C_4 L_3 L_4 R_3 R_4 s^5 + 2R_3 R_4 g_m + s^4 (2C_1 C_3 L_3 L_4 R_3 + C_1 C_3 L_3 L_4 R_4 + 2C_3 C_4 L_3 L_4 R_3 R_4 g_m) + s^3 (2C_1 C_3 L_3 R_3 R_4 + C_1 C_3 L_4 R_3 R_4 + 2C_1 C_4 L_4 R_3 R_4 + 2C_3 L_3 L_4 R_3 g_m + C_3 L_3 L_4 R_4 g_m) + s^2 (2C_1 L_4 R_3 + C_1 L_4 R_4 + 2C_3 L_3 R_3 R_4 g_m + C_3 L_4 R_3 R_4 g_m + 2C_4 L_4 R_3 R_4 g_m)}$$

$$\mathbf{9.255 \quad X-INVALID-ORDER-255} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_3 R_4 g_m s^4 + C_3 L_3 L_4 R_3 g_m s^3 + L_4 R_3 g_m s + R_3 R_4 g_m + s^2 (C_3 L_3 R_3 R_4 g_m + C_4 L_4 R_3 R_4 g_m)}{2R_3 g_m + R_4 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (C_1 C_3 C_4 L_4 R_3 R_4 + C_1 C_3 L_3 L_4 + 2C_3 C_4 L_3 L_4 R_3 g_m + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (2C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4 + C_1 C_3 L_4 R_3 + 2C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + C_3 C_4 L_4 R_3 R_4 g_m + C_3 L_3 L_4 g_m) + s^2 (C_1 C_3 R_3 R_4)}$$

$$\mathbf{9.256 \quad X-INVALID-ORDER-256} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_3 R_4 g_m s^4 + R_3 R_4 g_m + s^2 (C_3 L_3 R_3 R_4 g_m + C_4 L_4 R_3 R_4 g_m)}{2R_3 g_m + R_4 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (2C_1 C_3 C_4 L_3 R_3 R_4 + C_1 C_3 C_4 L_4 R_3 R_4 + 2C_3 C_4 L_3 L_4 R_3 g_m + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (2C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4 + 2C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + 2C_3 C_4 L_3 R_3 R_4 g_m + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (C_1 C_3 R_3 R_4)}$$

$$\mathbf{9.257 \quad X-INVALID-ORDER-257} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s (2C_1 R_1 R_3 + C_1 R_1 R_4)}$$

$$\mathbf{9.258 \quad X-INVALID-ORDER-258} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_3 g_m s^2 + R_1 R_3 g_m}{C_1 C_4 L_4 R_1 s^3 + R_1 g_m + s^2 (2C_1 C_4 R_1 R_3 + C_4 L_4 R_1 g_m + C_4 L_4) + s (C_1 R_1 + 2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

$$\mathbf{9.259 \quad X-INVALID-ORDER-259} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_4 R_1 R_3 g_m s}{2C_1 C_4 L_4 R_1 R_3 s^3 + 2R_1 R_3 g_m + 2R_3 + s^2 (C_1 L_4 R_1 + 2C_4 L_4 R_1 R_3 g_m + 2C_4 L_4 R_3) + s (2C_1 R_1 R_3 + L_4 R_1 g_m + L_4)}$$

$$\mathbf{9.260 \quad X-INVALID-ORDER-260} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_3 g_m s^2 + C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{C_1 C_4 L_4 R_1 s^3 + R_1 g_m + s^2 (2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 + C_4 L_4 R_1 g_m + C_4 L_4) + s (C_1 R_1 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4) + 1}$$

$$\mathbf{9.261 \quad X-INVALID-ORDER-261} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_4 R_1 R_3 R_4 g_m s}{2C_1 C_4 L_4 R_1 R_3 R_4 s^3 + 2R_1 R_3 R_4 g_m + 2R_3 R_4 + s^2 (2C_1 L_4 R_1 R_3 + C_1 L_4 R_1 R_4 + 2C_4 L_4 R_1 R_3 R_4 g_m + 2C_4 L_4 R_3 R_4) + s (2C_1 R_1 R_3 R_4 + 2L_4 R_1 R_3 g_m + L_4 R_1 R_4 g_m + 2L_4 R_3 + L_4 R_4)}$$

$$\mathbf{9.262 \quad X-INVALID-ORDER-262} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_3 R_4 g_m s^2 + L_4 R_1 R_3 g_m s + R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^3 (2 C_1 C_4 L_4 R_1 R_3 + C_1 C_4 L_4 R_1 R_4) + s^2 (C_1 L_4 R_1 + 2 C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_3 + C_4 L_4 R_4) + s (2 C_1 R_1 R_3 + C_1 R_1 R_4 + L_4 R_1 g_m + L_4)}$$

$$\mathbf{9.263 \quad X-INVALID-ORDER-263} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_3 R_4 g_m s^2 + R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^3 (2 C_1 C_4 L_4 R_1 R_3 + C_1 C_4 L_4 R_1 R_4) + s^2 (2 C_1 C_4 R_1 R_3 R_4 + 2 C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_3 + C_4 L_4 R_4) + s (2 C_1 R_1 R_3 + C_1 R_1 R_4 + 2 C_4 R_1 R_3 R_4 g_m + 2 C_4 R_3 R_4)}$$

$$\mathbf{9.264 \quad X-INVALID-ORDER-264} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{R_1 g_m}{s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.265 \quad X-INVALID-ORDER-265} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 R_1 R_4 g_m s + R_1 g_m}{C_1 C_3 C_4 R_1 R_4 s^3 + s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1 + C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.266 \quad X-INVALID-ORDER-266} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 g_m s^2 + R_1 g_m}{C_1 C_3 C_4 L_4 R_1 s^4 + s^3 (C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.267 \quad X-INVALID-ORDER-267} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{L_4 R_1 g_m s}{2 C_1 R_1 s + 2 R_1 g_m + s^3 (C_1 C_3 L_4 R_1 + 2 C_1 C_4 L_4 R_1) + s^2 (C_3 L_4 R_1 g_m + C_3 L_4 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4) + 2}$$

$$\mathbf{9.268 \quad X-INVALID-ORDER-268} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 g_m s^2 + C_4 R_1 R_4 g_m s + R_1 g_m}{C_1 C_3 C_4 L_4 R_1 s^4 + s^3 (C_1 C_3 C_4 R_1 R_4 + C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1 + C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.269 \quad X-INVALID-ORDER-269} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$$

$$H(s) = \frac{L_4 R_1 R_4 g_m s}{2 R_1 R_4 g_m + 2 R_4 + s^3 (C_1 C_3 L_4 R_1 R_4 + 2 C_1 C_4 L_4 R_1 R_4) + s^2 (2 C_1 L_4 R_1 + C_3 L_4 R_1 R_4 g_m + C_3 L_4 R_4 + 2 C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_4) + s (2 C_1 R_1 R_4 + 2 L_4 R_1 g_m + 2 L_4)}$$

$$\mathbf{9.270 \quad X-INVALID-ORDER-270} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_4 g_m s^2 + L_4 R_1 g_m s + R_1 R_4 g_m}{C_1 C_3 C_4 L_4 R_1 R_4 s^4 + 2 R_1 g_m + s^3 (C_1 C_3 L_4 R_1 + 2 C_1 C_4 L_4 R_1 + C_3 C_4 L_4 R_1 R_4 g_m + C_3 C_4 L_4 R_4) + s^2 (C_1 C_3 R_1 R_4 + C_3 L_4 R_1 g_m + C_3 L_4 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4) + s (2 C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4) + 2}$$

$$\mathbf{9.271 \quad X-INVALID-ORDER-271} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{1}{C_3 s}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_4 g_m s^2 + R_1 R_4 g_m}{C_1 C_3 C_4 L_4 R_1 R_4 s^4 + 2 R_1 g_m + s^3 (2 C_1 C_4 L_4 R_1 + C_3 C_4 L_4 R_1 R_4 g_m + C_3 C_4 L_4 R_4) + s^2 (C_1 C_3 R_1 R_4 + 2 C_1 C_4 R_1 R_4 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4) + s (2 C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_4) + 2}$$

$$\mathbf{9.272 \quad X-INVALID-ORDER-272} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{C_1 C_3 C_4 R_1 R_3 R_4 s^3 + R_1 g_m + s^2 (C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 + C_3 C_4 R_1 R_3 R_4 g_m + C_3 C_4 R_3 R_4) + s (C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4) + 1}$$

$$\mathbf{9.273 \quad X-INVALID-ORDER-273} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_3 g_m s^2 + R_1 R_3 g_m}{C_1 C_3 C_4 L_4 R_1 R_3 s^4 + R_1 g_m + s^3 (C_1 C_4 L_4 R_1 + C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_4 R_3) + s^2 (C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_4 L_4 R_1 g_m + C_4 L_4) + s (C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

$$\mathbf{9.274 \quad X-INVALID-ORDER-274} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_4 R_1 R_3 g_m s}{2R_1 R_3 g_m + 2R_3 + s^3 (C_1 C_3 L_4 R_1 R_3 + 2C_1 C_4 L_4 R_1 R_3) + s^2 (C_1 L_4 R_1 + C_3 L_4 R_1 R_3 g_m + C_3 L_4 R_3 + 2C_4 L_4 R_1 R_3 g_m + 2C_4 L_4 R_3) + s (2C_1 R_1 R_3 + L_4 R_1 g_m + L_4)}$$

$$\mathbf{9.275 \quad X-INVALID-ORDER-275} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_3 g_m s^2 + C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{C_1 C_3 C_4 L_4 R_1 R_3 s^4 + R_1 g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 + C_1 C_4 L_4 R_1 + C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_4 R_3) + s^2 (C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 + C_3 C_4 R_1 R_3 R_4 g_m + C_3 C_4 R_3 R_4 + C_4 L_4 R_1 g_m + C_4 L_4) + s (C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m)}$$

$$\mathbf{9.276 \quad X-INVALID-ORDER-276} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_4 R_1 R_3 R_4 g_m s}{2R_1 R_3 R_4 g_m + 2R_3 R_4 + s^3 (C_1 C_3 L_4 R_1 R_3 R_4 + 2C_1 C_4 L_4 R_1 R_3 R_4) + s^2 (2C_1 L_4 R_1 R_3 + C_1 L_4 R_1 R_4 + C_3 L_4 R_1 R_3 R_4 g_m + C_3 L_4 R_3 R_4 + 2C_4 L_4 R_1 R_3 R_4 g_m + 2C_4 L_4 R_3 R_4) + s (2C_1 R_1 R_3 R_4 + 2L_4 R_1 R_3 g_m + L_4 R_1 R_4 g_m + 2L_4 R_3 + L_4 R_4)}$$

$$\mathbf{9.277 \quad X-INVALID-ORDER-277} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_3 R_4 g_m s^2 + L_4 R_1 R_3 g_m s + R_1 R_3 R_4 g_m}{C_1 C_3 C_4 L_4 R_1 R_3 R_4 s^4 + 2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^3 (C_1 C_3 L_4 R_1 R_3 + 2C_1 C_4 L_4 R_1 R_3 + C_1 C_4 L_4 R_1 R_4 + C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_3 C_4 L_4 R_3 R_4) + s^2 (C_1 C_3 R_1 R_3 R_4 + C_1 L_4 R_1 + C_3 L_4 R_1 R_3 g_m + C_3 L_4 R_3 + 2C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_3 + C_4 L_4 R_4) + s (2C_1 R_1 R_3 R_4 + 2L_4 R_1 R_3 g_m + L_4 R_1 R_4 g_m)}$$

$$\mathbf{9.278 \quad X-INVALID-ORDER-278} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 R_3 R_4 g_m s^2 + R_1 R_3 R_4 g_m}{C_1 C_3 C_4 L_4 R_1 R_3 R_4 s^4 + 2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^3 (2C_1 C_4 L_4 R_1 R_3 + C_1 C_4 L_4 R_1 R_4 + C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_3 C_4 L_4 R_3 R_4) + s^2 (C_1 C_3 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4 + 2C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_3 + C_4 L_4 R_4) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m)}$$

$$\mathbf{9.279 \quad X-INVALID-ORDER-279} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 R_1 R_3 g_m s + R_1 g_m}{2C_1 C_3 C_4 R_1 R_3 s^3 + s^2 (C_1 C_3 R_1 + 2C_1 C_4 R_1 + 2C_3 C_4 R_1 R_3 g_m + 2C_3 C_4 R_3) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.280 \quad X-INVALID-ORDER-280} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 R_1 R_3 R_4 g_m s + R_1 R_4 g_m}{2C_1 C_3 C_4 R_1 R_3 R_4 s^3 + 2R_1 g_m + s^2 (2C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 + 2C_1 C_4 R_1 R_4 + 2C_3 C_4 R_1 R_3 R_4 g_m + 2C_3 C_4 R_3 R_4) + s (2C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4) + 2}$$

$$\mathbf{9.281 \quad X-INVALID-ORDER-281} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 R_1 R_3 R_4 g_m s^2 + R_1 g_m + s (C_3 R_1 R_3 g_m + C_4 R_1 R_4 g_m)}{s^3 (2C_1 C_3 C_4 R_1 R_3 + C_1 C_3 C_4 R_1 R_4) + s^2 (C_1 C_3 R_1 + 2C_1 C_4 R_1 + 2C_3 C_4 R_1 R_3 g_m + C_3 C_4 R_1 R_4 g_m + 2C_3 C_4 R_3 + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.282 \quad X-INVALID-ORDER-282} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_1 R_3 g_m s^3 + C_3 R_1 R_3 g_m s + C_4 L_4 R_1 g_m s^2 + R_1 g_m}{C_1 C_3 C_4 L_4 R_1 s^4 + s^3 (2C_1 C_3 C_4 R_1 R_3 + C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (C_1 C_3 R_1 + 2C_1 C_4 R_1 + 2C_3 C_4 R_1 R_3 g_m + 2C_3 C_4 R_3) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.283 \quad X-INVALID-ORDER-283} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_4 R_1 R_3 g_m s^2 + L_4 R_1 g_m s}{2C_1 C_3 C_4 L_4 R_1 R_3 s^4 + 2R_1 g_m + s^3 (C_1 C_3 L_4 R_1 + 2C_1 C_4 L_4 R_1 + 2C_3 C_4 L_4 R_1 R_3 g_m + 2C_3 C_4 L_4 R_3) + s^2 (2C_1 C_3 R_1 R_3 + C_3 L_4 R_1 g_m + C_3 L_4 + 2C_4 L_4 R_1 g_m + 2C_4 L_4) + s (2C_1 R_1 + 2C_3 R_1 R_3 g_m + 2C_3 R_3) + 2}$$

$$\mathbf{9.284 \quad X-INVALID-ORDER-284} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_1 R_3 g_m s^3 + R_1 g_m + s^2 (C_3 C_4 R_1 R_3 R_4 g_m + C_4 L_4 R_1 g_m) + s (C_3 R_1 R_3 g_m + C_4 R_1 R_4 g_m)}{C_1 C_3 C_4 L_4 R_1 s^4 + s^3 (2C_1 C_3 C_4 R_1 R_3 + C_1 C_3 C_4 R_1 R_4 + C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (C_1 C_3 R_1 + 2C_1 C_4 R_1 + 2C_3 C_4 R_1 R_3 g_m + C_3 C_4 R_1 R_4 g_m + 2C_3 C_4 R_3 + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.285 \quad X-INVALID-ORDER-285} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_4 R_1 R_3 R_4 g_m s^2 + L_4 R_1 R_4 g_m s}{2C_1 C_3 C_4 L_4 R_1 R_3 R_4 s^4 + 2R_1 R_4 g_m + 2R_4 + s^3 (2C_1 C_3 L_4 R_1 R_3 + C_1 C_3 L_4 R_1 R_4 + 2C_1 C_4 L_4 R_1 R_4 + 2C_3 C_4 L_4 R_1 R_3 R_4 g_m + 2C_3 C_4 L_4 R_3 R_4) + s^2 (2C_1 C_3 R_1 R_3 R_4 + 2C_1 L_4 R_1 + 2C_3 L_4 R_1 R_3 g_m + C_3 L_4 R_1 R_4 g_m + 2C_3 L_4 R_3 + C_3 L_4 R_4 + 2C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_4) + s (2C_1 R_1 + 2C_3 R_1 R_3 g_m + 2C_3 R_3) + 2}$$

$$\mathbf{9.286 \quad X-INVALID-ORDER-286} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_1 R_3 R_4 g_m s^3 + R_1 R_4 g_m + s^2 (C_3 L_4 R_1 R_3 g_m + C_4 L_4 R_1 R_4 g_m) + s (C_3 R_1 R_3 R_4 g_m + L_4 R_1 g_m)}{2R_1 g_m + s^4 (2C_1 C_3 C_4 L_4 R_1 R_3 + C_1 C_3 C_4 L_4 R_1 R_4) + s^3 (C_1 C_3 L_4 R_1 + 2C_1 C_4 L_4 R_1 + 2C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_4 R_1 R_4 g_m + 2C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4) + s^2 (2C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 + C_3 L_4 R_1 g_m + C_3 L_4 + 2C_4 L_4 R_1 g_m + 2C_4 L_4) + s (2C_1 R_1 + 2C_3 R_1 R_3 g_m + 2C_3 R_3) + 2}$$

$$\mathbf{9.287 \quad X-INVALID-ORDER-287} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_1 R_3 R_4 g_m s^3 + C_3 R_1 R_3 R_4 g_m s + C_4 L_4 R_1 R_4 g_m s^2 + R_1 R_4 g_m}{2R_1 g_m + s^4 (2C_1 C_3 C_4 L_4 R_1 R_3 + C_1 C_3 C_4 L_4 R_1 R_4) + s^3 (2C_1 C_3 C_4 R_1 R_3 R_4 + 2C_1 C_4 L_4 R_1 + 2C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_4 R_1 R_4 g_m + 2C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4) + s^2 (2C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 + 2C_1 C_4 R_1 R_4 + 2C_3 C_4 R_1 R_3 R_4 g_m + 2C_3 C_4 R_3 R_4 + 2C_4 L_4 R_1 g_m + 2C_4 L_4) + s (2C_1 R_1 + 2C_3 R_1 R_3 g_m + 2C_3 R_3) + 2}$$

$$\mathbf{9.288 \quad X-INVALID-ORDER-288} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_4 g_m s^2 + R_1 R_4 g_m}{2C_1 C_3 L_3 R_1 s^3 + 2R_1 g_m + s^2 (C_1 C_3 R_1 R_4 + 2C_3 L_3 R_1 g_m + 2C_3 L_3) + s (2C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4) + 2}$$

$$\mathbf{9.289 \quad X-INVALID-ORDER-289} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 g_m s^2 + R_1 g_m}{2C_1 C_3 C_4 L_3 R_1 s^4 + s^3 (2C_3 C_4 L_3 R_1 g_m + 2C_3 C_4 L_3) + s^2 (C_1 C_3 R_1 + 2C_1 C_4 R_1) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.290 \quad X-INVALID-ORDER-290} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_4 g_m s^2 + R_1 R_4 g_m}{2C_1 C_3 C_4 L_3 R_1 R_4 s^4 + 2R_1 g_m + s^3 (2C_1 C_3 L_3 R_1 + 2C_3 C_4 L_3 R_1 R_4 g_m + 2C_3 C_4 L_3 R_4) + s^2 (C_1 C_3 R_1 R_4 + 2C_1 C_4 R_1 R_4 + 2C_3 L_3 R_1 g_m + 2C_3 L_3) + s (2C_1 R_1 + C_3 R_1 R_4 g_m + C_3 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4) + 2}$$

$$\mathbf{9.291 \quad X-INVALID-ORDER-291} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 R_1 R_4 g_m s^3 + C_3 L_3 R_1 g_m s^2 + C_4 R_1 R_4 g_m s + R_1 g_m}{2 C_1 C_3 C_4 L_3 R_1 s^4 + s^3 (C_1 C_3 C_4 R_1 R_4 + 2 C_3 C_4 L_3 R_1 g_m + 2 C_3 C_4 L_3) + s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1 + C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.292 \quad X-INVALID-ORDER-292} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 g_m s^4 + R_1 g_m + s^2 (C_3 L_3 R_1 g_m + C_4 L_4 R_1 g_m)}{s^4 (2 C_1 C_3 C_4 L_3 R_1 + C_1 C_3 C_4 L_4 R_1) + s^3 (2 C_3 C_4 L_3 R_1 g_m + 2 C_3 C_4 L_3 + C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.293 \quad X-INVALID-ORDER-293} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_1 g_m s^3 + L_4 R_1 g_m s}{2 C_1 C_3 C_4 L_3 L_4 R_1 s^5 + 2 C_1 R_1 s + 2 R_1 g_m + s^4 (2 C_3 C_4 L_3 L_4 R_1 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (2 C_1 C_3 L_3 R_1 + C_1 C_3 L_4 R_1 + 2 C_1 C_4 L_4 R_1) + s^2 (2 C_3 L_3 R_1 g_m + 2 C_3 L_3 + C_3 L_4 R_1 g_m + C_3 L_4 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4) + 2}$$

$$\mathbf{9.294 \quad X-INVALID-ORDER-294} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 g_m s^4 + C_3 C_4 L_3 R_1 R_4 g_m s^3 + C_4 R_1 R_4 g_m s + R_1 g_m + s^2 (C_3 L_3 R_1 g_m + C_4 L_4 R_1 g_m)}{s^4 (2 C_1 C_3 C_4 L_3 R_1 + C_1 C_3 C_4 L_4 R_1) + s^3 (C_1 C_3 C_4 R_1 R_4 + 2 C_3 C_4 L_3 R_1 g_m + 2 C_3 C_4 L_3 + C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1 + C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.295 \quad X-INVALID-ORDER-295} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_1 R_4 g_m s^3 + L_4 R_1 R_4 g_m s}{2 C_1 C_3 C_4 L_3 L_4 R_1 R_4 s^5 + 2 R_1 R_4 g_m + 2 R_4 + s^4 (2 C_1 C_3 L_3 L_4 R_1 + 2 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2 C_3 C_4 L_3 L_4 R_4) + s^3 (2 C_1 C_3 L_3 R_1 R_4 + C_1 C_3 L_4 R_1 R_4 + 2 C_1 C_4 L_4 R_1 R_4 + 2 C_3 L_3 L_4 R_1 g_m + 2 C_3 L_3 L_4) + s^2 (2 C_1 L_4 R_1 + 2 C_3 L_3 R_1 R_4 g_m + 2 C_3 L_3 R_4 + C_3 L_4 R_1 R_4 g_m + C_3 L_4 R_4) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.296 \quad X-INVALID-ORDER-296} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_4 g_m s^4 + C_3 L_3 L_4 R_1 g_m s^3 + L_4 R_1 g_m s + R_1 R_4 g_m + s^2 (C_3 L_3 R_1 R_4 g_m + C_4 L_4 R_1 R_4 g_m)}{2 C_1 C_3 C_4 L_3 L_4 R_1 s^5 + 2 R_1 g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_4 + 2 C_3 C_4 L_3 L_4 R_1 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (2 C_1 C_3 L_3 R_1 + C_1 C_3 L_4 R_1 + 2 C_1 C_4 L_4 R_1 + C_3 C_4 L_4 R_1 R_4 g_m + C_3 C_4 L_4 R_4) + s^2 (C_1 C_3 R_1 R_4 + 2 C_3 L_3 R_1 g_m + 2 C_3 L_3 + C_3 L_4 R_1 g_m + C_3 L_4 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.297 \quad X-INVALID-ORDER-297} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_4 g_m s^4 + R_1 R_4 g_m + s^2 (C_3 L_3 R_1 R_4 g_m + C_4 L_4 R_1 R_4 g_m)}{2 C_1 C_3 C_4 L_3 L_4 R_1 s^5 + 2 R_1 g_m + s^4 (2 C_1 C_3 C_4 L_3 R_1 R_4 + C_1 C_3 C_4 L_4 R_1 R_4 + 2 C_3 C_4 L_3 L_4 R_1 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (2 C_1 C_3 L_3 R_1 + 2 C_1 C_4 L_4 R_1 + 2 C_3 C_4 L_3 R_1 R_4 g_m + 2 C_3 C_4 L_3 R_4 + C_3 C_4 L_4 R_1 R_4 g_m + C_3 C_4 L_4 R_4) + s^2 (C_1 C_3 R_1 R_4 + 2 C_1 C_4 R_1 R_4 + 2 C_3 L_3 R_1 g_m + 2 C_3 L_3 + C_3 L_4 R_1 g_m + C_3 L_4 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.298 \quad X-INVALID-ORDER-298} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, R_4, \infty, \infty \right)$$

$$H(s) = \frac{L_3 R_1 R_4 g_m s}{C_1 C_3 L_3 R_1 R_4 s^3 + R_1 R_4 g_m + R_4 + s^2 (2 C_1 L_3 R_1 + C_3 L_3 R_1 R_4 g_m + C_3 L_3 R_4) + s (C_1 R_1 R_4 + 2 L_3 R_1 g_m + 2 L_3)}$$

$$\mathbf{9.299 \quad X-INVALID-ORDER-299} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{L_3 R_1 g_m s}{C_1 R_1 s + R_1 g_m + s^3 (C_1 C_3 L_3 R_1 + 2 C_1 C_4 L_3 R_1) + s^2 (C_3 L_3 R_1 g_m + C_3 L_3 + 2 C_4 L_3 R_1 g_m + 2 C_4 L_3) + 1}$$

$$\mathbf{9.300 \quad X-INVALID-ORDER-300} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{L_3 R_1 R_4 g_m s}{R_1 R_4 g_m + R_4 + s^3 (C_1 C_3 L_3 R_1 R_4 + 2 C_1 C_4 L_3 R_1 R_4) + s^2 (2 C_1 L_3 R_1 + C_3 L_3 R_1 R_4 g_m + C_3 L_3 R_4 + 2 C_4 L_3 R_1 R_4 g_m + 2 C_4 L_3 R_4) + s (C_1 R_1 R_4 + 2 L_3 R_1 g_m + 2 L_3)}$$

$$\mathbf{9.301 \quad X-INVALID-ORDER-301} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 R_1 R_4 g_m s^2 + L_3 R_1 g_m s}{C_1 C_3 C_4 L_3 R_1 R_4 s^4 + R_1 g_m + s^3 (C_1 C_3 L_3 R_1 + 2C_1 C_4 L_3 R_1 + C_3 C_4 L_3 R_1 R_4 g_m + C_3 C_4 L_3 R_4) + s^2 (C_1 C_4 R_1 R_4 + C_3 L_3 R_1 g_m + C_3 L_3 + 2C_4 L_3 R_1 g_m + 2C_4 L_3) + s (C_1 R_1 + C_4 R_1 R_4 g_m + C_4 R_4) + 1}$$

$$\mathbf{9.302 \quad X-INVALID-ORDER-302} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 g_m s^3 + L_3 R_1 g_m s}{C_1 C_3 C_4 L_3 L_4 R_1 s^5 + C_1 R_1 s + R_1 g_m + s^4 (C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4) + s^3 (C_1 C_3 L_3 R_1 + 2C_1 C_4 L_3 R_1 + C_1 C_4 L_4 R_1) + s^2 (C_3 L_3 R_1 g_m + C_3 L_3 + 2C_4 L_3 R_1 g_m + 2C_4 L_3 + C_4 L_4 R_1 g_m + C_4 L_4) + 1}$$

$$\mathbf{9.303 \quad X-INVALID-ORDER-303} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 L_4 R_1 g_m s}{2L_3 R_1 g_m + 2L_3 + L_4 R_1 g_m + L_4 + s^3 (C_1 C_3 L_3 L_4 R_1 + 2C_1 C_4 L_3 L_4 R_1) + s^2 (C_3 L_3 L_4 R_1 g_m + C_3 L_3 L_4 + 2C_4 L_3 L_4 R_1 g_m + 2C_4 L_3 L_4) + s (2C_1 L_3 R_1 + C_1 L_4 R_1)}$$

$$\mathbf{9.304 \quad X-INVALID-ORDER-304} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 g_m s^3 + C_4 L_3 R_1 R_4 g_m s^2 + L_3 R_1 g_m s}{C_1 C_3 C_4 L_3 L_4 R_1 s^5 + R_1 g_m + s^4 (C_1 C_3 C_4 L_3 R_1 R_4 + C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4) + s^3 (C_1 C_3 L_3 R_1 + 2C_1 C_4 L_3 R_1 + C_1 C_4 L_4 R_1 + C_3 C_4 L_3 R_1 R_4 g_m + C_3 C_4 L_3 R_4) + s^2 (C_1 C_4 R_1 R_4 + C_3 L_3 R_1 g_m + C_3 L_3 + 2C_4 L_3 R_1 g_m + 2C_4 L_3 + C_4 L_4 R_1 g_m + C_4 L_4) + s (C_1 R_1 + C_4 R_1 R_4 g_m + C_4 R_4) + 1}$$

$$\mathbf{9.305 \quad X-INVALID-ORDER-305} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 L_4 R_1 R_4 g_m s}{2L_3 R_1 R_4 g_m + 2L_3 R_4 + L_4 R_1 R_4 g_m + L_4 R_4 + s^3 (C_1 C_3 L_3 L_4 R_1 R_4 + 2C_1 C_4 L_3 L_4 R_1 R_4) + s^2 (2C_1 L_3 L_4 R_1 + C_3 L_3 L_4 R_1 R_4 g_m + C_3 L_3 L_4 R_4 + 2C_4 L_3 L_4 R_1 R_4 g_m + 2C_4 L_3 L_4 R_4) + s (2C_1 L_3 R_1 R_4 + C_1 L_4 R_1 R_4 + 2L_3 L_4 R_1 g_m + 2L_3 L_4)}$$

$$\mathbf{9.306 \quad X-INVALID-ORDER-306} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 R_4 g_m s^3 + L_3 L_4 R_1 g_m s^2 + L_3 R_1 R_4 g_m s}{C_1 C_3 C_4 L_3 L_4 R_1 R_4 s^5 + R_1 R_4 g_m + R_4 + s^4 (C_1 C_3 L_3 L_4 R_1 + 2C_1 C_4 L_3 L_4 R_1 + C_3 C_4 L_3 L_4 R_1 R_4 g_m + C_3 C_4 L_3 L_4 R_4) + s^3 (C_1 C_3 L_3 R_1 R_4 + C_1 C_4 L_4 R_1 R_4 + C_3 L_3 L_4 R_1 g_m + C_3 L_3 L_4 + 2C_4 L_3 L_4 R_1 g_m + 2C_4 L_3 L_4) + s^2 (2C_1 L_3 R_1 + C_1 L_4 R_1 + C_3 L_3 R_1 R_4 g_m + C_3 L_3 R_4) + s (2C_1 L_3 R_1 R_4 + C_1 L_4 R_1 R_4 + 2L_3 L_4 R_1 g_m + 2L_3 L_4)}$$

$$\mathbf{9.307 \quad X-INVALID-ORDER-307} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 R_4 g_m s^3 + L_3 R_1 R_4 g_m s}{C_1 C_3 C_4 L_3 L_4 R_1 R_4 s^5 + R_1 R_4 g_m + R_4 + s^4 (2C_1 C_4 L_3 L_4 R_1 + C_3 C_4 L_3 L_4 R_1 R_4 g_m + C_3 C_4 L_3 L_4 R_4) + s^3 (C_1 C_3 L_3 R_1 R_4 + 2C_1 C_4 L_3 R_1 R_4 + C_1 C_4 L_4 R_1 R_4 + 2C_4 L_3 L_4 R_1 g_m + 2C_4 L_3 L_4) + s^2 (2C_1 L_3 R_1 + C_3 L_3 R_1 R_4 g_m + C_3 L_3 R_4 + 2C_4 L_3 R_1 R_4 g_m + 2C_4 L_3 R_4 + C_4 L_4 R_1 g_m + C_4 L_4) + s (2C_1 L_3 R_1 R_4 + C_1 L_4 R_1 R_4 + 2L_3 L_4 R_1 g_m + 2L_3 L_4)}$$

$$\mathbf{9.308 \quad X-INVALID-ORDER-308} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_4 g_m s^2 + C_3 R_1 R_3 R_4 g_m s + R_1 R_4 g_m}{2C_1 C_3 L_3 R_1 s^3 + 2R_1 g_m + s^2 (2C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 + 2C_3 L_3 R_1 g_m + 2C_3 L_3) + s (2C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + 2}$$

$$\mathbf{9.309 \quad X-INVALID-ORDER-309} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 g_m s^2 + C_3 R_1 R_3 g_m s + R_1 g_m}{2C_1 C_3 C_4 L_3 R_1 s^4 + s^3 (2C_1 C_3 C_4 R_1 R_3 + 2C_3 C_4 L_3 R_1 g_m + 2C_3 C_4 L_3) + s^2 (C_1 C_3 R_1 + 2C_1 C_4 R_1 + 2C_3 C_4 R_1 R_3 g_m + 2C_3 C_4 R_3) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.310 \quad X-INVALID-ORDER-310} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_4 g_m s^2 + C_3 R_1 R_3 R_4 g_m s + R_1 R_4 g_m}{2C_1 C_3 C_4 L_3 R_1 R_4 s^4 + 2R_1 g_m + s^3 (2C_1 C_3 C_4 R_1 R_3 R_4 + 2C_1 C_3 L_3 R_1 + 2C_3 C_4 L_3 R_1 R_4 g_m + 2C_3 C_4 L_3 R_4) + s^2 (2C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 + 2C_1 C_4 R_1 R_4 + 2C_3 C_4 R_1 R_3 R_4 g_m + 2C_3 C_4 R_3 R_4 + 2C_3 L_3 R_1 g_m + 2C_3 L_3) + s (2C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + 2}$$

$$\mathbf{9.311 \quad X-INVALID-ORDER-311} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 R_1 R_4 g_m s^3 + R_1 g_m + s^2 (C_3 C_4 R_1 R_3 R_4 g_m + C_3 L_3 R_1 g_m) + s (C_3 R_1 R_3 g_m + C_4 R_1 R_4 g_m)}{2 C_1 C_3 C_4 L_3 R_1 s^4 + s^3 (2 C_1 C_3 C_4 R_1 R_3 + C_1 C_3 C_4 R_1 R_4 + 2 C_3 C_4 L_3 R_1 g_m + 2 C_3 C_4 L_3) + s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1 + 2 C_3 C_4 R_1 R_3 g_m + C_3 C_4 R_1 R_4 g_m + 2 C_3 C_4 R_3 + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.312 \quad X-INVALID-ORDER-312} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 g_m s^4 + C_3 C_4 L_4 R_1 R_3 g_m s^3 + C_3 R_1 R_3 g_m s + R_1 g_m + s^2 (C_3 L_3 R_1 g_m + C_4 L_4 R_1 g_m)}{s^4 (2 C_1 C_3 C_4 L_3 R_1 + C_1 C_3 C_4 L_4 R_1) + s^3 (2 C_1 C_3 C_4 R_1 R_3 + 2 C_3 C_4 L_3 R_1 g_m + 2 C_3 C_4 L_3 + C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1 + 2 C_3 C_4 R_1 R_3 g_m + 2 C_3 C_4 R_3) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.313 \quad X-INVALID-ORDER-313} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_1 g_m s^3 + C_3 L_4 R_1 R_3 g_m s^2 + L_4 R_1 g_m s}{2 C_1 C_3 C_4 L_3 L_4 R_1 s^5 + 2 R_1 g_m + s^4 (2 C_1 C_3 C_4 L_4 R_1 R_3 + 2 C_3 C_4 L_3 L_4 R_1 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (2 C_1 C_3 L_3 R_1 + C_1 C_3 L_4 R_1 + 2 C_1 C_4 L_4 R_1 + 2 C_3 C_4 L_4 R_1 R_3 g_m + 2 C_3 C_4 L_4 R_3) + s^2 (2 C_1 C_3 R_1 R_3 + 2 C_3 L_3 R_1 g_m + 2 C_3 L_3 + C_3 L_4 R_1 g_m + C_3 L_4 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4)}$$

$$\mathbf{9.314 \quad X-INVALID-ORDER-314} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 g_m s^4 + R_1 g_m + s^3 (C_3 C_4 L_3 R_1 R_4 g_m + C_3 C_4 L_4 R_1 R_3 g_m) + s^2 (C_3 C_4 R_1 R_3 R_4 g_m + C_3 L_3 R_1 g_m + C_4 L_4 R_1 g_m) + s (C_3 R_1 R_3 g_m + C_4 R_1 R_4 g_m)}{s^4 (2 C_1 C_3 C_4 L_3 R_1 + C_1 C_3 C_4 L_4 R_1) + s^3 (2 C_1 C_3 C_4 R_1 R_3 + C_1 C_3 C_4 R_1 R_4 + 2 C_3 C_4 L_3 R_1 g_m + 2 C_3 C_4 L_3 + C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (C_1 C_3 R_1 + 2 C_1 C_4 R_1 + 2 C_3 C_4 R_1 R_3 g_m + C_3 C_4 R_1 R_4 g_m + 2 C_3 C_4 R_3 + C_3 C_4 R_4) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.315 \quad X-INVALID-ORDER-315} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_1 R_4 g_m s^3 + C_3 L_4 R_1 R_3 R_4 g_m s^2 + L_4 R_1 g_m s}{2 C_1 C_3 C_4 L_3 L_4 R_1 R_4 s^5 + 2 R_1 R_4 g_m + 2 R_4 + s^4 (2 C_1 C_3 C_4 L_4 R_1 R_3 R_4 + 2 C_1 C_3 L_3 L_4 R_1 + 2 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2 C_3 C_4 L_3 L_4 R_4) + s^3 (2 C_1 C_3 L_3 R_1 R_4 + 2 C_1 C_3 L_4 R_1 R_3 + C_1 C_3 L_4 R_1 R_4 + 2 C_1 C_4 L_4 R_1 R_4 + 2 C_3 C_4 L_4 R_1 R_3 R_4 g_m + 2 C_3 C_4 L_4 R_3 R_4 + 2 C_3 L_3 L_4 R_1 g_m + 2 C_3 L_3 L_4 R_4)}$$

$$\mathbf{9.316 \quad X-INVALID-ORDER-316} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_4 g_m s^4 + R_1 R_4 g_m + s^3 (C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_3 L_3 L_4 R_1 g_m) + s^2 (C_3 L_3 R_1 R_4 g_m + C_3 L_4 R_1 R_3 g_m + C_4 L_4 R_1 R_4 g_m) + s (C_3 R_1 R_3 R_4 g_m + C_4 R_1 R_4 g_m)}{2 C_1 C_3 C_4 L_3 L_4 R_1 s^5 + 2 R_1 g_m + s^4 (2 C_1 C_3 C_4 L_4 R_1 R_3 + C_1 C_3 C_4 L_4 R_1 R_4 + 2 C_3 C_4 L_3 L_4 R_1 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (2 C_1 C_3 L_3 R_1 + C_1 C_3 L_4 R_1 + 2 C_1 C_4 L_4 R_1 + 2 C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_4 R_1 R_4 g_m + 2 C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4) + s^2 (2 C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 + 2 C_3 L_3 R_1 g_m + 2 C_3 L_3 R_4 + 2 C_3 L_4 R_1 g_m + 2 C_3 L_4 R_4)}$$

$$\mathbf{9.317 \quad X-INVALID-ORDER-317} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_4 g_m s^4 + C_3 C_4 L_4 R_1 R_3 R_4 g_m s^3 + C_3 R_1 R_3 R_4 g_m s}{2 C_1 C_3 C_4 L_3 L_4 R_1 s^5 + 2 R_1 g_m + s^4 (2 C_1 C_3 C_4 L_3 R_1 R_4 + 2 C_1 C_3 C_4 L_4 R_1 R_3 + C_1 C_3 C_4 L_4 R_1 R_4 + 2 C_3 C_4 L_3 L_4 R_1 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (2 C_1 C_3 C_4 R_1 R_3 R_4 + 2 C_1 C_3 L_3 R_1 + 2 C_1 C_4 L_4 R_1 + 2 C_3 C_4 L_3 R_1 R_4 g_m + 2 C_3 C_4 L_3 R_4 + 2 C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_4 R_1 R_4 g_m + 2 C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4) + s^2 (2 C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 + 2 C_3 L_3 R_1 g_m + 2 C_3 L_3 R_4 + 2 C_3 L_4 R_1 g_m + 2 C_3 L_4 R_4)}$$

$$\mathbf{9.318 \quad X-INVALID-ORDER-318} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, R_4, \infty, \infty \right)$$

$$H(s) = \frac{L_3 R_1 R_3 R_4 g_m s}{C_1 C_3 L_3 R_1 R_3 R_4 s^3 + R_1 R_3 R_4 g_m + R_3 R_4 + s^2 (2 C_1 L_3 R_1 R_3 + C_1 L_3 R_1 R_4 + C_3 L_3 R_1 R_3 R_4 g_m + C_3 L_3 R_3 R_4) + s (C_1 R_1 R_3 R_4 + 2 L_3 R_1 R_3 g_m + L_3 R_1 R_4 g_m + 2 L_3 R_3 + L_3 R_4)}$$

$$\mathbf{9.319 \quad X-INVALID-ORDER-319} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{L_3 R_1 R_3 g_m s}{R_1 R_3 g_m + R_3 + s^3 (C_1 C_3 L_3 R_1 R_3 + 2 C_1 C_4 L_3 R_1 R_3) + s^2 (C_1 L_3 R_1 + C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2 C_4 L_3 R_1 R_3 g_m + 2 C_4 L_3 R_3) + s (C_1 R_1 R_3 + L_3 R_1 g_m + L_3)}$$

$$\mathbf{9.320 \quad X-INVALID-ORDER-320} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{L_3 R_1 R_3 R_4 g_m s}{R_1 R_3 R_4 g_m + R_3 R_4 + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 + 2 C_1 C_4 L_3 R_1 R_3 R_4) + s^2 (2 C_1 L_3 R_1 R_3 + C_1 L_3 R_1 R_4 + C_3 L_3 R_1 R_3 R_4 g_m + C_3 L_3 R_3 R_4 + 2 C_4 L_3 R_1 R_3 R_4 g_m + 2 C_4 L_3 R_3 R_4) + s (C_1 R_1 R_3 R_4 + 2 L_3 R_1 R_3 g_m + L_3 R_1 R_4 g_m + 2 L_3 R_3 + L_3 R_4)}$$

$$\mathbf{9.321 \quad X-INVALID-ORDER-321} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 R_1 R_3 R_4 g_m s^2 + L_3 R_1 R_3 g_m s}{C_1 C_3 C_4 L_3 R_1 R_3 R_4 s^4 + R_1 R_3 g_m + R_3 + s^3 (C_1 C_3 L_3 R_1 R_3 + 2C_1 C_4 L_3 R_1 R_3 + C_1 C_4 L_3 R_1 R_4 + C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_3 C_4 L_3 R_3 R_4) + s^2 (C_1 C_4 R_1 R_3 R_4 + C_1 L_3 R_1 + C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + C_4 L_3 R_1 R_4 g_m + 2C_4 L_3 R_3 + C_4 L_3 R_4) + s (C_1 R_1 R_3 R_4 + C_1 L_3 R_1 + C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + C_4 L_3 R_1 R_4 g_m + 2C_4 L_3 R_3 + C_4 L_3 R_4)}$$

$$\mathbf{9.322 \quad X-INVALID-ORDER-322} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 R_3 g_m s^3 + L_3 R_1 R_3 g_m s}{C_1 C_3 C_4 L_3 L_4 R_1 R_3 s^5 + R_1 R_3 g_m + R_3 + s^4 (C_1 C_4 L_3 L_4 R_1 + C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_3 L_4 R_3) + s^3 (C_1 C_3 L_3 R_1 R_3 + 2C_1 C_4 L_3 R_1 R_3 + C_1 C_4 L_4 R_1 R_3 + C_4 L_3 L_4 R_1 g_m + C_4 L_3 L_4) + s^2 (C_1 L_3 R_1 + C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + 2C_4 L_3 R_3 + C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_3) + s (C_1 L_3 R_1 + C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + 2C_4 L_3 R_3 + C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_3)}$$

$$\mathbf{9.323 \quad X-INVALID-ORDER-323} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 L_4 R_1 R_3 g_m s}{2L_3 R_1 R_3 g_m + 2L_3 R_3 + L_4 R_1 R_3 g_m + L_4 R_3 + s^3 (C_1 C_3 L_3 L_4 R_1 R_3 + 2C_1 C_4 L_3 L_4 R_1 R_3) + s^2 (C_1 L_3 L_4 R_1 + C_3 L_3 L_4 R_1 R_3 g_m + C_3 L_3 L_4 R_3 + 2C_4 L_3 L_4 R_1 R_3 g_m + 2C_4 L_3 L_4 R_3) + s (2C_1 L_3 R_1 R_3 + C_1 L_4 R_1 R_3 + L_3 L_4 R_1 g_m + L_3 L_4)}$$

$$\mathbf{9.324 \quad X-INVALID-ORDER-324} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 R_3 g_m s^3 + C_4 L_3 R_1 R_3 R_4 g_m s^2 + L_3 R_1 R_3 g_m s}{C_1 C_3 C_4 L_3 L_4 R_1 R_3 s^5 + R_1 R_3 g_m + R_3 + s^4 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 + C_1 C_4 L_3 L_4 R_1 + C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_3 L_4 R_3) + s^3 (C_1 C_3 L_3 R_1 R_3 + 2C_1 C_4 L_3 R_1 R_3 + C_1 C_4 L_3 R_1 R_4 + C_1 C_4 L_4 R_1 R_3 + C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_3 C_4 L_3 R_3 R_4 + C_4 L_3 L_4 R_1 g_m + C_4 L_3 L_4) + s^2 (C_1 L_3 R_1 + C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + 2C_4 L_3 R_3 + C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_3) + s (C_1 L_3 R_1 + C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + 2C_4 L_3 R_3 + C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_3)}$$

$$\mathbf{9.325 \quad X-INVALID-ORDER-325} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 L_4 R_1 R_3 R_4 g_m s}{2L_3 R_1 R_3 R_4 g_m + 2L_3 R_3 R_4 + L_4 R_1 R_3 R_4 g_m + L_4 R_3 R_4 + s^3 (C_1 C_3 L_3 L_4 R_1 R_3 R_4 + 2C_1 C_4 L_3 L_4 R_1 R_3 R_4) + s^2 (2C_1 L_3 L_4 R_1 R_3 + C_1 L_3 L_4 R_1 R_4 + C_3 L_3 L_4 R_1 R_3 R_4 g_m + C_3 L_3 L_4 R_3 R_4 + 2C_4 L_3 L_4 R_1 R_3 R_4 g_m + 2C_4 L_3 L_4 R_3 R_4) + s (2C_1 L_3 R_1 R_3 R_4 + C_1 L_4 R_1 R_3 R_4 + C_3 L_3 R_1 R_3 R_4 g_m + C_3 L_3 R_3 R_4 + 2C_4 L_3 R_1 R_3 R_4 g_m + 2C_4 L_3 R_3 R_4)}$$

$$\mathbf{9.326 \quad X-INVALID-ORDER-326} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 R_3 R_4 g_m s^3 + C_4 L_3 R_1 R_3 R_4 g_m s^2 + L_3 R_1 R_3 g_m s}{C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 s^5 + R_1 R_3 R_4 g_m + R_3 R_4 + s^4 (C_1 C_3 L_3 L_4 R_1 R_3 + 2C_1 C_4 L_3 L_4 R_1 R_3 + C_1 C_4 L_3 L_4 R_1 R_4 + C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + C_3 C_4 L_3 L_4 R_3 R_4) + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 + C_1 C_4 L_4 R_1 R_3 R_4 + C_1 L_3 L_4 R_1 + C_3 L_3 L_4 R_1 R_3 g_m + C_3 L_3 L_4 R_3 + 2C_4 L_3 L_4 R_1 R_3 g_m + 2C_4 L_3 L_4 R_3) + s^2 (C_1 L_3 R_1 + C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + 2C_4 L_3 R_3 + C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_3) + s (C_1 L_3 R_1 + C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + 2C_4 L_3 R_3 + C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_3)}$$

$$\mathbf{9.327 \quad X-INVALID-ORDER-327} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_1 R_3 R_4 g_m s^3 + L_3 R_1 R_3 R_4 g_m s}{C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 s^5 + R_1 R_3 R_4 g_m + R_3 R_4 + s^4 (2C_1 C_4 L_3 L_4 R_1 R_3 + C_1 C_4 L_3 L_4 R_1 R_4 + C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + C_3 C_4 L_3 L_4 R_3 R_4) + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 + 2C_1 C_4 L_3 R_1 R_3 R_4 + C_1 C_4 L_4 R_1 R_3 R_4 + 2C_4 L_3 L_4 R_1 R_3 g_m + C_4 L_3 L_4 R_1 R_4 g_m + 2C_4 L_3 L_4 R_3 + C_4 L_3 L_4 R_4) + s^2 (C_1 L_3 R_1 + C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + 2C_4 L_3 R_3 + C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_3) + s (C_1 L_3 R_1 + C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + 2C_4 L_3 R_3 + C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_3)}$$

$$\mathbf{9.328 \quad X-INVALID-ORDER-328} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_3 R_4 g_m s^2 + L_3 R_1 R_4 g_m s + R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^3 (2C_1 C_3 L_3 R_1 R_3 + C_1 C_3 L_3 R_1 R_4) + s^2 (2C_1 L_3 R_1 + 2C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_1 R_4 g_m + 2C_3 L_3 R_3 + C_3 L_3 R_4) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + 2L_3 R_1 g_m + 2L_3)}$$

$$\mathbf{9.329 \quad X-INVALID-ORDER-329} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_3 g_m s^2 + L_3 R_1 g_m s + R_1 R_3 g_m}{2C_1 C_3 C_4 L_3 R_1 R_3 s^4 + R_1 g_m + s^3 (C_1 C_3 L_3 R_1 + 2C_1 C_4 L_3 R_1 + 2C_3 C_4 L_3 R_1 R_3 g_m + 2C_3 C_4 L_3 R_3) + s^2 (2C_1 C_4 R_1 R_3 + C_3 L_3 R_1 g_m + C_3 L_3 + 2C_4 L_3 R_1 g_m + 2C_4 L_3) + s (C_1 R_1 + 2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

$$\mathbf{9.330 \quad X-INVALID-ORDER-330} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_3 R_4 g_m s^2 + L_3 R_1 R_4 g_m s + R_1 R_3 R_4 g_m}{2C_1 C_3 C_4 L_3 R_1 R_3 R_4 s^4 + 2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^3 (2C_1 C_3 L_3 R_1 R_3 + C_1 C_3 L_3 R_1 R_4 + 2C_1 C_4 L_3 R_1 R_4 + 2C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_3 C_4 L_3 R_3 R_4) + s^2 (2C_1 C_4 R_1 R_3 R_4 + 2C_1 L_3 R_1 + 2C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_1 R_4 g_m + 2C_3 L_3 R_3 + C_3 L_3 R_4 + 2C_4 L_3 R_1 R_3 g_m + 2C_4 L_3 R_3 + C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_3) + s (C_1 L_3 R_1 + C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + 2C_4 L_3 R_3 + C_4 L_4 R_1 R_3 g_m + C_4 L_4 R_3)}$$

$$\mathbf{9.331 \quad X-INVALID-ORDER-331} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 R_1 R_3 R_4 g_m s^3 + R_1 R_3 g_m + s^2 (C_3 L_3 R_1 R_3 g_m + C_4 L_3 R_1 R_4 g_m) + s (C_4 R_1 R_3 R_4 g_m + L_3 R_1 g_m)}{R_1 g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 + C_1 C_3 C_4 L_3 R_1 R_4) + s^3 (C_1 C_3 L_3 R_1 + 2C_1 C_4 L_3 R_1 + 2C_3 C_4 L_3 R_1 R_3 g_m + C_3 C_4 L_3 R_1 R_4 g_m + 2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4) + s^2 (2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 + C_3 L_3 R_1 g_m + C_3 L_3 + 2C_4 L_3 R_1 g_m + 2C_4 L_3) + s (C_1 R_1 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m)}$$

$$\mathbf{9.332 \quad X-INVALID-ORDER-332} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_3 g_m s^4 + C_4 L_3 L_4 R_1 g_m s^3 + L_3 R_1 g_m s + R_1 R_3 g_m + s^2 (C_3 L_3 R_1 R_3 g_m + C_4 L_4 R_1 R_3 g_m)}{C_1 C_3 C_4 L_3 L_4 R_1 s^5 + R_1 g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 + C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4) + s^3 (C_1 C_3 L_3 R_1 + 2C_1 C_4 L_3 R_1 + C_1 C_4 L_4 R_1 + 2C_3 C_4 L_3 R_1 R_3 g_m + 2C_3 C_4 L_3 R_3) + s^2 (2C_1 C_4 R_1 R_3 + C_3 L_3 R_1 g_m + C_3 L_3 + 2C_4 L_3 R_1 g_m + 2C_4 L_3 + C_4 L_4 R_1 g_m + C_4 L_4) + s (C_1 R_1 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m)}$$

$$\mathbf{9.333 \quad X-INVALID-ORDER-333} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_1 R_3 g_m s^3 + L_3 L_4 R_1 g_m s^2 + L_4 R_1 R_3 g_m s}{2C_1 C_3 C_4 L_3 L_4 R_1 R_3 s^5 + 2R_1 R_3 g_m + 2R_3 + s^4 (C_1 C_3 L_3 L_4 R_1 + 2C_1 C_4 L_3 L_4 R_1 + 2C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_3 C_4 L_3 L_4 R_3) + s^3 (2C_1 C_3 L_3 R_1 R_3 + 2C_1 C_4 L_4 R_1 R_3 + C_3 L_3 L_4 R_1 g_m + C_3 L_3 L_4 + 2C_4 L_3 L_4 R_1 g_m + 2C_4 L_3 L_4) + s^2 (2C_1 L_3 R_1 + C_1 L_4 R_1 + 2C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_1 R_4 g_m)}$$

$$\mathbf{9.334 \quad X-INVALID-ORDER-334} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_3 g_m s^4 + R_1 R_3 g_m + s^3 (C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_4 L_3 L_4 R_1 g_m) + s^2 (C_3 L_3 R_1 R_3 g_m + C_4 L_3 R_1 R_4 g_m + C_4 L_4 R_1 R_3 g_m) + s (C_4 R_1 R_3 R_4 g_m + L_3 R_1 g_m)}{C_1 C_3 C_4 L_3 L_4 R_1 s^5 + R_1 g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 + C_1 C_3 C_4 L_3 R_1 R_4 + C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4) + s^3 (C_1 C_3 L_3 R_1 + 2C_1 C_4 L_3 R_1 + C_1 C_4 L_4 R_1 + 2C_3 C_4 L_3 R_1 R_3 g_m + C_3 C_4 L_3 R_1 R_4 g_m + 2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4) + s^2 (2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 + C_3 L_3 R_1 g_m + C_3 L_3 R_1 R_4 g_m)}$$

$$\mathbf{9.335 \quad X-INVALID-ORDER-335} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_1 R_3 g_m s^3 + L_3 L_4 R_1 g_m s^2 + L_4 R_1 R_3 g_m s}{2C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 s^5 + 2R_1 R_3 R_4 g_m + 2R_3 R_4 + s^4 (2C_1 C_3 L_3 L_4 R_1 R_3 + C_1 C_3 L_3 L_4 R_1 R_4 + 2C_1 C_4 L_3 L_4 R_1 R_4 + 2C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + 2C_3 C_4 L_3 L_4 R_3 R_4) + s^3 (2C_1 C_3 L_3 R_1 R_3 R_4 + 2C_1 C_4 L_4 R_1 R_3 R_4 + 2C_1 L_3 L_4 R_1 + 2C_3 L_3 L_4 R_1 R_3 g_m + C_3 L_3 L_4 R_1 R_4 g_m + C_3 L_3 L_4 R_1 R_4 g_m)}$$

$$\mathbf{9.336 \quad X-INVALID-ORDER-336} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m s^4 + R_1 R_3 R_4 g_m + s^3 (C_3 L_3 L_4 R_1 R_3 g_m + C_4 L_3 L_4 R_1 R_4 g_m) + s^2 (C_3 L_3 R_1 R_3 g_m + C_4 L_3 R_1 R_4 g_m + C_4 L_4 R_1 R_3 g_m) + s (C_4 R_1 R_3 R_4 g_m + L_3 R_1 g_m)}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 R_3 + C_1 C_3 C_4 L_3 L_4 R_1 R_4) + s^4 (C_1 C_3 L_3 L_4 R_1 + 2C_1 C_4 L_3 L_4 R_1 + 2C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_3 C_4 L_3 L_4 R_3 + C_3 C_4 L_3 L_4 R_4) + s^3 (2C_1 C_3 L_3 R_1 R_3 + C_1 C_3 L_3 R_1 R_4 + 2C_1 C_4 L_4 R_1 R_3 + C_1 C_4 L_4 R_1 R_4 g_m)}$$

$$\mathbf{9.337 \quad X-INVALID-ORDER-337} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m s^4 + R_1 R_3 R_4 g_m + s^3 (C_3 L_3 L_4 R_1 R_3 g_m + C_4 L_3 L_4 R_1 R_4 g_m) + s^2 (C_3 L_3 R_1 R_3 g_m + C_4 L_3 R_1 R_4 g_m + C_4 L_4 R_1 R_3 g_m) + s (C_4 R_1 R_3 R_4 g_m + L_3 R_1 g_m)}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 R_3 + C_1 C_3 C_4 L_3 L_4 R_1 R_4) + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 R_4 + 2C_1 C_4 L_3 L_4 R_1 + 2C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_3 C_4 L_3 L_4 R_3 + C_3 C_4 L_3 L_4 R_4) + s^3 (2C_1 C_3 L_3 R_1 R_3 + C_1 C_3 L_3 R_1 R_4 + 2C_1 C_4 L_4 R_1 R_3 + C_1 C_4 L_4 R_1 R_4 g_m)}$$

$$\mathbf{9.338 \quad X-INVALID-ORDER-338} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, R_4, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_3 R_4 g_m s^2 + R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^3 (2C_1 C_3 L_3 R_1 R_3 + C_1 C_3 L_3 R_1 R_4) + s^2 (C_1 C_3 R_1 R_3 R_4 + 2C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_1 R_4 g_m + 2C_3 L_3 R_3 + C_3 L_3 R_4) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4)}$$

$$\mathbf{9.339 \quad X-INVALID-ORDER-339} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_3 g_m s^2 + R_1 R_3 g_m}{2C_1 C_3 C_4 L_3 R_1 R_3 s^4 + R_1 g_m + s^3 (C_1 C_3 L_3 R_1 + 2C_3 C_4 L_3 R_1 R_3 g_m + 2C_3 C_4 L_3 R_3) + s^2 (C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_3 L_3 R_1 g_m + C_3 L_3) + s (C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3 + 2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

$$\mathbf{9.340} \quad \mathbf{X-INVALID-ORDER-340} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_1 R_3 R_4 g_m s^2 + R_1 R_3 R_4 g_m}{2C_1 C_3 C_4 L_3 R_1 R_3 R_4 s^4 + 2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^3 (2C_1 C_3 L_3 R_1 R_3 + C_1 C_3 L_3 R_1 R_4 + 2C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_3 C_4 L_3 R_3 R_4) + s^2 (C_1 C_3 R_1 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4 + 2C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_1 R_4 g_m + 2C_3 L_3 R_3 + C_3 L_3 R_4) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_3 R_3 + C_3 R_4)}$$

$$\mathbf{9.341} \quad \mathbf{X-INVALID-ORDER-341} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 R_1 R_3 R_4 g_m s^3 + C_3 L_3 R_1 R_3 g_m s^2 + C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{R_1 g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 + C_1 C_3 C_4 L_3 R_1 R_4) + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 + C_1 C_3 L_3 R_1 + 2C_3 C_4 L_3 R_1 R_3 g_m + C_3 C_4 L_3 R_1 R_4 g_m + 2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4) + s^2 (C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 + C_3 C_4 R_1 R_3 R_4 g_m + C_3 C_4 R_3 R_4 + C_3 L_3 R_1 g_m + C_3 L_3) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_3 R_3 + C_3 R_4)}$$

$$\mathbf{9.342} \quad \mathbf{X-INVALID-ORDER-342} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_3 g_m s^4 + R_1 R_3 g_m + s^2 (C_3 L_3 R_1 R_3 g_m + C_4 L_4 R_1 R_3 g_m)}{C_1 C_3 C_4 L_3 L_4 R_1 s^5 + R_1 g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 + C_1 C_3 C_4 L_4 R_1 R_3 + C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4) + s^3 (C_1 C_3 L_3 R_1 + C_1 C_4 L_4 R_1 + 2C_3 C_4 L_3 R_1 R_3 g_m + 2C_3 C_4 L_3 R_3 + C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_4 R_3) + s^2 (C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_3 L_3 R_1 g_m + C_3 L_3) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_3 R_3 + C_3 R_4)}$$

$$\mathbf{9.343} \quad \mathbf{X-INVALID-ORDER-343} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_1 R_3 g_m s^3 + L_4 R_1 R_3 g_m s}{2C_1 C_3 C_4 L_3 L_4 R_1 R_3 s^5 + 2R_1 R_3 g_m + 2R_3 + s^4 (C_1 C_3 L_3 L_4 R_1 + 2C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_3 C_4 L_3 L_4 R_3) + s^3 (2C_1 C_3 L_3 R_1 R_3 + C_1 C_3 L_4 R_1 R_3 + 2C_1 C_4 L_4 R_1 R_3 + C_3 L_3 L_4 R_1 g_m + C_3 L_3 L_4) + s^2 (C_1 L_4 R_1 + 2C_3 L_3 R_1 R_3 g_m + 2C_3 L_3 R_3 + C_3 L_4 R_1 R_3 g_m + C_3 L_4 R_3) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_3 R_3 + C_3 R_4)}$$

$$\mathbf{9.344} \quad \mathbf{X-INVALID-ORDER-344} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_3 g_m s^4 + C_3 C_4 L_3 R_1 R_3 R_4 g_m s^3 + C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{C_1 C_3 C_4 L_3 L_4 R_1 s^5 + R_1 g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 + C_1 C_3 C_4 L_3 R_1 R_4 + C_1 C_3 C_4 L_4 R_1 R_3 + C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4) + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 + C_1 C_3 L_3 R_1 + C_1 C_4 L_4 R_1 + 2C_3 C_4 L_3 R_1 R_3 g_m + C_3 C_4 L_3 R_1 R_4 g_m + 2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4 + C_3 C_4 L_4 R_1 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_3 L_3 R_1 g_m + C_3 L_3) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_3 R_3 + C_3 R_4)}$$

$$\mathbf{9.345} \quad \mathbf{X-INVALID-ORDER-345} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_1 R_3 R_4 g_m s^3 + L_4 R_1 R_3 R_4 g_m s}{2C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 s^5 + 2R_1 R_3 R_4 g_m + 2R_3 R_4 + s^4 (2C_1 C_3 L_3 L_4 R_1 R_3 + C_1 C_3 L_3 L_4 R_1 R_4 + 2C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + 2C_3 C_4 L_3 L_4 R_3 R_4) + s^3 (2C_1 C_3 L_3 R_1 R_3 R_4 + C_1 C_3 L_4 R_1 R_3 R_4 + 2C_1 C_4 L_4 R_1 R_3 R_4 + 2C_3 L_3 L_4 R_1 R_3 g_m + C_3 L_3 L_4 R_1 R_4 g_m + 2C_3 L_3 L_4 R_3 + C_3 L_3 L_4 R_4) + s^2 (C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_3 L_3 R_1 g_m + C_3 L_3) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_3 R_3 + C_3 R_4)}$$

$$\mathbf{9.346} \quad \mathbf{X-INVALID-ORDER-346} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m s^3 + L_4 R_1 R_3 R_4 g_m s}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 R_3 + C_1 C_3 C_4 L_3 L_4 R_1 R_4) + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 + C_1 C_3 L_3 L_4 R_1 + 2C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_3 C_4 L_3 L_4 R_3 + C_3 C_4 L_3 L_4 R_4) + s^3 (2C_1 C_3 L_3 R_1 R_3 + C_1 C_3 L_3 R_1 R_4 + C_1 C_3 L_4 R_1 R_3 + C_3 L_3 L_4 R_1 R_3 g_m + C_3 L_3 L_4 R_1 R_4 g_m + 2C_3 L_3 L_4 R_3 + C_3 L_3 L_4 R_4) + s^2 (C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_3 L_3 R_1 g_m + C_3 L_3) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_3 R_3 + C_3 R_4)}$$

$$\mathbf{9.347} \quad \mathbf{X-INVALID-ORDER-347} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 R_3 + C_1 C_3 C_4 L_3 L_4 R_1 R_4) + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 R_4 + C_1 C_3 C_4 L_4 R_1 R_3 R_4 + 2C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_3 C_4 L_3 L_4 R_3 + C_3 C_4 L_3 L_4 R_4) + s^3 (2C_1 C_3 L_3 R_1 R_3 + C_1 C_3 L_3 R_1 R_4 + 2C_1 C_4 L_4 R_1 R_3 + C_3 L_3 L_4 R_1 R_3 g_m + C_3 L_3 L_4 R_1 R_4 g_m + 2C_3 L_3 L_4 R_3 + C_3 L_3 L_4 R_4) + s^2 (C_1 C_3 R_1 R_3 + 2C_1 C_4 R_1 R_3 + C_3 L_3 R_1 g_m + C_3 L_3) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_3 R_3 + C_3 R_4)}$$

$$\mathbf{9.348} \quad \mathbf{X-INVALID-ORDER-348} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s (2C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2C_1 R_3 + C_1 R_4)}$$

9.349 X-INVALID-ORDER-349 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_4 R_1 R_3 g_m s^3 + C_1 R_1 R_3 g_m s + C_4 L_4 R_3 g_m s^2 + R_3 g_m}{g_m + s^3 (C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4) + s^2 (2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_1 + 2C_4 R_3 g_m)}$$

9.350 X-INVALID-ORDER-350 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_4 R_1 R_3 g_m s^2 + L_4 R_3 g_m s}{2R_3 g_m + s^3 (2C_1 C_4 L_4 R_1 R_3 g_m + 2C_1 C_4 L_4 R_3) + s^2 (C_1 L_4 R_1 g_m + C_1 L_4 + 2C_4 L_4 R_3 g_m) + s (2C_1 R_1 R_3 g_m + 2C_1 R_3 + L_4 g_m)}$$

9.351 X-INVALID-ORDER-351 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_4 R_1 R_3 g_m s^3 + R_3 g_m + s^2 (C_1 C_4 R_1 R_3 R_4 g_m + C_4 L_4 R_3 g_m) + s (C_1 R_1 R_3 g_m + C_4 R_3 R_4 g_m)}{g_m + s^3 (C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4) + s^2 (2C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

9.352 X-INVALID-ORDER-352 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_4 R_1 R_3 R_4 g_m s^2 + L_4 R_3 R_4 g_m s}{2R_3 R_4 g_m + s^3 (2C_1 C_4 L_4 R_1 R_3 R_4 g_m + 2C_1 C_4 L_4 R_3 R_4) + s^2 (2C_1 L_4 R_1 R_3 g_m + C_1 L_4 R_1 R_4 g_m + 2C_1 L_4 R_3 + C_1 L_4 R_4 + 2C_4 L_4 R_3 R_4 g_m) + s (2C_1 R_1 R_3 R_4 g_m + 2C_1 R_3 R_4 + 2L_4 R_3 g_m + L_4 R_4 g_m)}$$

9.353 X-INVALID-ORDER-353 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_4 R_1 R_3 R_4 g_m s^3 + R_3 R_4 g_m + s^2 (C_1 L_4 R_1 R_3 g_m + C_4 L_4 R_3 R_4 g_m) + s (C_1 R_1 R_3 R_4 g_m + L_4 R_3 g_m)}{2R_3 g_m + R_4 g_m + s^3 (2C_1 C_4 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_4 g_m + 2C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4) + s^2 (C_1 L_4 R_1 g_m + C_1 L_4 + 2C_4 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (2C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2C_1 R_3 + C_1 R_4 + L_4 g_m)}$$

9.354 X-INVALID-ORDER-354 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_4 R_1 R_3 R_4 g_m s^3 + C_1 R_1 R_3 R_4 g_m s + C_4 L_4 R_3 R_4 g_m s^2 + R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s^3 (2C_1 C_4 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_4 g_m + 2C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4) + s^2 (2C_1 C_4 R_1 R_3 R_4 g_m + 2C_1 C_4 R_3 R_4 + 2C_4 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (2C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2C_1 R_3 + C_1 R_4 + 2C_4 R_3 R_4 g_m)}$$

9.355 X-INVALID-ORDER-355 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 R_1 g_m s + g_m}{s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4) + s (C_3 g_m + 2C_4 g_m)}$$

9.356 X-INVALID-ORDER-356 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 R_1 R_4 g_m s^2 + g_m + s (C_1 R_1 g_m + C_4 R_4 g_m)}{s^3 (C_1 C_3 C_4 R_1 R_4 g_m + C_1 C_3 C_4 R_4) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

9.357 X-INVALID-ORDER-357 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_4 R_1 g_m s^3 + C_1 R_1 g_m s + C_4 L_4 g_m s^2 + g_m}{C_3 C_4 L_4 g_m s^3 + s^4 (C_1 C_3 C_4 L_4 R_1 g_m + C_1 C_3 C_4 L_4) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4) + s (C_3 g_m + 2C_4 g_m)}$$

9.358 X-INVALID-ORDER-358 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_4 R_1 g_m s^2 + L_4 g_m s}{2g_m + s^3 (C_1 C_3 L_4 R_1 g_m + C_1 C_3 L_4 + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4) + s^2 (C_3 L_4 g_m + 2C_4 L_4 g_m) + s (2C_1 R_1 g_m + 2C_1)}$$

9.359 X-INVALID-ORDER-359 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_4 R_1 g_m s^3 + g_m + s^2 (C_1 C_4 R_1 R_4 g_m + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m)}{s^4 (C_1 C_3 C_4 L_4 R_1 g_m + C_1 C_3 C_4 L_4) + s^3 (C_1 C_3 C_4 R_1 R_4 g_m + C_1 C_3 C_4 R_4 + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

9.360 X-INVALID-ORDER-360 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_4 R_1 R_4 g_m s^2 + L_4 R_4 g_m s}{2R_4 g_m + s^3 (C_1 C_3 L_4 R_1 R_4 g_m + C_1 C_3 L_4 R_4 + 2C_1 C_4 L_4 R_1 R_4 g_m + 2C_1 C_4 L_4 R_4) + s^2 (2C_1 L_4 R_1 g_m + 2C_1 L_4 + C_3 L_4 R_4 g_m + 2C_4 L_4 R_4 g_m) + s (2C_1 R_1 R_4 g_m + 2C_1 R_4 + 2L_4 g_m)}$$

9.361 X-INVALID-ORDER-361 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_4 R_1 R_4 g_m s^3 + R_4 g_m + s^2 (C_1 L_4 R_1 g_m + C_4 L_4 R_4 g_m) + s (C_1 R_1 R_4 g_m + L_4 g_m)}{2g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_4 R_4) + s^3 (C_1 C_3 L_4 R_1 g_m + C_1 C_3 L_4 + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + C_3 C_4 L_4 R_4 g_m) + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + C_3 L_4 g_m + 2C_4 L_4 g_m) + s (2C_1 R_1 g_m + 2C_1 + C_3 R_4 g_m)}$$

9.362 X-INVALID-ORDER-362 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_4 R_1 R_4 g_m s^3 + C_1 R_1 R_4 g_m s + C_4 L_4 R_4 g_m s^2 + R_4 g_m}{2g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_4 R_4) + s^3 (2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + C_3 C_4 L_4 R_4 g_m) + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + 2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_4 L_4 g_m) + s (2C_1 R_1 g_m + 2C_1 + C_3 R_4 g_m + 2C_4 R_4 g_m)}$$

9.363 X-INVALID-ORDER-363 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 R_1 R_3 R_4 g_m s^2 + R_3 g_m + s (C_1 R_1 R_3 g_m + C_4 R_3 R_4 g_m)}{g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 R_3 R_4) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_3 C_4 R_3 R_4 g_m) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

9.364 X-INVALID-ORDER-364 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_4 R_1 R_3 g_m s^3 + C_1 R_1 R_3 g_m s + C_4 L_4 R_3 g_m s^2 + R_3 g_m}{g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_4 R_3) + s^3 (C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m)}$$

9.365 X-INVALID-ORDER-365 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_4 R_1 R_3 g_m s^2 + L_4 R_3 g_m s}{2R_3 g_m + s^3 (C_1 C_3 L_4 R_1 R_3 g_m + C_1 C_3 L_4 R_3 + 2C_1 C_4 L_4 R_1 R_3 g_m + 2C_1 C_4 L_4 R_3) + s^2 (C_1 L_4 R_1 g_m + C_1 L_4 + C_3 L_4 R_3 g_m + 2C_4 L_4 R_3 g_m) + s (2C_1 R_1 R_3 g_m + 2C_1 R_3 + L_4 g_m)}$$

9.366 X-INVALID-ORDER-366 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_4 R_1 R_3 g_m s^3 + R_3 g_m + s^2 (C_1 C_4 R_1 R_3 R_4 g_m + C_4 L_4 R_3 g_m) + s (C_1 R_1 R_3 g_m + C_4 R_3 R_4 g_m)}{g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_4 R_3) + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 R_3 R_4 + C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_3 C_4 R_3 R_4 g_m + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m)}$$

9.367 X-INVALID-ORDER-367 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_4 R_1 R_3 R_4 g_m s^2 + L_4 R_3 R_4 g_m s}{2R_3 R_4 g_m + s^3 (C_1 C_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 L_4 R_3 R_4 + 2C_1 C_4 L_4 R_1 R_3 R_4 g_m + 2C_1 C_4 L_4 R_3 R_4) + s^2 (2C_1 L_4 R_1 R_3 g_m + C_1 L_4 R_1 R_4 g_m + 2C_1 L_4 R_3 + C_1 L_4 R_4 + C_3 L_4 R_3 R_4 g_m + 2C_4 L_4 R_3 R_4 g_m) + s (2C_1 R_1 R_3 R_4 g_m + 2C_1 R_3 R_4 + 2L_4 R_3 g_m + L_4 R_4 g_m)}$$

$$\mathbf{9.368 \quad X-INVALID-ORDER-368} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_4 R_1 R_3 R_4 g_m s^3 + R_3 R_4 g_m + s^2 (C_1 L_4 R_1 R_3 g_m + C_4 L_4 R_3 R_4 g_m) + s (C_1 R_1 R_3 R_4 g_m + L_4 R_3 g_m)}{2 R_3 g_m + R_4 g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4) + s^3 (C_1 C_3 L_4 R_1 R_3 g_m + C_1 C_3 L_4 R_3 + 2 C_1 C_4 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + C_1 L_4 R_1 g_m + C_1 L_4 + C_3 L_4 R_3 g_m)}$$

$$\mathbf{9.369 \quad X-INVALID-ORDER-369} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_4 R_1 R_3 R_4 g_m s^3 + C_1 R_1 R_3 R_4 g_m s + C_4 L_4 R_3 R_4 g_m s^2 + R_3 R_4 g_m}{2 R_3 g_m + R_4 g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4) + s^3 (2 C_1 C_4 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4 + 2 C_4 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (C_1 R_1 R_3 R_4 g_m + L_4 R_3 g_m)}$$

$$\mathbf{9.370 \quad X-INVALID-ORDER-370} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 R_1 R_3 g_m s^2 + g_m + s (C_1 R_1 g_m + C_3 R_3 g_m)}{s^3 (2 C_1 C_3 C_4 R_1 R_3 g_m + 2 C_1 C_3 C_4 R_3) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_3 C_4 R_3 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.371 \quad X-INVALID-ORDER-371} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 R_1 R_3 R_4 g_m s^2 + R_4 g_m + s (C_1 R_1 R_4 g_m + C_3 R_3 R_4 g_m)}{2 g_m + s^3 (2 C_1 C_3 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 R_3 R_4) + s^2 (2 C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2 C_1 C_3 R_3 + C_1 C_3 R_4 + 2 C_1 C_4 R_1 R_4 g_m + 2 C_1 C_4 R_4 + 2 C_3 C_4 R_3 R_4 g_m) + s (2 C_1 R_1 g_m + 2 C_1 + 2 C_3 R_3 g_m + C_3 R_4 g_m + 2 C_4 R_4 g_m)}$$

$$\mathbf{9.372 \quad X-INVALID-ORDER-372} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 R_1 R_3 R_4 g_m s^3 + g_m + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + C_3 C_4 R_3 R_4 g_m) + s (C_1 R_1 g_m + C_3 R_3 g_m + C_4 R_4 g_m)}{s^3 (2 C_1 C_3 C_4 R_1 R_3 g_m + C_1 C_3 C_4 R_1 R_4 g_m + 2 C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.373 \quad X-INVALID-ORDER-373} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_4 R_1 R_3 g_m s^4 + g_m + s^3 (C_1 C_4 L_4 R_1 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_3 R_3 g_m)}{s^4 (C_1 C_3 C_4 L_4 R_1 g_m + C_1 C_3 C_4 L_4) + s^3 (2 C_1 C_3 C_4 R_1 R_3 g_m + 2 C_1 C_3 C_4 R_3 + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_3 C_4 R_3 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.374 \quad X-INVALID-ORDER-374} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_4 R_1 R_3 g_m s^3 + L_4 g_m s + s^2 (C_1 L_4 R_1 g_m + C_3 L_4 R_3 g_m)}{2 g_m + s^4 (2 C_1 C_3 C_4 L_4 R_1 R_3 g_m + 2 C_1 C_3 C_4 L_4 R_3) + s^3 (C_1 C_3 L_4 R_1 g_m + C_1 C_3 L_4 + 2 C_1 C_4 L_4 R_1 g_m + 2 C_1 C_4 L_4 + 2 C_3 C_4 L_4 R_3 g_m) + s^2 (2 C_1 C_3 R_1 R_3 g_m + 2 C_1 C_3 R_3 + C_3 L_4 g_m + 2 C_4 L_4 g_m) + s (2 C_1 R_1 g_m + 2 C_1 + 2 C_3 R_3 g_m)}$$

$$\mathbf{9.375 \quad X-INVALID-ORDER-375} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_4 R_1 R_3 g_m s^4 + g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_4 L_4 R_1 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + C_3 C_4 R_3 R_4 g_m + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_3 R_3 g_m + C_4 R_4 g_m)}{s^4 (C_1 C_3 C_4 L_4 R_1 g_m + C_1 C_3 C_4 L_4) + s^3 (2 C_1 C_3 C_4 R_1 R_3 g_m + C_1 C_3 C_4 R_1 R_4 g_m + 2 C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.376 \quad X-INVALID-ORDER-376} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_4 R_1 R_3 R_4 g_m s^3 + L_4 R_4 g_m s + s^2 (C_1 L_4 R_1 R_4 g_m + C_3 L_4 R_3 R_4 g_m)}{2 R_4 g_m + s^4 (2 C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_4 R_3 R_4) + s^3 (2 C_1 C_3 L_4 R_1 R_3 g_m + C_1 C_3 L_4 R_1 R_4 g_m + 2 C_1 C_3 L_4 R_3 + C_1 C_3 L_4 R_4 + 2 C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_4 + 2 C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (2 C_1 C_3 R_1 R_3 R_4 g_m + 2 C_1 C_3 R_3 R_4 + 2 C_1 L_4 R_1 g_m + 2 C_1 L_4 + 2 C_3 L_4 R_3 g_m)}$$

$$\mathbf{9.377 \quad X-INVALID-ORDER-377} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m s^4 + R_4 g_m + s^3 (C_1 C_3 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_4 g_m + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 L_4 R_1 g_m + C_3 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (C_1 R_1 R_4 g_m + C_3 R_3 R_4 g_m + L_4 g_m)}{2g_m + s^4 (2C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_4 R_3 + C_1 C_3 C_4 L_4 R_4) + s^3 (C_1 C_3 L_4 R_1 g_m + C_1 C_3 L_4 + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m + C_3 C_4 L_4 R_4 g_m) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4 + C_3 L_4 g_m + 2C_4 g_m) + s (C_1 R_1 g_m + C_3 R_3 g_m + L_4 g_m)}$$

$$\mathbf{9.378 \quad X-INVALID-ORDER-378} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m s^4 + R_4 g_m + s^3 (C_1 C_4 L_4 R_1 R_4 g_m + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_4 L_4 R_4 g_m) + s (C_1 R_1 R_4 g_m + C_3 R_3 R_4 g_m)}{2g_m + s^4 (2C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_4 R_3 + C_1 C_3 C_4 L_4 R_4) + s^3 (2C_1 C_3 C_4 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 R_3 R_4 + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m + C_3 C_4 L_4 R_4 g_m) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4 + C_3 L_4 g_m + 2C_4 g_m) + s (C_1 R_1 g_m + C_3 R_3 g_m + L_4 g_m)}$$

$$\mathbf{9.379 \quad X-INVALID-ORDER-379} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 R_1 R_4 g_m s^3 + C_1 R_1 R_4 g_m s + C_3 L_3 R_4 g_m s^2 + R_4 g_m}{2g_m + s^3 (2C_1 C_3 L_3 R_1 g_m + 2C_1 C_3 L_3) + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + 2C_3 L_3 g_m) + s (2C_1 R_1 g_m + 2C_1 + C_3 R_4 g_m)}$$

$$\mathbf{9.380 \quad X-INVALID-ORDER-380} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 R_1 g_m s^3 + C_1 R_1 g_m s + C_3 L_3 g_m s^2 + g_m}{2C_3 C_4 L_3 g_m s^3 + s^4 (2C_1 C_3 C_4 L_3 R_1 g_m + 2C_1 C_3 C_4 L_3) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.381 \quad X-INVALID-ORDER-381} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 R_1 R_4 g_m s^3 + C_1 R_1 R_4 g_m s + C_3 L_3 R_4 g_m s^2 + R_4 g_m}{2g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 R_4) + s^3 (2C_1 C_3 L_3 R_1 g_m + 2C_1 C_3 L_3 + 2C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + 2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_3 L_3 g_m) + s (2C_1 R_1 g_m + 2C_1 + C_3 R_4 g_m + 2C_4 R_4 g_m)}$$

$$\mathbf{9.382 \quad X-INVALID-ORDER-382} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 R_1 R_4 g_m s^4 + g_m + s^3 (C_1 C_3 L_3 R_1 g_m + C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 C_4 R_1 R_4 g_m + C_3 L_3 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m)}{s^4 (2C_1 C_3 C_4 L_3 R_1 g_m + 2C_1 C_3 C_4 L_3) + s^3 (C_1 C_3 C_4 R_1 R_4 g_m + C_1 C_3 C_4 R_4 + 2C_3 C_4 L_3 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.383 \quad X-INVALID-ORDER-383} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 g_m s^5 + C_1 R_1 g_m s + C_3 C_4 L_3 L_4 g_m s^4 + g_m + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_4 L_4 R_1 g_m) + s^2 (C_3 L_3 g_m + C_4 L_4 g_m)}{s^4 (2C_1 C_3 C_4 L_3 R_1 g_m + 2C_1 C_3 C_4 L_3 + C_1 C_3 C_4 L_4 R_1 g_m + C_1 C_3 C_4 L_4) + s^3 (2C_3 C_4 L_3 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.384 \quad X-INVALID-ORDER-384} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 L_4 R_1 g_m s^4 + C_1 L_4 R_1 g_m s^2 + C_3 L_3 L_4 g_m s^3 + L_4 g_m s}{2C_3 C_4 L_3 L_4 g_m s^4 + 2g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 g_m + 2C_1 C_3 C_4 L_3 L_4) + s^3 (2C_1 C_3 L_3 R_1 g_m + 2C_1 C_3 L_3 + C_1 C_3 L_4 R_1 g_m + C_1 C_3 L_4 + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4) + s^2 (2C_3 L_3 g_m + C_3 L_4 g_m + 2C_4 L_4 g_m) + s (2C_1 R_1 g_m + 2C_1)}$$

$$\mathbf{9.385 \quad X-INVALID-ORDER-385} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 g_m s^5 + g_m + s^4 (C_1 C_3 C_4 L_3 R_1 R_4 g_m + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_4 L_4 R_1 g_m + C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 C_4 R_1 R_4 g_m + C_3 L_3 g_m + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m)}{s^4 (2C_1 C_3 C_4 L_3 R_1 g_m + 2C_1 C_3 C_4 L_3 + C_1 C_3 C_4 L_4 R_1 g_m + C_1 C_3 C_4 L_4) + s^3 (C_1 C_3 C_4 R_1 R_4 g_m + C_1 C_3 C_4 R_4 + 2C_3 C_4 L_3 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.386 \quad X-INVALID-ORDER-386} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 L_4 R_1 R_4 g_m s^4 + C_1 L_4 R_1 R_4 g_m s^2 + C_3 L_3 L_4 R_4 g_m s^3 + L_4 R_4 g_m s}{2 R_4 g_m + s^5 (2 C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (2 C_1 C_3 L_3 L_4 R_1 g_m + 2 C_1 C_3 L_3 L_4 + 2 C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (2 C_1 C_3 L_3 R_1 R_4 g_m + 2 C_1 C_3 L_3 R_4 + C_1 C_3 L_4 R_1 R_4 g_m + C_1 C_3 L_4 R_4 + 2 C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_4 + 2 C_3 L_3 L_4 g_m) + s^2 (2 C_1 L_4 R_1 g_m +$$

$$\mathbf{9.387 \quad X-INVALID-ORDER-387} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m s^5 + R_4 g_m + s^4 (C_1 C_3 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (C_1 C_3 L_3 R_1 R_4 g_m + C_1 C_4 L_4 R_1 R_4 g_m + C_3 L_3 L_4 g_m) + s^2 (C_1 L_4 R_1 g_m + C_3 L_3 R_4 g_m + C_4 L_4 R_4 g_m) + s (C_1 R_1 R_4 g_m + L_4 g_m)}{2 g_m + s^5 (2 C_1 C_3 C_4 L_3 L_4 R_1 g_m + 2 C_1 C_3 C_4 L_3 L_4) + s^4 (C_1 C_3 C_4 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_4 R_4 + 2 C_3 C_4 L_3 L_4 g_m) + s^3 (2 C_1 C_3 L_3 R_1 g_m + 2 C_1 C_3 L_3 + C_1 C_3 L_4 R_1 g_m + C_1 C_3 L_4 + 2 C_1 C_4 L_4 R_1 g_m + 2 C_1 C_4 L_4 + C_3 C_4 L_4 R_4 g_m) + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + 2 C_3 L_3 g_m)}$$

$$\mathbf{9.388 \quad X-INVALID-ORDER-388} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m s^5 + C_1 R_1 R_4 g_m s + C_3 C_4 L_3 L_4 R_4 g_m s^4 + R_4 g_m + s^3 (C_1 C_3 L_3 R_1 R_4 g_m + C_1 C_4 L_4 R_1 R_4 g_m) + s^2 (C_3 L_3 R_4 g_m + C_4 L_4 R_4 g_m)}{2 g_m + s^5 (2 C_1 C_3 C_4 L_3 L_4 R_1 g_m + 2 C_1 C_3 C_4 L_3 L_4) + s^4 (2 C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_3 R_4 + C_1 C_3 C_4 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_4 R_4 + 2 C_3 C_4 L_3 L_4 g_m) + s^3 (2 C_1 C_3 L_3 R_1 g_m + 2 C_1 C_3 L_3 + 2 C_1 C_4 L_4 R_1 g_m + 2 C_1 C_4 L_4 + 2 C_3 C_4 L_3 R_4 g_m + C_3 C_4 L_4 R_4 g_m) + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + 2 C_3 L_3 g_m)}$$

$$\mathbf{9.389 \quad X-INVALID-ORDER-389} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_3 R_1 R_4 g_m s^2 + L_3 R_4 g_m s}{R_4 g_m + s^3 (C_1 C_3 L_3 R_1 R_4 g_m + C_1 C_3 L_3 R_4) + s^2 (2 C_1 L_3 R_1 g_m + 2 C_1 L_3 + C_3 L_3 R_4 g_m) + s (C_1 R_1 R_4 g_m + C_1 R_4 + 2 L_3 g_m)}$$

$$\mathbf{9.390 \quad X-INVALID-ORDER-390} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_3 R_1 g_m s^2 + L_3 g_m s}{g_m + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2 C_1 C_4 L_3 R_1 g_m + 2 C_1 C_4 L_3) + s^2 (C_3 L_3 g_m + 2 C_4 L_3 g_m) + s (C_1 R_1 g_m + C_1)}$$

$$\mathbf{9.391 \quad X-INVALID-ORDER-391} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_3 R_1 R_4 g_m s^2 + L_3 R_4 g_m s}{R_4 g_m + s^3 (C_1 C_3 L_3 R_1 R_4 g_m + C_1 C_3 L_3 R_4 + 2 C_1 C_4 L_3 R_1 R_4 g_m + 2 C_1 C_4 L_3 R_4) + s^2 (2 C_1 L_3 R_1 g_m + 2 C_1 L_3 + C_3 L_3 R_4 g_m + 2 C_4 L_3 R_4 g_m) + s (C_1 R_1 R_4 g_m + C_1 R_4 + 2 L_3 g_m)}$$

$$\mathbf{9.392 \quad X-INVALID-ORDER-392} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_3 R_1 R_4 g_m s^3 + L_3 g_m s + s^2 (C_1 L_3 R_1 g_m + C_4 L_3 R_4 g_m)}{g_m + s^4 (C_1 C_3 C_4 L_3 R_1 R_4 g_m + C_1 C_3 C_4 L_3 R_4) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2 C_1 C_4 L_3 R_1 g_m + 2 C_1 C_4 L_3 + C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 C_4 R_1 R_4 g_m + C_1 C_4 R_4 + C_3 L_3 g_m + 2 C_4 L_3 g_m) + s (C_1 R_1 g_m + C_1 + C_4 R_4 g_m)}$$

$$\mathbf{9.393 \quad X-INVALID-ORDER-393} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_3 L_4 R_1 g_m s^4 + C_1 L_3 R_1 g_m s^2 + C_4 L_3 L_4 g_m s^3 + L_3 g_m s}{C_3 C_4 L_3 L_4 g_m s^4 + g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_3 L_4) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2 C_1 C_4 L_3 R_1 g_m + 2 C_1 C_4 L_3 + C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4) + s^2 (C_3 L_3 g_m + 2 C_4 L_3 g_m + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_1)}$$

$$\mathbf{9.394 \quad X-INVALID-ORDER-394} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_3 L_4 R_1 g_m s^2 + L_3 L_4 g_m s}{2 L_3 g_m + L_4 g_m + s^3 (C_1 C_3 L_3 L_4 R_1 g_m + C_1 C_3 L_3 L_4 + 2 C_1 C_4 L_3 L_4 R_1 g_m + 2 C_1 C_4 L_3 L_4) + s^2 (C_3 L_3 L_4 g_m + 2 C_4 L_3 L_4 g_m) + s (2 C_1 L_3 R_1 g_m + 2 C_1 L_3 + C_1 L_4 R_1 g_m + C_1 L_4)}$$

$$\mathbf{9.395 \quad X-INVALID-ORDER-395} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_3 L_4 R_1 g_m s^4 + L_3 g_m s + s^3 (C_1 C_4 L_3 R_1 R_4 g_m + C_4 L_3 L_4 g_m) + s^2 (C_1 L_3 R_1 g_m + C_4 L_3 R_4 g_m)}{g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_3 L_4) + s^4 (C_1 C_3 C_4 L_3 R_1 R_4 g_m + C_1 C_3 C_4 L_3 R_4 + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2C_1 C_4 L_3 R_1 g_m + 2C_1 C_4 L_3 + C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 C_4 R_1 R_4 g_m + C_1 C_4 R_4 + C_3 L_3 g_m + 2C_4 L_3 R_1 g_m)}$$

$$\mathbf{9.396 \quad X-INVALID-ORDER-396} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_3 L_4 R_1 R_4 g_m s^2 + L_3 L_4 R_4 g_m s}{2L_3 R_4 g_m + L_4 R_4 g_m + s^3 (C_1 C_3 L_3 L_4 R_1 R_4 g_m + C_1 C_3 L_3 L_4 R_4 + 2C_1 C_4 L_3 L_4 R_1 R_4 g_m + 2C_1 C_4 L_3 L_4 R_4) + s^2 (2C_1 L_3 L_4 R_1 g_m + 2C_1 L_3 L_4 + C_3 L_3 L_4 R_4 g_m + 2C_4 L_3 L_4 R_4 g_m) + s (2C_1 L_3 R_1 R_4 g_m + 2C_1 L_3 R_4 + C_1 L_4 R_1 R_4 g_m + C_1 L_4 R_4 + 2L_3 L_4 g_m)}$$

$$\mathbf{9.397 \quad X-INVALID-ORDER-397} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_3 L_4 R_1 R_4 g_m s^4 + L_3 R_4 g_m s + s^3 (C_1 L_3 L_4 R_1 g_m + C_4 L_3 L_4 R_4 g_m) + s^2 (C_1 L_3 R_1 R_4 g_m + L_3 L_4 g_m)}{R_4 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (C_1 C_3 L_3 L_4 R_1 g_m + C_1 C_3 L_3 L_4 + 2C_1 C_4 L_3 L_4 R_1 g_m + 2C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (C_1 C_3 L_3 R_1 R_4 g_m + C_1 C_3 L_3 R_4 + C_1 C_4 L_4 R_1 R_4 g_m + C_1 C_4 L_4 R_4 + C_3 L_3 L_4 g_m + 2C_4 L_3 L_4 g_m) + s^2 (2C_1 L_3 R_1 g_m)}$$

$$\mathbf{9.398 \quad X-INVALID-ORDER-398} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_3 L_4 R_1 R_4 g_m s^4 + C_1 L_3 R_1 R_4 g_m s^2 + C_4 L_3 L_4 R_4 g_m s^3 + L_3 R_4 g_m s}{R_4 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (2C_1 C_4 L_3 L_4 R_1 g_m + 2C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (C_1 C_3 L_3 R_1 R_4 g_m + C_1 C_3 L_3 R_4 + 2C_1 C_4 L_3 R_1 R_4 g_m + 2C_1 C_4 L_3 R_4 + C_1 C_4 L_4 R_1 R_4 g_m + C_1 C_4 L_4 R_4 + 2C_4 L_3 L_4 g_m) + s^2 (2C_1 L_3 R_1 g_m + 2C_1 L_3 R_4)}$$

$$\mathbf{9.399 \quad X-INVALID-ORDER-399} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 R_1 R_4 g_m s^3 + R_4 g_m + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_3 L_3 R_4 g_m) + s (C_1 R_1 R_4 g_m + C_3 R_3 R_4 g_m)}{2g_m + s^3 (2C_1 C_3 L_3 R_1 g_m + 2C_1 C_3 L_3) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_3 L_3 g_m) + s (2C_1 R_1 g_m + 2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m)}$$

$$\mathbf{9.400 \quad X-INVALID-ORDER-400} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 R_1 g_m s^3 + g_m + s^2 (C_1 C_3 R_1 R_3 g_m + C_3 L_3 g_m) + s (C_1 R_1 g_m + C_3 R_3 g_m)}{s^4 (2C_1 C_3 C_4 L_3 R_1 g_m + 2C_1 C_3 C_4 L_3) + s^3 (2C_1 C_3 C_4 R_1 R_3 g_m + 2C_1 C_3 C_4 R_3 + 2C_3 C_4 L_3 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_3 C_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.401 \quad X-INVALID-ORDER-401} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 R_1 R_4 g_m s^3 + R_4 g_m + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_3 L_3 R_4 g_m) + s (C_1 R_1 R_4 g_m + C_3 R_3 R_4 g_m)}{2g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 R_4) + s^3 (2C_1 C_3 C_4 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 R_3 R_4 + 2C_1 C_3 L_3 R_1 g_m + 2C_1 C_3 L_3 + 2C_3 C_4 L_3 R_4 g_m) + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_3 C_4 R_3 R_4 g_m + 2C_3 L_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.402 \quad X-INVALID-ORDER-402} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 R_1 R_4 g_m s^4 + g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_3 L_3 R_1 g_m + C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + C_3 C_4 R_3 R_4 g_m + C_3 L_3 g_m) + s (C_1 R_1 g_m + C_3 R_3 g_m + C_4 R_4 g_m)}{s^4 (2C_1 C_3 C_4 L_3 R_1 g_m + 2C_1 C_3 C_4 L_3) + s^3 (2C_1 C_3 C_4 R_1 R_3 g_m + C_1 C_3 C_4 R_1 R_4 g_m + 2C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + 2C_3 C_4 L_3 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.403 \quad X-INVALID-ORDER-403} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 g_m s^5 + g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_4 L_4 R_1 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_3 L_3 g_m + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_3 R_3 g_m)}{s^4 (2C_1 C_3 C_4 L_3 R_1 g_m + 2C_1 C_3 C_4 L_3 + C_1 C_3 C_4 L_4 R_1 g_m + C_1 C_3 C_4 L_4) + s^3 (2C_1 C_3 C_4 R_1 R_3 g_m + 2C_1 C_3 C_4 R_3 + 2C_3 C_4 L_3 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_3 C_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.404 \quad X-INVALID-ORDER-404} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 L_4 R_1 g_m s^4 + L_4 g_m s + s^3 (C_1 C_3 L_4 R_1 R_3 g_m + C_3 L_3 L_4 g_m) + s^2 (C_1 L_4 R_1 g_m + C_3 L_4 R_3 g_m)}{2g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 g_m + 2C_1 C_3 C_4 L_3 L_4) + s^4 (2C_1 C_3 C_4 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_4 R_3 + 2C_3 C_4 L_3 L_4 g_m) + s^3 (2C_1 C_3 L_3 R_1 g_m + 2C_1 C_3 L_3 + C_1 C_3 L_4 R_1 g_m + C_1 C_3 L_4 + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m) + s^2 (2C_1 C_3 R_1 R_3 g_m + 2C_1 C_3 R_3 + 2C_3 C_4 L_4 R_3 g_m) + s (C_1 R_1 R_3 g_m + C_1 R_3 + C_3 L_4 R_3 g_m) + L_4 g_m}$$

$$\mathbf{9.405 \quad X-INVALID-ORDER-405} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 g_m s^5 + g_m + s^4 (C_1 C_3 C_4 L_3 R_1 R_4 g_m + C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_3 L_3 R_1 g_m + C_1 C_4 L_4 R_1 g_m + C_3 C_4 L_3 R_4 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + C_3 C_4 R_3 R_4 g_m + C_3 L_3 g_m + C_4 L_4 g_m)}{s^4 (2C_1 C_3 C_4 L_3 R_1 g_m + 2C_1 C_3 C_4 L_3 + C_1 C_3 C_4 L_4 R_1 g_m + C_1 C_3 C_4 L_4) + s^3 (2C_1 C_3 C_4 R_1 R_3 g_m + C_1 C_3 C_4 R_1 R_4 g_m + 2C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + 2C_3 C_4 L_3 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + C_3 + C_4 L_4 g_m) + L_4 g_m}$$

$$\mathbf{9.406 \quad X-INVALID-ORDER-406} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 L_4 R_1 R_4 g_m s^4 + L_4 R_4 g_m s + s^3 (C_1 C_3 L_4 R_1 R_3 R_4 g_m)}{2R_4 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (2C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_4 R_3 R_4 + 2C_1 C_3 L_3 L_4 R_1 g_m + 2C_1 C_3 L_3 L_4 + 2C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (2C_1 C_3 L_3 R_1 R_4 g_m + 2C_1 C_3 L_3 R_4 + 2C_1 C_3 L_4 R_1 R_3 g_m + C_1 C_3 L_4 R_1 R_4 g_m + 2C_1 C_3 L_4 R_3 + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + C_3 C_4 L_4 R_3 g_m) + s (C_1 R_1 R_3 g_m + C_1 R_3 + C_3 L_4 R_3 g_m) + L_4 R_4 g_m}$$

$$\mathbf{9.407 \quad X-INVALID-ORDER-407} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m s^5 + R_4 g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (C_1 C_3 L_3 R_1 R_4 g_m + C_1 C_3 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_4 g_m + C_3 C_4 L_4 R_3 R_4 g_m + C_3 L_3 L_4 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + C_3 C_4 R_3 R_4 g_m + C_3 L_3 g_m + C_4 L_4 g_m)}{2g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 g_m + 2C_1 C_3 C_4 L_3 L_4) + s^4 (2C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_4 R_3 + C_1 C_3 C_4 L_4 R_4 + 2C_3 C_4 L_3 L_4 g_m) + s^3 (2C_1 C_3 L_3 R_1 g_m + 2C_1 C_3 L_3 + C_1 C_3 L_4 R_1 g_m + C_1 C_3 L_4 + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m + C_3 C_4 L_4 R_4 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + C_3 + C_4 L_4 g_m) + L_4 g_m}$$

$$\mathbf{9.408 \quad X-INVALID-ORDER-408} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m s^5 + R_4 g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (C_1 C_3 L_3 R_1 R_4 g_m + C_1 C_3 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_4 g_m + C_3 C_4 L_4 R_3 R_4 g_m + C_3 L_3 L_4 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + C_3 C_4 R_3 R_4 g_m + C_3 L_3 g_m + C_4 L_4 g_m)}{2g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 g_m + 2C_1 C_3 C_4 L_3 L_4) + s^4 (2C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 R_4 + 2C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_4 R_3 + C_1 C_3 C_4 L_4 R_4 + 2C_3 C_4 L_3 L_4 g_m) + s^3 (2C_1 C_3 C_4 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 R_3 R_4 + 2C_1 C_3 L_3 R_1 g_m + 2C_1 C_3 L_3 + C_1 C_3 L_4 R_1 g_m + C_1 C_3 L_4 + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m + C_3 C_4 L_4 R_4 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + C_3 + C_4 L_4 g_m) + L_4 g_m}$$

$$\mathbf{9.409 \quad X-INVALID-ORDER-409} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_3 R_1 R_3 R_4 g_m s^2 + L_3 R_3 R_4 g_m s}{R_3 R_4 g_m + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_3 R_3 R_4) + s^2 (2C_1 L_3 R_1 R_3 g_m + C_1 L_3 R_1 R_4 g_m + 2C_1 L_3 R_3 + C_1 L_3 R_4 + C_3 L_3 R_3 R_4 g_m) + s (C_1 R_1 R_3 R_4 g_m + C_1 R_3 R_4 + 2L_3 R_3 g_m + L_3 R_4 g_m) + L_3 R_3 g_m}$$

$$\mathbf{9.410 \quad X-INVALID-ORDER-410} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_3 R_1 R_3 g_m s^2 + L_3 R_3 g_m s}{R_3 g_m + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_3 + 2C_1 C_4 L_3 R_1 R_3 g_m + 2C_1 C_4 L_3 R_3) + s^2 (C_1 L_3 R_1 g_m + C_1 L_3 + C_3 L_3 R_3 g_m + 2C_4 L_3 R_3 g_m) + s (C_1 R_1 R_3 g_m + C_1 R_3 + L_3 g_m) + L_3 g_m}$$

$$\mathbf{9.411 \quad X-INVALID-ORDER-411} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_3 R_1 R_3 R_4 g_m s^2 + L_3 R_3 R_4 g_m s}{R_3 R_4 g_m + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_3 R_3 R_4 + 2C_1 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_4 L_3 R_3 R_4) + s^2 (2C_1 L_3 R_1 R_3 g_m + C_1 L_3 R_1 R_4 g_m + 2C_1 L_3 R_3 + C_1 L_3 R_4 + C_3 L_3 R_3 R_4 g_m + 2C_4 L_3 R_3 R_4 g_m) + s (C_1 R_1 R_3 R_4 g_m + C_1 R_3 R_4 + 2L_3 R_3 g_m + L_3 R_4 g_m) + L_3 R_3 g_m}$$

$$\mathbf{9.412 \quad X-INVALID-ORDER-412} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_3 R_1 R_3 R_4 g_m s^3 + L_3 R_3 g_m s + s^2 (C_1 L_3 R_1 R_3 g_m + C_4 L_3 R_3 R_4 g_m)}{R_3 g_m + s^4 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_3 R_3 R_4) + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_3 + 2C_1 C_4 L_3 R_1 R_3 g_m + C_1 C_4 L_3 R_1 R_4 g_m + 2C_1 C_4 L_3 R_3 + C_1 C_4 L_3 R_4 + C_3 C_4 L_3 R_3 R_4 g_m) + s^2 (C_1 C_4 R_1 R_3 R_4 g_m + C_1 C_4 R_3 R_4 + C_1 L_3 R_1 g_m + C_1 L_3 + C_3 L_3 R_3 g_m + 2C_4 L_3 R_3 g_m) + s (C_1 R_1 R_3 g_m + C_1 R_3 + L_3 g_m) + L_3 g_m}$$

9.413 X-INVALID-ORDER-413 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_3 L_4 R_1 R_3 g_m s^4 + C_1 L_3 R_1 R_3 g_m s^2 + C_4 L_3 L_4 R_3 g_m s^3 + L_3 R_3 g_m s}{R_3 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_3) + s^4 (C_1 C_4 L_3 L_4 R_1 g_m + C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_3 + 2 C_1 C_4 L_3 R_1 R_3 g_m + 2 C_1 C_4 L_3 R_3 + C_1 C_4 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_3 + C_4 L_3 L_4 g_m) + s^2 (C_1 L_3 R_1 g_m + C_1 L_3 + C_3 L_3)}$$

9.414 X-INVALID-ORDER-414 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_3 L_4 R_1 R_3 g_m s^2 + L_3 L_4 R_3 g_m s}{2L_3 R_3 g_m + L_4 R_3 g_m + s^3 (C_1 C_3 L_3 L_4 R_1 R_3 g_m + C_1 C_3 L_3 L_4 R_3 + 2C_1 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_4 L_3 L_4 R_3) + s^2 (C_1 L_3 L_4 R_1 g_m + C_1 L_3 L_4 + C_3 L_3 L_4 R_3 g_m + 2C_4 L_3 L_4 R_3 g_m) + s (2C_1 L_3 R_1 R_3 g_m + 2C_1 L_3 R_3 + C_1 L_4 R_1 R_3 g_m + C_1 L_4 R_3 + L_3 L_4 g_m)}$$

9.415 X-INVALID-ORDER-415 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_3 L_4 R_1 R_3 g_m s^4 + L_3 R_3 g_m s + s^3 (C_1 C_4 L_3 R_1 R_3 R_4 g_m + C_4 L_3 L_4 R_3)}{R_3 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_3) + s^4 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_3 R_3 R_4 + C_1 C_4 L_3 L_4 R_1 g_m + C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_3 + 2 C_1 C_4 L_3 R_1 R_3 g_m + C_1 C_4 L_3 R_1 R_4 g_m + 2 C_1 C_4 L_3 R_3 + C_1 C_4 L_3 R_4 +$$

9.416 X-INVALID-ORDER-416 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_3 L_4 R_1 R_3 R_4 g_m s^2 + L_3 L_4 R_3 R_4 g_m s}{2L_3 R_3 R_4 g_m + L_4 R_3 R_4 g_m + s^3 (C_1 C_3 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 L_3 L_4 R_3 R_4 + 2C_1 C_4 L_3 L_4 R_1 R_3 R_4 g_m + 2C_1 C_4 L_3 L_4 R_3 R_4) + s^2 (2C_1 L_3 L_4 R_1 R_3 g_m + C_1 L_3 L_4 R_1 R_4 g_m + 2C_1 L_3 L_4 R_3 + C_1 L_3 L_4 R_4 + C_3 L_3 L_4 R_3 R_4 g_m + 2C_4 L_3 L_4 R_3 R_4 g_m) + s (2C_1 L_3 R_1 R_3 R_4 g_m + 2C_1 L_3 R_1 R_4 g_m + 2C_1 L_3 R_3 R_4 + 2C_1 L_3 R_4 + C_3 L_3 R_3 R_4 g_m + 2C_4 L_3 R_3 R_4 g_m) + C_3 L_3 R_3 R_4 + 2C_4 L_3 R_3 R_4}.$$

9.417 X-INVALID-ORDER-417 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_3 L_4 R_1 R_3 R_4 g_m s^4 + L_3 R_3 R_4 g_m}{R_3 R_4 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_3 R_4) + s^4 (C_1 C_3 L_3 L_4 R_1 R_3 g_m + C_1 C_3 L_3 L_4 R_3 + 2C_1 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_3 L_4 R_1 R_4 g_m + 2C_1 C_4 L_3 L_4 R_3 + C_1 C_4 L_3 L_4 R_4 + C_3 C_4 L_3 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_3 R_3 R_4 + C_1 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_4 L_4 R_3 R_4 + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (C_1 C_3 L_3 R_1 R_3 + C_1 C_3 L_3 R_3 + C_1 C_4 L_4 R_1 R_3 + C_1 C_4 L_4 R_3 + C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4 + C_3 C_4 L_4 R_4 g_m) + s (C_1 C_3 L_3 R_1 + C_1 C_3 L_3 R_3 + C_1 C_4 L_4 R_1 + C_1 C_4 L_4 R_3 + C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4) + C_1 C_4 L_4 R_1 + C_3 C_4 L_4 R_4}.$$

9.418 X-INVALID-ORDER-418 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_3 L_4 R_1 R_3 R_4 g_m s^4 + C_1 L_3 R_1 R_3 R_4 g_m s^2 + C_4 L_3 L_4 R_3 R_4 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_3 R_4) + s^4 (2C_1 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_3 L_4 R_1 R_4 g_m + 2C_1 C_4 L_3 L_4 R_3 + C_1 C_4 L_3 L_4 R_4 + C_3 C_4 L_3 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_3 R_3 R_4 + 2C_1 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_4 L_3 R_3 R_4 + C_1 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_4 L_4 R_3 R_4)}{R_3 R_4 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_3 R_4) + s^4 (2C_1 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_3 L_4 R_1 R_4 g_m + 2C_1 C_4 L_3 L_4 R_3 + C_1 C_4 L_3 L_4 R_4 + C_3 C_4 L_3 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_3 R_3 R_4 + 2C_1 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_4 L_3 R_3 R_4 + C_1 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_4 L_4 R_3 R_4)}$$

9.419 X-INVALID-ORDER-419 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_3 R_1 R_3 R_4 g_m s^3 + R_3 R_4 g_m + s^2 (C_1 L_3 R_1 R_4 g_m + C_3 L_3 R_3 R_4 g_m) + s (C_1 R_1 R_3 R_4 g_m + L_3 R_4 g_m)}{2 R_3 g_m + R_4 g_m + s^3 (2 C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_1 R_4 g_m + 2 C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4) + s^2 (2 C_1 L_3 R_1 g_m + 2 C_1 L_3 + 2 C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m) + s (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2 C_1 R_3 + C_1 R_4 + 2 L_3 g_m)}$$

9.420 X-INVALID-ORDER-420 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_3 R_1 R_3 g_m s^3 + R_3 g_m + s^2 (C_1 L_3 R_1 g_m + C_3 L_3 R_3 g_m) + s (C_1 R_1 R_3 g_m + L_3 g_m)}{g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 R_3) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2C_1 C_4 L_3 R_1 g_m + 2C_1 C_4 L_3 + 2C_3 C_4 L_3 R_3 g_m) + s^2 (2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_3 L_3 g_m + 2C_4 L_3 g_m) + s (C_1 R_1 g_m + C_1 + 2C_4 R_3 g_m)}$$

9.421 X-INVALID-ORDER-421 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_3 R_1 R_3 R_4 g_m s^3 + R_3 R_4 g_m + s^2 (C_1 L_3 R_1 R_4 g_m + C_3 L_3 R_3 R_4 g_m) + s (C_1 R_1 R_3 R_4 g_m + L_3 R_4 g_m)}{2 R_3 g_m + R_4 g_m + s^4 (2 C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_3 R_3 R_4) + s^3 (2 C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_1 R_4 g_m + 2 C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4 + 2 C_1 C_4 L_3 R_1 R_4 g_m + 2 C_1 C_4 L_3 R_4 + 2 C_3 C_4 L_3 R_3 R_4 g_m) + s^2 (2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4 + 2 C_1 L_3 R_1 g_m + 2 C_1 L_3 +$$

$$\mathbf{9.422 \quad X-INVALID-ORDER-422} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m s^4 + R_3 g_m + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_4 L_3 R_1 R_4 g_m + C_3 C_4 L_3 R_3 R_4 g_m) + s^2 (C_1 C_4 R_1 R_3 R_4 g_m + C_1 L_3 R_1 g_m + C_3 L_3 R_3 g_m + C_4 L_3 R_4 g_m) + s (C_1 R_1 R_3 g_m + C_4 R_3 R_4 g_m + L_3 g_m)}{g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2C_1 C_4 L_3 R_1 g_m + 2C_1 C_4 L_3 + 2C_3 C_4 L_3 R_3 g_m + C_3 C_4 L_3 R_4 g_m) + s^2 (2C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_3 L_3 g_m + 2C_4 L_3 g_m) + s (C_1 R_1 g_m + C_4 R_3 g_m + L_3 g_m)}$$

$$\mathbf{9.423 \quad X-INVALID-ORDER-423} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m s^5 + R_3 g_m + s^4 (C_1 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_3 g_m + C_4 L_3 L_4 g_m) + s^2 (C_1 L_3 R_1 g_m + C_3 L_3 R_3 g_m + C_4 L_4 R_3 g_m) + s (C_1 R_1 R_3 g_m + L_3 g_m)}{g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_3 L_4) + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 R_3 + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2C_1 C_4 L_3 R_1 g_m + 2C_1 C_4 L_3 + C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + 2C_3 C_4 L_3 R_3 g_m) + s^2 (2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_3 L_3 g_m + 2C_4 L_3 g_m) + s (C_1 R_1 g_m + C_4 R_3 g_m + L_3 g_m)}$$

$$\mathbf{9.424 \quad X-INVALID-ORDER-424} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 L_4 R_1 R_3 g_m s^4 + L_4 R_3 g_m s + s^3 (C_1 L_3 L_4 R_1 g_m + C_3 L_3 L_4 R_3 g_m) + s^2 (C_1 L_4 R_1 R_3 g_m + L_3 L_4 g_m)}{2R_3 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3) + s^4 (C_1 C_3 L_3 L_4 R_1 g_m + C_1 C_3 L_3 L_4 + 2C_1 C_4 L_3 L_4 R_1 g_m + 2C_1 C_4 L_3 L_4 + 2C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (2C_1 C_3 L_3 R_1 R_3 g_m + 2C_1 C_3 L_3 R_3 + 2C_1 C_4 L_4 R_1 R_3 g_m + 2C_1 C_4 L_4 R_3 + C_3 L_3 L_4 g_m + 2C_4 L_3 L_4 g_m) + s^2 (2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_3 L_3 g_m + 2C_4 L_3 g_m) + s (C_1 R_1 g_m + C_4 R_3 g_m + L_3 g_m)}$$

$$\mathbf{9.425 \quad X-INVALID-ORDER-425} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m s^5 + R_3 g_m + s^4 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_1 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_4 L_3 R_1 R_4 g_m + C_1 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_3 R_3 R_4 g_m + C_4 L_3 L_4 g_m) + s^2 (C_1 C_4 R_1 R_3 R_4 g_m + C_1 C_4 R_1 R_4 g_m + C_1 C_4 R_3 R_4 g_m + L_3 L_4 g_m)}{g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_3 L_4) + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2C_1 C_4 L_3 R_1 g_m + 2C_1 C_4 L_3 + C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + 2C_3 C_4 L_3 R_3 g_m + C_3 C_4 L_3 R_4 g_m) + s^2 (2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_3 L_3 g_m + 2C_4 L_3 g_m) + s (C_1 R_1 g_m + C_4 R_3 g_m + L_3 g_m)}$$

$$\mathbf{9.426 \quad X-INVALID-ORDER-426} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 L_4 R_1 R_3 R_4 g_m s^4 + L_4 R_3 g_m s + s^3 (C_1 L_3 L_4 R_1 g_m + C_3 L_3 L_4 R_3 g_m) + s^2 (C_1 L_4 R_1 R_3 g_m + L_3 L_4 g_m)}{2R_3 R_4 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 R_4) + s^4 (2C_1 C_3 L_3 L_4 R_1 R_3 g_m + C_1 C_3 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 L_3 L_4 R_3 + C_1 C_3 L_3 L_4 R_4 + 2C_1 C_4 L_3 L_4 R_1 R_4 g_m + 2C_1 C_4 L_3 L_4 R_4 + 2C_3 C_4 L_3 L_4 R_3 R_4 g_m) + s^3 (2C_1 C_3 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 L_3 R_3 R_4 + 2C_1 C_4 L_4 R_1 R_3 g_m + 2C_1 C_4 L_4 R_3 + C_3 L_3 L_4 g_m + 2C_4 L_3 L_4 g_m) + s^2 (2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_3 L_3 g_m + 2C_4 L_3 g_m) + s (C_1 R_1 g_m + C_4 R_3 g_m + L_3 g_m)}$$

$$\mathbf{9.427 \quad X-INVALID-ORDER-427} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m s^5 + R_3 R_4 g_m + s^4 (C_1 C_3 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_3 L_4 R_1 R_4 g_m + C_3 C_4 L_3 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 g_m + C_1 C_4 L_3 R_1 R_4 g_m + C_1 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_3 R_3 R_4 g_m + C_4 L_3 L_4 g_m) + s^2 (C_1 C_4 R_1 R_3 R_4 g_m + C_1 C_4 R_1 R_4 g_m + C_1 C_4 R_3 R_4 g_m + L_3 L_4 g_m)}{2R_3 g_m + R_4 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (C_1 C_3 L_3 L_4 R_1 g_m + C_1 C_3 L_3 L_4 + 2C_1 C_4 L_3 L_4 R_1 g_m + 2C_1 C_4 L_3 L_4 + 2C_3 C_4 L_3 L_4 R_3 g_m + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (2C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_1 R_4 g_m + 2C_1 C_4 L_4 R_1 R_3 g_m + 2C_1 C_4 L_4 R_3 + C_3 L_3 L_4 g_m + 2C_4 L_3 L_4 g_m) + s^2 (2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_3 L_3 g_m + 2C_4 L_3 g_m) + s (C_1 R_1 g_m + C_4 R_3 g_m + L_3 g_m)}$$

$$\mathbf{9.428 \quad X-INVALID-ORDER-428} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m s^5 + R_3 R_4 g_m + s^4 (C_1 C_4 L_3 L_4 R_1 R_4 g_m + C_3 C_4 L_3 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 g_m + C_1 C_4 L_3 R_1 R_4 g_m + C_1 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_3 R_3 R_4 g_m + C_4 L_3 L_4 g_m) + s^2 (C_1 C_4 R_1 R_3 R_4 g_m + C_1 C_4 R_1 R_4 g_m + C_1 C_4 R_3 R_4 g_m + L_3 L_4 g_m)}{2R_3 g_m + R_4 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 R_4 + 2C_1 C_4 L_3 L_4 R_1 g_m + 2C_1 C_4 L_3 L_4 + 2C_3 C_4 L_3 L_4 R_3 g_m + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (2C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_1 R_4 g_m + 2C_1 C_4 L_4 R_1 R_3 g_m + 2C_1 C_4 L_4 R_3 + C_3 L_3 L_4 g_m + 2C_4 L_3 L_4 g_m) + s^2 (2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_3 L_3 g_m + 2C_4 L_3 g_m) + s (C_1 R_1 g_m + C_4 R_3 g_m + L_3 g_m)}$$

$$\mathbf{9.429 \quad X-INVALID-ORDER-429} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 R_1 R_3 R_4 g_m s^3 + C_1 R_1 R_3 R_4 g_m s + C_3 L_3 R_3 R_4 g_m s^2 + R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s^3 (2C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_1 R_4 g_m + 2C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4) + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m) + s (2C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m)}$$

$$\mathbf{9.430 \quad X-INVALID-ORDER-430} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 R_1 R_3 g_m s^3 + C_1 R_1 R_3 g_m s + C_3 L_3 R_3 g_m s^2 + R_3 g_m}{g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 R_3) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2C_3 C_4 L_3 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_3 L_3 g_m) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m)}$$

$$\mathbf{9.431 \quad X-INVALID-ORDER-431} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 R_1 R_3 R_4 g_m s^3 + C_1 R_1 R_3 R_4 g_m s + C_3 L_3 R_3 R_4 g_m s^2 + R_3 R_4 g_m}{2 R_3 g_m + R_4 g_m + s^4 (2 C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_3 R_3 R_4) + s^3 (2 C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_1 R_4 g_m + 2 C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4 + 2 C_3 C_4 L_3 R_3 R_4 g_m) + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4 + 2 C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m)}$$

$$\mathbf{9.432 \quad X-INVALID-ORDER-432} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m s^4 + R_3 g_m + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_3 C_4 L_3 R_3 R_4 g_m) + s^2 (C_1 C_4 R_1 R_3 R_4 g_m + C_3 L_3 R_3 g_m) + s (C_1 R_1 R_3 g_m + C_4 R_3 R_4 g_m)}{g_m + s^4 (2 C_1 C_3 C_4 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4) + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 R_3 R_4 + C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2 C_3 C_4 L_3 R_3 g_m + C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2 C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2 C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m)}$$

$$\mathbf{9.433 \quad X-INVALID-ORDER-433} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m s^5 + C_1 R_1 R_3 g_m s + C_3 C_4 L_3 L_4 R_3 g_m s^4 + R_3 g_m + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_3 g_m) + s^2 (C_3 L_3 R_3 g_m + C_4 L_4 R_3 g_m)}{g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_3 L_4) + s^4 (2 C_1 C_3 C_4 L_3 R_1 R_3 g_m + 2 C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_4 R_3 + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + 2 C_3 C_4 L_3 R_3 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2 C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2 C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m)}$$

$$\mathbf{9.434 \quad X-INVALID-ORDER-434} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 L_4 R_1 R_3 g_m s^4 + C_1 L_4 R_1 R_3 g_m s^2 + C_3 L_3 L_4 R_3 g_m s^3 + L_4 R_3 g_m s}{2 R_3 g_m + s^5 (2 C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2 C_1 C_3 C_4 L_3 L_4 R_3) + s^4 (C_1 C_3 L_3 L_4 R_1 g_m + C_1 C_3 L_3 L_4 + 2 C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (2 C_1 C_3 L_3 R_1 R_3 g_m + 2 C_1 C_3 L_3 R_3 + C_1 C_3 L_4 R_1 R_3 g_m + C_1 C_3 L_4 R_3 + 2 C_1 C_4 L_4 R_1 R_3 g_m + 2 C_1 C_4 L_4 R_3 + C_3 L_3 L_4 g_m) + s^2 (C_1 L_4 R_1 g_m + C_1 L_4 R_3)}$$

$$\mathbf{9.435 \quad X-INVALID-ORDER-435} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m s^5 + R_3 g_m + s^4 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_3 g_m) + s^2 (C_3 L_3 R_3 g_m + C_4 L_4 R_3 g_m)}{g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_3 L_4) + s^4 (2 C_1 C_3 C_4 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_4 R_3 + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 R_3 R_4 + C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 R_3)}$$

$$\mathbf{9.436 \quad X-INVALID-ORDER-436} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_3 L_4 R_1 R_3 R_4 g_m s^4 + C_1 L_4 R_1 R_3 R_4 g_m s^2 + C_3 L_3 L_4 R_3 R_4 g_m s^3 + L_4 R_3 R_4 g_m s}{2 R_3 R_4 g_m + s^5 (2 C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_3 L_4 R_3 R_4) + s^4 (2 C_1 C_3 L_3 L_4 R_1 R_3 g_m + C_1 C_3 L_3 L_4 R_1 R_4 g_m + 2 C_1 C_3 L_3 L_4 R_3 + C_1 C_3 L_3 L_4 R_4 + 2 C_3 C_4 L_3 L_4 R_3 R_4 g_m) + s^3 (2 C_1 C_3 L_3 R_1 R_3 R_4 g_m + 2 C_1 C_3 L_3 R_3 R_4 + C_1 C_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 L_4 R_3 R_4 + 2 C_3 C_4 L_3 L_4 R_3 g_m + C_3 C_4 L_3 L_4 R_4 g_m)}$$

$$\mathbf{9.437 \quad X-INVALID-ORDER-437} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m s^5 + R_3 R_4 g_m + s^4 (C_1 C_3 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_3 g_m) + s^3 (C_3 L_3 R_3 g_m + C_4 L_4 R_3 g_m)}{2 R_3 g_m + R_4 g_m + s^5 (2 C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4 + C_1 C_3 L_3 L_4 R_1 g_m + C_1 C_3 L_3 L_4 + 2 C_3 C_4 L_3 L_4 R_3 g_m + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (2 C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_3 + 2 C_1 C_4 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_3)}$$

$$\mathbf{9.438 \quad X-INVALID-ORDER-438} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m s^5 + R_3 R_4 g_m + s^4 (C_1 C_3 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_3 g_m) + s^3 (C_3 L_3 R_3 g_m + C_4 L_4 R_3 g_m)}{2 R_3 g_m + R_4 g_m + s^5 (2 C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (2 C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_3 R_3 R_4 + C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4 + 2 C_3 C_4 L_3 L_4 R_3 g_m + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (2 C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_3 + 2 C_1 C_4 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_3)}$$

$$\mathbf{9.439 \quad X-INVALID-ORDER-439} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_3 g_m s^2 + R_3 g_m}{2 C_1 C_4 L_1 R_3 g_m s^3 + g_m + s^2 (2 C_1 C_4 R_3 + C_1 L_1 g_m) + s (C_1 + 2 C_4 R_3 g_m)}$$

$$\mathbf{9.440} \quad \mathbf{X-INVALID-ORDER-440} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_3 R_4 g_m s^2 + R_3 R_4 g_m}{2C_1 C_4 L_1 R_3 R_4 g_m s^3 + 2R_3 g_m + R_4 g_m + s^2 (2C_1 C_4 R_3 R_4 + 2C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m) + s (2C_1 R_3 + C_1 R_4 + 2C_4 R_3 R_4 g_m)}$$

$$\mathbf{9.441} \quad \mathbf{X-INVALID-ORDER-441} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 R_3 R_4 g_m s^3 + C_1 L_1 R_3 g_m s^2 + C_4 R_3 R_4 g_m s + R_3 g_m}{g_m + s^3 (2C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m) + s^2 (2C_1 C_4 R_3 + C_1 C_4 R_4 + C_1 L_1 g_m) + s (C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

$$\mathbf{9.442} \quad \mathbf{X-INVALID-ORDER-442} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 g_m s^4 + R_3 g_m + s^2 (C_1 L_1 R_3 g_m + C_4 L_4 R_3 g_m)}{C_1 C_4 L_1 L_4 g_m s^4 + g_m + s^3 (2C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_4) + s^2 (2C_1 C_4 R_3 + C_1 L_1 g_m + C_4 L_4 g_m) + s (C_1 + 2C_4 R_3 g_m)}$$

$$\mathbf{9.443} \quad \mathbf{X-INVALID-ORDER-443} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 R_3 g_m s^3 + L_4 R_3 g_m s}{2C_1 C_4 L_1 L_4 R_3 g_m s^4 + 2R_3 g_m + s^3 (2C_1 C_4 L_4 R_3 + C_1 L_1 L_4 g_m) + s^2 (2C_1 L_1 R_3 g_m + C_1 L_4 + 2C_4 L_4 R_3 g_m) + s (2C_1 R_3 + L_4 g_m)}$$

$$\mathbf{9.444} \quad \mathbf{X-INVALID-ORDER-444} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 g_m s^4 + C_1 C_4 L_1 R_3 R_4 g_m s^3 + C_4 R_3 R_4 g_m s + R_3 g_m + s^2 (C_1 L_1 R_3 g_m + C_4 L_4 R_3 g_m)}{C_1 C_4 L_1 L_4 g_m s^4 + g_m + s^3 (2C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m + C_1 C_4 L_4) + s^2 (2C_1 C_4 R_3 + C_1 C_4 R_4 + C_1 L_1 g_m + C_4 L_4 g_m) + s (C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

$$\mathbf{9.445} \quad \mathbf{X-INVALID-ORDER-445} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 R_3 R_4 g_m s^3 + L_4 R_3 R_4 g_m s}{2C_1 C_4 L_1 L_4 R_3 R_4 g_m s^4 + 2R_3 R_4 g_m + s^3 (2C_1 C_4 L_4 R_3 R_4 + 2C_1 L_1 L_4 R_3 g_m + C_1 L_1 L_4 R_4 g_m) + s^2 (2C_1 L_1 R_3 R_4 g_m + 2C_1 L_4 R_3 + C_1 L_4 R_4 + 2C_4 L_4 R_3 R_4 g_m) + s (2C_1 R_3 R_4 + 2L_4 R_3 g_m + L_4 R_4 g_m)}$$

$$\mathbf{9.446} \quad \mathbf{X-INVALID-ORDER-446} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 R_4 g_m s^4 + C_1 L_1 L_4 R_3 g_m s^3 + L_4 R_3 g_m s + R_3 R_4 g_m + s^2 (C_1 L_1 R_3 R_4 g_m + C_4 L_4 R_3 R_4 g_m)}{2R_3 g_m + R_4 g_m + s^4 (2C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (2C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + C_1 L_1 L_4 g_m) + s^2 (2C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m + C_1 L_4 + 2C_4 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (2C_1 R_3 + C_1 R_4 + L_4 g_m)}$$

$$\mathbf{9.447} \quad \mathbf{X-INVALID-ORDER-447} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 R_4 g_m s^4 + R_3 R_4 g_m + s^2 (C_1 L_1 R_3 R_4 g_m + C_4 L_4 R_3 R_4 g_m)}{2R_3 g_m + R_4 g_m + s^4 (2C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (2C_1 C_4 L_1 R_3 R_4 g_m + 2C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4) + s^2 (2C_1 C_4 R_3 R_4 + 2C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m + 2C_4 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (2C_1 R_3 + C_1 R_4 + 2C_4 R_3 R_4 g_m)}$$

$$\mathbf{9.448} \quad \mathbf{X-INVALID-ORDER-448} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_4 g_m s^2 + R_4 g_m}{C_1 C_3 L_1 R_4 g_m s^3 + 2g_m + s^2 (C_1 C_3 R_4 + 2C_1 L_1 g_m) + s (2C_1 + C_3 R_4 g_m)}$$

9.449 X-INVALID-ORDER-449 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 g_m s^2 + g_m}{s^3 (C_1 C_3 L_1 g_m + 2C_1 C_4 L_1 g_m) + s^2 (C_1 C_3 + 2C_1 C_4) + s (C_3 g_m + 2C_4 g_m)}$$

9.450 X-INVALID-ORDER-450 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 R_4 g_m s^2 + R_4 g_m}{2g_m + s^3 (C_1 C_3 L_1 R_4 g_m + 2C_1 C_4 L_1 R_4 g_m) + s^2 (C_1 C_3 R_4 + 2C_1 C_4 R_4 + 2C_1 L_1 g_m) + s (2C_1 + C_3 R_4 g_m + 2C_4 R_4 g_m)}$$

9.451 X-INVALID-ORDER-451 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 R_4 g_m s^3 + C_1 L_1 g_m s^2 + C_4 R_4 g_m s + g_m}{C_1 C_3 C_4 L_1 R_4 g_m s^4 + s^3 (C_1 C_3 C_4 R_4 + C_1 C_3 L_1 g_m + 2C_1 C_4 L_1 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

9.452 X-INVALID-ORDER-452 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 g_m s^4 + g_m + s^2 (C_1 L_1 g_m + C_4 L_4 g_m)}{C_1 C_3 C_4 L_1 L_4 g_m s^5 + C_1 C_3 C_4 L_4 s^4 + s^3 (C_1 C_3 L_1 g_m + 2C_1 C_4 L_1 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 + 2C_1 C_4) + s (C_3 g_m + 2C_4 g_m)}$$

9.453 X-INVALID-ORDER-453 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_4 g_m s^3 + L_4 g_m s}{2C_1 s + 2g_m + s^4 (C_1 C_3 L_1 L_4 g_m + 2C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_4 + 2C_1 C_4 L_4) + s^2 (2C_1 L_1 g_m + C_3 L_4 g_m + 2C_4 L_4 g_m)}$$

9.454 X-INVALID-ORDER-454 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 g_m s^4 + C_1 C_4 L_1 R_4 g_m s^3 + C_4 R_4 g_m s + g_m + s^2 (C_1 L_1 g_m + C_4 L_4 g_m)}{C_1 C_3 C_4 L_1 L_4 g_m s^5 + s^4 (C_1 C_3 C_4 L_1 R_4 g_m + C_1 C_3 C_4 L_4) + s^3 (C_1 C_3 C_4 R_4 + C_1 C_3 L_1 g_m + 2C_1 C_4 L_1 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

9.455 X-INVALID-ORDER-455 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_4 R_4 g_m s^3 + L_4 R_4 g_m s}{2R_4 g_m + s^4 (C_1 C_3 L_1 L_4 R_4 g_m + 2C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (C_1 C_3 L_4 R_4 + 2C_1 C_4 L_4 R_4 + 2C_1 L_1 L_4 g_m) + s^2 (2C_1 L_1 R_4 g_m + 2C_1 L_4 + C_3 L_4 R_4 g_m + 2C_4 L_4 R_4 g_m) + s (2C_1 R_4 + 2L_4 g_m)}$$

9.456 X-INVALID-ORDER-456 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_4 g_m s^4 + C_1 L_1 L_4 g_m s^3 + L_4 g_m s + R_4 g_m + s^2 (C_1 L_1 R_4 g_m + C_4 L_4 R_4 g_m)}{C_1 C_3 C_4 L_1 L_4 R_4 g_m s^5 + 2g_m + s^4 (C_1 C_3 C_4 L_4 R_4 + C_1 C_3 L_1 L_4 g_m + 2C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_4 g_m + C_1 C_3 L_4 + 2C_1 C_4 L_4 + C_3 C_4 L_4 R_4 g_m) + s^2 (C_1 C_3 R_4 + 2C_1 L_1 g_m + C_3 L_4 g_m + 2C_4 L_4 g_m) + s (2C_1 + C_3 R_4 g_m)}$$

9.457 X-INVALID-ORDER-457 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{1}{C_3 s}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_4 g_m s^4 + R_4 g_m + s^2 (C_1 L_1 R_4 g_m + C_4 L_4 R_4 g_m)}{C_1 C_3 C_4 L_1 L_4 R_4 g_m s^5 + 2g_m + s^4 (C_1 C_3 C_4 L_4 R_4 + 2C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_4 g_m + 2C_1 C_4 L_1 R_4 g_m + 2C_1 C_4 L_4 + C_3 C_4 L_4 R_4 g_m) + s^2 (C_1 C_3 R_4 + 2C_1 C_4 R_4 + 2C_1 L_1 g_m + 2C_4 L_4 g_m) + s (2C_1 + C_3 R_4 g_m + 2C_4 R_4 g_m)}$$

$$\mathbf{9.458 \quad X-INVALID-ORDER-458} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_3 R_4 g_m s^2 + R_3 R_4 g_m}{C_1 C_3 L_1 R_3 R_4 g_m s^3 + 2 R_3 g_m + R_4 g_m + s^2 (C_1 C_3 R_3 R_4 + 2 C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m) + s (2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m)}$$

$$\mathbf{9.459 \quad X-INVALID-ORDER-459} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_3 g_m s^2 + R_3 g_m}{g_m + s^3 (C_1 C_3 L_1 R_3 g_m + 2 C_1 C_4 L_1 R_3 g_m) + s^2 (C_1 C_3 R_3 + 2 C_1 C_4 R_3 + C_1 L_1 g_m) + s (C_1 + C_3 R_3 g_m + 2 C_4 R_3 g_m)}$$

$$\mathbf{9.460 \quad X-INVALID-ORDER-460} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_3 R_4 g_m s^2 + R_3 R_4 g_m}{2 R_3 g_m + R_4 g_m + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + 2 C_1 C_4 L_1 R_3 R_4 g_m) + s^2 (C_1 C_3 R_3 R_4 + 2 C_1 C_4 R_3 R_4 + 2 C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m) + s (2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2 C_4 R_3 R_4 g_m)}$$

$$\mathbf{9.461 \quad X-INVALID-ORDER-461} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 R_3 R_4 g_m s^3 + C_1 L_1 R_3 g_m s^2 + C_4 R_3 R_4 g_m s + R_3 g_m}{C_1 C_3 C_4 L_1 R_3 R_4 g_m s^4 + g_m + s^3 (C_1 C_3 C_4 R_3 R_4 + C_1 C_3 L_1 R_3 g_m + 2 C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m) + s^2 (C_1 C_3 R_3 + 2 C_1 C_4 R_3 + C_1 C_4 R_4 + C_1 L_1 g_m + C_3 C_4 R_3 R_4 g_m) + s (C_1 + C_3 R_3 g_m + 2 C_4 R_3 g_m + C_4 R_4 g_m)}$$

$$\mathbf{9.462 \quad X-INVALID-ORDER-462} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 g_m s^4 + R_3 g_m + s^2 (C_1 L_1 R_3 g_m + C_4 L_4 R_3 g_m)}{C_1 C_3 C_4 L_1 L_4 R_3 g_m s^5 + g_m + s^4 (C_1 C_3 C_4 L_4 R_3 + C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_3 g_m + 2 C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_4 + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_3 + 2 C_1 C_4 R_3 + C_1 L_1 g_m + C_4 L_4 g_m) + s (C_1 + C_3 R_3 g_m + 2 C_4 R_3 g_m)}$$

$$\mathbf{9.463 \quad X-INVALID-ORDER-463} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 R_3 g_m s^3 + L_4 R_3 g_m s}{2 R_3 g_m + s^4 (C_1 C_3 L_1 L_4 R_3 g_m + 2 C_1 C_4 L_1 L_4 R_3 g_m) + s^3 (C_1 C_3 L_4 R_3 + 2 C_1 C_4 L_4 R_3 + C_1 L_1 L_4 g_m) + s^2 (2 C_1 L_1 R_3 g_m + C_1 L_4 + C_3 L_4 R_3 g_m + 2 C_4 L_4 R_3 g_m) + s (2 C_1 R_3 + L_4 g_m)}$$

$$\mathbf{9.464 \quad X-INVALID-ORDER-464} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 g_m s^4 + C_1 C_4 L_1 R_3 R_4 g_m s^3 + C_4 R_3 R_4 g_m s + R_3 g_m + s^2 (C_1 L_1 R_3 g_m + C_4 L_4 R_3 g_m)}{C_1 C_3 C_4 L_1 L_4 R_3 g_m s^5 + g_m + s^4 (C_1 C_3 C_4 L_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 + C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 C_4 R_3 R_4 + C_1 C_3 L_1 R_3 g_m + 2 C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m + C_1 C_4 L_4 + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_3 + 2 C_1 C_4 R_3 + C_1 C_4 R_4 + C_1 L_1 g_m + C_3 C_4 R_3 R_4 g_m + C_4 L_4 g_m)}$$

$$\mathbf{9.465 \quad X-INVALID-ORDER-465} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 R_3 R_4 g_m s^3 + L_4 R_3 R_4 g_m s}{2 R_3 R_4 g_m + s^4 (C_1 C_3 L_1 L_4 R_3 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_4 R_3 R_4 + 2 C_1 C_4 L_4 R_3 R_4 + 2 C_1 L_1 L_4 R_3 g_m + C_1 L_1 L_4 R_4 g_m) + s^2 (2 C_1 L_1 R_3 R_4 g_m + 2 C_1 L_4 R_3 + C_1 L_4 R_4 + C_3 L_4 R_3 R_4 g_m + 2 C_4 L_4 R_3 R_4 g_m) + s (2 C_1 R_3 R_4 + 2 L_4 R_3 g_m + L_4 R_4 g_m)}$$

$$\mathbf{9.466 \quad X-INVALID-ORDER-466} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 R_4 g_m s^4 + C_1 L_1 L_4 R_3 g_m s^3 + L_4 R_3 g_m s + R_3 R_4 g_m + s^2 (C_1 L_1 R_3 R_4 g_m + C_4 L_4 R_3 R_4 g_m)}{C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m s^5 + 2 R_3 g_m + R_4 g_m + s^4 (C_1 C_3 C_4 L_4 R_3 R_4 + C_1 C_3 L_1 L_4 R_3 g_m + 2 C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + C_1 C_3 L_4 R_3 + 2 C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + C_1 L_1 L_4 g_m + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (C_1 C_3 R_3 R_4 + 2 C_1 L_1 R_3 g_m + C_1 L_1 R_4 + C_3 R_3 R_4 g_m + 2 C_4 R_3 R_4 g_m) + s (2 C_1 R_3 R_4 + 2 L_4 R_3 g_m + L_4 R_4 g_m)}$$

$$\mathbf{9.467 \quad X-INVALID-ORDER-467} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 R_4 g_m s^4 + R_3 R_4 g_m + s^2 (C_1 L_1 R_3 R_4 g_m + C_4 L_4 R_3 R_4 g_m)}{C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m s^5 + 2 R_3 g_m + R_4 g_m + s^4 (C_1 C_3 C_4 L_4 R_3 R_4 + 2 C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + 2 C_1 C_4 L_1 R_3 R_4 g_m + 2 C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (C_1 C_3 R_3 R_4 + 2 C_1 C_4 R_3 R_4 + 2 C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m + C_3 R_3 R_4 g_m + 2 C_4 R_3 R_4 g_m) + s (2 C_1 R_3 R_4 + 2 L_4 R_3 g_m + L_4 R_4 g_m)}$$

$$\mathbf{9.468 \quad X-INVALID-ORDER-468} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 R_3 R_4 g_m s^3 + C_1 L_1 R_4 g_m s^2 + C_3 R_3 R_4 g_m s + R_4 g_m}{2g_m + s^3 (2C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_1 R_4 g_m) + s^2 (2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 L_1 g_m) + s (2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m)}$$

$$\mathbf{9.469 \quad X-INVALID-ORDER-469} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 R_3 g_m s^3 + C_1 L_1 g_m s^2 + C_3 R_3 g_m s + g_m}{2C_1 C_3 C_4 L_1 R_3 g_m s^4 + s^3 (2C_1 C_3 C_4 R_3 + C_1 C_3 L_1 g_m + 2C_1 C_4 L_1 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.470 \quad X-INVALID-ORDER-470} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 R_3 R_4 g_m s^3 + C_1 L_1 R_4 g_m s^2 + C_3 R_3 R_4 g_m s + R_4 g_m}{2C_1 C_3 C_4 L_1 R_3 R_4 g_m s^4 + 2g_m + s^3 (2C_1 C_3 C_4 R_3 R_4 + 2C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_1 R_4 g_m + 2C_1 C_4 L_1 R_4 g_m) + s^2 (2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 C_4 R_4 + 2C_1 L_1 g_m + 2C_3 C_4 R_3 R_4 g_m) + s (2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m + 2C_4 R_4 g_m)}$$

$$\mathbf{9.471 \quad X-INVALID-ORDER-471} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 R_3 R_4 g_m s^4 + g_m + s^3 (C_1 C_3 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m) + s^2 (C_1 L_1 g_m + C_3 C_4 R_3 R_4 g_m) + s (C_3 R_3 g_m + C_4 R_4 g_m)}{s^4 (2C_1 C_3 C_4 L_1 R_3 g_m + C_1 C_3 C_4 L_1 R_4 g_m) + s^3 (2C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + C_1 C_3 L_1 g_m + 2C_1 C_4 L_1 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.472 \quad X-INVALID-ORDER-472} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_4 R_3 g_m s^5 + C_1 C_4 L_1 L_4 g_m s^4 + C_3 R_3 g_m s + g_m + s^3 (C_1 C_3 L_1 R_3 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 L_1 g_m + C_4 L_4 g_m)}{C_1 C_3 C_4 L_1 L_4 g_m s^5 + s^4 (2C_1 C_3 C_4 L_1 R_3 g_m + C_1 C_3 C_4 L_4) + s^3 (2C_1 C_3 C_4 R_3 + C_1 C_3 L_1 g_m + 2C_1 C_4 L_1 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.473 \quad X-INVALID-ORDER-473} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_4 R_3 g_m s^4 + C_1 L_1 L_4 g_m s^3 + C_3 L_4 R_3 g_m s^2 + L_4 g_m s}{2C_1 C_3 C_4 L_1 L_4 R_3 g_m s^5 + 2g_m + s^4 (2C_1 C_3 C_4 L_4 R_3 + C_1 C_3 L_1 L_4 g_m + 2C_1 C_4 L_1 L_4 g_m) + s^3 (2C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_4 + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m) + s^2 (2C_1 C_3 R_3 + 2C_1 L_1 g_m + C_3 L_4 g_m + 2C_4 L_4 g_m) + s (2C_1 + 2C_3 R_3 g_m)}$$

$$\mathbf{9.474 \quad X-INVALID-ORDER-474} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_4 R_3 g_m s^5 + g_m + s^4 (C_1 C_3 C_4 L_1 R_3 R_4 g_m + C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 L_1 g_m + C_3 C_4 R_3 R_4 g_m + C_4 L_4 g_m) + s (C_3 R_3 g_m + C_4 R_4 g_m)}{C_1 C_3 C_4 L_1 L_4 g_m s^5 + s^4 (2C_1 C_3 C_4 L_1 R_3 g_m + C_1 C_3 C_4 L_1 R_4 g_m + C_1 C_3 C_4 L_4) + s^3 (2C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + C_1 C_3 L_1 g_m + 2C_1 C_4 L_1 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.475 \quad X-INVALID-ORDER-475} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_4 R_3 R_4 g_m s^4 + C_1 L_1 L_4 R_4 g_m s^3 + C_3 L_4 R_3 R_4 g_m s^2 + L_4 R_4 g_m s}{2C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m s^5 + 2R_4 g_m + s^4 (2C_1 C_3 C_4 L_4 R_3 R_4 + 2C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_1 L_4 R_4 g_m + 2C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (2C_1 C_3 L_1 R_3 R_4 g_m + 2C_1 C_3 L_4 R_3 + C_1 C_3 L_4 R_4 + 2C_1 C_4 L_4 R_4 + 2C_1 L_1 L_4 g_m + 2C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (2C_1 C_3 R_3 R_4 + 2C_1 L_1 R_4 g_m + 2C_3 R_3 R_4 g_m) + s (2C_1 + 2C_3 R_3 g_m)}$$

$$\mathbf{9.476 \quad X-INVALID-ORDER-476} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m s^5 + R_4 g_m + s^4 (C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + C_1 L_1 L_4 g_m + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (C_1 L_1 R_4 g_m + C_3 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (C_3 R_3 R_4 g_m + L_4 g_m)}{2g_m + s^5 (2C_1 C_3 C_4 L_1 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_4 g_m) + s^4 (2C_1 C_3 C_4 L_4 R_3 + C_1 C_3 C_4 L_4 R_4 + C_1 C_3 L_1 L_4 g_m + 2C_1 C_4 L_1 L_4 g_m) + s^3 (2C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_1 R_4 g_m + C_1 C_3 L_4 + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m + C_3 C_4 L_4 R_4 g_m) + s^2 (2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 L_1 g_m)}$$

$$\mathbf{9.477 \quad X-INVALID-ORDER-477} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m s^5 + C_1 C_4 L_1 L_4 R_4 g_m s^4 + C_3 R_3 R_4 g_m s + R_4 g_m + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (C_1 L_1 R_4 g_m + C_4 L_4 R_4 g_m)}{2g_m + s^5 (2C_1 C_3 C_4 L_1 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_4 g_m) + s^4 (2C_1 C_3 C_4 L_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_4 R_3 + C_1 C_3 C_4 L_4 R_4 + 2C_1 C_4 L_1 L_4 g_m) + s^3 (2C_1 C_3 C_4 R_3 R_4 + 2C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_1 R_4 g_m + 2C_1 C_4 L_1 R_4 g_m + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m + C_3 C_4 L_4 R_4 g_m) + s^2 (2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 L_1 g_m)}$$

9.478 X-INVALID-ORDER-478 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, L_3s + \frac{1}{C_3s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1C_3L_1L_3R_4g_ms^4 + R_4g_m + s^2(C_1L_1R_4g_m + C_3L_3R_4g_m)}{2C_1C_3L_1L_3g_ms^4 + 2g_m + s^3(C_1C_3L_1R_4g_m + 2C_1C_3L_3) + s^2(C_1C_3R_4 + 2C_1L_1g_m + 2C_3L_3g_m) + s(2C_1 + C_3R_4g_m)}$$

9.479 X-INVALID-ORDER-479 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, L_3s + \frac{1}{C_3s}, \frac{1}{C_4s}, \infty, \infty \right)$

$$H(s) = \frac{C_1C_3L_1L_3g_ms^4 + g_m + s^2(C_1L_1g_m + C_3L_3g_m)}{2C_1C_3C_4L_1L_3g_ms^5 + 2C_1C_3C_4L_3s^4 + s^3(C_1C_3L_1g_m + 2C_1C_4L_1g_m + 2C_3C_4L_3g_m) + s^2(C_1C_3 + 2C_1C_4) + s(C_3g_m + 2C_4g_m)}$$

9.480 X-INVALID-ORDER-480 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, L_3s + \frac{1}{C_3s}, \frac{R_4}{C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{C_1C_3L_1L_3R_4g_ms^4 + R_4g_m + s^2(C_1L_1R_4g_m + C_3L_3R_4g_m)}{2C_1C_3C_4L_1L_3R_4g_ms^5 + 2g_m + s^4(2C_1C_3C_4L_3R_4 + 2C_1C_3L_1L_3g_m) + s^3(C_1C_3L_1R_4g_m + 2C_1C_3L_3 + 2C_1C_4L_1R_4g_m + 2C_3C_4L_3R_4g_m) + s^2(C_1C_3R_4 + 2C_1C_4R_4 + 2C_1L_1g_m + 2C_3L_3g_m) + s(2C_1 + C_3R_4g_m + 2C_4R_4g_m)}$$

9.481 X-INVALID-ORDER-481 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, L_3s + \frac{1}{C_3s}, R_4 + \frac{1}{C_4s}, \infty, \infty \right)$

$$H(s) = \frac{C_1C_3C_4L_1L_3R_4g_ms^5 + C_1C_3L_1L_3g_ms^4 + C_4R_4g_ms + g_m + s^3(C_1C_4L_1R_4g_m + C_3C_4L_3R_4g_m) + s^2(C_1L_1g_m + C_3L_3g_m)}{2C_1C_3C_4L_1L_3g_ms^5 + s^4(C_1C_3C_4L_1R_4g_m + 2C_1C_3C_4L_3) + s^3(C_1C_3C_4R_4 + C_1C_3L_1g_m + 2C_1C_4L_1g_m + 2C_3C_4L_3g_m) + s^2(C_1C_3 + 2C_1C_4 + C_3C_4R_4g_m) + s(C_3g_m + 2C_4g_m)}$$

9.482 X-INVALID-ORDER-482 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, L_3s + \frac{1}{C_3s}, L_4s + \frac{1}{C_4s}, \infty, \infty \right)$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4g_ms^6 + g_m + s^4(C_1C_3L_1L_3g_m + C_1C_4L_1L_4g_m + C_3C_4L_3L_4g_m) + s^2(C_1L_1g_m + C_3L_3g_m + C_4L_4g_m)}{s^5(2C_1C_3C_4L_1L_3g_m + C_1C_3C_4L_1L_4g_m) + s^4(2C_1C_3C_4L_3 + C_1C_3C_4L_4) + s^3(C_1C_3L_1g_m + 2C_1C_4L_1g_m + 2C_3C_4L_3g_m + C_3C_4L_4g_m) + s^2(C_1C_3 + 2C_1C_4) + s(C_3g_m + 2C_4g_m)}$$

9.483 X-INVALID-ORDER-483 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, L_3s + \frac{1}{C_3s}, \frac{L_4s}{C_4L_4s^2+1}, \infty, \infty \right)$

$$H(s) = \frac{C_1C_3L_1L_3L_4g_ms^5 + L_4g_ms + s^3(C_1L_1L_4g_m + C_3L_3L_4g_m)}{2C_1C_3C_4L_1L_3L_4g_ms^6 + 2C_1C_3C_4L_3L_4s^5 + 2C_1s + 2g_m + s^4(2C_1C_3L_1L_3g_m + C_1C_3L_1L_4g_m + 2C_1C_4L_1L_4g_m + 2C_3C_4L_3L_4g_m) + s^3(2C_1C_3L_3 + C_1C_3L_4 + 2C_1C_4L_4) + s^2(2C_1L_1g_m + 2C_3L_3g_m + C_3L_4g_m + 2C_4L_4g_m)}$$

9.484 X-INVALID-ORDER-484 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, L_3s + \frac{1}{C_3s}, L_4s + R_4 + \frac{1}{C_4s}, \infty, \infty \right)$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4g_ms^6 + C_1C_3C_4L_1L_3R_4g_ms^5 + C_4R_4g_ms + g_m + s^4(C_1C_3L_1L_3g_m + C_1C_4L_1L_4g_m + C_3C_4L_3L_4g_m) + s^3(C_1C_4L_1R_4g_m + C_3C_4L_3R_4g_m) + s^2(C_1L_1g_m + C_3L_3g_m + C_4L_4g_m)}{s^5(2C_1C_3C_4L_1L_3g_m + C_1C_3C_4L_1L_4g_m) + s^4(C_1C_3C_4L_1R_4g_m + 2C_1C_3C_4L_3 + C_1C_3C_4L_4) + s^3(C_1C_3C_4R_4 + C_1C_3L_1g_m + 2C_1C_4L_1g_m + 2C_3C_4L_3g_m + C_3C_4L_4g_m) + s^2(C_1C_3 + 2C_1C_4 + C_3C_4R_4g_m) + s(C_3g_m + 2C_4g_m)}$$

9.485 X-INVALID-ORDER-485 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, L_3s + \frac{1}{C_3s}, \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_1C_3L_1L_3L_4R_4g_ms^5 + L_4R_4g_ms + s^3(C_1L_1L_4R_4g_m + C_3L_3L_4R_4g_m)}{2C_1C_3C_4L_1L_3L_4R_4g_ms^6 + 2R_4g_m + s^5(2C_1C_3C_4L_3L_4R_4 + 2C_1C_3L_1L_3L_4g_m) + s^4(2C_1C_3L_1L_3R_4g_m + C_1C_3L_1L_4R_4g_m + 2C_1C_3L_3L_4 + 2C_1C_4L_1L_4R_4g_m + 2C_3C_4L_3L_4R_4g_m) + s^3(2C_1C_3L_3R_4 + C_1C_3L_4R_4 + 2C_1C_4L_4R_4 + 2C_1L_1L_4g_m + 2C_3L_3L_4g_m) + s^2(2C_1L_1R_4g_m + 2C_3L_3R_4g_m + C_4L_4R_4g_m)}$$

9.486 X-INVALID-ORDER-486 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, L_3s + \frac{1}{C_3s}, \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1}, \infty, \infty \right)$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4R_4g_ms^6 + C_1C_3L_1L_3L_4g_ms^5 + L_4g_ms + R_4g_m + s^4(C_1C_3L_1L_3R_4g_m + C_1C_4L_1L_4R_4g_m + C_3C_4L_3L_4R_4g_m) + s^3(C_1L_1L_4g_m + C_3L_3L_4g_m) + s^2(C_1L_1R_4g_m + C_3L_3R_4g_m + C_4L_4R_4g_m)}{2C_1C_3C_4L_1L_3L_4g_ms^6 + 2g_m + s^5(C_1C_3C_4L_1L_4R_4g_m + 2C_1C_3C_4L_3L_4) + s^4(C_1C_3C_4L_4R_4 + 2C_1C_3L_1L_3g_m + C_1C_3L_1L_4g_m + 2C_1C_4L_1L_4g_m + 2C_3C_4L_3L_4g_m) + s^3(C_1C_3L_1R_4g_m + 2C_1C_3L_3 + C_1C_3L_4 + 2C_1C_4L_4 + C_3C_4L_4R_4g_m) + s^2(C_1C_3R_4 + 2C_1L_1g_m)}$$

9.487 X-INVALID-ORDER-487 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, L_3s + \frac{1}{C_3s}, \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4R_4g_ms^6 + R_4g_m + s^4(C_1C_3L_1L_3R_4g_m + C_1C_4L_1L_4R_4g_m + C_3C_4L_3L_4R_4g_m) + s^2(C_1L_1R_4g_m + C_3L_3R_4g_m + C_4L_4R_4g_m)}{2C_1C_3C_4L_1L_3L_4g_ms^6 + 2g_m + s^5(2C_1C_3C_4L_1L_3R_4g_m + C_1C_3C_4L_1L_4R_4g_m + 2C_1C_3C_4L_3L_4) + s^4(2C_1C_3C_4L_3R_4 + C_1C_3C_4L_4R_4 + 2C_1C_3L_1L_3g_m + 2C_1C_4L_1L_4g_m + 2C_3C_4L_3L_4g_m) + s^3(C_1C_3L_1R_4g_m + 2C_1C_3L_3 + 2C_1C_4L_1R_4g_m + 2C_1C_4L_4 + 2C_3C_4L_3)}$$

9.488 X-INVALID-ORDER-488 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 R_4 g_m s^3 + L_3 R_4 g_m s}{C_1 C_3 L_1 L_3 R_4 g_m s^4 + R_4 g_m + s^3 (C_1 C_3 L_3 R_4 + 2 C_1 L_1 L_3 g_m) + s^2 (C_1 L_1 R_4 g_m + 2 C_1 L_3 + C_3 L_3 R_4 g_m) + s (C_1 R_4 + 2 L_3 g_m)}$$

9.489 X-INVALID-ORDER-489 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 g_m s^3 + L_3 g_m s}{C_1 s + g_m + s^4 (C_1 C_3 L_1 L_3 g_m + 2 C_1 C_4 L_1 L_3 g_m) + s^3 (C_1 C_3 L_3 + 2 C_1 C_4 L_3) + s^2 (C_1 L_1 g_m + C_3 L_3 g_m + 2 C_4 L_3 g_m)}$$

9.490 X-INVALID-ORDER-490 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 R_4 g_m s^3 + L_3 R_4 g_m s}{R_4 g_m + s^4 (C_1 C_3 L_1 L_3 R_4 g_m + 2 C_1 C_4 L_1 L_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_4 + 2 C_1 C_4 L_3 R_4 + 2 C_1 L_1 L_3 g_m) + s^2 (C_1 L_1 R_4 g_m + 2 C_1 L_3 + C_3 L_3 R_4 g_m + 2 C_4 L_3 R_4 g_m) + s (C_1 R_4 + 2 L_3 g_m)}$$

9.491 X-INVALID-ORDER-491 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_3 R_4 g_m s^4 + C_1 L_1 L_3 g_m s^3 + C_4 L_3 R_4 g_m s^2 + L_3 g_m s}{C_1 C_3 C_4 L_1 L_3 R_4 g_m s^5 + g_m + s^4 (C_1 C_3 C_4 L_3 R_4 + C_1 C_3 L_1 L_3 g_m + 2 C_1 C_4 L_1 L_3 g_m) + s^3 (C_1 C_3 L_3 + C_1 C_4 L_1 R_4 g_m + 2 C_1 C_4 L_3 + C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 C_4 R_4 + C_1 L_1 g_m + C_3 L_3 g_m + 2 C_4 L_3 g_m) + s (C_1 + C_4 R_4 g_m)}$$

9.492 X-INVALID-ORDER-492 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 g_m s^5 + L_3 g_m s + s^3 (C_1 L_1 L_3 g_m + C_4 L_3 L_4 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 g_m s^6 + C_1 C_3 C_4 L_3 L_4 s^5 + C_1 s + g_m + s^4 (C_1 C_3 L_1 L_3 g_m + 2 C_1 C_4 L_1 L_3 g_m + C_1 C_4 L_1 L_4 g_m + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 + 2 C_1 C_4 L_3 + C_1 C_4 L_4) + s^2 (C_1 L_1 g_m + C_3 L_3 g_m + 2 C_4 L_3 g_m + C_4 L_4 g_m)}$$

9.493 X-INVALID-ORDER-493 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 L_4 g_m s^3 + L_3 L_4 g_m s}{2 L_3 g_m + L_4 g_m + s^4 (C_1 C_3 L_1 L_3 L_4 g_m + 2 C_1 C_4 L_1 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 L_4 + 2 C_1 C_4 L_3 L_4) + s^2 (2 C_1 L_1 L_3 g_m + C_1 L_1 L_4 g_m + C_3 L_3 L_4 g_m + 2 C_4 L_3 L_4 g_m) + s (2 C_1 L_3 + C_1 L_4)}$$

9.494 X-INVALID-ORDER-494 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 g_m s^5 + C_1 C_4 L_1 L_3 R_4 g_m s^4 + C_4 L_3 R_4 g_m s^2 + L_3 g_m s + s^3 (C_1 L_1 L_3 g_m + C_4 L_3 L_4 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 g_m s^6 + g_m + s^5 (C_1 C_3 C_4 L_1 L_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4) + s^4 (C_1 C_3 C_4 L_3 R_4 + C_1 C_3 L_1 L_3 g_m + 2 C_1 C_4 L_1 L_3 g_m + C_1 C_4 L_1 L_4 g_m + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 + C_1 C_4 L_1 R_4 g_m + 2 C_1 C_4 L_3 + C_1 C_4 L_4 + C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 C_4 R_4 + C_1 L_1 g_m + C_3 L_3 g_m + 2 C_4 L_3 g_m)}$$

9.495 X-INVALID-ORDER-495 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 L_4 R_4 g_m s^3 + L_3 L_4 R_4 g_m s}{2 L_3 R_4 g_m + L_4 R_4 g_m + s^4 (C_1 C_3 L_1 L_3 L_4 R_4 g_m + 2 C_1 C_4 L_1 L_3 L_4 R_4 g_m) + s^3 (C_1 C_3 L_3 L_4 R_4 + 2 C_1 C_4 L_3 L_4 R_4 + 2 C_1 L_1 L_3 L_4 g_m) + s^2 (2 C_1 L_1 L_3 R_4 g_m + C_1 L_1 L_4 R_4 g_m + 2 C_1 L_3 L_4 + C_3 L_3 L_4 R_4 g_m + 2 C_4 L_3 L_4 R_4 g_m) + s (2 C_1 L_3 R_4 + C_1 L_4 R_4 + 2 L_3 L_4 g_m)}$$

9.496 X-INVALID-ORDER-496 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_4 g_m s^5 + C_1 L_1 L_3 L_4 g_m s^4 + L_3 L_4 g_m s^2 + L_3 R_4 g_m s + s^3 (C_1 L_1 L_3 R_4 g_m + C_4 L_3 L_4 R_4 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m s^6 + R_4 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_4 + C_1 C_3 L_1 L_3 L_4 g_m + 2 C_1 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_4 g_m + C_1 C_3 L_3 L_4 + C_1 C_4 L_1 L_4 R_4 g_m + 2 C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (C_1 C_3 L_3 R_4 + C_1 C_4 L_4 R_4 + 2 C_1 L_1 L_3 g_m + C_1 L_1 L_4 g_m + C_3 L_3 L_4 g_m) + s^2 (C_1 C_4 R_4 + C_1 L_1 g_m + C_3 L_3 g_m + 2 C_4 L_3 g_m)}$$

9.497 X-INVALID-ORDER-497 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_4 g_m s^5 + L_3 R_4 g_m s + s^3 (C_1 L_1 L_3 R_4 g_m + C_4 L_3 L_4 R_4 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m s^6 + R_4 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_4 + 2 C_1 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_4 g_m + 2 C_1 C_4 L_1 L_3 R_4 g_m + C_1 C_4 L_1 L_4 R_4 g_m + 2 C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (C_1 C_3 L_3 R_4 + 2 C_1 C_4 L_3 R_4 + C_1 C_4 L_4 R_4 + 2 C_1 L_1 L_3 g_m + 2 C_4 L_3 L_4 g_m) + s^2 (C_1 L_1 g_m + C_3 L_3 g_m + 2 C_4 L_3 g_m)}$$

$$\mathbf{9.498 \quad X-INVALID-ORDER-498} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_4 g_m s^4 + C_1 C_3 L_1 R_3 R_4 g_m s^3 + C_3 R_3 R_4 g_m s + R_4 g_m + s^2 (C_1 L_1 R_4 g_m + C_3 L_3 R_4 g_m)}{2C_1 C_3 L_1 L_3 g_m s^4 + 2g_m + s^3 (2C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_1 R_4 g_m + 2C_1 C_3 L_3) + s^2 (2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 L_1 g_m + 2C_3 L_3 g_m) + s (2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m)}$$

$$\mathbf{9.499 \quad X-INVALID-ORDER-499} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 g_m s^4 + C_1 C_3 L_1 R_3 g_m s^3 + C_3 R_3 g_m s + g_m + s^2 (C_1 L_1 g_m + C_3 L_3 g_m)}{2C_1 C_3 C_4 L_1 L_3 g_m s^5 + s^4 (2C_1 C_3 C_4 L_1 R_3 g_m + 2C_1 C_3 C_4 L_3) + s^3 (2C_1 C_3 C_4 R_3 + C_1 C_3 L_1 g_m + 2C_1 C_4 L_1 g_m + 2C_3 C_4 L_3 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.500 \quad X-INVALID-ORDER-500} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_4 g_m s^4 + C_1 C_3 L_1 R_3 R_4 g_m s^3 + C_3 R_3 R_4 g_m s + R_4 g_m + s^2 (C_1 L_1 R_4 g_m + C_3 L_3 R_4 g_m)}{2C_1 C_3 C_4 L_1 L_3 R_4 g_m s^5 + 2g_m + s^4 (2C_1 C_3 C_4 L_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_4 + 2C_1 C_3 L_1 L_3 g_m) + s^3 (2C_1 C_3 C_4 R_3 R_4 + 2C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_1 R_4 g_m + 2C_1 C_3 L_3 + 2C_1 C_4 L_1 R_4 g_m + 2C_3 C_4 L_3 R_4 g_m) + s^2 (2C_1 C_3 R_3 + C_1 C_3 R_4 + 2C_1 C_4 R_4 + 2C_1 L_1 g_m + 2C_3 C_4 R_3 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.501 \quad X-INVALID-ORDER-501} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 R_4 g_m s^5 + g_m + s^4 (C_1 C_3 C_4 L_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 g_m) + s^3 (C_1 C_3 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m + C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 L_1 g_m + C_3 C_4 R_3 R_4 g_m + C_3 L_3 g_m) + s (C_3 R_3 g_m + C_4 R_4 g_m)}{2C_1 C_3 C_4 L_1 L_3 g_m s^5 + s^4 (2C_1 C_3 C_4 L_1 R_3 g_m + C_1 C_3 C_4 L_1 R_4 g_m + 2C_1 C_3 C_4 L_3) + s^3 (2C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + C_1 C_3 L_1 g_m + 2C_1 C_4 L_1 g_m + 2C_3 C_4 L_3 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.502 \quad X-INVALID-ORDER-502} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 g_m s^6 + C_1 C_3 C_4 L_1 L_4 R_3 g_m s^5 + C_3 R_3 g_m s + g_m + s^4 (C_1 C_3 L_1 L_3 g_m + C_1 C_4 L_1 L_4 g_m + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_1 R_3 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 L_1 g_m + C_3 L_3 g_m + C_4 L_4 g_m)}{s^5 (2C_1 C_3 C_4 L_1 L_3 g_m + C_1 C_3 C_4 L_1 L_4 g_m) + s^4 (2C_1 C_3 C_4 L_1 R_3 g_m + 2C_1 C_3 C_4 L_3 + C_1 C_3 C_4 L_4) + s^3 (2C_1 C_3 C_4 R_3 + C_1 C_3 L_1 g_m + 2C_1 C_4 L_1 g_m + 2C_3 C_4 L_3 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.503 \quad X-INVALID-ORDER-503} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 L_4 g_m s^5 + C_1 C_3 L_1 L_4 R_3 g_m s^4 + C_3 L_4 R_3 g_m s^2 + L_4 g_m s + s^3 (C_1 L_1 L_4 g_m + C_3 L_3 L_4 g_m)}{2C_1 C_3 C_4 L_1 L_3 L_4 g_m s^6 + 2g_m + s^5 (2C_1 C_3 C_4 L_1 L_4 R_3 g_m + 2C_1 C_3 C_4 L_3 L_4) + s^4 (2C_1 C_3 C_4 L_4 R_3 + 2C_1 C_3 L_1 L_3 g_m + C_1 C_3 L_1 L_4 g_m + 2C_1 C_4 L_1 L_4 g_m + 2C_3 C_4 L_3 L_4 g_m) + s^3 (2C_1 C_3 L_1 R_3 g_m + 2C_1 C_3 L_3 + C_1 C_3 L_4 + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m) + s^2 (2C_1 C_3 R_3 + 2C_1 C_4 R_4 + 2C_3 C_4 L_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.504 \quad X-INVALID-ORDER-504} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 g_m s^6 + g_m + s^5 (C_1 C_3 C_4 L_1 L_3 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_3 g_m) + s^4 (C_1 C_3 C_4 L_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 g_m + C_1 C_4 L_1 L_4 g_m + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m + C_3 C_4 L_3 R_4 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 L_1 g_m + C_3 C_4 R_3 R_4 g_m + C_3 L_3 g_m + C_4 L_4 g_m)}{s^5 (2C_1 C_3 C_4 L_1 L_3 g_m + C_1 C_3 C_4 L_1 L_4 g_m) + s^4 (2C_1 C_3 C_4 L_1 R_3 g_m + C_1 C_3 C_4 L_1 R_4 g_m + 2C_1 C_3 C_4 L_3 + C_1 C_3 C_4 L_4) + s^3 (2C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + C_1 C_3 L_1 g_m + 2C_1 C_4 L_1 g_m + 2C_3 C_4 L_3 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m + C_3 C_4 L_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.505 \quad X-INVALID-ORDER-505} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 L_4 R_4 g_m s^5 + C_1 C_3 L_1 L_4 R_3 R_4 g_m s^4 + C_3 L_4 R_3 R_4 g_m s^2 + L_4 g_m s + s^3 (C_1 L_1 L_4 g_m + C_3 L_3 L_4 g_m)}{2C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m s^6 + 2R_4 g_m + s^5 (2C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_4 + 2C_1 C_3 L_1 L_3 L_4 g_m) + s^4 (2C_1 C_3 C_4 L_4 R_3 R_4 + 2C_1 C_3 L_1 L_3 R_4 g_m + 2C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_1 L_4 R_4 g_m + 2C_1 C_3 L_3 L_4 + 2C_1 C_4 L_1 L_4 R_4 g_m + 2C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (2C_1 C_3 L_1 R_3 g_m + 2C_1 C_3 L_3 + C_1 C_3 L_4 + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m) + s^2 (2C_1 C_3 R_3 + 2C_1 C_4 R_4 + 2C_3 C_4 L_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.506 \quad X-INVALID-ORDER-506} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m s^6 + R_4 g_m + s^5 (C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m + C_1 C_3 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_4 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + C_1 L_1 L_4 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 L_1 g_m + C_3 C_4 R_3 R_4 g_m + C_3 L_3 g_m + C_4 L_4 g_m)}{2C_1 C_3 C_4 L_1 L_3 L_4 g_m s^6 + 2g_m + s^5 (2C_1 C_3 C_4 L_1 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4) + s^4 (2C_1 C_3 C_4 L_4 R_3 + C_1 C_3 C_4 L_4 R_4 + 2C_1 C_3 L_1 L_3 g_m + C_1 C_3 L_1 L_4 g_m + 2C_1 C_4 L_1 L_4 g_m + 2C_3 C_4 L_3 L_4 g_m) + s^3 (2C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_1 R_4 g_m + 2C_1 C_3 L_3 + C_1 C_3 L_4 + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m + C_3 C_4 L_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.507 \quad X-INVALID-ORDER-507} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m s^6 + C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m s^5 + C_3 R_3 R_4 g_m s + R_4 g_m + s^4 (C_1 C_3 L_1 L_3 R_4 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m + C_3 C_4 L_3 L_4 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + C_1 L_1 L_4 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 L_1 g_m + C_3 C_4 R_3 R_4 g_m + C_3 L_3 g_m + C_4 L_4 g_m)}{2C_1 C_3 C_4 L_1 L_3 L_4 g_m s^6 + 2g_m + s^5 (2C_1 C_3 C_4 L_1 L_3 R_4 g_m + 2C_1 C_3 C_4 L_1 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4) + s^4 (2C_1 C_3 C_4 L_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_4 + 2C_1 C_3 C_4 L_4 R_3 + C_1 C_3 C_4 L_4 R_4 + 2C_1 C_3 L_1 L_3 g_m + 2C_1 C_4 L_1 L_4 g_m + 2C_3 C_4 L_3 L_4 g_m) + s^3 (2C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_1 R_4 g_m + 2C_1 C_3 L_3 + C_1 C_3 L_4 + 2C_1 C_4 L_4 + 2C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 + 2C_1 C_4 + 2C_3 C_4 R_3 g_m + C_3 C_4 L_4 R_3 g_m) + s (C_3 g_m + 2C_4 g_m)}$$

$$\mathbf{9.508 \quad X-INVALID-ORDER-508} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_3 R_3 R_4 g_m s^3 + L_3 R_3 R_4 g_m s}{C_1 C_3 L_1 L_3 R_3 R_4 g_m s^4 + R_3 R_4 g_m + s^3 (C_1 C_3 L_3 R_3 R_4 + 2C_1 L_1 L_3 R_3 g_m + C_1 L_1 L_3 R_4 g_m) + s^2 (C_1 L_1 R_3 R_4 g_m + 2C_1 L_3 R_3 + C_1 L_3 R_4 + C_3 L_3 R_3 R_4 g_m) + s (C_1 R_3 R_4 + 2L_3 R_3 g_m + L_3 R_4 g_m)}$$

$$\mathbf{9.509 \quad X-INVALID-ORDER-509} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_3 R_3 g_m s^3 + L_3 R_3 g_m s}{R_3 g_m + s^4 (C_1 C_3 L_1 L_3 R_3 g_m + 2C_1 C_4 L_1 L_3 R_3 g_m) + s^3 (C_1 C_3 L_3 R_3 + 2C_1 C_4 L_3 R_3 + C_1 L_1 L_3 g_m) + s^2 (C_1 L_1 R_3 g_m + C_1 L_3 + C_3 L_3 R_3 g_m + 2C_4 L_3 R_3 g_m) + s (C_1 R_3 + L_3 g_m)}$$

$$\mathbf{9.510 \quad X-INVALID-ORDER-510} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_3 R_3 R_4 g_m s^3 + L_3 R_3 R_4 g_m s}{R_3 R_4 g_m + s^4 (C_1 C_3 L_1 L_3 R_3 R_4 g_m + 2C_1 C_4 L_1 L_3 R_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_3 R_4 + 2C_1 C_4 L_3 R_3 R_4 + 2C_1 L_1 L_3 R_3 g_m + C_1 L_1 L_3 R_4 g_m) + s^2 (C_1 L_1 R_3 R_4 g_m + 2C_1 L_3 R_3 + C_1 L_3 R_4 + C_3 L_3 R_3 R_4 g_m + 2C_4 L_3 R_3 R_4 g_m) + s (C_1 R_3 R_4 + 2L_3 R_3 g_m + L_3 R_4 g_m)}$$

$$\mathbf{9.511 \quad X-INVALID-ORDER-511} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_3 R_3 R_4 g_m s^4 + C_1 L_1 L_3 R_3 g_m s^3 + C_4 L_3 R_3 R_4 g_m s^2 + L_3 R_3 g_m s}{C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m s^5 + R_3 g_m + s^4 (C_1 C_3 C_4 L_3 R_3 R_4 + C_1 C_3 L_1 L_3 R_3 g_m + 2C_1 C_4 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_3 + C_1 C_4 L_1 R_3 R_4 g_m + 2C_1 C_4 L_3 R_3 + C_1 C_4 L_3 R_4 + C_1 L_1 L_3 g_m + C_3 C_4 L_3 R_3 R_4 g_m) + s^2 (C_1 C_4 R_3 R_4 + C_1 L_1 R_3 g_m + C_1 L_3 + C_3 L_3 R_3 R_4 g_m) + s (C_1 R_3 R_4 + 2L_3 R_3 g_m + L_3 R_4 g_m)}$$

$$\mathbf{9.512 \quad X-INVALID-ORDER-512} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_3 g_m s^5 + L_3 R_3 g_m s + s^3 (C_1 L_1 L_3 R_3 g_m + C_4 L_3 L_4 R_3 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + R_3 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 g_m + 2C_1 C_4 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (C_1 C_3 L_3 R_3 + 2C_1 C_4 L_3 R_3 + C_1 C_4 L_4 R_3 + C_1 L_1 L_3 g_m + C_4 L_3 L_4 g_m) + s^2 (C_1 L_1 R_3 g_m + C_1 L_3 + C_3 L_3 R_3 R_4 g_m) + s (C_1 R_3 R_4 + 2L_3 R_3 g_m + L_3 R_4 g_m)}$$

$$\mathbf{9.513 \quad X-INVALID-ORDER-513} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_3 L_4 R_3 g_m s^3 + L_3 L_4 R_3 g_m s}{2L_3 R_3 g_m + L_4 R_3 g_m + s^4 (C_1 C_3 L_1 L_3 L_4 R_3 g_m + 2C_1 C_4 L_1 L_3 L_4 R_3 g_m) + s^3 (C_1 C_3 L_3 L_4 R_3 + 2C_1 C_4 L_3 L_4 R_3 + C_1 L_1 L_3 L_4 g_m) + s^2 (2C_1 L_1 L_3 R_3 g_m + C_1 L_1 L_4 R_3 g_m + C_1 L_3 L_4 + C_3 L_3 L_4 R_3 g_m + 2C_4 L_3 L_4 R_3 g_m) + s (2C_1 L_3 R_3 + C_1 L_4 R_3 + L_3 L_4 g_m)}$$

$$\mathbf{9.514 \quad X-INVALID-ORDER-514} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_3 g_m s^5 + C_1 C_4 L_1 L_3 R_3 R_4 g_m s^4 + C_4 L_3 R_3 R_4 g_m s^2 + L_3 R_3 g_m s}{C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + R_3 g_m + s^5 (C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 C_4 L_3 R_3 R_4 + C_1 C_3 L_1 L_3 R_3 g_m + 2C_1 C_4 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_3 R_4 g_m + C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (C_1 C_3 L_3 R_3 + C_1 C_4 L_3 R_3 + C_1 C_4 L_4 R_3 + C_1 L_1 L_3 g_m + C_4 L_3 L_4 g_m) + s^2 (C_1 L_1 R_3 g_m + C_1 L_3 + C_3 L_3 R_3 R_4 g_m) + s (C_1 R_3 R_4 + 2L_3 R_3 g_m + L_3 R_4 g_m)}$$

$$\mathbf{9.515 \quad X-INVALID-ORDER-515} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_3 L_4 R_3 R_4 g_m s^3 + L_3 L_4 R_3 R_4 g_m s}{2L_3 R_3 R_4 g_m + L_4 R_3 R_4 g_m + s^4 (C_1 C_3 L_1 L_3 L_4 R_3 R_4 g_m + 2C_1 C_4 L_1 L_3 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_3 L_4 R_3 R_4 + 2C_1 C_4 L_3 L_4 R_3 R_4 + 2C_1 L_1 L_3 L_4 R_3 g_m + C_1 L_1 L_3 L_4 R_4 g_m) + s^2 (2C_1 L_1 L_3 R_3 R_4 g_m + C_1 L_1 L_4 R_3 R_4 g_m + 2C_1 L_3 L_4 R_3 + C_1 L_3 L_4 R_4 + C_3 L_3 L_4 R_3 R_4 g_m + 2C_4 L_3 L_4 R_3 R_4 g_m) + s (C_1 R_3 R_4 + 2L_3 R_3 g_m + L_3 R_4 g_m)}$$

$$\mathbf{9.516 \quad X-INVALID-ORDER-516} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^5 + C_1 L_1 L_3 L_4 R_3 g_m s}{C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^6 + R_3 R_4 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_3 R_4 + C_1 C_3 L_1 L_3 L_4 R_3 g_m + 2C_1 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 R_4 g_m + C_1 C_3 L_3 L_4 R_3 + C_1 C_4 L_1 L_4 R_3 R_4 g_m + 2C_1 C_4 L_3 L_4 R_3 + C_1 C_4 L_3 L_4 R_4 + C_1 L_1 L_3 L_4 g_m + C_3 C_4 L_3 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_3 R_4 + C_1 C_4 L_3 R_3 R_4 + C_1 C_4 L_4 R_3 + C_1 L_1 L_3 g_m + C_4 L_3 L_4 g_m) + s^2 (C_1 L_1 R_3 g_m + C_1 L_3 + C_3 L_3 R_3 R_4 g_m) + s (C_1 R_3 R_4 + 2L_3 R_3 g_m + L_3 R_4 g_m)}$$

$$\mathbf{9.517 \quad X-INVALID-ORDER-517} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^5 + L_3 R_3 R_4 g_m s + s^3 (C_1 L_1 L_3 R_3 g_m + C_4 L_3 L_4 R_3 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^6 + R_3 R_4 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_3 R_4 + 2C_1 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 R_4 g_m + 2C_1 C_4 L_1 L_3 R_3 R_4 g_m + C_1 C_4 L_1 L_4 R_3 R_4 g_m + 2C_1 C_4 L_3 L_4 R_3 + C_1 C_4 L_3 L_4 R_4 + C_3 C_4 L_3 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_3 R_4 + C_1 C_4 L_3 R_3 R_4 + C_1 C_4 L_4 R_3 + C_1 L_1 L_3 g_m + C_4 L_3 L_4 g_m) + s^2 (C_1 L_1 R_3 g_m + C_1 L_3 + C_3 L_3 R_3 R_4 g_m) + s (C_1 R_3 R_4 + 2L_3 R_3 g_m + L_3 R_4 g_m)}$$

9.518 X-INVALID-ORDER-518 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 R_4 g_m s^4 + C_1 L_1 L_3 R_4 g_m s^3 + L_3 R_4 g_m s + R_3 R_4 g_m + s^2 (C_1 L_1 R_3 R_4 g_m + C_3 L_3 R_3 R_4 g_m)}{2 R_3 g_m + R_4 g_m + s^4 (2 C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_3 R_4 g_m) + s^3 (2 C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4 + 2 C_1 L_1 L_3 g_m) + s^2 (2 C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m + 2 C_1 L_3 + 2 C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m) + s (2 C_1 R_3 + C_1 R_4 + 2 L_3 g_m)}$$

9.519 X-INVALID-ORDER-519 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 g_m s^4 + C_1 L_1 L_3 g_m s^3 + L_3 g_m s + R_3 g_m + s^2 (C_1 L_1 R_3 g_m + C_3 L_3 R_3 g_m)}{2C_1 C_3 C_4 L_1 L_3 R_3 g_m s^5 + g_m + s^4 (2C_1 C_3 C_4 L_3 R_3 + C_1 C_3 L_1 L_3 g_m + 2C_1 C_4 L_1 L_3 g_m) + s^3 (C_1 C_3 L_3 + 2C_1 C_4 L_1 R_3 g_m + 2C_1 C_4 L_3 + 2C_3 C_4 L_3 R_3 g_m) + s^2 (2C_1 C_4 R_3 + C_1 L_1 g_m + C_3 L_3 g_m + 2C_4 L_3 g_m) + s (C_1 + 2C_4 R_3 g_m)}$$

9.520 X-INVALID-ORDER-520 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 R_4 g_m s^4 + C_1 L_1 L_3 R_4 g_m s^3 + L_3 R_4 g_m s + R_3 R_4 g_m + s^2 (C_1 L_1 R_3 R_4 g_m + C_3 L_3 R_3 R_4 g_m)}{2C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m s^5 + 2R_3 g_m + R_4 g_m + s^4 (2C_1 C_3 C_4 L_3 R_3 R_4 + 2C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_3 R_4 g_m + 2C_1 C_4 L_1 L_3 R_4 g_m) + s^3 (2C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4 + 2C_1 C_4 L_1 R_3 R_4 g_m + 2C_1 C_4 L_3 R_4 + 2C_1 L_1 L_3 g_m + 2C_3 C_4 L_3 R_3 R_4 g_m) + s^2 (2C_1 C_4 R_3 R_4 + 2C_1 L_1 R_3 R_4 g_m + 2C_3 C_4 R_3 R_4 g_m) + s (C_1 C_4 L_1 R_3 R_4 g_m + C_3 C_4 L_3 R_3 R_4 g_m) + R_3 R_4 g_m}$$

9.521 X-INVALID-ORDER-521 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m s^5 + R_3 g_m + s^4 (C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_3 R_4 g_m) + s^3 (C_1 C_4 L_1 R_3 R_4 g_m + C_1 L_1 L_3 g_m + C_3 C_4 L_3 R_3 R_4 g_m) + s^2 (C_1 L_1 R_3 g_m + C_3 L_3 R_3 g_m + C_4 L_3 R_4 g_m) + s (C_4 R_3 R_4 g_m + L_3 g_m)}{g_m + s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 g_m + C_1 C_3 C_4 L_1 L_3 R_4 g_m) + s^4 (2C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_1 C_3 L_1 L_3 g_m + 2C_1 C_4 L_1 L_3 g_m) + s^3 (C_1 C_3 L_3 + 2C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m + 2C_1 C_4 L_3 + 2C_3 C_4 L_3 R_3 g_m + C_3 C_4 L_3 R_4 g_m) + s^2 (2C_1 C_4 R_3 + C_1 C_4 R_4 + C_1 L_1 g_m + 2C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_4 R_3 + C_4 R_4 + L_1 g_m) + 1}.$$

9.522 X-INVALID-ORDER-522 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + C_1 C_4 L_1 L_3 L_4 g_m s^5 + L_3 g_m s + R_3 g_m + s^4 (C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (C_1 L_1 L_3 g_m + C_4 L_3 L_4 g_m) + s^2 (C_1 L_1 R_3 g_m + C_3 L_3 R_3 g_m + C_4 L_4 R_3 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 g_m s^6 + g_m + s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 g_m + C_1 C_3 C_4 L_3 L_4) + s^4 (2C_1 C_3 C_4 L_3 R_3 + C_1 C_3 L_1 L_3 g_m + 2C_1 C_4 L_1 L_3 g_m + C_1 C_4 L_1 L_4 g_m + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 + 2C_1 C_4 L_1 R_3 g_m + 2C_1 C_4 L_3 + C_1 C_4 L_4 + 2C_3 C_4 L_3 R_3 g_m) + s^2 (2C_1 C_4 R_3 + C_1 L_1 g_m +$$

9.523 X-INVALID-ORDER-523 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 L_4 R_3 g_m s^5 + C_1 L_1 L_3 L_4 g_m s^4 + L_3 L_4 g_m s^2 + L_4 R_3 g_m s + s^3 (C_1 L_1 L_4 R_3 g_m + C_3 L_3 L_4 R_3 g_m)}{2C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + 2R_3 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 L_1 L_3 L_4 g_m + 2C_1 C_4 L_1 L_3 L_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_3 L_4 + 2C_1 C_4 L_1 L_4 R_3 g_m + 2C_1 C_4 L_3 L_4 + 2C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (2C_1 C_3 L_3 R_3 + 2C_1 C_4 L_4 R_3 + 2C_1 L_1 L_3 g_m + C_1 L_1 L_4 g_m + C_3 L_3 L_4 R_3 g_m)}$$

9.524 X-INVALID-ORDER-524 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + R_3 g_m + s^5 (C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_1 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_3 R_4 g_m + C_1 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (C_1 C_4 L_1 R_3 R_4 g_m + C_1 L_1 L_3 g_m + C_3 C_4 L_3 R_3 R_4 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 g_m s^6 + g_m + s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 g_m + C_1 C_3 C_4 L_1 L_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4) + s^4 (2C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_1 C_3 L_1 L_3 g_m + 2C_1 C_4 L_1 L_3 g_m + C_1 C_4 L_1 L_4 g_m + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 + 2C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m + 2C_1 C_4 L_3 + C_3 C_4 L_3 R_3 g_m)}$$

9.525 X-INVALID-ORDER-525 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 L_4 R_3 R_4 g_m s^5 + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^6 + 2R_3 R_4 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_3 R_4 + 2C_1 C_3 L_1 L_3 L_4 R_3 g_m + C_1 C_3 L_1 L_3 L_4 R_4 g_m + 2C_1 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_3 R_4 g_m + 2C_1 C_3 L_3 L_4 R_3 + C_1 C_3 L_3 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_3 R_4 g_m + 2C_1 C_4 L_3 L_4 R_4 + 2C_1 L_1 L_3 L_4 g_m + 2C_3 L_1 L_3 L_4 R_3 R_4 g_m + 2C_3 L_1 L_3 L_4 R_4 g_m + 2C_3 L_1 L_3 L_4 R_3 R_4 g_m + 2C_3 L_1 L_3 L_4 R_4 g_m + 2C_3 L_1 L_3 L_4 R_3 R_4 g_m + 2C_3 L_1 L_3 L_4 R_4 g_m)}{2C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^6 + 2R_3 R_4 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_3 R_4 + 2C_1 C_3 L_1 L_3 L_4 R_3 g_m + C_1 C_3 L_1 L_3 L_4 R_4 g_m + 2C_1 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_3 R_4 g_m + 2C_1 C_3 L_3 L_4 R_3 + C_1 C_3 L_3 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_3 R_4 g_m + 2C_1 C_4 L_3 L_4 R_4 + 2C_1 L_1 L_3 L_4 g_m + 2C_3 L_1 L_3 L_4 R_3 R_4 g_m + 2C_3 L_1 L_3 L_4 R_4 g_m + 2C_3 L_1 L_3 L_4 R_3 R_4 g_m + 2C_3 L_1 L_3 L_4 R_4 g_m + 2C_3 L_1 L_3 L_4 R_3 R_4 g_m + 2C_3 L_1 L_3 L_4 R_4 g_m)}$$

9.526 X-INVALID-ORDER-526 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^6 + R_3 R_4 g_m + s^5 (C_1 C_3 L_1 L_3 L_4 R_3 g_m + C_1 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 R_4 g_m + C_1 C_4 L_1 L_4 R_3 R_4 g_m + C_1 L_1 L_3 R_3 R_4 g_m)}{2 R_3 g_m + R_4 g_m + s^6 (2 C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^5 (2 C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4 + C_1 C_3 L_1 L_3 L_4 g_m + 2 C_1 C_4 L_1 L_3 L_4 g_m) + s^4 (2 C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_3 R_4 g_m + C_1 C_3 L_3 L_4 + 2 C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m + 2 C_1 C_4 L_3 L_4)}$$

9.527 X-INVALID-ORDER-527 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^6 + C_1 C_4 L_1 L_3 L_4 R_4 g_m}{2 R_3 g_m + R_4 g_m + s^6 (2 C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4 + 2 C_1 C_4 L_1 L_3 L_4 g_m) + s^4 (2 C_1 C_3 C_4 L_3 R_3 R_4 + 2 C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_3 R_4 g_m + 2 C_1 C_4 L_1 L_3 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_3)}$$

$$\mathbf{9.528 \quad X-INVALID-ORDER-528} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 R_4 g_m s^4 + R_3 R_4 g_m + s^2 (C_1 L_1 R_3 R_4 g_m + C_3 L_3 R_3 R_4 g_m)}{2 R_3 g_m + R_4 g_m + s^4 (2 C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + 2 C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4) + s^2 (C_1 C_3 R_3 R_4 + 2 C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m + 2 C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m) + s (2 C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m)}$$

$$\mathbf{9.529 \quad X-INVALID-ORDER-529} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 g_m s^4 + R_3 g_m + s^2 (C_1 L_1 R_3 g_m + C_3 L_3 R_3 g_m)}{2 C_1 C_3 C_4 L_1 L_3 R_3 g_m s^5 + g_m + s^4 (2 C_1 C_3 C_4 L_3 R_3 + C_1 C_3 L_1 L_3 g_m) + s^3 (C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_3 + 2 C_1 C_4 L_1 R_3 g_m + 2 C_3 C_4 L_3 R_3 g_m) + s^2 (C_1 C_3 R_3 + 2 C_1 C_4 R_3 + C_1 L_1 g_m + C_3 L_3 g_m) + s (C_1 + C_3 R_3 g_m + 2 C_4 R_3 g_m)}$$

$$\mathbf{9.530 \quad X-INVALID-ORDER-530} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 R_4 g_m s^4 + R_3 R_4 g_m + s^2 (C_1 L_1 R_3 R_4 g_m + C_3 L_3 R_3 R_4 g_m)}{2 C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m s^5 + 2 R_3 g_m + R_4 g_m + s^4 (2 C_1 C_3 C_4 L_3 R_3 R_4 + 2 C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + 2 C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4 + 2 C_1 C_4 L_1 R_3 R_4 g_m + 2 C_3 C_4 L_3 R_3 R_4 g_m) + s^2 (C_1 C_3 R_3 R_4 + 2 C_1 C_4 R_3 R_4 + 2 C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m)}$$

$$\mathbf{9.531 \quad X-INVALID-ORDER-531} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m s^5 + C_1 C_3 L_1 L_3 R_3 g_m s^4 + C_4 R_3 R_4 g_m s + R_3 g_m + s^3 (C_1 C_4 L_1 R_3 R_4 g_m + C_3 C_4 L_3 R_3 R_4 g_m) + s^2 (C_1 L_1 R_3 g_m + C_3 L_3 R_3 g_m)}{g_m + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_3 g_m + C_1 C_3 C_4 L_1 L_3 R_4 g_m) + s^4 (C_1 C_3 C_4 L_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_1 C_3 L_1 L_3 g_m) + s^3 (C_1 C_3 C_4 R_3 R_4 + C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_3 + 2 C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m + 2 C_3 C_4 L_3 R_3 g_m + C_3 C_4 L_3 R_4 g_m) + s^2 (C_1 C_3 R_3)}$$

$$\mathbf{9.532 \quad X-INVALID-ORDER-532} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + R_3 g_m + s^4 (C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_3 L_4 R_3 g_m) + s^2 (C_1 L_1 R_3 g_m + C_3 L_3 R_3 g_m + C_4 L_4 R_3 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 g_m s^6 + g_m + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_3 g_m + C_1 C_3 C_4 L_3 L_4) + s^4 (2 C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_4 R_3 + C_1 C_3 L_1 L_3 g_m + C_1 C_4 L_1 L_4 g_m + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_3 + 2 C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_4 + 2 C_3 C_4 L_3 R_3 g_m + C_3 C_4 L_3 R_4 g_m)}$$

$$\mathbf{9.533 \quad X-INVALID-ORDER-533} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 L_4 R_3 g_m s^5 + L_4 R_3 g_m s + s^3 (C_1 L_1 L_4 R_3 g_m + C_3 L_3 L_4 R_3 g_m)}{2 C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + 2 R_3 g_m + s^5 (2 C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 L_1 L_3 L_4 g_m) + s^4 (2 C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_3 L_4 + 2 C_1 C_4 L_1 L_4 R_3 g_m + 2 C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (2 C_1 C_3 L_3 R_3 + C_1 C_3 L_4 R_3 + 2 C_1 C_4 L_4 R_3 + C_1 L_1 L_4 g_m + C_3 L_3 L_4 g_m) + s^2 (2 C_1 C_4 L_1 L_4 R_3 g_m + 2 C_3 C_4 L_3 L_4 R_3 g_m)}$$

$$\mathbf{9.534 \quad X-INVALID-ORDER-534} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m s^5 + C_4 R_3 R_4 g_m s + R_3 g_m + s^4 (C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_4 R_3 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 g_m s^6 + g_m + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_3 g_m + C_1 C_3 C_4 L_1 L_3 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_3 g_m + C_1 C_3 C_4 L_3 L_4) + s^4 (C_1 C_3 C_4 L_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_1 C_3 C_4 L_4 R_3 + C_1 C_3 L_1 L_3 g_m + C_1 C_4 L_1 L_4 g_m + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 C_4 R_3 R_4 + C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_1 C_3 C_4 L_4 R_3)}$$

$$\mathbf{9.535 \quad X-INVALID-ORDER-535} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 L_4 R_3 R_4 g_m s^5 + L_4 R_3 R_4 g_m s + s^3 (C_1 L_1 L_4 R_3 g_m + C_3 L_3 L_4 R_3 g_m)}{2 C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^6 + 2 R_3 R_4 g_m + s^5 (2 C_1 C_3 C_4 L_3 L_4 R_3 R_4 + 2 C_1 C_3 L_1 L_3 L_4 R_3 g_m + C_1 C_3 L_1 L_3 L_4 R_4 g_m) + s^4 (2 C_1 C_3 L_1 L_3 R_3 R_4 g_m + C_1 C_3 L_1 L_4 R_3 R_4 g_m + 2 C_1 C_3 L_3 L_4 R_3 + C_1 C_3 L_3 L_4 R_4 + 2 C_1 C_4 L_1 L_4 R_3 R_4 g_m + 2 C_3 C_4 L_3 L_4 R_3 R_4 g_m) + s^3 (2 C_1 C_3 L_3 R_3 + C_1 C_3 L_4 R_3 + 2 C_1 C_4 L_4 R_3 + C_1 L_1 L_4 g_m + C_3 L_3 L_4 g_m)}$$

$$\mathbf{9.536 \quad X-INVALID-ORDER-536} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^6 + C_1 C_3 L_1 L_3 L_4 R_3 g_m s^5 + L_4 R_3 R_4 g_m s}{2 R_3 g_m + R_4 g_m + s^6 (2 C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^5 (C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4 + C_1 C_3 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 C_4 L_4 R_3 R_4 + 2 C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_3 R_4 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_3 L_4 + 2 C_1 C_4 L_1 L_4 R_3 R_4 g_m + 2 C_3 C_4 L_3 L_4 R_3 R_4 g_m)}$$

$$\mathbf{9.537 \quad X-INVALID-ORDER-537} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m}{2 R_3 g_m + R_4 g_m + s^6 (2 C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (2 C_1 C_3 C_4 L_3 R_3 R_4 + C_1 C_3 C_4 L_4 R_3 R_4 + 2 C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_3 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (2 C_1 C_4 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_3 R_4 g_m + 2 C_1 C_4 L_3 L_4 R_3 + C_1 C_4 L_3 L_4 R_4) + s^2 (2 C_1 C_4 L_3 R_3 R_4 + C_1 C_4 L_3 R_4 R_4 + 2 C_1 L_1 L_3 R_3 g_m + C_1 L_1 L_3 R_4 g_m + 2 C_1 L_4 R_3 g_m + C_1 L_4 R_4 g_m) + s (2 C_1 L_3 R_3 R_4 + C_1 L_3 R_4 R_4 + 2 C_1 L_4 R_3 R_4 + C_1 L_4 R_4 R_4) + 1}$$

$$\mathbf{9.538 \quad X-INVALID-ORDER-538} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad R_3, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_3 g_m s}{2 C_1 C_4 L_1 R_3 s^3 + s^2 (C_1 L_1 + 2 C_4 L_1 R_3 g_m) + s (2 C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.539 \quad X-INVALID-ORDER-539} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad R_3, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_3 R_4 g_m s}{2 C_1 C_4 L_1 R_3 R_4 s^3 + 2 R_3 + R_4 + s^2 (2 C_1 L_1 R_3 + C_1 L_1 R_4 + 2 C_4 L_1 R_3 R_4 g_m) + s (2 C_4 R_3 R_4 + 2 L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.540 \quad X-INVALID-ORDER-540} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad R_3, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 R_3 R_4 g_m s^2 + L_1 R_3 g_m s}{s^3 (2 C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4) + s^2 (C_1 L_1 + 2 C_4 L_1 R_3 g_m + C_4 L_1 R_4 g_m) + s (2 C_4 R_3 + C_4 R_4 + L_1 g_m) + 1}$$

$$\mathbf{9.541 \quad X-INVALID-ORDER-541} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad R_3, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_3 g_m s^3 + L_1 R_3 g_m s}{C_1 C_4 L_1 L_4 s^4 + s^3 (2 C_1 C_4 L_1 R_3 + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 + 2 C_4 L_1 R_3 g_m + C_4 L_4) + s (2 C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.542 \quad X-INVALID-ORDER-542} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad R_3, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_3 g_m s^2}{2 C_1 C_4 L_1 L_4 R_3 s^4 + 2 R_3 + s^3 (C_1 L_1 L_4 + 2 C_4 L_1 L_4 R_3 g_m) + s^2 (2 C_1 L_1 R_3 + 2 C_4 L_4 R_3 + L_1 L_4 g_m) + s (2 L_1 R_3 g_m + L_4)}$$

$$\mathbf{9.543 \quad X-INVALID-ORDER-543} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad R_3, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_3 g_m s^3 + C_4 L_1 R_3 R_4 g_m s^2 + L_1 R_3 g_m s}{C_1 C_4 L_1 L_4 s^4 + s^3 (2 C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4 + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 + 2 C_4 L_1 R_3 g_m + C_4 L_1 R_4 g_m + C_4 L_4) + s (2 C_4 R_3 + C_4 R_4 + L_1 g_m) + 1}$$

$$\mathbf{9.544 \quad X-INVALID-ORDER-544} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad R_3, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_3 R_4 g_m s^2}{2 C_1 C_4 L_1 L_4 R_3 R_4 s^4 + 2 R_3 R_4 + s^3 (2 C_1 L_1 L_4 R_3 + C_1 L_1 L_4 R_4 + 2 C_4 L_1 L_4 R_3 R_4 g_m) + s^2 (2 C_1 L_1 R_3 R_4 + 2 C_4 L_4 R_3 R_4 + 2 L_1 L_4 R_3 g_m + L_1 L_4 R_4 g_m) + s (2 L_1 R_3 R_4 g_m + 2 L_4 R_3 + L_4 R_4)}$$

$$\mathbf{9.545 \quad X-INVALID-ORDER-545} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad R_3, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_3 R_4 g_m s^3 + L_1 L_4 R_3 g_m s^2 + L_1 R_3 R_4 g_m s}{2 R_3 + R_4 + s^4 (2 C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4) + s^3 (C_1 L_1 L_4 + 2 C_4 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (2 C_1 L_1 R_3 + C_1 L_1 R_4 + 2 C_4 L_4 R_3 + C_4 L_4 R_4 + L_1 L_4 g_m) + s (2 L_1 R_3 g_m + L_1 R_4 g_m + L_4)}$$

$$\mathbf{9.546 \quad X-INVALID-ORDER-546} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad R_3, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_3 R_4 g_m s^3 + L_1 R_3 R_4 g_m s}{2 R_3 + R_4 + s^4 (2 C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4) + s^3 (2 C_1 C_4 L_1 R_3 R_4 + 2 C_4 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (2 C_1 L_1 R_3 + C_1 L_1 R_4 + 2 C_4 L_1 R_3 R_4 g_m + 2 C_4 L_4 R_3 + C_4 L_4 R_4) + s (2 C_4 R_3 R_4 + 2 L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.547 \quad X-INVALID-ORDER-547} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_4 g_m s}{C_1 C_3 L_1 R_4 s^3 + s^2 (2 C_1 L_1 + C_3 L_1 R_4 g_m) + s (C_3 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.548 \quad X-INVALID-ORDER-548} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_4 g_m s}{s^3 (C_1 C_3 L_1 R_4 + 2 C_1 C_4 L_1 R_4) + s^2 (2 C_1 L_1 + C_3 L_1 R_4 g_m + 2 C_4 L_1 R_4 g_m) + s (C_3 R_4 + 2 C_4 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.549 \quad X-INVALID-ORDER-549} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 R_4 g_m s + L_1 g_m}{C_1 C_3 C_4 L_1 R_4 s^3 + C_3 + 2 C_4 + s^2 (C_1 C_3 L_1 + 2 C_1 C_4 L_1 + C_3 C_4 L_1 R_4 g_m) + s (C_3 C_4 R_4 + C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

$$\mathbf{9.550 \quad X-INVALID-ORDER-550} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 g_m s^2 + L_1 g_m}{C_1 C_3 C_4 L_1 L_4 s^4 + C_3 C_4 L_1 L_4 g_m s^3 + C_3 + 2 C_4 + s^2 (C_1 C_3 L_1 + 2 C_1 C_4 L_1 + C_3 C_4 L_4) + s (C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

$$\mathbf{9.551 \quad X-INVALID-ORDER-551} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 g_m s^2}{2 L_1 g_m s + s^4 (C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4) + s^3 (C_3 L_1 L_4 g_m + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_1 L_1 + C_3 L_4 + 2 C_4 L_4) + 2}$$

$$\mathbf{9.552 \quad X-INVALID-ORDER-552} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 g_m s^2 + C_4 L_1 R_4 g_m s + L_1 g_m}{C_1 C_3 C_4 L_1 L_4 s^4 + C_3 + 2 C_4 + s^3 (C_1 C_3 C_4 L_1 R_4 + C_3 C_4 L_1 L_4 g_m) + s^2 (C_1 C_3 L_1 + 2 C_1 C_4 L_1 + C_3 C_4 L_1 R_4 g_m + C_3 C_4 L_4) + s (C_3 C_4 R_4 + C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

$$\mathbf{9.553 \quad X-INVALID-ORDER-553} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_4 g_m s^2}{2 R_4 + s^4 (C_1 C_3 L_1 L_4 R_4 + 2 C_1 C_4 L_1 L_4 R_4) + s^3 (2 C_1 L_1 L_4 + C_3 L_1 L_4 R_4 g_m + 2 C_4 L_1 L_4 R_4 g_m) + s^2 (2 C_1 L_1 R_4 + C_3 L_4 R_4 + 2 C_4 L_4 R_4 + 2 L_1 L_4 g_m) + s (2 L_1 R_4 g_m + 2 L_4)}$$

$$\mathbf{9.554 \quad X-INVALID-ORDER-554} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_4 g_m s^3 + L_1 L_4 g_m s^2 + L_1 R_4 g_m s}{C_1 C_3 C_4 L_1 L_4 R_4 s^5 + s^4 (C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4 + C_3 C_4 L_1 L_4 R_4 g_m) + s^3 (C_1 C_3 L_1 R_4 + C_3 C_4 L_4 R_4 + C_3 L_1 L_4 g_m + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_1 L_1 + C_3 L_1 R_4 g_m + C_3 L_4 + 2 C_4 L_4) + s (C_3 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.555 \quad X-INVALID-ORDER-555} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_4 g_m s^3 + L_1 R_4 g_m s}{C_1 C_3 C_4 L_1 L_4 R_4 s^5 + s^4 (2 C_1 C_4 L_1 L_4 + C_3 C_4 L_1 L_4 R_4 g_m) + s^3 (C_1 C_3 L_1 R_4 + 2 C_1 C_4 L_1 R_4 + C_3 C_4 L_4 R_4 + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_1 L_1 + C_3 L_1 R_4 g_m + 2 C_4 L_1 R_4 g_m + 2 C_4 L_4) + s (C_3 R_4 + 2 C_4 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.556 \quad X-INVALID-ORDER-556} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_3 R_4 g_m s}{C_1 C_3 L_1 R_3 R_4 s^3 + 2R_3 + R_4 + s^2 (2C_1 L_1 R_3 + C_1 L_1 R_4 + C_3 L_1 R_3 R_4 g_m) + s (C_3 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.557 \quad X-INVALID-ORDER-557} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_3 g_m s}{s^3 (C_1 C_3 L_1 R_3 + 2C_1 C_4 L_1 R_3) + s^2 (C_1 L_1 + C_3 L_1 R_3 g_m + 2C_4 L_1 R_3 g_m) + s (C_3 R_3 + 2C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.558 \quad X-INVALID-ORDER-558} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_3 R_4 g_m s}{2R_3 + R_4 + s^3 (C_1 C_3 L_1 R_3 R_4 + 2C_1 C_4 L_1 R_3 R_4) + s^2 (2C_1 L_1 R_3 + C_1 L_1 R_4 + C_3 L_1 R_3 R_4 g_m + 2C_4 L_1 R_3 R_4 g_m) + s (C_3 R_3 R_4 + 2C_4 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.559 \quad X-INVALID-ORDER-559} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 R_3 R_4 g_m s^2 + L_1 R_3 g_m s}{C_1 C_3 C_4 L_1 R_3 R_4 s^4 + s^3 (C_1 C_3 L_1 R_3 + 2C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4 + C_3 C_4 L_1 R_3 R_4 g_m) + s^2 (C_1 L_1 + C_3 C_4 R_3 R_4 + C_3 L_1 R_3 g_m + 2C_4 L_1 R_3 g_m + C_4 L_1 R_4 g_m) + s (C_3 R_3 + 2C_4 R_3 + C_4 R_4 + L_1 g_m) + 1}$$

$$\mathbf{9.560 \quad X-INVALID-ORDER-560} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_3 g_m s^3 + L_1 R_3 g_m s}{C_1 C_3 C_4 L_1 L_4 R_3 s^5 + s^4 (C_1 C_4 L_1 L_4 + C_3 C_4 L_1 L_4 R_3 g_m) + s^3 (C_1 C_3 L_1 R_3 + 2C_1 C_4 L_1 R_3 + C_3 C_4 L_4 R_3 + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 + C_3 L_1 R_3 g_m + 2C_4 L_1 R_3 g_m + C_4 L_4) + s (C_3 R_3 + 2C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.561 \quad X-INVALID-ORDER-561} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_3 g_m s^2}{2R_3 + s^4 (C_1 C_3 L_1 L_4 R_3 + 2C_1 C_4 L_1 L_4 R_3) + s^3 (C_1 L_1 L_4 + C_3 L_1 L_4 R_3 g_m + 2C_4 L_1 L_4 R_3 g_m) + s^2 (2C_1 L_1 R_3 + C_3 L_4 R_3 + 2C_4 L_4 R_3 + L_1 L_4 g_m) + s (2L_1 R_3 g_m + L_4)}$$

$$\mathbf{9.562 \quad X-INVALID-ORDER-562} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_3 g_m s^3 + C_4 L_1 R_3 R_4 g_m s^2 + L_1 R_3 g_m s}{C_1 C_3 C_4 L_1 L_4 R_3 s^5 + s^4 (C_1 C_3 C_4 L_1 R_3 R_4 + C_1 C_4 L_1 L_4 + C_3 C_4 L_1 L_4 R_3 g_m) + s^3 (C_1 C_3 L_1 R_3 + 2C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4 + C_3 C_4 L_1 R_3 R_4 g_m + C_3 C_4 L_4 R_3 + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 + C_3 C_4 R_3 R_4 + C_3 L_1 R_3 g_m + 2C_4 L_1 R_3 g_m + C_4 L_1 R_4 g_m + C_4 L_4) + s (C_3 R_3 + 2C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.563 \quad X-INVALID-ORDER-563} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_3 R_4 g_m s^2}{2R_3 R_4 + s^4 (C_1 C_3 L_1 L_4 R_3 R_4 + 2C_1 C_4 L_1 L_4 R_3 R_4) + s^3 (2C_1 L_1 L_4 R_3 + C_1 L_1 L_4 R_4 + C_3 L_1 L_4 R_3 R_4 g_m + 2C_4 L_1 L_4 R_3 R_4 g_m) + s^2 (2C_1 L_1 R_3 R_4 + C_3 L_4 R_3 R_4 + 2C_4 L_4 R_3 R_4 + 2L_1 L_4 R_3 g_m + L_1 L_4 R_4 g_m) + s (2L_1 R_3 R_4 g_m + 2L_4 R_3 + L_4 R_4)}$$

$$\mathbf{9.564 \quad X-INVALID-ORDER-564} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_3 R_4 g_m s^3 + L_1 L_4 R_3 g_m s^2 + L_1 R_3 R_4 g_m s}{C_1 C_3 C_4 L_1 L_4 R_3 R_4 s^5 + 2R_3 + R_4 + s^4 (C_1 C_3 L_1 L_4 R_3 + 2C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4 + C_3 C_4 L_1 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 + C_1 L_1 L_4 + C_3 C_4 L_4 R_3 R_4 + C_3 L_1 L_4 R_3 g_m + 2C_4 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (2C_1 L_1 R_3 + C_1 L_1 R_4 + C_3 L_1 R_3 R_4 g_m + C_3 L_4 R_3 + 2C_4 L_4 R_3 + L_1 L_4 g_m) + s (2L_1 R_3 g_m + L_4)}$$

$$\mathbf{9.565 \quad X-INVALID-ORDER-565} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_3 R_4 g_m s^3 + L_1 R_3 R_4 g_m s}{C_1 C_3 C_4 L_1 L_4 R_3 R_4 s^5 + 2R_3 + R_4 + s^4 (2C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4 + C_3 C_4 L_1 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 + 2C_1 C_4 L_1 R_3 R_4 + C_3 C_4 L_4 R_3 R_4 + 2C_4 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (2C_1 L_1 R_3 + C_1 L_1 R_4 + C_3 L_1 R_3 R_4 g_m + 2C_4 L_1 R_3 R_4 g_m + 2C_4 L_4 R_3 + C_4 L_4 g_m) + s (2L_1 R_3 g_m + L_4)}$$

9.566 X-INVALID-ORDER-566 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 R_3 R_4 g_m s^2 + L_1 R_4 g_m s}{s^3 (2C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4) + s^2 (2C_1 L_1 + 2C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m) + s (2C_3 R_3 + C_3 R_4 + 2L_1 g_m) + 2}$$

9.567 X-INVALID-ORDER-567 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 R_3 g_m s + L_1 g_m}{2C_1 C_3 C_4 L_1 R_3 s^3 + C_3 + 2C_4 + s^2 (C_1 C_3 L_1 + 2C_1 C_4 L_1 + 2C_3 C_4 L_1 R_3 g_m) + s (2C_3 C_4 R_3 + C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

9.568 X-INVALID-ORDER-568 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 R_3 R_4 g_m s^2 + L_1 R_4 g_m s}{2C_1 C_3 C_4 L_1 R_3 R_4 s^4 + s^3 (2C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4 + 2C_1 C_4 L_1 R_4 + 2C_3 C_4 L_1 R_3 R_4 g_m) + s^2 (2C_1 L_1 + 2C_3 C_4 R_3 R_4 + 2C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m + 2C_4 L_1 R_4 g_m) + s (2C_3 R_3 + C_3 R_4 + 2C_4 R_4 + 2L_1 g_m) + 2}$$

9.569 X-INVALID-ORDER-569 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 R_3 R_4 g_m s^2 + L_1 g_m + s (C_3 L_1 R_3 g_m + C_4 L_1 R_4 g_m)}{C_3 + 2C_4 + s^3 (2C_1 C_3 C_4 L_1 R_3 + C_1 C_3 C_4 L_1 R_4) + s^2 (C_1 C_3 L_1 + 2C_1 C_4 L_1 + 2C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_1 R_4 g_m) + s (2C_3 C_4 R_3 + C_3 C_4 R_4 + C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

9.570 X-INVALID-ORDER-570 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_4 R_3 g_m s^3 + C_3 L_1 R_3 g_m s + C_4 L_1 L_4 g_m s^2 + L_1 g_m}{C_1 C_3 C_4 L_1 L_4 s^4 + C_3 + 2C_4 + s^3 (2C_1 C_3 C_4 L_1 R_3 + C_3 C_4 L_1 L_4 g_m) + s^2 (C_1 C_3 L_1 + 2C_1 C_4 L_1 + 2C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_4) + s (2C_3 C_4 R_3 + C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

9.571 X-INVALID-ORDER-571 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_4 R_3 g_m s^3 + L_1 L_4 g_m s^2}{2C_1 C_3 C_4 L_1 L_4 R_3 s^5 + s^4 (C_1 C_3 L_1 L_4 + 2C_1 C_4 L_1 L_4 + 2C_3 C_4 L_1 L_4 R_3 g_m) + s^3 (2C_1 C_3 L_1 R_3 + 2C_3 C_4 L_4 R_3 + C_3 L_1 L_4 g_m + 2C_4 L_1 L_4 g_m) + s^2 (2C_1 L_1 + 2C_3 L_1 R_3 g_m + C_3 L_4 + 2C_4 L_4) + s (2C_3 R_3 + 2L_1 g_m) + 2}$$

9.572 X-INVALID-ORDER-572 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_4 R_3 g_m s^3 + L_1 g_m + s^2 (C_3 C_4 L_1 R_3 R_4 g_m + C_4 L_1 L_4 g_m) + s (C_3 L_1 R_3 g_m + C_4 L_1 R_4 g_m)}{C_1 C_3 C_4 L_1 L_4 s^4 + C_3 + 2C_4 + s^3 (2C_1 C_3 C_4 L_1 R_3 + C_1 C_3 C_4 L_1 R_4 + C_3 C_4 L_1 L_4 g_m) + s^2 (C_1 C_3 L_1 + 2C_1 C_4 L_1 + 2C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_1 R_4 g_m + C_3 C_4 L_4) + s (2C_3 C_4 R_3 + C_3 C_4 R_4 + C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

9.573 X-INVALID-ORDER-573 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_4 R_3 R_4 g_m s^3 + L_1 L_4 R_4 g_m s^2}{2C_1 C_3 C_4 L_1 L_4 R_3 R_4 s^5 + 2R_4 + s^4 (2C_1 C_3 L_1 L_4 R_3 + C_1 C_3 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_4 + 2C_3 C_4 L_1 L_4 R_3 R_4 g_m) + s^3 (2C_1 C_3 L_1 R_3 R_4 + 2C_1 L_1 L_4 + 2C_3 C_4 L_4 R_3 R_4 + 2C_3 L_1 L_4 R_3 g_m + C_3 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 R_4 g_m) + s^2 (2C_1 L_1 R_4 + 2C_3 L_1 R_3 R_4 g_m + 2C_3 L_4 R_3 + C_3 L_4 R_4) + s (2C_3 R_3 + 2L_1 g_m) + 2}$$

9.574 X-INVALID-ORDER-574 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_4 R_3 R_4 g_m s^4 + L_1 R_4 g_m s + s^3 (C_3 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (C_3 L_1 R_3 R_4 g_m + L_1 L_4 g_m)}{s^5 (2C_1 C_3 C_4 L_1 L_4 R_3 + C_1 C_3 C_4 L_1 L_4 R_4) + s^4 (C_1 C_3 L_1 L_4 + 2C_1 C_4 L_1 L_4 + 2C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_1 L_4 R_4 g_m) + s^3 (2C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4 + 2C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4 + C_3 L_1 L_4 g_m + 2C_4 L_1 L_4 g_m) + s^2 (2C_1 L_1 + 2C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m + C_3 L_4 + 2C_4 L_4) + s (2C_3 R_3 + 2L_1 g_m) + 2}$$

9.575 X-INVALID-ORDER-575 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_4 R_3 R_4 g_m s^4 + C_3 L_1 R_3 R_4 g_m s^2 + C_4 L_1 L_4 R_4 g_m s^3 + L_1 R_4 g_m s}{s^5 (2C_1 C_3 C_4 L_1 L_4 R_3 + C_1 C_3 C_4 L_1 L_4 R_4) + s^4 (2C_1 C_3 C_4 L_1 R_3 R_4 + 2C_1 C_4 L_1 L_4 + 2C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_1 L_4 R_4 g_m) + s^3 (2C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4 + 2C_1 C_4 L_1 R_4 + 2C_3 C_4 L_1 R_3 R_4 g_m + 2C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4 + 2C_4 L_1 L_4 g_m) + s^2 (2C_1 L_1 + 2C_3 C_4 R_3 R_4 + C_3 L_1 R_4 g_m + L_1 L_4 g_m) + s (2C_3 R_3 + 2L_1 g_m) + 2}$$

9.576 X-INVALID-ORDER-576 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, L_3 s + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 R_4 g_m s^3 + L_1 R_4 g_m s}{2 C_1 C_3 L_1 L_3 s^4 + s^3 (C_1 C_3 L_1 R_4 + 2 C_3 L_1 L_3 g_m) + s^2 (2 C_1 L_1 + C_3 L_1 R_4 g_m + 2 C_3 L_3) + s (C_3 R_4 + 2 L_1 g_m) + 2}$$

9.577 X-INVALID-ORDER-577 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 g_m s^2 + L_1 g_m}{2 C_1 C_3 C_4 L_1 L_3 s^4 + 2 C_3 C_4 L_1 L_3 g_m s^3 + C_3 + 2 C_4 + s^2 (C_1 C_3 L_1 + 2 C_1 C_4 L_1 + 2 C_3 C_4 L_3) + s (C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

9.578 X-INVALID-ORDER-578 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 R_4 g_m s^3 + L_1 R_4 g_m s}{2 C_1 C_3 C_4 L_1 L_3 R_4 s^5 + s^4 (2 C_1 C_3 L_1 L_3 + 2 C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_4 + 2 C_1 C_4 L_1 R_4 + 2 C_3 C_4 L_3 R_4 + 2 C_3 L_1 L_3 g_m) + s^2 (2 C_1 L_1 + C_3 L_1 R_4 g_m + 2 C_3 L_3 + 2 C_4 L_1 R_4 g_m) + s (C_3 R_4 + 2 C_4 R_4 + 2 L_1 g_m) + 2}$$

9.579 X-INVALID-ORDER-579 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, L_3 s + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 R_4 g_m s^3 + C_3 L_1 L_3 g_m s^2 + C_4 L_1 R_4 g_m s + L_1 g_m}{2 C_1 C_3 C_4 L_1 L_3 s^4 + C_3 + 2 C_4 + s^3 (C_1 C_3 C_4 L_1 R_4 + 2 C_3 C_4 L_1 L_3 g_m) + s^2 (C_1 C_3 L_1 + 2 C_1 C_4 L_1 + C_3 C_4 L_1 R_4 g_m + 2 C_3 C_4 L_3) + s (C_3 C_4 R_4 + C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

9.580 X-INVALID-ORDER-580 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, L_3 s + \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 g_m s^4 + L_1 g_m + s^2 (C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m)}{C_3 + 2 C_4 + s^4 (2 C_1 C_3 C_4 L_1 L_3 + C_1 C_3 C_4 L_1 L_4) + s^3 (2 C_3 C_4 L_1 L_3 g_m + C_3 C_4 L_1 L_4 g_m) + s^2 (C_1 C_3 L_1 + 2 C_1 C_4 L_1 + 2 C_3 C_4 L_3 + C_3 C_4 L_4) + s (C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

9.581 X-INVALID-ORDER-581 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 L_4 g_m s^4 + L_1 L_4 g_m s^2}{2 C_1 C_3 C_4 L_1 L_3 L_4 s^6 + 2 C_3 C_4 L_1 L_3 L_4 g_m s^5 + 2 L_1 g_m s + s^4 (2 C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_3 L_4) + s^3 (2 C_3 L_1 L_3 g_m + C_3 L_1 L_4 g_m + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_1 L_1 + 2 C_3 L_3 + C_3 L_4 + 2 C_4 L_4) + 2}$$

9.582 X-INVALID-ORDER-582 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, L_3 s + \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 g_m s^4 + C_3 C_4 L_1 L_3 R_4 g_m s^3 + C_4 L_1 R_4 g_m s + L_1 g_m + s^2 (C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m)}{C_3 + 2 C_4 + s^4 (2 C_1 C_3 C_4 L_1 L_3 + C_1 C_3 C_4 L_1 L_4) + s^3 (C_1 C_3 C_4 L_1 R_4 + 2 C_3 C_4 L_1 L_3 g_m + C_3 C_4 L_1 L_4 g_m) + s^2 (C_1 C_3 L_1 + 2 C_1 C_4 L_1 + C_3 C_4 L_1 R_4 g_m + 2 C_3 C_4 L_3 + C_3 C_4 L_4) + s (C_3 C_4 R_4 + C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

9.583 X-INVALID-ORDER-583 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_4 g_m s^4 + L_1 L_4 R_4 g_m s^2}{2 C_1 C_3 C_4 L_1 L_3 L_4 R_4 s^6 + 2 R_4 + s^5 (2 C_1 C_3 L_1 L_3 L_4 + 2 C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2 C_1 C_3 L_1 L_3 R_4 + C_1 C_3 L_1 L_4 R_4 + 2 C_1 C_4 L_1 L_4 R_4 + 2 C_3 C_4 L_3 L_4 R_4 + 2 C_3 L_1 L_3 L_4 g_m) + s^3 (2 C_1 L_1 L_4 + 2 C_3 L_1 L_3 R_4 g_m + C_3 L_1 L_4 R_4 g_m + 2 C_3 L_3 L_4 + 2 C_4 L_1 L_4 R_4 g_m) + s^2 (2 C_1 L_1 R_4 + 2 C_3 L_3 R_4 + 2 C_4 L_1 R_4 g_m) + s (C_3 R_4 + 2 L_1 g_m) + 2}$$

9.584 X-INVALID-ORDER-584 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_4 g_m s^5 + C_3 L_1 L_3 L_4 g_m s^4 + L_1 L_4 g_m s^2 + L_1 R_4 g_m s + s^3 (C_3 L_1 L_3 R_4 g_m + C_4 L_1 L_4 R_4 g_m)}{2 C_1 C_3 C_4 L_1 L_3 L_4 s^6 + s^5 (C_1 C_3 C_4 L_1 L_4 R_4 + 2 C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (2 C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4 + C_3 C_4 L_1 L_4 R_4 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (C_1 C_3 L_1 R_4 + C_3 C_4 L_4 R_4 + 2 C_3 L_1 L_3 g_m + C_3 L_1 L_4 g_m + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_1 L_1 + C_3 L_1 R_4 g_m + 2 C_3 L_3 + C_3 L_4) + s (C_3 R_4 + 2 L_1 g_m) + 2}$$

9.585 X-INVALID-ORDER-585 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_4 g_m s^5 + L_1 R_4 g_m s + s^3 (C_3 L_1 L_3 R_4 g_m + C_4 L_1 L_4 R_4 g_m)}{2 C_1 C_3 C_4 L_1 L_3 L_4 s^6 + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_4 + 2 C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (2 C_1 C_3 L_1 L_3 + 2 C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_1 L_3 R_4 g_m + C_3 C_4 L_1 L_4 R_4 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (C_1 C_3 L_1 R_4 + 2 C_1 C_4 L_1 R_4 + 2 C_3 C_4 L_3 R_4 + C_3 C_4 L_4 R_4 + 2 C_3 L_1 L_3 g_m + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_1 L_1 + C_3 L_1 R_4 g_m + 2 C_3 L_3 + C_3 L_4) + s (C_3 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.586 \quad X-INVALID-ORDER-586} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 R_4 g_m s^2}{C_1 C_3 L_1 L_3 R_4 s^4 + R_4 + s^3 (2C_1 L_1 L_3 + C_3 L_1 L_3 R_4 g_m) + s^2 (C_1 L_1 R_4 + C_3 L_3 R_4 + 2L_1 L_3 g_m) + s (L_1 R_4 g_m + 2L_3)}$$

$$\mathbf{9.587 \quad X-INVALID-ORDER-587} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 g_m s^2}{L_1 g_m s + s^4 (C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3) + s^3 (C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s^2 (C_1 L_1 + C_3 L_3 + 2C_4 L_3) + 1}$$

$$\mathbf{9.588 \quad X-INVALID-ORDER-588} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 R_4 g_m s^2}{R_4 + s^4 (C_1 C_3 L_1 L_3 R_4 + 2C_1 C_4 L_1 L_3 R_4) + s^3 (2C_1 L_1 L_3 + C_3 L_1 L_3 R_4 g_m + 2C_4 L_1 L_3 R_4 g_m) + s^2 (C_1 L_1 R_4 + C_3 L_3 R_4 + 2C_4 L_3 R_4 + 2L_1 L_3 g_m) + s (L_1 R_4 g_m + 2L_3)}$$

$$\mathbf{9.589 \quad X-INVALID-ORDER-589} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 R_4 g_m s^3 + L_1 L_3 g_m s^2}{C_1 C_3 C_4 L_1 L_3 R_4 s^5 + s^4 (C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 + C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (C_1 C_4 L_1 R_4 + C_3 C_4 L_3 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s^2 (C_1 L_1 + C_3 L_3 + C_4 L_1 R_4 g_m + 2C_4 L_3) + s (C_4 R_4 + L_1 g_m) + 1}$$

$$\mathbf{9.590 \quad X-INVALID-ORDER-590} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 g_m s^4 + L_1 L_3 g_m s^2}{C_1 C_3 C_4 L_1 L_3 L_4 s^6 + C_3 C_4 L_1 L_3 L_4 g_m s^5 + L_1 g_m s + s^4 (C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 + C_1 C_4 L_1 L_4 + C_3 C_4 L_3 L_4) + s^3 (C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 + C_3 L_3 + 2C_4 L_3 + C_4 L_4) + 1}$$

$$\mathbf{9.591 \quad X-INVALID-ORDER-591} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 L_4 g_m s^2}{2L_3 + L_4 + s^4 (C_1 C_3 L_1 L_3 L_4 + 2C_1 C_4 L_1 L_3 L_4) + s^3 (C_3 L_1 L_3 L_4 g_m + 2C_4 L_1 L_3 L_4 g_m) + s^2 (2C_1 L_1 L_3 + C_1 L_1 L_4 + C_3 L_3 L_4 + 2C_4 L_3 L_4) + s (2L_1 L_3 g_m + L_1 L_4 g_m)}$$

$$\mathbf{9.592 \quad X-INVALID-ORDER-592} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 g_m s^4 + C_4 L_1 L_3 R_4 g_m s^3 + L_1 L_3 g_m s^2}{C_1 C_3 C_4 L_1 L_3 L_4 s^6 + s^5 (C_1 C_3 C_4 L_1 L_3 R_4 + C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 + C_1 C_4 L_1 L_4 + C_3 C_4 L_1 L_3 R_4 g_m + C_3 C_4 L_3 L_4) + s^3 (C_1 C_4 L_1 R_4 + C_3 C_4 L_3 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 + C_3 L_3 + C_4 L_1 R_4 g_m + 2C_4 L_3 + C_4 L_4) + 1}$$

$$\mathbf{9.593 \quad X-INVALID-ORDER-593} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 L_4 R_4 g_m s^2}{2L_3 R_4 + L_4 R_4 + s^4 (C_1 C_3 L_1 L_3 L_4 R_4 + 2C_1 C_4 L_1 L_3 L_4 R_4) + s^3 (2C_1 L_1 L_3 L_4 + C_3 L_1 L_3 L_4 R_4 g_m + 2C_4 L_1 L_3 L_4 R_4 g_m) + s^2 (2C_1 L_1 L_3 R_4 + C_1 L_1 L_4 R_4 + C_3 L_3 L_4 R_4 + 2C_4 L_3 L_4 R_4 + 2L_1 L_3 L_4 g_m) + s (2L_1 L_3 R_4 g_m + L_1 L_4 R_4 g_m + 2L_3 L_4)}$$

$$\mathbf{9.594 \quad X-INVALID-ORDER-594} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_4 g_m s^4 + L_1 L_3 L_4 g_m s^3 + L_1 L_3 R_4 g_m s^2}{C_1 C_3 C_4 L_1 L_3 L_4 R_4 s^6 + R_4 + s^5 (C_1 C_3 L_1 L_3 L_4 + 2C_1 C_4 L_1 L_3 L_4 + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_4 + C_1 C_4 L_1 L_4 R_4 + C_3 C_4 L_3 L_4 R_4 + C_3 L_1 L_3 L_4 g_m + 2C_4 L_1 L_3 L_4 g_m) + s^3 (2C_1 L_1 L_3 + C_1 L_1 L_4 + C_3 L_1 L_3 R_4 g_m + C_3 L_3 L_4 + C_4 L_1 L_4 R_4 g_m + 2C_4 L_3 L_4) + s^2 (C_1 L_1 R_4 + C_3 L_3 R_4 + 2C_4 L_3 R_4) + s (L_1 R_4 g_m + 2L_3)}$$

$$\mathbf{9.595 \quad X-INVALID-ORDER-595} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_4 g_m s^4 + L_1 L_3 R_4 g_m s^2}{C_1 C_3 C_4 L_1 L_3 L_4 R_4 s^6 + R_4 + s^5 (2C_1 C_4 L_1 L_3 L_4 + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_4 + 2C_1 C_4 L_1 L_3 R_4 + C_1 C_4 L_1 L_4 R_4 + C_3 C_4 L_3 L_4 R_4 + 2C_4 L_1 L_3 L_4 g_m) + s^3 (2C_1 L_1 L_3 + C_3 L_1 L_3 R_4 g_m + 2C_4 L_1 L_3 R_4 g_m + C_4 L_1 L_4 R_4 g_m + 2C_4 L_3 L_4) + s^2 (C_1 L_1 R_4 + C_3 L_3 R_4 + 2C_4 L_3 R_4) + s (L_1 R_4 g_m + 2L_3)}$$

$$\mathbf{9.596 \quad X-INVALID-ORDER-596} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_4 g_m s^3 + C_3 L_1 R_3 R_4 g_m s^2 + L_1 R_4 g_m s}{2 C_1 C_3 L_1 L_3 s^4 + s^3 (2 C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4 + 2 C_3 L_1 L_3 g_m) + s^2 (2 C_1 L_1 + 2 C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m + 2 C_3 L_3) + s (2 C_3 R_3 + C_3 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.597 \quad X-INVALID-ORDER-597} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 g_m s^2 + C_3 L_1 R_3 g_m s + L_1 g_m}{2 C_1 C_3 C_4 L_1 L_3 s^4 + C_3 + 2 C_4 + s^3 (2 C_1 C_3 C_4 L_1 R_3 + 2 C_3 C_4 L_1 L_3 g_m) + s^2 (C_1 C_3 L_1 + 2 C_1 C_4 L_1 + 2 C_3 C_4 L_1 R_3 g_m + 2 C_3 C_4 L_3) + s (2 C_3 C_4 R_3 + C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

$$\mathbf{9.598 \quad X-INVALID-ORDER-598} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_4 g_m s^3 + C_3 L_1 R_3 R_4 g_m s^2 + L_1 R_4 g_m s}{2 C_1 C_3 C_4 L_1 L_3 R_4 s^5 + s^4 (2 C_1 C_3 C_4 L_1 R_3 R_4 + 2 C_1 C_3 L_1 L_3 + 2 C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (2 C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4 + 2 C_1 C_4 L_1 R_4 + 2 C_3 C_4 L_1 R_3 R_4 g_m + 2 C_3 C_4 L_3 R_4 + 2 C_3 L_1 L_3 g_m) + s^2 (2 C_1 L_1 + 2 C_3 C_4 R_3 R_4 + 2 C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m + 2 C_3 L_3 + 2 C_4 L_1 R_4 g_m) + s (2 C_3 C_4 R_3 + C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

$$\mathbf{9.599 \quad X-INVALID-ORDER-599} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 R_4 g_m s^3 + L_1 g_m + s^2 (C_3 C_4 L_1 R_3 R_4 g_m + C_3 L_1 L_3 g_m) + s (C_3 L_1 R_3 g_m + C_4 L_1 R_4 g_m)}{2 C_1 C_3 C_4 L_1 L_3 s^4 + C_3 + 2 C_4 + s^3 (2 C_1 C_3 C_4 L_1 R_3 + C_1 C_3 C_4 L_1 R_4 + 2 C_3 C_4 L_1 L_3 g_m) + s^2 (C_1 C_3 L_1 + 2 C_1 C_4 L_1 + 2 C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_1 R_4 g_m + 2 C_3 C_4 L_3) + s (2 C_3 C_4 R_3 + C_3 C_4 R_4 + C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

$$\mathbf{9.600 \quad X-INVALID-ORDER-600} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 g_m s^4 + C_3 C_4 L_1 L_4 R_3 g_m s^3 + C_3 L_1 R_3 g_m s + L_1 g_m + s^2 (C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m)}{C_3 + 2 C_4 + s^4 (2 C_1 C_3 C_4 L_1 L_3 + C_1 C_3 C_4 L_1 L_4) + s^3 (2 C_1 C_3 C_4 L_1 R_3 + 2 C_3 C_4 L_1 L_3 g_m + C_3 C_4 L_1 L_4 g_m) + s^2 (C_1 C_3 L_1 + 2 C_1 C_4 L_1 + 2 C_3 C_4 L_1 R_3 g_m + 2 C_3 C_4 L_3 + C_3 C_4 L_4) + s (2 C_3 C_4 R_3 + C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

$$\mathbf{9.601 \quad X-INVALID-ORDER-601} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 g_m s^4 + C_3 L_1 L_4 R_3 g_m s^3 + L_1 L_4 g_m s^2}{2 C_1 C_3 C_4 L_1 L_3 L_4 s^6 + s^5 (2 C_1 C_3 C_4 L_1 L_4 R_3 + 2 C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (2 C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_1 L_4 R_3 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (2 C_1 C_3 L_1 R_3 + 2 C_3 C_4 L_4 R_3 + 2 C_3 L_1 L_3 g_m + C_3 L_1 L_4 g_m + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_1 L_1 + 2 C_3 L_1 R_3 g_m + 2 C_3 L_3)}$$

$$\mathbf{9.602 \quad X-INVALID-ORDER-602} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 g_m s^4 + L_1 g_m + s^3 (C_3 C_4 L_1 L_3 R_4 g_m + C_3 C_4 L_1 L_4 R_3 g_m) + s^2 (C_3 C_4 L_1 R_3 R_4 g_m + C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s (C_3 L_1 R_3 g_m + C_4 L_1 R_4 g_m)}{C_3 + 2 C_4 + s^4 (2 C_1 C_3 C_4 L_1 L_3 + C_1 C_3 C_4 L_1 L_4) + s^3 (2 C_1 C_3 C_4 L_1 R_3 + C_1 C_3 C_4 L_1 R_4 + 2 C_3 C_4 L_1 L_3 g_m + C_3 C_4 L_1 L_4 g_m) + s^2 (C_1 C_3 L_1 + 2 C_1 C_4 L_1 + 2 C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_1 R_4 g_m + 2 C_3 C_4 L_3 + C_3 C_4 L_4) + s (2 C_3 C_4 R_3 + C_3 C_4 R_4 + C_3 L_1 g_m + 2 C_4 L_1 g_m)}$$

$$\mathbf{9.603 \quad X-INVALID-ORDER-603} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_4 g_m s^4 + C_3 L_1 L_4 R_3 R_4 g_m s^3 + L_1 L_4 g_m s^2}{2 C_1 C_3 C_4 L_1 L_3 L_4 R_4 s^6 + 2 R_4 + s^5 (2 C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2 C_1 C_3 L_1 L_3 L_4 + 2 C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2 C_1 C_3 L_1 L_3 R_4 + 2 C_1 C_3 L_1 L_4 R_3 + C_1 C_3 L_1 L_4 R_4 + 2 C_1 C_4 L_1 L_4 R_4 + 2 C_3 C_4 L_1 L_4 R_3 R_4 g_m + 2 C_3 C_4 L_3 L_4 R_4 + 2 C_3 L_1 L_3 L_4 g_m) + s^3 (2 C_1 C_3 L_1 R_3 R_4 + 2 C_1 L_1 L_4)}$$

$$\mathbf{9.604 \quad X-INVALID-ORDER-604} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_4 g_m s^5 + L_1 R_4 g_m s + s^4 (C_3 C_4 L_1 L_4 R_3 R_4 g_m + C_3 L_1 L_3 L_4 g_m) + s^3 (C_3 L_1 L_3 R_4 g_m + C_3 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (C_3 L_1 R_3 R_4 g_m + C_4 L_1 R_4 R_3 g_m) + s (C_3 L_1 R_3 g_m + C_4 L_1 R_4 g_m)}{2 C_1 C_3 C_4 L_1 L_3 L_4 s^6 + s^5 (2 C_1 C_3 C_4 L_1 L_4 R_3 + C_1 C_3 C_4 L_1 L_4 R_4 + 2 C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (2 C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_1 L_4 R_4 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (2 C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4 + 2 C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4 + 2 C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s^2 (2 C_1 L_1 + 2 C_3 L_1 R_3 g_m + 2 C_3 L_3)}$$

$$\mathbf{9.605 \quad X-INVALID-ORDER-605} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_4 g_m s^5 + C_3 C_4 L_1 L_4 R_3 R_4 g_m s^4 + C_3 L_1 R_3 R_4 g_m s^2 + L_1 R_4 g_m s}{2 C_1 C_3 C_4 L_1 L_3 L_4 s^6 + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_4 + 2 C_1 C_3 C_4 L_1 L_4 R_3 + C_1 C_3 C_4 L_1 L_4 R_4 + 2 C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (2 C_1 C_3 C_4 L_1 R_3 R_4 + 2 C_1 C_3 L_1 L_3 + 2 C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_1 L_3 R_4 g_m + 2 C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_1 L_4 R_4 g_m + 2 C_3 C_4 L_3 L_4) + s^3 (2 C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4 + 2 C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4 + 2 C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s^2 (2 C_1 L_1 + 2 C_3 L_1 R_3 g_m + 2 C_3 L_3)}$$

$$\mathbf{9.606 \quad X-INVALID-ORDER-606} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 R_3 R_4 g_m s^2}{C_1 C_3 L_1 L_3 R_3 R_4 s^4 + R_3 R_4 + s^3 (2C_1 L_1 L_3 R_3 + C_1 L_1 L_3 R_4 + C_3 L_1 L_3 R_3 R_4 g_m) + s^2 (C_1 L_1 R_3 R_4 + C_3 L_3 R_3 R_4 + 2L_1 L_3 R_3 g_m + L_1 L_3 R_4 g_m) + s (L_1 R_3 R_4 g_m + 2L_3 R_3 + L_3 R_4)}$$

$$\mathbf{9.607 \quad X-INVALID-ORDER-607} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 R_3 g_m s^2}{R_3 + s^4 (C_1 C_3 L_1 L_3 R_3 + 2C_1 C_4 L_1 L_3 R_3) + s^3 (C_1 L_1 L_3 + C_3 L_1 L_3 R_3 g_m + 2C_4 L_1 L_3 R_3 g_m) + s^2 (C_1 L_1 R_3 + C_3 L_3 R_3 + 2C_4 L_3 R_3 + L_1 L_3 g_m) + s (L_1 R_3 g_m + L_3)}$$

$$\mathbf{9.608 \quad X-INVALID-ORDER-608} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 R_3 R_4 g_m s^2}{R_3 R_4 + s^4 (C_1 C_3 L_1 L_3 R_3 R_4 + 2C_1 C_4 L_1 L_3 R_3 R_4) + s^3 (2C_1 L_1 L_3 R_3 + C_1 L_1 L_3 R_4 + C_3 L_1 L_3 R_3 R_4 g_m + 2C_4 L_1 L_3 R_3 R_4 g_m) + s^2 (C_1 L_1 R_3 R_4 + C_3 L_3 R_3 R_4 + 2C_4 L_3 R_3 R_4 + 2L_1 L_3 R_3 g_m + L_1 L_3 R_4 g_m) + s (L_1 R_3 R_4 g_m + 2L_3 R_3 + L_3 R_4)}$$

$$\mathbf{9.609 \quad X-INVALID-ORDER-609} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 R_3 R_4 g_m s^3 + L_1 L_3 R_3 g_m s^2}{C_1 C_3 C_4 L_1 L_3 R_3 R_4 s^5 + R_3 + s^4 (C_1 C_3 L_1 L_3 R_3 + 2C_1 C_4 L_1 L_3 R_3 + C_1 C_4 L_1 L_3 R_4 + C_3 C_4 L_1 L_3 R_3 R_4 g_m) + s^3 (C_1 C_4 L_1 R_3 R_4 + C_1 L_1 L_3 + C_3 C_4 L_3 R_3 R_4 + C_3 L_1 L_3 R_3 g_m + 2C_4 L_1 L_3 R_3 g_m + C_4 L_1 L_3 R_4 g_m) + s^2 (C_1 L_1 R_3 + C_3 L_3 R_3 + C_4 L_1 R_3 R_4 g_m + 2C_4 L_3 R_3 + C_4 L_3 R_4) + s (L_1 R_3 g_m + L_3)}$$

$$\mathbf{9.610 \quad X-INVALID-ORDER-610} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_3 g_m s^4 + L_1 L_3 R_3 g_m s^2}{C_1 C_3 C_4 L_1 L_3 L_4 R_3 s^6 + R_3 + s^5 (C_1 C_4 L_1 L_3 L_4 + C_3 C_4 L_1 L_3 L_4 R_3 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 + 2C_1 C_4 L_1 L_3 R_3 + C_1 C_4 L_1 L_4 R_3 + C_3 C_4 L_3 L_4 R_3 + C_4 L_1 L_3 L_4 g_m) + s^3 (C_1 L_1 L_3 + C_3 L_1 L_3 R_3 g_m + 2C_4 L_1 L_3 R_3 g_m + C_4 L_1 L_4 R_3 g_m + C_4 L_3 L_4) + s^2 (C_1 L_1 R_3 + C_3 L_3 R_3 + 2C_4 L_3 R_3 + C_4 L_3 R_4) + s (L_1 R_3 g_m + L_3)}$$

$$\mathbf{9.611 \quad X-INVALID-ORDER-611} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 L_4 R_3 g_m s^2}{2L_3 R_3 + L_4 R_3 + s^4 (C_1 C_3 L_1 L_3 L_4 R_3 + 2C_1 C_4 L_1 L_3 L_4 R_3) + s^3 (C_1 L_1 L_3 L_4 + C_3 L_1 L_3 L_4 R_3 g_m + 2C_4 L_1 L_3 L_4 R_3 g_m) + s^2 (2C_1 L_1 L_3 R_3 + C_1 L_1 L_4 R_3 + C_3 L_3 L_4 R_3 + 2C_4 L_3 L_4 R_3 + L_1 L_3 L_4 g_m) + s (2L_1 L_3 R_3 g_m + L_1 L_4 R_3 g_m + L_3 L_4)}$$

$$\mathbf{9.612 \quad X-INVALID-ORDER-612} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_3 g_m s^4 + C_4 L_1 L_3 R_3 R_4 g_m s^3 + L_1 L_3 R_3 g_m s^2}{C_1 C_3 C_4 L_1 L_3 L_4 R_3 s^6 + R_3 + s^5 (C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_4 L_1 L_3 L_4 + C_3 C_4 L_1 L_3 L_4 R_3 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 + 2C_1 C_4 L_1 L_3 R_3 + C_1 C_4 L_1 L_3 R_4 + C_1 C_4 L_1 L_4 R_3 + C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_3 C_4 L_3 L_4 R_3 + C_4 L_1 L_3 L_4 g_m) + s^3 (C_1 C_4 L_1 R_3 R_4 + C_1 L_1 L_3 + C_3 C_4 L_3 R_3 + C_4 L_1 L_3 R_4 g_m + L_1 L_3 R_4 g_m) + s^2 (C_1 L_1 R_3 + C_3 L_3 R_3 + 2C_4 L_3 R_3 + C_4 L_3 R_4) + s (L_1 R_3 g_m + L_3)}$$

$$\mathbf{9.613 \quad X-INVALID-ORDER-613} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 L_4 R_3 R_4 g_m s^2}{2L_3 R_3 R_4 + L_4 R_3 R_4 + s^4 (C_1 C_3 L_1 L_3 L_4 R_3 R_4 + 2C_1 C_4 L_1 L_3 L_4 R_3 R_4) + s^3 (2C_1 L_1 L_3 L_4 R_3 + C_1 L_1 L_3 L_4 R_4 + C_3 L_1 L_3 L_4 R_3 R_4 g_m + 2C_4 L_1 L_3 L_4 R_3 R_4 g_m) + s^2 (2C_1 L_1 L_3 R_3 R_4 + C_1 L_1 L_4 R_3 R_4 + C_3 L_3 L_4 R_3 R_4 + 2C_4 L_3 L_4 R_3 R_4 + 2L_1 L_3 L_4 R_3 g_m + L_1 L_3 L_4 R_4 g_m) + s (L_1 R_3 R_4 g_m + L_3)}$$

$$\mathbf{9.614 \quad X-INVALID-ORDER-614} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_3 R_4 g_m s^4 + C_4 L_1 L_3 R_3 R_4 g_m s^3 + L_1 L_3 R_3 g_m s^2}{C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 s^6 + R_3 R_4 + s^5 (C_1 C_3 L_1 L_3 L_4 R_3 + 2C_1 C_4 L_1 L_3 L_4 R_3 + C_1 C_4 L_1 L_3 L_4 R_4 + C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 R_4 + C_1 C_4 L_1 L_4 R_3 R_4 + C_1 L_1 L_3 L_4 + C_3 C_4 L_3 L_4 R_3 R_4 + C_3 L_1 L_3 L_4 R_3 g_m + 2C_4 L_1 L_3 L_4 R_3 g_m + C_4 L_1 L_3 L_4 R_4 g_m) + s^3 (2C_1 L_1 L_3 R_3 + C_1 L_1 L_3 R_4 + C_3 L_3 R_3 + C_4 L_3 R_3 + C_4 L_3 R_4) + s^2 (C_1 L_1 R_3 + C_3 L_3 R_3 + 2C_4 L_3 R_3 + C_4 L_3 R_4) + s (L_1 R_3 g_m + L_3)}$$

$$\mathbf{9.615 \quad X-INVALID-ORDER-615} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_3 R_4 g_m s^4 + L_1 L_3 R_3 R_4 g_m s^2}{C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 s^6 + R_3 R_4 + s^5 (2C_1 C_4 L_1 L_3 L_4 R_3 + C_1 C_4 L_1 L_3 L_4 R_4 + C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 R_4 + 2C_1 C_4 L_1 L_3 R_3 R_4 + C_1 C_4 L_1 L_4 R_3 R_4 + C_3 C_4 L_3 L_4 R_3 R_4 + 2C_4 L_1 L_3 L_4 R_3 g_m + C_4 L_1 L_3 L_4 R_4 g_m) + s^3 (2C_1 L_1 L_3 R_3 + C_1 L_1 L_3 R_4 + C_3 L_3 R_3 + C_4 L_3 R_3 + C_4 L_3 R_4) + s^2 (C_1 L_1 R_3 + C_3 L_3 R_3 + 2C_4 L_3 R_3 + C_4 L_3 R_4) + s (L_1 R_3 g_m + L_3)}$$

$$\mathbf{9.616 \quad X-INVALID-ORDER-616} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_3 R_4 g_m s^3 + L_1 L_3 R_4 g_m s^2 + L_1 R_3 R_4 g_m s}{2R_3 + R_4 + s^4 (2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_3 R_4) + s^3 (2C_1 L_1 L_3 + 2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 g_m) + s^2 (2C_1 L_1 R_3 + C_1 L_1 R_4 + 2C_3 L_3 R_3 + C_3 L_3 R_4 + 2L_1 L_3 g_m) + s (2L_1 R_3 g_m + L_1 R_4 g_m + 2L_3)}$$

$$\mathbf{9.617 \quad X-INVALID-ORDER-617} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_3 g_m s^3 + L_1 L_3 g_m s^2 + L_1 R_3 g_m s}{2C_1 C_3 C_4 L_1 L_3 R_3 s^5 + s^4 (C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 + 2C_3 C_4 L_1 L_3 R_3 g_m) + s^3 (2C_1 C_4 L_1 R_3 + 2C_3 C_4 L_3 R_3 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s^2 (C_1 L_1 + C_3 L_3 + 2C_4 L_1 R_3 g_m + 2C_4 L_3) + s (2C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.618 \quad X-INVALID-ORDER-618} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_3 R_4 g_m s^3 + L_1 L_3 R_4 g_m s^2 + L_1 R_3 R_4 g_m s}{2C_1 C_3 C_4 L_1 L_3 R_3 R_4 s^5 + 2R_3 + R_4 + s^4 (2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_3 R_4 + 2C_1 C_4 L_1 L_3 R_4 + 2C_3 C_4 L_1 L_3 R_3 R_4 g_m) + s^3 (2C_1 C_4 L_1 R_3 R_4 + 2C_1 L_1 L_3 + 2C_3 C_4 L_3 R_3 R_4 + 2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 g_m + 2C_4 L_1 L_3 R_4 g_m) + s^2 (2C_1 L_1 R_3 + C_1 L_1 R_4 + 2C_3 L_3 R_3 + C_3 L_3 R_4 + 2L_1 L_3 g_m) + s (2L_1 R_3 g_m + L_1 R_4 g_m + 2L_3)}$$

$$\mathbf{9.619 \quad X-INVALID-ORDER-619} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 R_3 R_4 g_m s^4 + L_1 R_3 g_m s + s^3 (C_3 L_1 L_3 R_3 g_m + C_4 L_1 L_3 R_4 g_m) + s^2 (C_4 L_1 R_3 R_4 g_m + L_1 L_3 g_m)}{s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 + C_1 C_3 C_4 L_1 L_3 R_4) + s^4 (C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 + 2C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (2C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4 + 2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s^2 (C_1 L_1 + C_3 L_3 + 2C_4 L_1 R_3 g_m + C_4 L_1 R_4 g_m + 2C_4 L_3)}$$

$$\mathbf{9.620 \quad X-INVALID-ORDER-620} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 g_m s^5 + C_4 L_1 L_3 L_4 g_m s^4 + L_1 L_3 g_m s^2 + L_1 R_3 g_m s + s^3 (C_3 L_1 L_3 R_3 g_m + C_4 L_1 L_4 R_3 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 s^6 + s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 + C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 + C_1 C_4 L_1 L_4 + 2C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_3 L_4) + s^3 (2C_1 C_4 L_1 R_3 + 2C_3 C_4 L_3 R_3 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 + C_3 L_3 + 2C_4 L_1 R_3 g_m + 2C_4 L_4)}$$

$$\mathbf{9.621 \quad X-INVALID-ORDER-621} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_3 g_m s^4 + L_1 L_3 L_4 g_m s^3 + L_1 L_4 R_3 g_m s^2}{2C_1 C_3 C_4 L_1 L_3 L_4 R_3 s^6 + 2R_3 + s^5 (C_1 C_3 L_1 L_3 L_4 + 2C_1 C_4 L_1 L_3 L_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_3 + 2C_1 C_4 L_1 L_4 R_3 + 2C_3 C_4 L_3 L_4 R_3 + C_3 L_1 L_3 L_4 g_m + 2C_4 L_1 L_3 L_4 g_m) + s^3 (2C_1 L_1 L_3 + C_1 L_1 L_4 + 2C_3 L_1 L_3 R_3 g_m + C_3 L_3 L_4 + 2C_4 L_1 L_4 R_3 g_m + 2C_4 L_4)}$$

$$\mathbf{9.622 \quad X-INVALID-ORDER-622} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 g_m s^5 + L_1 R_3 g_m s + s^4 (C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_4 L_1 L_3 L_4 g_m) + s^3 (C_3 L_1 L_3 R_3 g_m + C_4 L_1 L_3 R_4 g_m + C_4 L_1 L_4 R_3 g_m) + s^2 (C_4 L_1 R_3 R_4 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 s^6 + s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 + C_1 C_3 C_4 L_1 L_3 R_4 + C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 + C_1 C_4 L_1 L_4 + 2C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_3 R_4 g_m + C_3 C_4 L_3 L_4) + s^3 (2C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4 + 2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m + 2C_4 L_4)}$$

$$\mathbf{9.623 \quad X-INVALID-ORDER-623} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_3 g_m s^4 + L_1 L_3 L_4 g_m s^3 + L_1 L_4 R_3 g_m s^2}{2C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 s^6 + 2R_3 R_4 + s^5 (2C_1 C_3 L_1 L_3 L_4 R_3 + C_1 C_3 L_1 L_3 L_4 R_4 + 2C_1 C_4 L_1 L_3 L_4 R_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_3 R_4 + 2C_1 C_4 L_1 L_4 R_3 R_4 + 2C_1 L_1 L_3 L_4 + 2C_3 C_4 L_3 L_4 R_3 R_4 + 2C_3 L_1 L_3 L_4 R_3 g_m + C_3 L_1 L_3 L_4 R_4 g_m + 2C_4 L_1 L_3 L_4 R_4 g_m)}$$

$$\mathbf{9.624 \quad X-INVALID-ORDER-624} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^5 + L_1 R_3 R_4 g_m s + s^4 (C_3 L_1 L_3 L_4 R_3 g_m + C_4 L_1 L_3 L_4 R_4 g_m)}{2R_3 + R_4 + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (C_1 C_3 L_1 L_3 L_4 + 2C_1 C_4 L_1 L_3 L_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_3 R_4 + 2C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4 + 2C_3 C_4 L_3 L_4 R_3 + C_3 C_4 L_3 L_4 R_4 + C_3 L_1 L_3 L_4 g_m)}$$

$$\mathbf{9.625 \quad X-INVALID-ORDER-625} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^5 + L_1 R_3 R_4 g_m s + s^4 (C_3 L_1 L_3 L_4 R_3 g_m + C_4 L_1 L_3 L_4 R_4 g_m)}{2R_3 + R_4 + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + 2C_1 C_4 L_1 L_3 L_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_3 R_4 + 2C_1 C_4 L_1 L_3 R_4 + 2C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4 + 2C_3 C_4 L_1 L_3 R_3 R_4 g_m)}$$

$$\mathbf{9.626 \quad X-INVALID-ORDER-626} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_3 R_4 g_m s^3 + L_1 R_3 R_4 g_m s}{2R_3 + R_4 + s^4 (2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_3 R_4) + s^3 (C_1 C_3 L_1 R_3 R_4 + 2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 g_m) + s^2 (2C_1 L_1 R_3 + C_1 L_1 R_4 + C_3 L_1 R_3 R_4 g_m + 2C_3 L_3 R_3 + C_3 L_3 R_4) + s (C_3 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.627 \quad X-INVALID-ORDER-627} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_3 g_m s^3 + L_1 R_3 g_m s}{2C_1 C_3 C_4 L_1 L_3 R_3 s^5 + s^4 (C_1 C_3 L_1 L_3 + 2C_3 C_4 L_1 L_3 R_3 g_m) + s^3 (C_1 C_3 L_1 R_3 + 2C_1 C_4 L_1 R_3 + 2C_3 C_4 L_3 R_3 + C_3 L_1 L_3 g_m) + s^2 (C_1 L_1 + C_3 L_1 R_3 g_m + C_3 L_3 + 2C_4 L_1 R_3 g_m) + s (C_3 R_3 + 2C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.628 \quad X-INVALID-ORDER-628} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_3 R_4 g_m s^3 + L_1 R_3 R_4 g_m s}{2C_1 C_3 C_4 L_1 L_3 R_3 R_4 s^5 + 2R_3 + R_4 + s^4 (2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_3 R_4 + 2C_3 C_4 L_1 L_3 R_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 + 2C_1 C_4 L_1 R_3 R_4 + 2C_3 C_4 L_3 R_3 R_4 + 2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 g_m) + s^2 (2C_1 L_1 R_3 + C_1 L_1 R_4 + C_3 L_1 R_3 R_4 g_m + 2C_3 L_3 R_3 + C_3 L_3 R_4 + 2C_4 L_1 R_3 g_m) + s (C_3 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.629 \quad X-INVALID-ORDER-629} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 R_3 R_4 g_m s^4 + C_3 L_1 L_3 R_3 g_m s^3 + C_4 L_1 R_3 R_4 g_m s^2 + L_1 R_3 g_m s}{s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 + C_1 C_3 C_4 L_1 L_3 R_4) + s^4 (C_1 C_3 C_4 L_1 R_3 R_4 + C_1 C_3 L_1 L_3 + 2C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 + 2C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4 + C_3 C_4 L_1 R_3 R_4 g_m + 2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4 + C_3 L_1 L_3 g_m) + s^2 (C_1 L_1 + C_3 C_4 R_3 R_4 + C_3 L_1 R_3 R_4 g_m) + s (C_3 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.630 \quad X-INVALID-ORDER-630} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 g_m s^5 + L_1 R_3 g_m s + s^3 (C_3 L_1 L_3 R_3 g_m + C_4 L_1 L_4 R_3 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 s^6 + s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 + C_1 C_3 C_4 L_1 L_4 R_3 + C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 + C_1 C_4 L_1 L_4 + 2C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_3 L_4) + s^3 (C_1 C_3 L_1 R_3 + 2C_1 C_4 L_1 R_3 + 2C_3 C_4 L_3 R_3 + C_3 C_4 L_4 R_3 + C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s^2 (C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_4 R_3 + C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_4 R_3 g_m) + s (C_3 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.631 \quad X-INVALID-ORDER-631} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_3 g_m s^4 + L_1 L_4 R_3 g_m s^2}{2C_1 C_3 C_4 L_1 L_3 L_4 R_3 s^6 + 2R_3 + s^5 (C_1 C_3 L_1 L_3 L_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_4 R_3 + 2C_1 C_4 L_1 L_4 R_3 + 2C_3 C_4 L_3 L_4 R_3 + C_3 L_1 L_3 L_4 g_m) + s^3 (C_1 L_1 L_4 + 2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_4 R_3 g_m + C_3 L_3 L_4 + 2C_4 L_1 L_4 R_3 g_m) + s^2 (2C_1 L_1 R_3 + 2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_4 R_3 g_m) + s (C_3 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.632 \quad X-INVALID-ORDER-632} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 g_m s^5 + C_3 C_4 L_1 L_3 R_3 R_4 g_m s^4 + C_4 L_1 R_3 R_4 g_m s^2 + L_1 R_3 g_m s}{C_1 C_3 C_4 L_1 L_3 L_4 s^6 + s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 + C_1 C_3 C_4 L_1 L_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_3 + C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 C_4 L_1 R_3 R_4 + C_1 C_3 L_1 L_3 + C_1 C_4 L_1 L_4 + 2C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_3 R_4 g_m + C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_3 L_4) + s^3 (C_1 C_3 L_1 R_3 + 2C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4 + 2C_3 C_4 L_3 R_3 + C_3 C_4 L_4 R_3 + C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s^2 (2C_1 L_1 R_3 + 2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_4 R_3 g_m) + s (C_3 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.633 \quad X-INVALID-ORDER-633} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_3 R_4 g_m s^4 + L_1 L_4 R_3 R_4 g_m s^2}{2C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 s^6 + 2R_3 R_4 + s^5 (2C_1 C_3 L_1 L_3 L_4 R_3 + C_1 C_3 L_1 L_3 L_4 R_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_3 R_4 + C_1 C_3 L_1 L_4 R_3 R_4 + 2C_1 C_4 L_1 L_4 R_3 R_4 + 2C_3 C_4 L_3 L_4 R_3 R_4 + 2C_3 L_1 L_3 L_4 R_3 g_m + C_3 L_1 L_3 L_4 R_4 g_m) + s^3 (2C_1 L_1 L_4 R_3 + C_1 L_1 L_4 R_4 + 2C_3 L_1 L_3 L_4 R_3 g_m + C_3 L_1 L_3 L_4 R_4 g_m) + s^2 (2C_1 L_1 R_3 + 2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_4 R_3 g_m) + s (C_3 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.634 \quad X-INVALID-ORDER-634} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^4 + L_1 L_4 R_3 R_4 g_m s^2}{2R_3 + R_4 + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (C_1 C_3 C_4 L_1 L_4 R_3 R_4 + C_1 C_3 L_1 L_3 L_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_3 R_4 + C_1 C_3 L_1 L_4 R_3 + 2C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4 + C_3 C_4 L_1 L_4 R_3 R_4 g_m + 2C_3 L_1 L_3 L_4 R_3 g_m + C_3 L_1 L_3 L_4 R_4 g_m) + s^3 (2C_1 L_1 L_4 R_3 + C_1 L_1 L_4 R_4 + 2C_3 L_1 L_3 L_4 R_3 g_m + C_3 L_1 L_3 L_4 R_4 g_m) + s^2 (2C_1 L_1 R_3 + 2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_4 R_3 g_m) + s (C_3 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m)}$$

$$\mathbf{9.635 \quad X-INVALID-ORDER-635} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_3 R_4 g_m s^2 + C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m}{2R_3 + R_4 + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_3 R_4 + 2C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4 + 2C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_3 C_4 L_1 L_4 R_3 R_4 g_m) + s^3 (2C_1 C_4 L_1 R_3 R_4 g_m + C_1 C_4 L_1 R_4 g_m) + s^2 (2C_1 C_4 R_1 R_3 R_4 g_m + 2C_1 C_4 R_3 R_4 + 2C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m) + s (C_1 R_1 g_m + C_1 + 2C_4 R_3 g_m)}$$

$$\mathbf{9.636 \quad X-INVALID-ORDER-636} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_3 g_m s^2 + C_1 R_1 R_3 g_m s + R_3 g_m}{2C_1 C_4 L_1 R_3 g_m s^3 + g_m + s^2 (2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_1 L_1 g_m) + s (C_1 R_1 g_m + C_1 + 2C_4 R_3 g_m)}$$

$$\mathbf{9.637 \quad X-INVALID-ORDER-637} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_3 R_4 g_m s^2 + C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m}{2C_1 C_4 L_1 R_3 R_4 g_m s^3 + 2R_3 g_m + R_4 g_m + s^2 (2C_1 C_4 R_1 R_3 R_4 g_m + 2C_1 C_4 R_3 R_4 + 2C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m) + s (2C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2C_1 R_3 + C_1 R_4 + 2C_4 R_3 R_4 g_m)}$$

$$\mathbf{9.638 \quad X-INVALID-ORDER-638} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 R_3 R_4 g_m s^3 + R_3 g_m + s^2 (C_1 C_4 R_1 R_3 R_4 g_m + C_1 L_1 R_3 g_m) + s (C_1 R_1 R_3 g_m + C_4 R_3 R_4 g_m)}{g_m + s^3 (2C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m) + s^2 (2C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_1 L_1 g_m) + s (C_1 R_1 g_m + C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

$$\mathbf{9.639 \quad X-INVALID-ORDER-639} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 g_m s^4 + C_1 C_4 L_4 R_1 R_3 g_m s^3 + C_1 R_1 R_3 g_m s + R_3 g_m + s^2 (C_1 L_1 R_3 g_m + C_4 L_4 R_3 g_m)}{C_1 C_4 L_1 L_4 g_m s^4 + g_m + s^3 (2C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4) + s^2 (2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_1 L_1 g_m + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_1 + 2C_4 R_3 g_m)}$$

$$\mathbf{9.640 \quad X-INVALID-ORDER-640} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 R_3 g_m s^3 + C_1 L_4 R_1 R_3 g_m s^2 + L_4 R_3 g_m s}{2C_1 C_4 L_1 L_4 R_3 g_m s^4 + 2R_3 g_m + s^3 (2C_1 C_4 L_4 R_1 R_3 g_m + 2C_1 C_4 L_4 R_3 + C_1 L_1 L_4 g_m) + s^2 (2C_1 L_1 R_3 g_m + C_1 L_4 R_1 g_m + C_1 L_4 + 2C_4 L_4 R_3 g_m) + s (2C_1 R_1 R_3 g_m + 2C_1 R_3 + L_4 g_m)}$$

$$\mathbf{9.641 \quad X-INVALID-ORDER-641} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 g_m s^4 + R_3 g_m + s^3 (C_1 C_4 L_1 R_3 R_4 g_m + C_1 C_4 L_4 R_1 R_3 g_m) + s^2 (C_1 C_4 R_1 R_3 R_4 g_m + C_1 L_1 R_3 g_m + C_4 L_4 R_3 g_m) + s (C_1 R_1 R_3 g_m + C_4 R_3 R_4 g_m)}{C_1 C_4 L_1 L_4 g_m s^4 + g_m + s^3 (2C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m + C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4) + s^2 (2C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_1 L_1 g_m + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

$$\mathbf{9.642 \quad X-INVALID-ORDER-642} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 R_3 R_4 g_m s^3 + C_1 L_4 R_1 R_3 R_4 g_m s^2 + L_4 R_3 R_4 g_m s}{2C_1 C_4 L_1 L_4 R_3 R_4 g_m s^4 + 2R_3 R_4 g_m + s^3 (2C_1 C_4 L_4 R_1 R_3 R_4 g_m + 2C_1 C_4 L_4 R_3 R_4 + 2C_1 L_1 L_4 R_3 g_m + C_1 L_1 L_4 R_4 g_m) + s^2 (2C_1 L_1 R_3 R_4 g_m + 2C_1 L_4 R_1 R_3 g_m + C_1 L_4 R_1 R_4 g_m + 2C_1 L_4 R_3 + C_1 L_4 R_4 + 2C_4 L_4 R_3 R_4 g_m) + s (2C_1 R_1 R_3 R_4 g_m + 2C_1 R_3 R_4 + 2L_4 R_3 g_m + L_4 R_4 g_m)}$$

$$\mathbf{9.643 \quad X-INVALID-ORDER-643} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 R_4 g_m s^4 + R_3 R_4 g_m + s^3 (C_1 C_4 L_4 R_1 R_3 R_4 g_m + C_1 L_1 L_4 R_3 g_m) + s^2 (C_1 L_1 R_3 R_4 g_m + C_1 L_4 R_1 R_3 g_m + C_4 L_4 R_3 R_4 g_m) + s (C_1 R_1 R_3 R_4 g_m + L_4 R_3 g_m)}{2R_3 g_m + R_4 g_m + s^4 (2C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (2C_1 C_4 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_4 g_m + 2C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + C_1 L_1 L_4 g_m) + s^2 (2C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m + C_1 L_4 R_1 g_m + C_1 L_4 + 2C_4 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (2C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m)}$$

$$\mathbf{9.644 \quad X-INVALID-ORDER-644} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 R_4 g_m s^4 + C_1 C_4 L_4 R_1 R_3 R_4 g_m s^3 + C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m + s^2 (C_1 L_1 R_3 R_4 g_m + C_4 L_4 R_3 R_4 g_m)}{2 R_3 g_m + R_4 g_m + s^4 (2 C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (2 C_1 C_4 L_1 R_3 R_4 g_m + 2 C_1 C_4 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4) + s^2 (2 C_1 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 R_3 R_4 + 2 C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m + 2 C_4 L_4 R_3 g_m + C_4 L_4 R_4 g_m) + s (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + R_3 R_4 g_m)}$$

$$\mathbf{9.645 \quad X-INVALID-ORDER-645} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_4 g_m s^2 + C_1 R_1 R_4 g_m s + R_4 g_m}{C_1 C_3 L_1 R_4 g_m s^3 + 2 g_m + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + 2 C_1 L_1 g_m) + s (2 C_1 R_1 g_m + 2 C_1 + C_3 R_4 g_m)}$$

$$\mathbf{9.646 \quad X-INVALID-ORDER-646} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 g_m s^2 + C_1 R_1 g_m s + g_m}{s^3 (C_1 C_3 L_1 g_m + 2 C_1 C_4 L_1 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.647 \quad X-INVALID-ORDER-647} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_4 g_m s^2 + C_1 R_1 R_4 g_m s + R_4 g_m}{2 g_m + s^3 (C_1 C_3 L_1 R_4 g_m + 2 C_1 C_4 L_1 R_4 g_m) + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + 2 C_1 C_4 R_1 R_4 g_m + 2 C_1 C_4 R_4 + 2 C_1 L_1 g_m) + s (2 C_1 R_1 g_m + 2 C_1 + C_3 R_4 g_m + 2 C_4 R_4 g_m)}$$

$$\mathbf{9.648 \quad X-INVALID-ORDER-648} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 R_4 g_m s^3 + g_m + s^2 (C_1 C_4 R_1 R_4 g_m + C_1 L_1 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m)}{C_1 C_3 C_4 L_1 R_4 g_m s^4 + s^3 (C_1 C_3 C_4 R_1 R_4 g_m + C_1 C_3 C_4 R_4 + C_1 C_3 L_1 g_m + 2 C_1 C_4 L_1 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.649 \quad X-INVALID-ORDER-649} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 g_m s^4 + C_1 C_4 L_4 R_1 g_m s^3 + C_1 R_1 g_m s + g_m + s^2 (C_1 L_1 g_m + C_4 L_4 g_m)}{C_1 C_3 C_4 L_1 L_4 g_m s^5 + s^4 (C_1 C_3 C_4 L_4 R_1 g_m + C_1 C_3 C_4 L_4) + s^3 (C_1 C_3 L_1 g_m + 2 C_1 C_4 L_1 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.650 \quad X-INVALID-ORDER-650} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 g_m s^3 + C_1 L_4 R_1 g_m s^2 + L_4 g_m s}{2 g_m + s^4 (C_1 C_3 L_1 L_4 g_m + 2 C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_4 R_1 g_m + C_1 C_3 L_4 + 2 C_1 C_4 L_4 R_1 g_m + 2 C_1 C_4 L_4) + s^2 (2 C_1 L_1 g_m + C_3 L_4 g_m + 2 C_4 L_4 g_m) + s (2 C_1 R_1 g_m + 2 C_1)}$$

$$\mathbf{9.651 \quad X-INVALID-ORDER-651} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 g_m s^4 + g_m + s^3 (C_1 C_4 L_1 R_4 g_m + C_1 C_4 L_4 R_1 g_m) + s^2 (C_1 C_4 R_1 R_4 g_m + C_1 L_1 g_m + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m)}{C_1 C_3 C_4 L_1 L_4 g_m s^5 + s^4 (C_1 C_3 C_4 L_1 R_4 g_m + C_1 C_3 C_4 L_4 R_1 g_m + C_1 C_3 C_4 L_4) + s^3 (C_1 C_3 C_4 R_1 R_4 g_m + C_1 C_3 C_4 R_4 + C_1 C_3 L_1 g_m + 2 C_1 C_4 L_1 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.652 \quad X-INVALID-ORDER-652} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 R_4 g_m s^3 + C_1 L_4 R_1 R_4 g_m s^2 + L_4 R_4 g_m s}{2 R_4 g_m + s^4 (C_1 C_3 L_1 L_4 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (C_1 C_3 L_4 R_1 R_4 g_m + C_1 C_3 L_4 R_4 + 2 C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_4 + 2 C_1 L_1 L_4 g_m) + s^2 (2 C_1 L_1 R_4 g_m + 2 C_1 L_4 R_1 g_m + 2 C_1 L_4 + C_3 L_4 R_4 g_m + 2 C_4 L_4 R_4 g_m) + s (2 C_1 R_1 R_4 g_m + 2 C_1 R_4 + 2 L_4 g_m)}$$

$$\mathbf{9.653 \quad X-INVALID-ORDER-653} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_4 g_m s^4 + R_4 g_m + s^3 (C_1 C_4 L_4 R_1 R_4 g_m + C_1 L_1 L_4 g_m) + s^2 (C_1 L_1 R_4 g_m + C_1 L_4 R_1 g_m + C_4 L_4 R_4 g_m) + s (C_1 R_1 R_4 g_m + L_4 g_m)}{C_1 C_3 C_4 L_1 L_4 R_4 g_m s^5 + 2 g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_4 R_4 + C_1 C_3 L_1 L_4 g_m + 2 C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_4 g_m + C_1 C_3 L_4 R_1 g_m + C_1 C_3 L_4 + 2 C_1 C_4 L_4 R_1 g_m + 2 C_1 C_4 L_4 + C_3 C_4 L_4 R_4 g_m) + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + 2 C_1 L_1 g_m + C_3 L_4 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.654 \quad X-INVALID-ORDER-654} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_4 g_m s^4 + C_1 C_4 L_4 R_1 R_4 g_m s^3 + C_1 R_1 R_4 g_m s + R_4 g_m + s^2 (C_1 L_1 R_4 g_m + C_4 L_4 R_4 g_m)}{C_1 C_3 C_4 L_1 L_4 R_4 g_m s^5 + 2g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_4 R_4 + 2C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_4 g_m + 2C_1 C_4 L_1 R_4 g_m + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + C_3 C_4 L_4 R_4 g_m) + s^2 (C_1 C_3 R_1 R_4 g_m + C_1 C_3 R_4 + 2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_1 L_1 g_m + 2C_4 L_4 g_m)}$$

$$\mathbf{9.655 \quad X-INVALID-ORDER-655} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_3 R_4 g_m s^2 + C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m}{C_1 C_3 L_1 R_3 R_4 g_m s^3 + 2R_3 g_m + R_4 g_m + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m) + s (2C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m)}$$

$$\mathbf{9.656 \quad X-INVALID-ORDER-656} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_3 g_m s^2 + C_1 R_1 R_3 g_m s + R_3 g_m}{g_m + s^3 (C_1 C_3 L_1 R_3 g_m + 2C_1 C_4 L_1 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_1 L_1 g_m) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m)}$$

$$\mathbf{9.657 \quad X-INVALID-ORDER-657} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_3 R_4 g_m s^2 + C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + 2C_1 C_4 L_1 R_3 R_4 g_m) + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4 g_m + 2C_1 C_4 R_3 R_4 + 2C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m) + s (2C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m + 2C_1 R_3 + C_1 R_4 + C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m)}$$

$$\mathbf{9.658 \quad X-INVALID-ORDER-658} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 R_3 R_4 g_m s^3 + R_3 g_m + s^2 (C_1 C_4 R_1 R_3 R_4 g_m + C_1 L_1 R_3 g_m) + s (C_1 R_1 R_3 g_m + C_4 R_3 R_4 g_m)}{C_1 C_3 C_4 L_1 R_3 R_4 g_m s^4 + g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 R_3 R_4 + C_1 C_3 L_1 R_3 g_m + 2C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_3 + C_1 C_4 R_4 + C_1 L_1 g_m + C_3 C_4 R_3 R_4 g_m) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m)}$$

$$\mathbf{9.659 \quad X-INVALID-ORDER-659} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 g_m s^4 + C_1 C_4 L_4 R_1 R_3 g_m s^3 + C_1 R_1 R_3 g_m s + R_3 g_m + s^2 (C_1 L_1 R_3 g_m + C_4 L_4 R_3 g_m)}{C_1 C_3 C_4 L_1 L_4 R_3 g_m s^5 + g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_4 R_3 + C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_3 g_m + 2C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_1 L_1 g_m + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m)}$$

$$\mathbf{9.660 \quad X-INVALID-ORDER-660} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 R_3 g_m s^3 + C_1 L_4 R_1 R_3 g_m s^2 + L_4 R_3 g_m s}{2R_3 g_m + s^4 (C_1 C_3 L_1 L_4 R_3 g_m + 2C_1 C_4 L_1 L_4 R_3 g_m) + s^3 (C_1 C_3 L_4 R_1 R_3 g_m + C_1 C_3 L_4 R_3 + 2C_1 C_4 L_4 R_1 R_3 g_m + 2C_1 C_4 L_4 R_3 + C_1 L_1 L_4 g_m) + s^2 (2C_1 L_1 R_3 g_m + C_1 L_4 R_1 g_m + C_1 L_4 + C_3 L_4 R_3 g_m + 2C_4 L_4 R_3 g_m) + s (2C_1 R_1 R_3 g_m + 2C_1 R_3 + L_4 g_m)}$$

$$\mathbf{9.661 \quad X-INVALID-ORDER-661} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 g_m s^4 + R_3 g_m + s^3 (C_1 C_4 L_1 R_3 R_4 g_m + C_1 C_4 L_4 R_1 R_3 g_m) + s^2 (C_1 C_4 R_1 R_3 R_4 g_m + C_1 L_1 R_3 g_m + C_4 L_4 R_3 g_m)}{C_1 C_3 C_4 L_1 L_4 R_3 g_m s^5 + g_m + s^4 (C_1 C_3 C_4 L_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_4 R_3 + C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 R_3 R_4 + C_1 C_3 L_1 R_3 g_m + 2C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m + C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_1 L_1 g_m + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_1 + C_3 R_3 g_m + 2C_4 R_3 g_m)}$$

$$\mathbf{9.662 \quad X-INVALID-ORDER-662} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 R_3 R_4 g_m s^3 + C_1 L_4 R_1 R_3 R_4 g_m s^2 + L_4 R_3 R_4 g_m s}{2R_3 R_4 g_m + s^4 (C_1 C_3 L_1 L_4 R_3 R_4 g_m + 2C_1 C_4 L_1 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 L_4 R_3 R_4 + 2C_1 C_4 L_4 R_1 R_3 R_4 g_m + 2C_1 C_4 L_4 R_3 R_4 + 2C_1 L_1 L_4 R_3 g_m + C_1 L_1 L_4 R_4 g_m) + s^2 (2C_1 L_1 R_3 R_4 g_m + 2C_1 L_4 R_1 R_3 g_m + C_1 L_4 R_1 R_4 g_m + 2C_1 L_4 R_3 + C_1 L_4 R_4 + C_3 L_4 R_3 R_4 g_m) + s (2C_1 R_1 R_3 R_4 g_m + 2C_1 R_3 R_4 + L_4 R_3 R_4 g_m)}$$

$$\mathbf{9.663 \quad X-INVALID-ORDER-663} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 R_4 g_m s^4 + R_3 R_4 g_m + s^3 (C_1 C_4 L_4 R_1 R_3 R_4 g_m + C_1 L_1 L_4 R_3 g_m) + s^2 (C_1 L_1 R_3 R_4 g_m + C_4 L_4 R_3 g_m)}{C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m s^5 + 2R_3 g_m + R_4 g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4 + C_1 C_3 L_1 L_4 R_3 g_m + 2C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + C_1 C_3 L_4 R_1 R_3 g_m + C_1 C_3 L_4 R_3 + 2C_1 C_4 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_4 g_m + 2C_1 C_4 L_4 R_3 + C_1 L_1 L_4 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2C_1 C_4 R_1 R_3 R_4 g_m + 2C_1 C_4 R_3 R_4 + C_1 L_1 L_4 R_3 g_m + C_1 L_1 L_4 R_4 g_m) + s (C_1 R_1 R_3 R_4 g_m + C_1 R_3 R_4 + L_4 R_3 R_4 g_m)}$$

$$\mathbf{9.664 \quad X-INVALID-ORDER-664} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_3 R_4 g_m s^4 + C_1 C_4 L_4 R_1 R_3 R_4 g_m s^3 + C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m + s^2 (C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m s^5 + 2 R_3 g_m + R_4 g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4 + 2 C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + 2 C_1 C_4 L_1 R_3 R_4 g_m + 2 C_1 C_4 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + C_3 C_4 L_4 R_3 R_4 g_m))}{C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m s^5 + 2 R_3 g_m + R_4 g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4 + 2 C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + 2 C_1 C_4 L_1 R_3 R_4 g_m + 2 C_1 C_4 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + C_3 C_4 L_4 R_3 R_4 g_m)}$$

$$\mathbf{9.665 \quad X-INVALID-ORDER-665} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 R_3 R_4 g_m s^3 + R_4 g_m + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 L_1 R_4 g_m) + s (C_1 R_1 R_4 g_m + C_3 R_3 R_4 g_m)}{2 g_m + s^3 (2 C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_1 R_4 g_m) + s^2 (2 C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2 C_1 C_3 R_3 + C_1 C_3 R_4 + 2 C_1 L_1 g_m) + s (2 C_1 R_1 g_m + 2 C_1 + 2 C_3 R_3 g_m + C_3 R_4 g_m)}$$

$$\mathbf{9.666 \quad X-INVALID-ORDER-666} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 R_3 g_m s^3 + g_m + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 L_1 g_m) + s (C_1 R_1 g_m + C_3 R_3 g_m)}{2 C_1 C_3 C_4 L_1 R_3 g_m s^4 + s^3 (2 C_1 C_3 C_4 R_1 R_3 g_m + 2 C_1 C_3 C_4 R_3 + C_1 C_3 L_1 g_m + 2 C_1 C_4 L_1 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_3 C_4 R_3 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.667 \quad X-INVALID-ORDER-667} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 R_3 R_4 g_m s^3 + R_4 g_m + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 L_1 R_4 g_m) + s (C_1 R_1 R_4 g_m + C_3 R_3 R_4 g_m)}{2 C_1 C_3 C_4 L_1 R_3 R_4 g_m s^4 + 2 g_m + s^3 (2 C_1 C_3 C_4 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 R_3 R_4 + 2 C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_1 R_4 g_m + 2 C_1 C_4 L_1 R_4 g_m) + s^2 (2 C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2 C_1 C_3 R_3 + C_1 C_3 R_4 + 2 C_1 C_4 R_1 R_4 g_m + 2 C_1 C_4 R_4 + 2 C_1 L_1 g_m + 2 C_3 C_4 R_3 R_4 g_m) + s (2 C_1 R_1 g_m + 2 C_1 + 2 C_3 R_3 g_m + C_3 R_4 g_m)}$$

$$\mathbf{9.668 \quad X-INVALID-ORDER-668} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 R_3 R_4 g_m s^4 + g_m + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_3 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + C_1 L_1 g_m + C_3 C_4 R_3 R_4 g_m) + s (C_1 R_1 g_m + C_3 R_3 g_m + C_4 R_4 g_m)}{s^4 (2 C_1 C_3 C_4 L_1 R_3 g_m + C_1 C_3 C_4 L_1 R_4 g_m) + s^3 (2 C_1 C_3 C_4 R_1 R_3 g_m + C_1 C_3 C_4 R_1 R_4 g_m + 2 C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + C_1 C_3 L_1 g_m + 2 C_1 C_4 L_1 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_3 C_4 R_3 g_m + C_3 C_4 R_4 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.669 \quad X-INVALID-ORDER-669} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_4 R_3 g_m s^5 + g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_3 g_m + C_1 C_4 L_4 R_1 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 L_1 g_m + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_3 R_3 g_m)}{C_1 C_3 C_4 L_1 L_4 g_m s^5 + s^4 (2 C_1 C_3 C_4 L_1 R_3 g_m + C_1 C_3 C_4 L_4 R_1 g_m + C_1 C_3 C_4 L_4) + s^3 (2 C_1 C_3 C_4 R_1 R_3 g_m + 2 C_1 C_3 C_4 R_3 + C_1 C_3 L_1 g_m + 2 C_1 C_4 L_1 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_3 C_4 R_3 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.670 \quad X-INVALID-ORDER-670} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_4 R_3 g_m s^4 + L_4 g_m s + s^3 (C_1 C_3 L_4 R_1 R_3 g_m + C_1 L_1 L_4 g_m) + s^2 (C_1 L_4 R_1 g_m + C_3 L_4 R_3 g_m)}{2 C_1 C_3 C_4 L_1 L_4 R_3 g_m s^5 + 2 g_m + s^4 (2 C_1 C_3 C_4 L_4 R_1 R_3 g_m + 2 C_1 C_3 C_4 L_4 R_3 + C_1 C_3 L_1 L_4 g_m + 2 C_1 C_4 L_1 L_4 g_m) + s^3 (2 C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_4 R_1 g_m + C_1 C_3 L_4 + 2 C_1 C_4 L_4 R_1 g_m + 2 C_1 C_4 L_4 + 2 C_3 C_4 L_4 R_3 g_m) + s^2 (2 C_1 C_3 R_1 R_3 g_m + 2 C_1 C_3 R_3 + 2 C_1 L_1 g_m + C_3 L_4 g_m)}$$

$$\mathbf{9.671 \quad X-INVALID-ORDER-671} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_4 R_3 g_m s^5 + g_m + s^4 (C_1 C_3 C_4 L_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 C_4 R_1 R_3 R_4 g_m + C_1 C_3 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m + C_1 C_4 L_4 R_1 g_m + C_3 C_4 L_4 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + C_1 L_1 g_m + C_3 C_4 R_3 R_4 g_m + C_4 R_4 g_m) + s (C_3 R_3 g_m + C_4 R_4 g_m)}{C_1 C_3 C_4 L_1 L_4 g_m s^5 + s^4 (2 C_1 C_3 C_4 L_1 R_3 g_m + C_1 C_3 C_4 L_1 R_4 g_m + C_1 C_3 C_4 L_4 R_1 g_m + C_1 C_3 C_4 L_4) + s^3 (2 C_1 C_3 C_4 R_1 R_3 g_m + C_1 C_3 C_4 R_1 R_4 g_m + 2 C_1 C_3 C_4 R_3 + C_1 C_3 C_4 R_4 + C_1 C_3 L_1 g_m + 2 C_1 C_4 L_1 g_m + C_3 C_4 L_4 g_m) + s^2 (C_1 C_3 R_1 g_m + C_1 C_3 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_3 C_4 R_3 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.672 \quad X-INVALID-ORDER-672} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_4 R_3 R_4 g_m s^4 + L_4 R_4 g_m s + s^3 (C_1 C_3 L_4 R_1 R_3 R_4 g_m + C_1 L_1 L_4 R_4 g_m)}{2 C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m s^5 + 2 R_4 g_m + s^4 (2 C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_4 R_3 R_4 + 2 C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_1 L_4 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (2 C_1 C_3 L_1 R_3 R_4 g_m + 2 C_1 C_3 L_4 R_1 R_3 g_m + C_1 C_3 L_4 R_1 R_4 g_m + 2 C_1 C_3 L_4 R_3 + C_1 C_3 L_4 R_4 + 2 C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (2 C_1 C_3 R_1 R_3 g_m + 2 C_1 C_3 R_3 + 2 C_1 L_1 g_m + C_3 L_4 g_m)}$$

$$\mathbf{9.673 \quad X-INVALID-ORDER-673} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m s^5 + R_4 g_m + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + C_1 C_3 L_4 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_4 g_m + C_1 L_1 L_4 g_m + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + C_1 L_1 g_m + C_3 C_4 R_3 R_4 g_m + C_4 R_4 g_m) + s (C_3 R_3 g_m + C_4 R_4 g_m)}{2 g_m + s^5 (2 C_1 C_3 C_4 L_1 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_4 g_m) + s^4 (2 C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_4 R_3 + C_1 C_3 C_4 L_4 R_4 + C_1 C_3 L_1 L_4 g_m + 2 C_1 C_4 L_1 L_4 g_m) + s^3 (2 C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_1 R_4 g_m + C_1 C_3 L_4 R_1 g_m + C_1 C_3 L_4 + 2 C_1 C_4 L_4 R_1 g_m + 2 C_1 C_4 L_4 R_3 + C_1 C_4 L_4 R_4 + C_3 C_4 L_4 R_3 R_4 g_m) + s^2 (2 C_1 C_3 R_1 R_3 g_m + 2 C_1 C_3 R_3 + 2 C_1 L_1 g_m + C_3 L_4 g_m) + s (C_3 g_m + 2 C_4 g_m)}$$

$$\mathbf{9.684 \quad X-INVALID-ORDER-684} \quad Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \quad \infty, \quad L_3s + \frac{1}{C_3s}, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4R_4g_ms^6 + C_1C_3C_4L_3L_4R_1R_4g_ms^5 + C_1R_1R_4g_ms + R_4g_m + s^4(C_1C_3L_1L_3R_4g_m + C_1C_4L_1L_3R_4g_m + C_1C_3C_4L_1L_3L_4g_ms^6 + 2g_m + s^5(2C_1C_3C_4L_1L_3R_4g_m + C_1C_3C_4L_1L_4R_4g_m + 2C_1C_3C_4L_3L_4R_1g_m + 2C_1C_3C_4L_3L_4) + s^4(2C_1C_3C_4L_3R_1R_4g_m + 2C_1C_3C_4L_3R_4 + C_1C_3C_4L_4R_1R_4g_m + C_1C_3C_4L_4R_4 + 2C_1C_3L_1L_3g_m + 2C_1C_4L_1L_4g_m + 2C_3C_4L_3L_4g_m) + s^3($$

$$\mathbf{9.685 \quad X-INVALID-ORDER-685} \quad Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \quad \infty, \quad \frac{L_3s}{C_3L_3s^2+1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1L_3R_4g_ms^3 + C_1L_3R_1R_4g_ms^2 + L_3R_4g_ms}{C_1C_3L_1L_3R_4g_ms^4 + R_4g_m + s^3(C_1C_3L_3R_1R_4g_m + C_1C_3L_3R_4 + 2C_1L_1L_3g_m) + s^2(C_1L_1R_4g_m + 2C_1L_3R_1g_m + 2C_1L_3 + C_3L_3R_4g_m) + s(C_1R_1R_4g_m + C_1R_4 + 2L_3g_m)}$$

$$\mathbf{9.686 \quad X-INVALID-ORDER-686} \quad Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \quad \infty, \quad \frac{L_3s}{C_3L_3s^2+1}, \quad \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1L_3g_ms^3 + C_1L_3R_1g_ms^2 + L_3g_ms}{g_m + s^4(C_1C_3L_1L_3g_m + 2C_1C_4L_1L_3g_m) + s^3(C_1C_3L_3R_1g_m + C_1C_3L_3 + 2C_1C_4L_3R_1g_m + 2C_1C_4L_3) + s^2(C_1L_1g_m + C_3L_3g_m + 2C_4L_3g_m) + s(C_1R_1g_m + C_1)}$$

$$\mathbf{9.687 \quad X-INVALID-ORDER-687} \quad Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \quad \infty, \quad \frac{L_3s}{C_3L_3s^2+1}, \quad \frac{R_4}{C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1L_3R_4g_ms^3 + C_1L_3R_1R_4g_ms^2 + L_3R_4g_ms}{R_4g_m + s^4(C_1C_3L_1L_3R_4g_m + 2C_1C_4L_1L_3R_4g_m) + s^3(C_1C_3L_3R_1R_4g_m + C_1C_3L_3R_4 + 2C_1C_4L_3R_1R_4g_m + 2C_1C_4L_3R_4 + 2C_1L_1L_3g_m) + s^2(C_1L_1R_4g_m + 2C_1L_3R_1g_m + 2C_1L_3 + C_3L_3R_4g_m + 2C_4L_3R_4g_m) + s(C_1R_1R_4g_m + C_1R_4 + 2L_3g_m)}$$

$$\mathbf{9.688 \quad X-INVALID-ORDER-688} \quad Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \quad \infty, \quad \frac{L_3s}{C_3L_3s^2+1}, \quad R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1L_3R_4g_ms^4 + L_3g_ms + s^3(C_1C_4L_3R_1R_4g_m + C_1L_1L_3g_m) + s^2(C_1L_3R_1g_m + C_4L_3R_4g_m)}{C_1C_3C_4L_1L_3R_4g_ms^5 + g_m + s^4(C_1C_3C_4L_3R_1R_4g_m + C_1C_3C_4L_3R_4 + C_1C_3L_1L_3g_m + 2C_1C_4L_1L_3g_m) + s^3(C_1C_3L_3R_1g_m + C_1C_3L_3 + C_1C_4L_1R_4g_m + 2C_1C_4L_3R_1g_m + 2C_1C_4L_3 + C_3C_4L_3R_4g_m) + s^2(C_1C_4R_1R_4g_m + C_1C_4R_4 + C_1L_1g_m + C_3L_3g_m + 2C_4L_3R_4g_m)}$$

$$\mathbf{9.689 \quad X-INVALID-ORDER-689} \quad Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \quad \infty, \quad \frac{L_3s}{C_3L_3s^2+1}, \quad L_4s + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1L_3L_4g_ms^5 + C_1C_4L_3L_4R_1g_ms^4 + C_1L_3R_1g_ms^2 + L_3g_ms + s^3(C_1L_1L_3g_m + C_4L_3L_4g_m)}{C_1C_3C_4L_1L_3L_4g_ms^6 + g_m + s^5(C_1C_3C_4L_3L_4R_1g_m + C_1C_3C_4L_3L_4) + s^4(C_1C_3L_1L_3g_m + 2C_1C_4L_1L_3g_m + C_1C_4L_1L_4g_m + C_3C_4L_3L_4g_m) + s^3(C_1C_3L_3R_1g_m + C_1C_3L_3 + 2C_1C_4L_3R_1g_m + 2C_1C_4L_3 + C_1C_4L_4R_1g_m + C_1C_4L_4) + s^2(C_1L_1g_m + C_3L_3g_m + 2C_4L_3R_4g_m)}$$

$$\mathbf{9.690 \quad X-INVALID-ORDER-690} \quad Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \quad \infty, \quad \frac{L_3s}{C_3L_3s^2+1}, \quad \frac{L_4s}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1L_3L_4g_ms^3 + C_1L_3L_4R_1g_ms^2 + L_3L_4g_ms}{2L_3g_m + L_4g_m + s^4(C_1C_3L_1L_3L_4g_m + 2C_1C_4L_1L_3L_4g_m) + s^3(C_1C_3L_3L_4R_1g_m + C_1C_3L_3L_4 + 2C_1C_4L_3L_4R_1g_m + 2C_1C_4L_3L_4) + s^2(2C_1L_1L_3g_m + C_1L_1L_4g_m + C_3L_3L_4g_m + 2C_4L_3L_4g_m) + s(2C_1L_3R_1g_m + 2C_1L_3 + C_1L_4R_1g_m + C_1L_4)}$$

$$\mathbf{9.691 \quad X-INVALID-ORDER-691} \quad Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \quad \infty, \quad \frac{L_3s}{C_3L_3s^2+1}, \quad L_4s + R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1L_3L_4g_ms^5 + L_3g_ms + s^4(C_1C_4L_1L_3R_4g_m + C_1C_4L_3L_4R_1g_m) + s^3(C_1C_4L_3R_1R_4g_m + C_1L_1L_3g_m + C_4L_3L_4g_m)}{C_1C_3C_4L_1L_3L_4g_ms^6 + g_m + s^5(C_1C_3C_4L_1L_3R_4g_m + C_1C_3C_4L_3L_4R_1g_m + C_1C_3C_4L_3L_4) + s^4(C_1C_3C_4L_3R_1R_4g_m + C_1C_3C_4L_3R_4 + C_1C_3L_1L_3g_m + 2C_1C_4L_1L_3g_m + C_1C_4L_1L_4g_m + C_3C_4L_3L_4g_m) + s^3(C_1C_3L_3R_1g_m + C_1C_3L_3 + C_1C_4L_1R_4g_m + 2C_1C_4L_3R_4)}$$

$$\mathbf{9.692 \quad X-INVALID-ORDER-692} \quad Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \quad \infty, \quad \frac{L_3s}{C_3L_3s^2+1}, \quad \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1L_3L_4R_4g_ms^3 + C_1L_3L_4R_1R_4g_ms^2 + L_3L_4R_4g_ms}{2L_3R_4g_m + L_4R_4g_m + s^4(C_1C_3L_1L_3L_4R_4g_m + 2C_1C_4L_1L_3L_4R_4g_m) + s^3(C_1C_3L_3L_4R_1R_4g_m + C_1C_3L_3L_4R_4 + 2C_1C_4L_3L_4R_1R_4g_m + 2C_1C_4L_3L_4R_4 + 2C_1L_1L_3L_4g_m) + s^2(2C_1L_1L_3R_4g_m + C_1L_1L_4R_4g_m + 2C_1L_3L_4R_1g_m + 2C_1L_3L_4 + C_3L_3L_4R_4g_m + 2C_4L_3L_4R_4g_m)}$$

$$\mathbf{9.693 \quad X-INVALID-ORDER-693} \quad Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \quad \infty, \quad \frac{L_3s}{C_3L_3s^2+1}, \quad \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1L_3L_4R_4g_ms^5 + L_3R_4g_ms + s^4(C_1C_4L_3L_4R_1R_4g_m + C_1L_1L_3L_4g_m) + s^3(C_1C_4L_3R_1R_4g_m + C_1L_1L_3g_m + C_4L_3L_4g_m)}{C_1C_3C_4L_1L_3L_4R_4g_ms^6 + R_4g_m + s^5(C_1C_3C_4L_3L_4R_1R_4g_m + C_1C_3C_4L_3L_4R_4 + C_1C_3L_1L_3L_4g_m + 2C_1C_4L_1L_3L_4g_m) + s^4(C_1C_3L_1L_3R_4g_m + C_1C_3L_3L_4R_1g_m + C_1C_3L_3L_4 + C_1C_4L_1L_4R_4g_m + 2C_1C_4L_3L_4R_1g_m + 2C_1C_4L_3L_4 + C_3C_4L_3L_4R_4g_m) + s^3(C_1C_3L_3R_1g_m + C_1C_3L_3 + C_1C_4L_1R_4g_m + 2C_1C_4L_3R_4)}$$

9.704 X-INVALID-ORDER-704 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m s^6 + R_4 g_m + s^5 (C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_3 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_4 g_m + 2C_1 C_3 C_4 L_1 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_1 g_m + 2C_1 C_3 C_4 L_3 L_4) + s^4 (2C_1 C_3 C_4 L_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 R_4 + 2C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_4 R_3 R_4 g_m)}{2C_1 C_3 C_4 L_1 L_3 L_4 g_m s^6 + 2g_m + s^5 (2C_1 C_3 C_4 L_1 L_3 R_4 g_m + 2C_1 C_3 C_4 L_1 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_1 g_m + 2C_1 C_3 C_4 L_3 L_4) + s^4 (2C_1 C_3 C_4 L_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 R_4 + 2C_1 C_3 C_4 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_4 R_3 R_4 g_m)}$$

9.705 X-INVALID-ORDER-705 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 R_3 R_4 g_m s^3 + C_1 L_3 R_1 R_3 R_4 g_m s^2 + L_3 R_3 R_4 g_m s}{C_1 C_3 L_1 L_3 R_3 R_4 g_m s^4 + R_3 R_4 g_m + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_3 R_3 R_4 + 2 C_1 L_1 L_3 R_3 g_m + C_1 L_1 L_3 R_4 g_m) + s^2 (C_1 L_1 R_3 R_4 g_m + 2 C_1 L_3 R_1 R_3 g_m + C_1 L_3 R_1 R_4 g_m + 2 C_1 L_3 R_3 + C_1 L_3 R_4 + C_3 L_3 R_3 R_4 g_m) + s (C_1 R_1 R_3 R_4 g_m + C_1 R_3 R_4 + 2 L_3 R_3 g_m + L_3 R_4 g_m)}$$

9.706 X-INVALID-ORDER-706 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 R_3 g_m s^3 + C_1 L_3 R_1 R_3 g_m s^2 + L_3 R_3 g_m s}{R_3 g_m + s^4 (C_1 C_3 L_1 L_3 R_3 g_m + 2 C_1 C_4 L_1 L_3 R_3 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_3 + 2 C_1 C_4 L_3 R_1 R_3 g_m + 2 C_1 C_4 L_3 R_3 + C_1 L_1 L_3 g_m) + s^2 (C_1 L_1 R_3 g_m + C_1 L_3 R_1 g_m + C_1 L_3 + C_3 L_3 R_3 g_m + 2 C_4 L_3 R_3 g_m) + s (C_1 R_1 R_3 g_m + C_1 R_3 + L_3 g_m)}$$

9.707 X-INVALID-ORDER-707 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 R_3 R_4 g_m s^3 + C_1 L_3 R_1 R_3 R_4 g_m s^2 + L_3 R_3 R_4 g_m s}{R_3 R_4 g_m + s^4 (C_1 C_3 L_1 L_3 R_3 R_4 g_m + 2C_1 C_4 L_1 L_3 R_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_3 R_3 R_4 + 2C_1 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_4 L_3 R_3 R_4 + 2C_1 L_1 L_3 R_3 g_m + C_1 L_1 L_3 R_4 g_m) + s^2 (C_1 L_1 R_3 R_4 g_m + 2C_1 L_3 R_1 R_3 g_m + C_1 L_3 R_1 R_4 g_m + 2C_1 L_3 R_3 + C_1 L_3 R_4 + C_3 L_1 L_3 R_3 R_4 g_m) + s (C_1 L_1 R_3 R_4 + C_1 L_3 R_1 R_3 R_4 + C_3 L_1 L_3 R_3 R_4) + C_3 L_1 L_3 R_3 R_4}$$

9.708 X-INVALID-ORDER-708 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_3 R_3 R_4 g_m s^4 + L_3 R_3 g_m s + s^3 (C_1 C_4 L_3 R_1 R_3 R_4 g_m + C_1 L_1 L_3 R_3 g_m) + s^2 (C_1 L_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m s^5 + R_3 g_m + s^4 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_3 R_3 R_4 + C_1 C_3 L_1 L_3 R_3 g_m + 2 C_1 C_4 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_3 + C_1 C_4 L_1 R_3 R_4 g_m + 2 C_1 C_4 L_3 R_1 R_3 g_m + C_1 C_4 L_3 R_1 R_4 g_m + 2 C_1 C_4 L_3 R_3 + C_1 C_4 L_3 g_m))}{C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m s^5 + R_3 g_m + s^4 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_3 R_3 R_4 + C_1 C_3 L_1 L_3 R_3 g_m + 2 C_1 C_4 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_3 + C_1 C_4 L_1 R_3 R_4 g_m + 2 C_1 C_4 L_3 R_1 R_3 g_m + C_1 C_4 L_3 R_1 R_4 g_m + 2 C_1 C_4 L_3 R_3 + C_1 C_4 L_3 g_m)}$$

9.709 X-INVALID-ORDER-709 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_3 g_m s^5 + C_1 C_4 L_3 L_4 R_1 R_3 g_m s^4 + C_1 L_3 R_1 R_3 g_m s^2 + L_3 R_3 g_m s + s^3 (C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + R_3 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 g_m + 2 C_1 C_4 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_3 L_4 R_1 g_m + C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_3 + 2 C_1 C_4 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_3 L_4 R_1 g_m + C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_3 g_m) + s^2 (C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m) + s (C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m) + s^0 (C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m))}{C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + R_3 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 g_m + 2 C_1 C_4 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_3 L_4 R_1 g_m + C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_3 + 2 C_1 C_4 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_3 L_4 R_1 g_m + C_1 C_4 L_3 L_4 + C_3 C_4 L_3 L_4 R_3 g_m) + s^2 (C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m) + s (C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m) + s^0 (C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m)}$$

9.710 X-INVALID-ORDER-710 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 L_4 R_3 g_m s^3 + C_1 L_3 L_4 R_1 R_3 g_m s^2 + L_3 L_4 R_3 g_m s}{2L_3 R_3 g_m + L_4 R_3 g_m + s^4 (C_1 C_3 L_1 L_3 L_4 R_3 g_m + 2C_1 C_4 L_1 L_3 L_4 R_3 g_m) + s^3 (C_1 C_3 L_3 L_4 R_1 R_3 g_m + C_1 C_3 L_3 L_4 R_3 + 2C_1 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_4 L_3 L_4 R_3 + C_1 L_1 L_3 L_4 g_m) + s^2 (2C_1 L_1 L_3 R_3 g_m + C_1 L_1 L_4 R_3 g_m + C_1 L_3 L_4 R_1 g_m + C_1 L_3 L_4 + C_3 L_3 L_4 R_3 g_m + 2C_4 L_3 L_4 R_3 g_m) + s (C_1 L_1 L_3 L_4 R_3 g_m + C_1 L_1 L_4 R_3 g_m + C_1 L_3 L_4 R_1 g_m + C_1 L_3 L_4 + C_3 L_3 L_4 R_3 g_m + 2C_4 L_3 L_4 R_3 g_m) + C_1 L_1 L_3 L_4 R_3 g_m}$$

9.711 X-INVALID-ORDER-711 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1}{C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + R_3 g_m + s^5 (C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_4 L_1 L_3 L_4 g_m)} + s^4 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_3 R_3 R_4 + C_1 C_3 L_1 L_3 R_3 g_m + 2 C_1 C_4 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_3 R_4 g_m + C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_3$$

9.712 X-INVALID-ORDER-712 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 L_4 R_3 R_4 g_m s^3 + C_1 L_3 L_4 R_1 R_3 R_4 g_m s^2 + L_3 L_4 R_3 R_4 g_m s + L_4 R_3 R_4 g_m}{2L_3 R_3 R_4 g_m + L_4 R_3 R_4 g_m + s^4 (C_1 C_3 L_1 L_3 L_4 R_3 R_4 g_m + 2C_1 C_4 L_1 L_3 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 L_3 L_4 R_3 R_4 + 2C_1 C_4 L_3 L_4 R_1 R_3 R_4 g_m + 2C_1 C_4 L_3 L_4 R_3 R_4 + 2C_1 L_1 L_3 L_4 R_3 g_m + C_1 L_1 L_3 L_4 R_4 g_m) + s^2 (2C_1 L_1 L_3 R_3 R_4 g_m + C_1 L_1 L_4 R_3 R_4 g_m +$$

9.713 X-INVALID-ORDER-713 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^6 + R_3 R_4 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_3 R_4 + C_1 C_3 L_1 L_3 L_4 R_3 g_m + 2 C_1 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_4 L_1 L_3 L_4 R_4 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^6 + R_3 R_4 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_3 R_4 + C_1 C_3 L_1 L_3 L_4 R_3 g_m + 2 C_1 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 R_4 g_m + C_1 C_3 L_3 L_4 R_1 R_3 g_m + C_1 C_3 L_3 L_4 R_3 + C_1 C_4 L_1 L_4 R_3 R_4 g_m + 2 C_1 C_4 L_3 L_4 R_3)}$$

9.714 X-INVALID-ORDER-714 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^6 + R_3 R_4 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_3 R_4 + 2 C_1 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 R_4 g_m + 2 C_1 C_4 L_1 L_3 R_3 R_4 g_m + C_1 C_4 L_1 L_4 R_3 R_4 g_m + 2 C_1 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_3 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_3 L_4 R_3 R_4 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^6 + R_3 R_4 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_3 R_4 + 2 C_1 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 R_4 g_m + 2 C_1 C_4 L_1 L_3 R_3 R_4 g_m + C_1 C_4 L_1 L_4 R_3 R_4 g_m + 2 C_1 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_3 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_3 L_4 R_3 R_4 g_m)}$$

9.715 X-INVALID-ORDER-715 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 R_4 g_m s^4 + R_3 R_4 g_m + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 g_m + C_1 L_1 L_3 R_4 g_m) + s^2 (C_1 L_1 R_3 R_4 g_m + C_1 L_3 R_1 R_4 g_m + C_3 L_3 R_3 R_4 g_m) + s (C_1 R_1 R_3 R_4 g_m + L_3 R_4 g_m)}{2 R_3 g_m + R_4 g_m + s^4 (2 C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_3 R_4 g_m) + s^3 (2 C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_1 R_4 g_m + 2 C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4 + 2 C_1 L_1 L_3 g_m) + s^2 (2 C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m + 2 C_1 L_3 R_1 g_m + 2 C_1 L_3 + 2 C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m) + s (2 C_1 R_1 R_3 g_m + C_1 R_1 R_4 g_m)}$$

9.716 X-INVALID-ORDER-716 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 g_m s^4 + R_3 g_m + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 L_1 L_3 g_m) + s^2 (C_1 L_1 R_3 g_m + C_1 L_3 R_1 g_m + C_3 L_3 R_3 g_m) + s (C_1 R_1 R_3 g_m + L_3 g_m)}{2C_1 C_3 C_4 L_1 L_3 R_3 g_m s^5 + g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 R_3 + C_1 C_3 L_1 L_3 g_m + 2C_1 C_4 L_1 L_3 g_m) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2C_1 C_4 L_1 R_3 g_m + 2C_1 C_4 L_3 R_1 g_m + 2C_1 C_4 L_3 + 2C_3 C_4 L_3 R_3 g_m) + s^2 (2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_1 L_1 g_m + C_3 L_3 g_m)}$$

9.717 X-INVALID-ORDER-717 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 R_4 g_m s^4 + R_3 R_4 g_m + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 g_m + C_1 L_1 L_3 R_4 g_m)}{2C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m s^5 + 2R_3 g_m + R_4 g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 R_4 + 2C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_3 R_4 g_m + 2C_1 C_4 L_1 L_3 R_4 g_m) + s^3 (2C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_1 R_4 g_m + 2C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4 + 2C_1 C_4 L_1 R_3 R_4 g_m + 2C_1 C_4 L_3 R_1 R_3 g_m)}$$

9.718 X-INVALID-ORDER-718 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m s^5 + R_3 g_m + s^4 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_3 R_4 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_4 L_1 R_3 R_4 g_m + C_1 C_4 L_3 R_1 R_4 g_m + C_1 L_1 L_3 g_m + C_3 C_4 L_3 R_3 R_4 g_m) + g_m + s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 g_m + C_1 C_3 C_4 L_1 L_3 R_4 g_m) + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_1 C_3 L_1 L_3 g_m + 2C_1 C_4 L_1 L_3 g_m) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m + 2C_1 C_4 L_3 R_1 g_m + 2C_1 C_4 L_3 R_3 R_4 g_m) + s^2 (C_1 C_3 L_3 R_1 + C_1 C_3 L_3 + C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4 + C_1 C_4 L_3 R_1 + C_1 C_4 L_3 R_3 + C_1 L_1 L_3) + s (C_1 C_3 L_3 + C_1 C_4 L_1 + C_1 C_4 L_3) + C_1}{g_m + s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 g_m + C_1 C_3 C_4 L_1 L_3 R_4 g_m) + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_1 C_3 L_1 L_3 g_m + 2C_1 C_4 L_1 L_3 g_m) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m + 2C_1 C_4 L_3 R_1 g_m + 2C_1 C_4 L_3 R_3 R_4 g_m) + s^2 (C_1 C_3 L_3 R_1 + C_1 C_3 L_3 + C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4 + C_1 C_4 L_3 R_1 + C_1 C_4 L_3 R_3 + C_1 L_1 L_3) + s (C_1 C_3 L_3 + C_1 C_4 L_1 + C_1 C_4 L_3) + C_1}.$$

9.719 X-INVALID-ORDER-719 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + R_3 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_4 L_4 R_1 R_3 g_m)}{C_1 C_3 C_4 L_1 L_3 L_4 g_m s^6 + g_m + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_3 L_4) + s^4 (2 C_1 C_3 C_4 L_3 R_1 R_3 g_m + 2 C_1 C_3 C_4 L_3 R_3 + C_1 C_3 L_1 L_3 g_m + 2 C_1 C_4 L_1 L_3 g_m + C_1 C_4 L_1 L_4 g_m + C_3 C_4 L_3 L_4 g_m) + s^3 (C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2 C_1 C_4 L_1 R_3 g_m + 2 C_1 C_4 L_1 R_3)}$$

9.720 X-INVALID-ORDER-720 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 L_4 R_3 g_m s^5 + L_4 R_3 g_m s + s^4 (C_1 C_3 L_3 L_4 R_1 R_3 g_m + C_1 L_1 L_3 L_4 g_m)}{2C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + 2R_3 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 L_1 L_3 L_4 g_m + 2C_1 C_4 L_1 L_3 L_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_3 L_4 R_1 g_m + C_1 C_3 L_3 L_4 + 2C_1 C_4 L_1 L_4 R_3 g_m + 2C_1 C_4 L_3 L_4 R_1 g_m + 2C_1 C_4 L_3 L_4 + 2C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (2C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + C_1 C_3 L_1 L_3 L_4 g_m + 2C_1 C_4 L_1 L_3 L_4 g_m) + s^2 (2C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + C_1 C_3 L_1 L_3 L_4 g_m + 2C_1 C_4 L_1 L_3 L_4 g_m) + s (2C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + C_1 C_3 L_1 L_3 L_4 g_m + 2C_1 C_4 L_1 L_3 L_4 g_m) + 2R_3 g_m}$$

9.721 X-INVALID-ORDER-721 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + R_3 g_m + s^5 (C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_3 R_4 g_m + C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4 g_m) + s^3 (C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_3 L_4 g_m) + s^2 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_3 R_4 g_m + C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4 g_m) + s (C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_3 R_4 g_m + C_1 C_4 L_1 L_4 R_3 g_m + C_1 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4 g_m) + C_3 C_4 L_1 L_3 L_4 g_m}{C_1 C_3 C_4 L_1 L_3 L_4 g_m s^6 + g_m + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_3 g_m + C_1 C_3 C_4 L_1 L_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_3 L_4) + s^4 (2 C_1 C_3 C_4 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_1 C_3 L_1 L_3 g_m + 2 C_1 C_4 L_1 L_3 g_m + C_1 C_4 L_1 L_4 g_m + C_3 C_4 L_3 L_4) + s^3 (2 C_1 C_3 C_4 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_1 C_3 L_1 L_3 g_m + 2 C_1 C_4 L_1 L_3 g_m + C_1 C_4 L_1 L_4 g_m + C_3 C_4 L_3 L_4) + s^2 (2 C_1 C_3 C_4 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_1 C_3 L_1 L_3 g_m + 2 C_1 C_4 L_1 L_3 g_m + C_1 C_4 L_1 L_4 g_m + C_3 C_4 L_3 L_4) + s (2 C_1 C_3 C_4 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_3 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_1 C_3 L_1 L_3 g_m + 2 C_1 C_4 L_1 L_3 g_m + C_1 C_4 L_1 L_4 g_m + C_3 C_4 L_3 L_4) + C_3 C_4 L_1 L_3 L_4 g_m}.$$

9.722 X-INVALID-ORDER-722 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{2C_1C_3C_4L_1L_3L_4R_3R_4q_ms^6 + 2R_3R_4q_ms^5 + s^5(2C_1C_3C_4L_3L_4R_1R_3R_4q_m + 2C_1C_3C_4L_3L_4R_3R_4 + 2C_1C_3L_1L_3L_4R_3q_m + C_1C_3L_1L_3L_4R_4q_m + 2C_1C_4L_1L_3L_4R_4q_m) + s^4(2C_1C_3L_1L_3R_3R_4q_m + 2C_1C_3L_3L_4R_1R_3q_m + C_1C_3L_3L_4R_1R_4q_m + 2C_1C_3L_3L_4R_3 + C_1C_3L_3L_4R_4)}{s^6 + 2C_1C_3C_4L_1L_3L_4R_3R_4q_ms^6 + 2R_3R_4q_ms^5 + s^5(2C_1C_3C_4L_3L_4R_1R_3R_4q_m + 2C_1C_3C_4L_3L_4R_3R_4 + 2C_1C_3L_1L_3L_4R_3q_m + C_1C_3L_1L_3L_4R_4q_m + 2C_1C_4L_1L_3L_4R_4q_m) + s^4(2C_1C_3L_1L_3R_3R_4q_m + 2C_1C_3L_3L_4R_1R_3q_m + C_1C_3L_3L_4R_1R_4q_m + 2C_1C_3L_3L_4R_3 + C_1C_3L_3L_4R_4)}$$

9.723 X-INVALID-ORDER-723 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^6 + R_3 R_4 g_m + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 L_4 R_3 g_m + C_1 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2 C_1 C_2 L_1 L_3 R_3 q_m + C_1 C_2 L_1 L_3 R_4 q_m + C_1 C_2 L_3 L_4 R_3 q_m + C_1 C_2 L_3 L_4 R_4 q_m) + s^3 (2 C_1 C_2 C_4 L_1 L_3 L_4 B_3 q_m + C_1 C_2 C_4 L_1 L_3 L_4 B_4 q_m) + s^2 (2 C_1 C_2 C_4 L_3 L_4 B_1 R_3 q_m + C_1 C_2 C_4 L_3 L_4 B_1 R_4 q_m + 2 C_1 C_2 C_4 L_3 L_4 B_3 + C_1 C_2 C_4 L_3 L_4 B_4 + C_1 C_2 L_1 L_3 L_4 q_m + 2 C_1 C_4 L_1 L_3 L_4 q_m) + s^4 (2 C_1 C_2 L_1 L_3 R_3 q_m + C_1 C_2 L_1 L_3 R_4 q_m + C_1 C_2 L_3 L_4 R_3 q_m + C_1 C_2 L_3 L_4 R_4 q_m)}{2 R_3 q_m + R_4 q_m + s^6 (2 C_1 C_3 C_4 L_1 L_3 L_4 R_3 q_m + C_1 C_3 C_4 L_1 L_3 L_4 R_4 q_m) + s^5 (2 C_1 C_3 C_4 L_3 L_4 R_1 R_3 q_m + C_1 C_3 C_4 L_3 L_4 R_1 R_4 q_m + 2 C_1 C_2 C_4 L_3 L_4 B_3 + C_1 C_2 C_4 L_3 L_4 B_4 + C_1 C_2 L_1 L_3 L_4 q_m + 2 C_1 C_4 L_1 L_3 L_4 q_m) + s^4 (2 C_1 C_2 L_1 L_3 R_3 q_m + C_1 C_2 L_1 L_3 R_4 q_m + C_1 C_2 L_3 L_4 R_3 q_m + C_1 C_2 L_3 L_4 R_4 q_m)}$$

$$\mathbf{9.724 \quad X-INVALID-ORDER-724} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{2R_3 g_m + R_4 g_m + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4 + 2C_1 C_4 L_1 L_3 L_4 g_m) + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 R_4 -$$

$$\mathbf{9.725 \quad X-INVALID-ORDER-725} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 R_4 g_m s^4 + C_1 C_3 L_3 R_1 R_3 R_4 g_m s^3 + C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m + s^2 (C_1 L_1 R_3 R_4 g_m + C_3 L_3 R_3 R_4 g_m)}{2R_3 g_m + R_4 g_m + s^4 (2C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + 2C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_1 R_4 g_m + 2C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4) + s^2 (C_1 C_3 R_1 R_3 R_4 g_m + C_1 C_3 R_3 R_4 + 2C_1 L_1 R_3 g_m + C_1 L_1 R_4 g_m + 2C_3 L_3 R_3 g_m + C_3 L_3 R_4 g_m) + s (2C_1 R_1 R_3 -$$

$$\mathbf{9.726 \quad X-INVALID-ORDER-726} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 g_m s^4 + C_1 C_3 L_3 R_1 R_3 g_m s^3 + C_1 R_1 R_3 g_m s + R_3 g_m + s^2 (C_1 L_1 R_3 g_m + C_3 L_3 R_3 g_m)}{2C_1 C_3 C_4 L_1 L_3 R_3 g_m s^5 + g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 R_3 + C_1 C_3 L_1 L_3 g_m) + s^3 (C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3 + 2C_1 C_4 L_1 R_3 g_m + 2C_3 C_4 L_3 R_3 g_m) + s^2 (C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + 2C_1 C_4 R_1 R_3 g_m + 2C_1 C_4 R_3 + C_1 L_1 g_m + C_3 L_3 g_m) + s (2C_1 R_1 R_3 -$$

$$\mathbf{9.727 \quad X-INVALID-ORDER-727} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 R_4 g_m s^4 + C_1 C_3 L_3 R_1 R_3 R_4 g_m s^3 + C_1 R_1 R_3 R_4 g_m s + R_3 R_4 g_m + s^2 (C_1 L_1 R_3 R_4 g_m + C_3 L_3 R_3 R_4 g_m)}{2C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m s^5 + 2R_3 g_m + R_4 g_m + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 R_4 + 2C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_3 R_4 g_m + 2C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_1 R_4 g_m + 2C_1 C_3 L_3 R_3 + C_1 C_3 L_3 R_4 + 2C_1 C_4 L_1 R_3 R_4 g_m + 2C_3 C_4 L_3 R_3 R_4 -$$

$$\mathbf{9.728 \quad X-INVALID-ORDER-728} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m s^5 + R_3 g_m + s^4 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 R_3 g_m) + s^3 (C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_4 L_1 R_3 -$$

$$\mathbf{9.729 \quad X-INVALID-ORDER-729} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m s^5 + C_1 R_1 R_3 g_m s + R_3 g_m + s^4 (C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_4 L_1 L_4 R_3 g_m -$$

$$\mathbf{9.730 \quad X-INVALID-ORDER-730} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 L_4 R_3 g_m s^5 + C_1 C_3 L_3 L_4 R_1 R_3 g_m s^4 + C_1 L_4 R_1 R_3 g_m s^2 + L_4 R_3 g_m s + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + 2R_3 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 L_1 L_3 L_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_3 g_m + C_1 C_3 L_1 L_4 R_3 g_m + C_1 C_3 L_3 L_4 R_1 g_m + C_1 C_3 L_3 L_4 + 2C_1 C_4 L_1 L_4 R_3 g_m + 2C_3 C_4 L_3 L_4 R_3 g_m) + s^3 (2C_1 C_3 L_3 R_1 R_3 g_m + 2C_1 C_3 L_3 R_3 -$$

$$\mathbf{9.731 \quad X-INVALID-ORDER-731} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m s^6 + R_3 g_m + s^5 (C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 + C_1 C_3 C_4 L_3 R_4 + C_1 C_3 C_4 L_4 R_1 R_3 g_m -$$

$$\mathbf{9.732 \quad X-INVALID-ORDER-732} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{2C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m s^6 + 2R_3 R_4 g_m + s^5 (2C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 R_4 + 2C_1 C_3 L_1 L_3 L_4 R_3 g_m + C_1 C_3 L_1 L_3 L_4 R_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_3 R_4 g_m + C_1 C_3 L_1 L_4 R_3 R_4 g_m + 2C_1 C_3 L_3 L_4 R_1 R_3 g_m + C_1 C_3 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 L_3 L_4 R_3 + C_1 C_3 L_3 L_4 R_4 -$$

$$\mathbf{9.733 \quad X-INVALID-ORDER-733} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{2R_3 g_m + R_4 g_m + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^5 (C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4 + C_1 C_3 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m) + s^3 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m) + s^2 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m) + s (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m) + C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^5 (C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4 + C_1 C_3 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m) + s^3 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m) + s^2 (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m) + s (C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m) + C_1 C_3 C_4 L_4 R_1 R_3 R_4 g_m}$$

$$\mathbf{9.734 \quad X-INVALID-ORDER-734} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{2R_3 g_m + R_4 g_m + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 R_4 + 2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m) + s^3 (2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 R_4 + 2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m) + s^2 (2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 R_4 + 2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m) + s (2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 R_4 + 2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m) + C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^5 (2C_1 C_3 C_4 L_1 L_3 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 R_4 + 2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m) + s^3 (2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 R_4 + 2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m) + s^2 (2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 R_4 + 2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m) + s (2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_3 R_3 R_4 + 2C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m) + C_1 C_3 C_4 L_3 R_1 R_3 R_4 g_m}$$

$$\mathbf{9.735 \quad X-INVALID-ORDER-735} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_1 R_3 g_m s}{2C_1 C_4 L_1 R_1 R_3 s^3 + R_1 + s^2 (C_1 L_1 R_1 + 2C_4 L_1 R_1 R_3 g_m + 2C_4 L_1 R_3) + s (2C_4 R_1 R_3 + L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.736 \quad X-INVALID-ORDER-736} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_1 R_3 R_4 g_m s}{2C_1 C_4 L_1 R_1 R_3 R_4 s^3 + 2R_1 R_3 + R_1 R_4 + s^2 (2C_1 L_1 R_1 R_3 + C_1 L_1 R_1 R_4 + 2C_4 L_1 R_1 R_3 R_4 g_m + 2C_4 L_1 R_3 R_4) + s (2C_4 R_1 R_3 R_4 + 2L_1 R_1 R_3 g_m + L_1 R_1 R_4 g_m + 2L_1 R_3 + L_1 R_4)}$$

$$\mathbf{9.737 \quad X-INVALID-ORDER-737} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 R_1 R_3 R_4 g_m s^2 + L_1 R_1 R_3 g_m s}{R_1 + s^3 (2C_1 C_4 L_1 R_1 R_3 + C_1 C_4 L_1 R_1 R_4) + s^2 (C_1 L_1 R_1 + 2C_4 L_1 R_1 R_3 g_m + C_4 L_1 R_1 R_4 g_m + 2C_4 L_1 R_3 + C_4 L_1 R_4) + s (2C_4 R_1 R_3 + C_4 R_1 R_4 + L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.738 \quad X-INVALID-ORDER-738} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_1 R_3 g_m s^3 + L_1 R_1 R_3 g_m s}{C_1 C_4 L_1 L_4 R_1 s^4 + R_1 + s^3 (2C_1 C_4 L_1 R_1 R_3 + C_4 L_1 L_4 R_1 g_m + C_4 L_1 L_4) + s^2 (C_1 L_1 R_1 + 2C_4 L_1 R_1 R_3 g_m + 2C_4 L_1 R_3 + C_4 L_4 R_1) + s (2C_4 R_1 R_3 + L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.739 \quad X-INVALID-ORDER-739} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_1 R_3 g_m s^2}{2C_1 C_4 L_1 L_4 R_1 R_3 s^4 + 2R_1 R_3 + s^3 (C_1 L_1 L_4 R_1 + 2C_4 L_1 L_4 R_1 R_3 g_m + 2C_4 L_1 L_4 R_3) + s^2 (2C_1 L_1 R_1 R_3 + 2C_4 L_4 R_1 R_3 + L_1 L_4 R_1 g_m + L_1 L_4) + s (2L_1 R_1 R_3 g_m + 2L_1 R_3 + L_4 R_1)}$$

$$\mathbf{9.740 \quad X-INVALID-ORDER-740} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_1 R_3 g_m s^3 + C_4 L_1 R_1 R_3 R_4 g_m s^2 + L_1 R_1 R_3 g_m s}{C_1 C_4 L_1 L_4 R_1 s^4 + R_1 + s^3 (2C_1 C_4 L_1 R_1 R_3 + C_1 C_4 L_1 R_1 R_4 + C_4 L_1 L_4 R_1 g_m + C_4 L_1 L_4) + s^2 (C_1 L_1 R_1 + 2C_4 L_1 R_1 R_3 g_m + C_4 L_1 R_1 R_4 g_m + 2C_4 L_1 R_3 + C_4 L_1 R_4 + C_4 L_4 R_1) + s (2C_4 R_1 R_3 + C_4 R_1 R_4 + L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.741 \quad X-INVALID-ORDER-741} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_1 R_3 R_4 g_m s^2}{2C_1 C_4 L_1 L_4 R_1 R_3 R_4 s^4 + 2R_1 R_3 R_4 + s^3 (2C_1 L_1 L_4 R_1 R_3 + C_1 L_1 L_4 R_1 R_4 + 2C_4 L_1 L_4 R_1 R_3 R_4 g_m + 2C_4 L_1 L_4 R_3 R_4) + s^2 (2C_1 L_1 R_1 R_3 R_4 + 2C_4 L_4 R_1 R_3 R_4 + 2L_1 L_4 R_1 R_3 g_m + L_1 L_4 R_1 R_4 g_m + 2L_1 L_4 R_3 + L_1 L_4 R_4) + s (2L_1 R_1 R_3 R_4 g_m + 2L_1 R_3 R_4 + 2L_4 R_1 R_3 + L_4 R_1)}$$

$$\mathbf{9.742 \quad X-INVALID-ORDER-742} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_1 R_3 R_4 g_m s^3 + L_1 L_4 R_1 R_3 g_m s^2 + L_1 R_1 R_3 R_4 g_m s}{2 R_1 R_3 + R_1 R_4 + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_3 + C_1 C_4 L_1 L_4 R_1 R_4) + s^3 (C_1 L_1 L_4 R_1 + 2 C_4 L_1 L_4 R_1 R_3 g_m + C_4 L_1 L_4 R_1 R_4 g_m + 2 C_4 L_1 L_4 R_3 + C_4 L_1 L_4 R_4) + s^2 (2 C_1 L_1 R_1 R_3 + C_1 L_1 R_1 R_4 + 2 C_4 L_4 R_1 R_3 + C_4 L_4 R_1 R_4 + L_1 L_4 R_1 g_m + L_1 L_4) + s (2 L_1 R_1 R_3 g_m + L_1 R_1 R_4 g_m)}$$

$$\mathbf{9.743 \quad X-INVALID-ORDER-743} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_1 R_3 R_4 g_m s^3 + L_1 R_1 R_3 R_4 g_m s}{2 R_1 R_3 + R_1 R_4 + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_3 + C_1 C_4 L_1 L_4 R_1 R_4) + s^3 (2 C_1 C_4 L_1 R_1 R_3 R_4 + 2 C_4 L_1 L_4 R_1 R_3 g_m + C_4 L_1 L_4 R_1 R_4 g_m + 2 C_4 L_1 L_4 R_3 + C_4 L_1 L_4 R_4) + s^2 (2 C_1 L_1 R_1 R_3 + C_1 L_1 R_1 R_4 + 2 C_4 L_4 R_1 R_3 + C_4 L_4 R_1 R_4 + L_1 L_4 R_1 g_m + L_1 L_4) + s (2 L_1 R_1 R_3 g_m + L_1 R_1 R_4 g_m)}$$

$$\mathbf{9.744 \quad X-INVALID-ORDER-744} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_1 R_4 g_m s}{C_1 C_3 L_1 R_1 R_4 s^3 + 2 R_1 + s^2 (2 C_1 L_1 R_1 + C_3 L_1 R_1 R_4 g_m + C_3 L_1 R_4) + s (C_3 R_1 R_4 + 2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.745 \quad X-INVALID-ORDER-745} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_1 R_4 g_m s}{2 R_1 + s^3 (C_1 C_3 L_1 R_1 R_4 + 2 C_1 C_4 L_1 R_1 R_4) + s^2 (2 C_1 L_1 R_1 + C_3 L_1 R_1 R_4 g_m + C_3 L_1 R_4 + 2 C_4 L_1 R_1 R_4 g_m + 2 C_4 L_1 R_4) + s (C_3 R_1 R_4 + 2 C_4 R_1 R_4 + 2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.746 \quad X-INVALID-ORDER-746} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 R_1 R_4 g_m s + L_1 R_1 g_m}{C_1 C_3 C_4 L_1 R_1 R_4 s^3 + C_3 R_1 + 2 C_4 R_1 + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + C_3 C_4 L_1 R_1 R_4 g_m + C_3 C_4 L_1 R_4) + s (C_3 C_4 R_1 R_4 + C_3 L_1 R_1 g_m + C_3 L_1 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1)}$$

$$\mathbf{9.747 \quad X-INVALID-ORDER-747} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_1 g_m s^2 + L_1 R_1 g_m}{C_1 C_3 C_4 L_1 L_4 R_1 s^4 + C_3 R_1 + 2 C_4 R_1 + s^3 (C_3 C_4 L_1 L_4 R_1 g_m + C_3 C_4 L_1 L_4) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + C_3 C_4 L_4 R_1) + s (C_3 L_1 R_1 g_m + C_3 L_1 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1)}$$

$$\mathbf{9.748 \quad X-INVALID-ORDER-748} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_1 g_m s^2}{2 R_1 + s^4 (C_1 C_3 L_1 L_4 R_1 + 2 C_1 C_4 L_1 L_4 R_1) + s^3 (C_3 L_1 L_4 R_1 g_m + C_3 L_1 L_4 + 2 C_4 L_1 L_4 R_1 g_m + 2 C_4 L_1 L_4) + s^2 (2 C_1 L_1 R_1 + C_3 L_4 R_1 + 2 C_4 L_4 R_1) + s (2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.749 \quad X-INVALID-ORDER-749} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_1 g_m s^2 + C_4 L_1 R_1 R_4 g_m s + L_1 R_1 g_m}{C_1 C_3 C_4 L_1 L_4 R_1 s^4 + C_3 R_1 + 2 C_4 R_1 + s^3 (C_1 C_3 C_4 L_1 R_1 R_4 + C_3 C_4 L_1 L_4 R_1 g_m + C_3 C_4 L_1 L_4) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + C_3 C_4 L_1 R_1 R_4 g_m + C_3 C_4 L_1 R_4 + C_3 C_4 L_4 R_1) + s (C_3 C_4 R_1 R_4 + C_3 L_1 R_1 g_m + C_3 L_1 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1)}$$

$$\mathbf{9.750 \quad X-INVALID-ORDER-750} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_1 R_4 g_m s^2}{2 R_1 R_4 + s^4 (C_1 C_3 L_1 L_4 R_1 R_4 + 2 C_1 C_4 L_1 L_4 R_1 R_4) + s^3 (2 C_1 L_1 L_4 R_1 + C_3 L_1 L_4 R_1 R_4 g_m + C_3 L_1 L_4 R_4 + 2 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_4 L_1 L_4 R_4) + s^2 (2 C_1 L_1 R_1 R_4 + C_3 L_4 R_1 R_4 + 2 C_4 L_4 R_1 R_4 + 2 L_1 L_4 R_1 g_m + 2 L_1 L_4) + s (2 L_1 R_1 R_4 g_m + 2 L_1 R_4 + 2 L_4 R_1)}$$

$$\mathbf{9.751 \quad X-INVALID-ORDER-751} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_1 R_4 g_m s^3 + L_1 L_4 R_1 g_m s^2 + L_1 R_1 R_4 g_m s}{C_1 C_3 C_4 L_1 L_4 R_1 R_4 s^5 + 2 R_1 + s^4 (C_1 C_3 L_1 L_4 R_1 + 2 C_1 C_4 L_1 L_4 R_1 + C_3 C_4 L_1 L_4 R_1 R_4 g_m + C_3 C_4 L_1 L_4 R_4) + s^3 (C_1 C_3 L_1 R_1 R_4 + C_3 C_4 L_4 R_1 R_4 + C_3 L_1 L_4 R_1 g_m + C_3 L_1 L_4 + 2 C_4 L_1 L_4 R_1 g_m + 2 C_4 L_1 L_4) + s^2 (2 C_1 L_1 R_1 + C_3 L_1 R_1 R_4 g_m + C_3 L_1 R_4 + C_3 L_4 R_1 + 2 C_4 L_1)}$$

$$\mathbf{9.752 \quad X-INVALID-ORDER-752} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_1 R_4 g_m s^3 + L_1 R_1 R_4 g_m s}{C_1 C_3 C_4 L_1 L_4 R_1 R_4 s^5 + 2R_1 + s^4 (2C_1 C_4 L_1 L_4 R_1 + C_3 C_4 L_1 L_4 R_1 R_4 g_m + C_3 C_4 L_1 L_4 R_4) + s^3 (C_1 C_3 L_1 R_1 R_4 + 2C_1 C_4 L_1 R_1 R_4 + C_3 C_4 L_4 R_1 R_4 + 2C_4 L_1 L_4 R_1 g_m + 2C_4 L_1 L_4) + s^2 (2C_1 L_1 R_1 + C_3 L_1 R_1 R_4 g_m + C_3 L_1 R_4 + 2C_4 L_1 R_1 R_4 g_m + 2C_4 L_1 R_4 + 2C_4 L_4 R_1) + s (C_3 R_1 R_3 R_4 + 2L_1 R_1 R_3 g_m + L_1 R_1 R_4 g_m + 2L_1 R_3 + L_1 R_4)}$$

$$\mathbf{9.753 \quad X-INVALID-ORDER-753} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_1 R_3 R_4 g_m s}{C_1 C_3 L_1 R_1 R_3 R_4 s^3 + 2R_1 R_3 + R_1 R_4 + s^2 (2C_1 L_1 R_1 R_3 + C_1 L_1 R_1 R_4 + C_3 L_1 R_1 R_3 R_4 g_m + C_3 L_1 R_3 R_4) + s (C_3 R_1 R_3 R_4 + 2L_1 R_1 R_3 g_m + L_1 R_1 R_4 g_m + 2L_1 R_3 + L_1 R_4)}$$

$$\mathbf{9.754 \quad X-INVALID-ORDER-754} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_1 R_3 g_m s}{R_1 + s^3 (C_1 C_3 L_1 R_1 R_3 + 2C_1 C_4 L_1 R_1 R_3) + s^2 (C_1 L_1 R_1 + C_3 L_1 R_1 R_3 g_m + C_3 L_1 R_3 + 2C_4 L_1 R_1 R_3 g_m + 2C_4 L_1 R_3) + s (C_3 R_1 R_3 + 2C_4 R_1 R_3 + L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.755 \quad X-INVALID-ORDER-755} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_1 R_3 R_4 g_m s}{2R_1 R_3 + R_1 R_4 + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 + 2C_1 C_4 L_1 R_1 R_3 R_4) + s^2 (2C_1 L_1 R_1 R_3 + C_1 L_1 R_1 R_4 + C_3 L_1 R_1 R_3 R_4 g_m + C_3 L_1 R_3 R_4 + 2C_4 L_1 R_1 R_3 R_4 g_m + 2C_4 L_1 R_3 R_4) + s (C_3 R_1 R_3 R_4 + 2C_4 R_1 R_3 R_4 + 2L_1 R_1 R_3 g_m + L_1 R_1 R_4 g_m + 2L_1 R_3 + L_1 R_4)}$$

$$\mathbf{9.756 \quad X-INVALID-ORDER-756} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 R_1 R_3 R_4 g_m s^2 + L_1 R_1 R_3 g_m s}{C_1 C_3 C_4 L_1 R_1 R_3 R_4 s^4 + R_1 + s^3 (C_1 C_3 L_1 R_1 R_3 + 2C_1 C_4 L_1 R_1 R_3 + C_1 C_4 L_1 R_1 R_4 + C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_3 C_4 L_1 R_3 R_4) + s^2 (C_1 L_1 R_1 + C_3 C_4 R_1 R_3 R_4 + C_3 L_1 R_1 R_3 g_m + C_3 L_1 R_3 + 2C_4 L_1 R_1 R_3 g_m + C_4 L_1 R_1 R_4 g_m + 2C_4 L_1 R_3 + C_4 L_1 R_4) + s (C_3 R_1 R_3 + 2C_4 R_1 R_3)}$$

$$\mathbf{9.757 \quad X-INVALID-ORDER-757} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_1 R_3 g_m s^3 + L_1 R_1 R_3 g_m s}{C_1 C_3 C_4 L_1 L_4 R_1 R_3 s^5 + R_1 + s^4 (C_1 C_4 L_1 L_4 R_1 + C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_3 C_4 L_1 L_4 R_3) + s^3 (C_1 C_3 L_1 R_1 R_3 + 2C_1 C_4 L_1 R_1 R_3 + C_3 C_4 L_4 R_1 R_3 + C_4 L_1 L_4 R_1 g_m + C_4 L_1 L_4) + s^2 (C_1 L_1 R_1 + C_3 L_1 R_1 R_3 g_m + C_3 L_1 R_3 + 2C_4 L_1 R_1 R_3 g_m + 2C_4 L_1 R_3 + C_4 L_4 R_1) + s (C_3 R_1 R_3 R_4 + 2L_1 R_1 R_3 g_m + L_1 R_1 R_4 g_m + 2L_1 R_3 + L_1 R_4)}$$

$$\mathbf{9.758 \quad X-INVALID-ORDER-758} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_1 R_3 g_m s^2}{2R_1 R_3 + s^4 (C_1 C_3 L_1 L_4 R_1 R_3 + 2C_1 C_4 L_1 L_4 R_1 R_3) + s^3 (C_1 L_1 L_4 R_1 + C_3 L_1 L_4 R_1 R_3 g_m + C_3 L_1 L_4 R_3 + 2C_4 L_1 L_4 R_1 R_3 g_m + 2C_4 L_1 L_4 R_3) + s^2 (2C_1 L_1 R_1 R_3 + C_3 L_4 R_1 R_3 + 2C_4 L_4 R_1 R_3 + L_1 L_4 R_1 g_m + L_1 L_4) + s (2L_1 R_1 R_3 g_m + 2L_1 R_3 + L_4 R_1)}$$

$$\mathbf{9.759 \quad X-INVALID-ORDER-759} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_1 R_3 g_m s^3 + C_4 L_1 R_1 R_3 R_4 g_m s^2 + L_1 R_1 R_3 g_m s}{C_1 C_3 C_4 L_1 L_4 R_1 R_3 s^5 + R_1 + s^4 (C_1 C_3 C_4 L_1 R_1 R_3 R_4 + C_1 C_4 L_1 L_4 R_1 + C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_3 C_4 L_1 L_4 R_3) + s^3 (C_1 C_3 L_1 R_1 R_3 + 2C_1 C_4 L_1 R_1 R_3 + C_1 C_4 L_1 R_1 R_4 + C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_3 C_4 L_1 R_3 R_4 + C_3 C_4 L_4 R_1 R_3 + C_4 L_1 L_4 R_1 g_m + C_4 L_1 L_4) + s^2 (C_1 L_1 R_1 + C_3 L_1 R_1 R_3 g_m + C_3 L_1 R_3 + 2C_4 L_1 R_1 R_3 g_m + 2C_4 L_1 R_3 + C_4 L_4 R_1) + s (C_3 R_1 R_3 R_4 + 2L_1 R_1 R_3 g_m + L_1 R_1 R_4 g_m + 2L_1 R_3 + L_1 R_4)}$$

$$\mathbf{9.760 \quad X-INVALID-ORDER-760} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_1 R_3 R_4 g_m s^2}{2R_1 R_3 R_4 + s^4 (C_1 C_3 L_1 L_4 R_1 R_3 R_4 + 2C_1 C_4 L_1 L_4 R_1 R_3 R_4) + s^3 (2C_1 L_1 L_4 R_1 R_3 + C_1 L_1 L_4 R_1 R_4 + C_3 L_1 L_4 R_1 R_3 R_4 g_m + C_3 L_1 L_4 R_3 R_4 + 2C_4 L_1 L_4 R_1 R_3 R_4 g_m + 2C_4 L_1 L_4 R_3 R_4) + s^2 (2C_1 L_1 R_1 R_3 R_4 + C_3 L_4 R_1 R_3 R_4 + 2C_4 L_4 R_1 R_3 R_4 + 2L_1 L_4 R_1 R_3 g_m + L_1 L_4 R_1) + s (C_3 R_1 R_3 R_4 + 2L_1 R_1 R_3 g_m + L_1 R_1 R_4 g_m + 2L_1 R_3 + L_1 R_4)}$$

$$\mathbf{9.761 \quad X-INVALID-ORDER-761} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_1 R_3 R_4 g_m s^3 + L_1 L_4 R_1 R_3 g_m s}{C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 s^5 + 2R_1 R_3 + R_1 R_4 + s^4 (C_1 C_3 L_1 L_4 R_1 R_3 + 2C_1 C_4 L_1 L_4 R_1 R_3 + C_1 C_4 L_1 L_4 R_1 R_4 + C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + C_3 C_4 L_1 L_4 R_3 R_4) + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 + C_1 L_1 L_4 R_1 + C_3 C_4 L_4 R_1 R_3 R_4 + C_3 L_1 L_4 R_1 R_3 g_m + C_3 L_1 L_4 R_3 + 2C_4 L_1 L_4 R_1 R_3 g_m + C_4 L_1 L_4 R_3) + s^2 (C_1 L_1 R_1 + C_3 L_1 R_1 R_3 g_m + C_3 L_1 R_3 + 2C_4 L_1 R_1 R_3 g_m + 2C_4 L_1 R_3 + C_4 L_4 R_1) + s (C_3 R_1 R_3 R_4 + 2L_1 R_1 R_3 g_m + L_1 R_1 R_4 g_m + 2L_1 R_3 + L_1 R_4)}$$

$$\mathbf{9.762 \quad X-INVALID-ORDER-762} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_1 R_3 R_4 g_m s^3 + L_1 R_1 R_3 R_4 g_m s}{C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 s^5 + 2 R_1 R_3 + R_1 R_4 + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_3 + C_1 C_4 L_1 L_4 R_1 R_4 + C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + C_3 C_4 L_1 L_4 R_3 R_4) + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 + 2 C_1 C_4 L_1 R_1 R_3 R_4 + C_3 C_4 L_4 R_1 R_3 R_4 + 2 C_4 L_1 L_4 R_1 R_3 g_m + C_4 L_1 L_4 R_1 R_4 g_m + 2 C_4 L_1 L_4 R_3 + C_4 L_1 L_4 R_4)}$$

$$\mathbf{9.763 \quad X-INVALID-ORDER-763} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 R_1 R_3 R_4 g_m s^2 + L_1 R_1 R_4 g_m s}{2 R_1 + s^3 (2 C_1 C_3 L_1 R_1 R_3 + C_1 C_3 L_1 R_1 R_4) + s^2 (2 C_1 L_1 R_1 + 2 C_3 L_1 R_1 R_3 g_m + C_3 L_1 R_1 R_4 g_m + 2 C_3 L_1 R_3 + C_3 L_1 R_4) + s (2 C_3 R_1 R_3 + C_3 R_1 R_4 + 2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.764 \quad X-INVALID-ORDER-764} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 R_1 R_3 g_m s + L_1 R_1 g_m}{2 C_1 C_3 C_4 L_1 R_1 R_3 s^3 + C_3 R_1 + 2 C_4 R_1 + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + 2 C_3 C_4 L_1 R_1 R_3 g_m + 2 C_3 C_4 L_1 R_3) + s (2 C_3 C_4 R_1 R_3 + C_3 L_1 R_1 g_m + C_3 L_1 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1)}$$

$$\mathbf{9.765 \quad X-INVALID-ORDER-765} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 R_1 R_3 R_4 g_m s^2 + L_1 R_1 R_4 g_m s}{2 C_1 C_3 C_4 L_1 R_1 R_3 R_4 s^4 + 2 R_1 + s^3 (2 C_1 C_3 L_1 R_1 R_3 + C_1 C_3 L_1 R_1 R_4 + 2 C_1 C_4 L_1 R_1 R_4 + 2 C_3 C_4 L_1 R_1 R_3 R_4 g_m + 2 C_3 C_4 L_1 R_3 R_4) + s^2 (2 C_1 L_1 R_1 + 2 C_3 C_4 R_1 R_3 R_4 + 2 C_3 L_1 R_1 R_3 g_m + C_3 L_1 R_1 R_4 g_m + 2 C_3 L_1 R_3 + C_3 L_1 R_4 + 2 C_4 L_1 R_1 R_4 g_m + 2 C_4 L_1 R_4) + s (2 C_3 R_1 R_3 + C_3 R_1 R_4 + 2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.766 \quad X-INVALID-ORDER-766} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 R_1 R_3 R_4 g_m s^2 + L_1 R_1 g_m + s (C_3 L_1 R_1 R_3 g_m + C_4 L_1 R_1 R_4 g_m)}{C_3 R_1 + 2 C_4 R_1 + s^3 (2 C_1 C_3 C_4 L_1 R_1 R_3 + C_1 C_3 C_4 L_1 R_1 R_4) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + 2 C_3 C_4 L_1 R_1 R_3 g_m + C_3 C_4 L_1 R_1 R_4 g_m + 2 C_3 C_4 L_1 R_3 + C_3 C_4 L_1 R_4) + s (2 C_3 C_4 R_1 R_3 + C_3 C_4 R_1 R_4 + C_3 L_1 R_1 g_m + C_3 L_1 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1)}$$

$$\mathbf{9.767 \quad X-INVALID-ORDER-767} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_4 R_1 R_3 g_m s^3 + C_3 L_1 R_1 R_3 g_m s + C_4 L_1 L_4 R_1 g_m s^2 + L_1 R_1 g_m}{C_1 C_3 C_4 L_1 L_4 R_1 s^4 + C_3 R_1 + 2 C_4 R_1 + s^3 (2 C_1 C_3 C_4 L_1 R_1 R_3 + C_3 C_4 L_1 L_4 R_1 g_m + C_3 C_4 L_1 L_4) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + 2 C_3 C_4 L_1 R_1 R_3 g_m + 2 C_3 C_4 L_1 R_3 + C_3 C_4 L_4 R_1) + s (2 C_3 C_4 R_1 R_3 + C_3 L_1 R_1 g_m + C_3 L_1 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1)}$$

$$\mathbf{9.768 \quad X-INVALID-ORDER-768} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_4 R_1 R_3 g_m s^3 + L_1 L_4 R_1 g_m s^2}{2 C_1 C_3 C_4 L_1 L_4 R_1 R_3 s^5 + 2 R_1 + s^4 (C_1 C_3 L_1 L_4 R_1 + 2 C_1 C_4 L_1 L_4 R_1 + 2 C_3 C_4 L_1 L_4 R_1 R_3 g_m + 2 C_3 C_4 L_1 L_4 R_3) + s^3 (2 C_1 C_3 L_1 R_1 R_3 + 2 C_3 C_4 L_4 R_1 R_3 + C_3 L_1 L_4 R_1 g_m + C_3 L_1 L_4 + 2 C_4 L_1 L_4 R_1 g_m + 2 C_4 L_1 L_4) + s^2 (2 C_1 L_1 R_1 + 2 C_3 L_1 R_1 R_3 g_m + 2 C_3 L_1 R_3 + C_3 L_4 R_1)}$$

$$\mathbf{9.769 \quad X-INVALID-ORDER-769} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_4 R_1 R_3 g_m s^3 + L_1 R_1 g_m + s^2 (C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_4 L_1 L_4 R_1 g_m) + s (C_3 L_1 R_1 R_3 g_m + C_4 L_1 R_1 R_4 g_m)}{C_1 C_3 C_4 L_1 L_4 R_1 s^4 + C_3 R_1 + 2 C_4 R_1 + s^3 (2 C_1 C_3 C_4 L_1 R_1 R_3 + C_1 C_3 C_4 L_1 R_1 R_4 + C_3 C_4 L_1 L_4 R_1 g_m + C_3 C_4 L_1 L_4) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + 2 C_3 C_4 L_1 R_1 R_3 g_m + C_3 C_4 L_1 R_1 R_4 g_m + 2 C_3 C_4 L_1 R_3 + C_3 C_4 L_1 R_4 + C_3 C_4 L_4 R_1) + s (2 C_3 C_4 R_1 R_3 + C_3 C_4 R_1 R_4 + C_3 L_1 R_1 g_m + C_3 L_1)}$$

$$\mathbf{9.770 \quad X-INVALID-ORDER-770} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_4 R_1 R_3 R_4 g_m s^3 + L_1 L_4 R_1 R_4 g_m s^2}{2 C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 s^5 + 2 R_1 R_4 + s^4 (2 C_1 C_3 L_1 L_4 R_1 R_3 + C_1 C_3 L_1 L_4 R_1 R_4 + 2 C_1 C_4 L_1 L_4 R_1 R_4 + 2 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + 2 C_3 C_4 L_1 L_4 R_3 R_4) + s^3 (2 C_1 C_3 L_1 R_1 R_3 R_4 + 2 C_1 L_1 L_4 R_1 + 2 C_3 C_4 L_4 R_1 R_3 R_4 + 2 C_3 L_1 L_4 R_1 R_3 g_m + C_3 L_1 L_4 R_1 R_4 g_m + 2 C_3 L_1 L_4 R_3 + C_3 L_1 L_4 R_4)}$$

$$\mathbf{9.771 \quad X-INVALID-ORDER-771} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m s^4 + L_1 R_1 R_4 g_m s + s^3 (C_3 L_1 L_4 R_1 R_3 g_m + C_4 L_1 L_4 R_1 R_4 g_m) + s^2 (C_3 L_1 R_1 R_3 + C_3 L_1 L_4 R_1 R_3 g_m + C_3 L_1 L_4 R_1 R_4 g_m + C_3 L_1 L_4 R_3 + C_3 L_1 L_4 R_4) + s (2 C_1 C_3 C_4 L_1 L_4 R_1 R_3 + C_1 C_3 C_4 L_1 L_4 R_1 R_4) + s^4 (C_1 C_3 L_1 L_4 R_1 + 2 C_1 C_4 L_1 L_4 R_1 + 2 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_3 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_3 C_4 L_1 L_4 R_3 + C_3 C_4 L_1 L_4 R_4) + s^3 (2 C_1 C_3 L_1 R_1 R_3 + C_1 C_3 L_1 R_1 R_4 + 2 C_3 C_4 L_4 R_1 R_3 + C_3 C_4 L_4 R_1 R_4 + C_3 L_1 L_4 R_1 g_m + C_3 L_1 L_4 R_4)}$$

$$\mathbf{9.772 \quad X-INVALID-ORDER-772} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad R_3 + \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m s^4 + C_3 L_1 R_1 R_4 g_m s}{2 R_1 + s^5 (2 C_1 C_3 C_4 L_1 L_4 R_1 R_3 + C_1 C_3 C_4 L_1 L_4 R_1 R_4) + s^4 (2 C_1 C_3 C_4 L_1 R_1 R_3 R_4 + 2 C_1 C_4 L_1 L_4 R_1 + 2 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_3 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_3 C_4 L_1 L_4 R_3 + C_3 C_4 L_1 L_4 R_4) + s^3 (2 C_1 C_3 L_1 R_1 R_3 + C_1 C_3 L_1 R_1 R_4 + 2 C_1 C_4 L_1 R_1 R_4 + 2 C_3 C_4 L_1 R_1 R_3 R_4 g_m + 2 C_3 C_4 L_1 R_1 R_4 g_m) + s^2 (2 C_1 C_3 L_1 R_1 + C_3 L_1 R_1 R_4 g_m + C_3 L_1 R_4 + 2 C_3 L_3 R_1) + s (C_3 R_1 R_4 + 2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.773 \quad X-INVALID-ORDER-773} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_1 R_4 g_m s^3 + L_1 R_1 R_4 g_m s}{2 C_1 C_3 L_1 L_3 R_1 s^4 + 2 R_1 + s^3 (C_1 C_3 L_1 R_1 R_4 + 2 C_3 L_1 L_3 R_1 g_m + 2 C_3 L_1 L_3) + s^2 (2 C_1 L_1 R_1 + C_3 L_1 R_1 R_4 g_m + C_3 L_1 R_4 + 2 C_3 L_3 R_1) + s (C_3 R_1 R_4 + 2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.774 \quad X-INVALID-ORDER-774} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_1 g_m s^2 + L_1 R_1 g_m}{2 C_1 C_3 C_4 L_1 L_3 R_1 s^4 + C_3 R_1 + 2 C_4 R_1 + s^3 (2 C_3 C_4 L_1 L_3 R_1 g_m + 2 C_3 C_4 L_1 L_3) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + 2 C_3 C_4 L_3 R_1) + s (C_3 L_1 R_1 g_m + C_3 L_1 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1)}$$

$$\mathbf{9.775 \quad X-INVALID-ORDER-775} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_1 R_4 g_m s^3 + L_1 R_1 R_4 g_m s}{2 C_1 C_3 C_4 L_1 L_3 R_1 R_4 s^5 + 2 R_1 + s^4 (2 C_1 C_3 L_1 L_3 R_1 + 2 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_3 C_4 L_1 L_3 R_4) + s^3 (C_1 C_3 L_1 R_1 R_4 + 2 C_1 C_4 L_1 R_1 R_4 + 2 C_3 C_4 L_3 R_1 R_4 + 2 C_3 L_1 L_3 R_1 g_m + 2 C_3 L_1 L_3) + s^2 (2 C_1 L_1 R_1 + C_3 L_1 R_1 R_4 g_m + C_3 L_1 R_4 + 2 C_3 L_3 R_1 + 2 C_4 L_1 R_1 R_4 g_m + 2 C_4 L_1 R_4) + s (C_3 R_1 R_4 + 2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.776 \quad X-INVALID-ORDER-776} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 R_1 R_4 g_m s^3 + C_3 L_1 L_3 R_1 g_m s^2 + C_4 L_1 R_1 R_4 g_m s + L_1 R_1 g_m}{2 C_1 C_3 C_4 L_1 L_3 R_1 s^4 + C_3 R_1 + 2 C_4 R_1 + s^3 (C_1 C_3 C_4 L_1 R_1 R_4 + 2 C_3 C_4 L_1 L_3 R_1 g_m + 2 C_3 C_4 L_1 L_3) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + C_3 C_4 L_1 R_1 R_4 g_m + C_3 C_4 L_1 R_4 + 2 C_3 C_4 L_3 R_1) + s (C_3 C_4 R_1 R_4 + C_3 L_1 R_1 g_m + C_3 L_1 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1)}$$

$$\mathbf{9.777 \quad X-INVALID-ORDER-777} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_1 g_m s^4 + L_1 R_1 g_m + s^2 (C_3 L_1 L_3 R_1 g_m + C_4 L_1 L_4 R_1 g_m)}{C_3 R_1 + 2 C_4 R_1 + s^4 (2 C_1 C_3 C_4 L_1 L_3 R_1 + C_1 C_3 C_4 L_1 L_4 R_1) + s^3 (2 C_3 C_4 L_1 L_3 R_1 g_m + 2 C_3 C_4 L_1 L_3 + C_3 C_4 L_1 L_4 R_1 g_m + C_3 C_4 L_1 L_4) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + 2 C_3 C_4 L_3 R_1 + C_3 C_4 L_4 R_1) + s (C_3 L_1 R_1 g_m + C_3 L_1 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1)}$$

$$\mathbf{9.778 \quad X-INVALID-ORDER-778} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_1 g_m s^4 + L_1 L_4 R_1 g_m s^2}{2 C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + 2 R_1 + s^5 (2 C_3 C_4 L_1 L_3 L_4 R_1 g_m + 2 C_3 C_4 L_1 L_3 L_4) + s^4 (2 C_1 C_3 L_1 L_3 R_1 + C_1 C_3 L_1 L_4 R_1 + 2 C_1 C_4 L_1 L_4 R_1 + 2 C_3 C_4 L_3 L_4 R_1) + s^3 (2 C_3 L_1 L_3 R_1 g_m + 2 C_3 L_1 L_3 + C_3 L_1 L_4 R_1 g_m + C_3 L_1 L_4 + 2 C_4 L_1 L_4 R_1 g_m + 2 C_4 L_1 L_4) + s^2 (2 C_1 L_1 R_1 + 2 C_3 L_1 L_3 R_1 g_m + 2 C_3 L_1 L_3 R_1) + s (C_3 R_1 R_4 + 2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.779 \quad X-INVALID-ORDER-779} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_1 g_m s^4 + C_3 C_4 L_1 L_3 R_1 R_4 g_m s^3 + C_4 L_1 R_1 R_4 g_m s + L_1 R_1 g_m + s^2 (C_3 L_1 L_3 R_1 g_m + C_4 L_1 L_4 R_1 g_m)}{C_3 R_1 + 2 C_4 R_1 + s^4 (2 C_1 C_3 C_4 L_1 L_3 R_1 + C_1 C_3 C_4 L_1 L_4 R_1) + s^3 (C_1 C_3 C_4 L_1 R_1 R_4 + 2 C_3 C_4 L_1 L_3 R_1 g_m + 2 C_3 C_4 L_1 L_3 + C_3 C_4 L_1 L_4 R_1 g_m + C_3 C_4 L_1 L_4) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + C_3 C_4 L_1 R_1 R_4 g_m + C_3 C_4 L_1 R_4 + 2 C_3 C_4 L_3 R_1 + C_3 C_4 L_4 R_1) + s (C_3 C_4 R_1 R_4 + 2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.780 \quad X-INVALID-ORDER-780} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_1 R_4 g_m s^4 + L_1 L_4 R_1 R_4 g_m s^2}{2 C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 s^6 + 2 R_1 R_4 + s^5 (2 C_1 C_3 L_1 L_3 L_4 R_1 + 2 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2 C_3 C_4 L_1 L_3 L_4 R_4) + s^4 (2 C_1 C_3 L_1 L_3 R_1 R_4 + C_1 C_3 L_1 L_4 R_1 R_4 + 2 C_1 C_4 L_1 L_4 R_1 R_4 + 2 C_3 C_4 L_3 L_4 R_1 R_4 + 2 C_3 L_1 L_3 L_4 R_1 g_m + 2 C_3 L_1 L_3 L_4) + s^3 (2 C_1 L_1 L_4 R_1 + 2 C_3 L_1 L_3 R_1 R_4 g_m + 2 C_3 L_1 L_3 R_1 R_4) + s^2 (2 C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + C_3 C_4 L_1 R_1 R_4 g_m + C_3 C_4 L_1 R_4 + 2 C_3 C_4 L_3 R_1 + C_3 C_4 L_4 R_1) + s (C_3 C_4 R_1 R_4 + 2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.781 \quad X-INVALID-ORDER-781} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m s^5 + C_3 L_1 L_3 L_4 R_1 g_m s^4 + L_1 L_4 R_1 g_m s^2 + L_1 R_1 R_4 g_m s + s^3 (C_3 L_1 L_3 R_1 R_4 g_m + C_3 L_1 L_3 R_1 R_4) + s^2 (C_3 L_1 L_3 R_1 g_m + C_4 L_1 L_4 R_1 g_m)}{2 C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + 2 R_1 + s^5 (C_1 C_3 C_4 L_1 L_4 R_1 R_4 + 2 C_3 C_4 L_1 L_3 L_4 R_1 g_m + 2 C_3 C_4 L_1 L_3 L_4) + s^4 (2 C_1 C_3 L_1 L_3 R_1 + C_1 C_3 L_1 L_4 R_1 + 2 C_1 C_4 L_1 L_4 R_1 + C_3 C_4 L_1 L_4 R_1 R_4 g_m + C_3 C_4 L_1 L_4 R_4 + 2 C_3 C_4 L_3 L_4 R_1) + s^3 (C_1 C_3 L_1 R_1 R_4 + C_3 C_4 L_4 R_1 R_4 + 2 C_3 L_1 L_3 R_1 g_m + 2 C_3 L_1 L_3 R_1 R_4) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + C_3 C_4 L_1 R_1 R_4 g_m + C_3 C_4 L_1 R_4 + 2 C_3 C_4 L_3 R_1 + C_3 C_4 L_4 R_1) + s (C_3 C_4 R_1 R_4 + 2 L_1 R_1 g_m + 2 L_1)}$$

9.782 X-INVALID-ORDER-782 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m s^5 + L_1 R_1 R_4 g_m}{2C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + 2R_1 + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_4 + 2C_3 C_4 L_1 L_3 L_4 R_1 g_m + 2C_3 C_4 L_1 L_3 L_4)} + s^4 (2C_1 C_3 L_1 L_3 R_1 + 2C_1 C_4 L_1 L_4 R_1 + 2C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2C_3 C_4 L_1 L_3 R_4 + C_3 C_4 L_1 L_4 R_1 R_4 g_m + C_3 C_4 L_1 L_4 R_4 + 2C_3 C_4 L_3 L_4 R_1) + s^3 ($$

9.783 X-INVALID-ORDER-783 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{L_1 L_3 R_1 R_4 g_m s^2}{C_1 C_3 L_1 L_3 R_1 R_4 s^4 + R_1 R_4 + s^3 (2C_1 L_1 L_3 R_1 + C_3 L_1 L_3 R_1 R_4 g_m + C_3 L_1 L_3 R_4) + s^2 (C_1 L_1 R_1 R_4 + C_3 L_3 R_1 R_4 + 2L_1 L_3 R_1 g_m + 2L_1 L_3) + s (L_1 R_1 R_4 g_m + L_1 R_4 + 2L_3 R_1)}$$

9.784 X-INVALID-ORDER-784 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{L_1 L_3 R_1 g_m s^2}{R_1 + s^4 (C_1 C_3 L_1 L_3 R_1 + 2 C_1 C_4 L_1 L_3 R_1) + s^3 (C_3 L_1 L_3 R_1 g_m + C_3 L_1 L_3 + 2 C_4 L_1 L_3 R_1 g_m + 2 C_4 L_1 L_3) + s^2 (C_1 L_1 R_1 + C_3 L_3 R_1 + 2 C_4 L_3 R_1) + s (L_1 R_1 g_m + L_1)}$$

9.785 X-INVALID-ORDER-785 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{L_1 L_3 R_1 R_4 g_m s^2}{R_1 R_4 + s^4 (C_1 C_3 L_1 L_3 R_1 R_4 + 2 C_1 C_4 L_1 L_3 R_1 R_4) + s^3 (2 C_1 L_1 L_3 R_1 + C_3 L_1 L_3 R_1 R_4 g_m + C_3 L_1 L_3 R_4 + 2 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_4 L_1 L_3 R_4) + s^2 (C_1 L_1 R_1 R_4 + C_3 L_3 R_1 R_4 + 2 C_4 L_3 R_1 R_4 + 2 L_1 L_3 R_1 g_m + 2 L_1 L_3) + s (L_1 R_1 R_4 g_m + L_1 R_4 + 2 L_3 R_1)}$$

9.786 X-INVALID-ORDER-786 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_1 L_3 R_1 R_4 g_m s^3 + L_1 L_3 R_1 g_m s^2}{C_1 C_3 C_4 L_1 L_3 R_1 R_4 s^5 + R_1 + s^4 (C_1 C_3 L_1 L_3 R_1 + 2 C_1 C_4 L_1 L_3 R_1 + C_3 C_4 L_1 L_3 R_1 R_4 g_m + C_3 C_4 L_1 L_3 R_4) + s^3 (C_1 C_4 L_1 R_1 R_4 + C_3 C_4 L_3 R_1 R_4 + C_3 L_1 L_3 R_1 g_m + C_3 L_1 L_3 + 2 C_4 L_1 L_3 R_1 g_m + 2 C_4 L_1 L_3) + s^2 (C_1 L_1 R_1 + C_3 L_3 R_1 + C_4 L_1 R_1 R_4 g_m + C_4 L_1 R_4 + 2 C_4 L_3 R_1)}$$

9.787 X-INVALID-ORDER-787 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_1 g_m s^4 + L_1 L_3 R_1 g_m s^2}{C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + R_1 + s^5 (C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 L_1 L_3 R_1 + 2 C_1 C_4 L_1 L_3 R_1 + C_1 C_4 L_1 L_4 R_1 + C_3 C_4 L_3 L_4 R_1) + s^3 (C_3 L_1 L_3 R_1 g_m + C_3 L_1 L_3 + 2 C_4 L_1 L_3 R_1 g_m + 2 C_4 L_1 L_3 + C_4 L_1 L_4 R_1 g_m + C_4 L_1 L_4) + s^2 (C_1 L_1 R_1 + C_3 L_3 R_1 + 2 C_4 L_4 R_1)}$$

9.788 X-INVALID-ORDER-788 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{L_1 L_3 L_4 R_1 g_m s^2}{2L_3 R_1 + L_4 R_1 + s^4 (C_1 C_3 L_1 L_3 L_4 R_1 + 2C_1 C_4 L_1 L_3 L_4 R_1) + s^3 (C_3 L_1 L_3 L_4 R_1 g_m + C_3 L_1 L_3 L_4 + 2C_4 L_1 L_3 L_4 R_1 g_m + 2C_4 L_1 L_3 L_4) + s^2 (2C_1 L_1 L_3 R_1 + C_1 L_1 L_4 R_1 + C_3 L_3 L_4 R_1 + 2C_4 L_3 L_4 R_1) + s (2L_1 L_3 R_1 g_m + 2L_1 L_3 + L_1 L_4 R_1 g_m + L_1 L_4)}$$

9.789 X-INVALID-ORDER-789 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_1 g_m s^4 + C_4 L_1 L_3 R_1 R_4 g_m s^3 + L_1 L_3 R_1 g_m s^2}{C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + R_1 + s^5 (C_1 C_3 C_4 L_1 L_3 R_1 R_4 + C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 L_1 L_3 R_1 + 2C_1 C_4 L_1 L_3 R_1 + C_1 C_4 L_1 L_4 R_1 + C_3 C_4 L_1 L_3 R_1 R_4 g_m + C_3 C_4 L_1 L_3 R_4 + C_3 C_4 L_3 L_4 R_1) + s^3 (C_1 C_4 L_1 R_1 R_4 + C_3 C_4 L_3 R_1 R_4 + C_3 L_1 L_3 R_1 g_m + C_3 L_1 L_3 R_1 R_4 g_m + C_3 L_1 L_3 R_1 R_4) + s^2 (C_1 C_4 L_1 R_1 R_4 + C_3 C_4 L_3 R_1 R_4 + C_3 L_1 L_3 R_1 g_m + C_3 L_1 L_3 R_1 R_4) + s (C_1 C_4 L_1 R_1 R_4 + C_3 C_4 L_3 R_1 R_4 + C_3 L_1 L_3 R_1 g_m + C_3 L_1 L_3 R_1 R_4) + C_1 C_4 L_1 R_1 R_4}.$$

9.790 X-INVALID-ORDER-790 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{L_1 L_3 L_4 R_1 R_4 g_m s^2}{2L_3 R_1 R_4 + L_4 R_1 R_4 + s^4 (C_1 C_3 L_1 L_3 L_4 R_1 R_4 + 2C_1 C_4 L_1 L_3 L_4 R_1 R_4) + s^3 (2C_1 L_1 L_3 L_4 R_1 + C_3 L_1 L_3 L_4 R_1 R_4 g_m + C_3 L_1 L_3 L_4 R_4 + 2C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_4 L_1 L_3 L_4 R_4) + s^2 (2C_1 L_1 L_3 R_1 R_4 + C_1 L_1 L_4 R_1 R_4 + C_3 L_3 L_4 R_1 R_4 + 2C_4 L_3 L_4 R_1 R_4 + 2L_1 L_3 L_4 R_1 g_m +$$

9.791 X-INVALID-ORDER-791 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_1 R_4 g_m s^4 + L_1 L_3 L_4 R_1 R_4}{C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 s^6 + R_1 R_4 + s^5 (C_1 C_3 L_1 L_3 L_4 R_1 + 2 C_1 C_4 L_1 L_3 L_4 R_1 + C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + C_3 C_4 L_1 L_3 L_4 R_4) + s^4 (C_1 C_3 L_1 L_3 R_1 R_4 + C_1 C_4 L_1 L_4 R_1 R_4 + C_3 C_4 L_3 L_4 R_1 R_4 + C_3 L_1 L_3 L_4 R_1 g_m + C_3 L_1 L_3 L_4 + 2 C_4 L_1 L_3 L_4 R_1 g_m + 2 C_4 L_1 L_3 L_4) + s^3 (2 C_1 L_1$$

$$\mathbf{9.792 \quad X-INVALID-ORDER-792} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_1 R_4 g_m s^4 + L_1 L_3 R_1 R_4 g_m s^2}{C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 s^6 + R_1 R_4 + s^5 (2 C_1 C_4 L_1 L_3 L_4 R_1 + C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + C_3 C_4 L_1 L_3 L_4 R_4) + s^4 (C_1 C_3 L_1 L_3 R_1 R_4 + 2 C_1 C_4 L_1 L_3 R_1 R_4 + C_1 C_4 L_1 L_4 R_1 R_4 + C_3 C_4 L_3 L_4 R_1 R_4 + 2 C_4 L_1 L_3 L_4 R_1 g_m + 2 C_4 L_1 L_3 L_4) + s^3 (2 C_1 L_1 L_3 R_1 + C_3 L_1 L_3 R_1 R_4 g_m + C_3 L_1 L_3 R_1 R_4) + s^2 (C_1 C_3 L_1 L_3 R_1 R_4 + 2 C_1 C_4 L_1 L_3 R_1 R_4 + C_1 C_4 L_1 L_4 R_1 R_4 + C_3 C_4 L_3 L_4 R_1 R_4 + 2 C_4 L_1 L_3 L_4 R_1 g_m + 2 C_4 L_1 L_3 L_4) + s (2 C_1 L_1 L_3 R_1 + C_3 L_1 L_3 R_1 R_4 g_m + C_3 L_1 L_3 R_1 R_4) + R_1 R_4}$$

$$\mathbf{9.793 \quad X-INVALID-ORDER-793} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_1 R_4 g_m s^3 + C_3 L_1 R_1 R_3 R_4 g_m s^2 + L_1 R_1 R_4 g_m s}{2 C_1 C_3 L_1 L_3 R_1 s^4 + 2 R_1 + s^3 (2 C_1 C_3 L_1 R_1 R_3 + C_1 C_3 L_1 R_1 R_4 + 2 C_3 L_1 L_3 R_1 g_m + 2 C_3 L_1 L_3) + s^2 (2 C_1 L_1 R_1 + 2 C_3 L_1 R_1 R_3 g_m + C_3 L_1 R_1 R_4 g_m + 2 C_3 L_1 R_3 + C_3 L_1 R_4 + 2 C_3 L_3 R_1) + s (2 C_3 R_1 R_3 + C_3 R_1 R_4 + 2 L_1 R_1 g_m + 2 L_1) + R_1 R_4}$$

$$\mathbf{9.794 \quad X-INVALID-ORDER-794} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_1 g_m s^2 + C_3 L_1 R_1 R_3 g_m s + L_1 R_1 g_m}{2 C_1 C_3 C_4 L_1 L_3 R_1 s^4 + C_3 R_1 + 2 C_4 R_1 + s^3 (2 C_1 C_3 C_4 L_1 R_1 R_3 + 2 C_3 C_4 L_1 L_3 R_1 g_m + 2 C_3 C_4 L_1 L_3) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + 2 C_3 C_4 L_1 R_1 R_3 g_m + 2 C_3 C_4 L_1 R_3 + 2 C_3 C_4 L_3 R_1) + s (2 C_3 C_4 R_1 R_3 + C_3 L_1 R_1 g_m + C_3 L_1 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1) + R_1 R_4}$$

$$\mathbf{9.795 \quad X-INVALID-ORDER-795} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 R_1 R_4 g_m s^3 + C_3 L_1 R_1 R_3 R_4 g_m s^2 + L_1 R_1 R_4 g_m s}{2 C_1 C_3 C_4 L_1 L_3 R_1 R_4 s^5 + 2 R_1 + s^4 (2 C_1 C_3 C_4 L_1 R_1 R_3 R_4 + 2 C_1 C_3 L_1 L_3 R_1 + 2 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_3 C_4 L_1 L_3 R_4) + s^3 (2 C_1 C_3 L_1 R_1 R_3 + C_1 C_3 L_1 R_1 R_4 + 2 C_1 C_4 L_1 R_1 R_4 + 2 C_3 C_4 L_1 R_1 R_3 R_4 g_m + 2 C_3 C_4 L_1 R_3 R_4 + 2 C_3 C_4 L_3 R_1 R_4 + 2 C_3 L_1 L_3 R_1 g_m + 2 C_3 L_1 L_3) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + 2 C_3 C_4 L_1 R_1 R_3 g_m + 2 C_3 C_4 L_1 R_3 + 2 C_3 C_4 L_3 R_1) + s (2 C_3 C_4 R_1 R_3 + C_3 L_1 R_1 g_m + C_3 L_1 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1) + R_1 R_4}$$

$$\mathbf{9.796 \quad X-INVALID-ORDER-796} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 R_1 R_4 g_m s^3 + L_1 R_1 g_m + s^2 (C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_3 L_1 L_3 R_1 g_m) + s (C_3 L_1 R_1 R_3 g_m + C_4 L_1 R_1 R_4 g_m)}{2 C_1 C_3 C_4 L_1 L_3 R_1 s^4 + C_3 R_1 + 2 C_4 R_1 + s^3 (2 C_1 C_3 C_4 L_1 R_1 R_3 + C_1 C_3 C_4 L_1 R_1 R_4 + 2 C_3 C_4 L_1 L_3 R_1 g_m + 2 C_3 C_4 L_1 L_3) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + 2 C_3 C_4 L_1 R_1 R_3 g_m + C_3 C_4 L_1 R_1 R_4 g_m + 2 C_3 C_4 L_1 R_3 + C_3 C_4 L_1 R_4 + 2 C_3 C_4 L_3 R_1) + s (2 C_3 C_4 R_1 R_3 + C_3 C_4 R_1 R_4 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1) + R_1 R_4}$$

$$\mathbf{9.797 \quad X-INVALID-ORDER-797} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_1 g_m s^4 + C_3 C_4 L_1 L_4 R_1 R_3 g_m s^3 + C_3 L_1 R_1 R_3 g_m s + L_1 R_1 g_m + s^2 (C_3 L_1 L_3 R_1 g_m + C_4 L_1 L_4 R_1 g_m)}{C_3 R_1 + 2 C_4 R_1 + s^4 (2 C_1 C_3 C_4 L_1 L_3 R_1 + C_1 C_3 C_4 L_1 L_4 R_1) + s^3 (2 C_1 C_3 C_4 L_1 R_1 R_3 + 2 C_3 C_4 L_1 L_3 R_1 g_m + 2 C_3 C_4 L_1 L_3 + C_3 C_4 L_1 L_4 R_1 g_m + C_3 C_4 L_1 L_4) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + 2 C_3 C_4 L_1 R_1 R_3 g_m + 2 C_3 C_4 L_1 R_3 + 2 C_3 C_4 L_3 R_1 + C_3 C_4 L_4 R_1) + s (2 C_3 C_4 R_1 R_3 + C_3 C_4 R_1 R_4 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1) + R_1 R_4}$$

$$\mathbf{9.798 \quad X-INVALID-ORDER-798} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_1 g_m s^4 + C_3 L_1 L_4 R_1 R_3 g_m s^3 + L_1 L_4 R_1 g_m s^2}{2 C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + 2 R_1 + s^5 (2 C_1 C_3 C_4 L_1 L_4 R_1 R_3 + 2 C_3 C_4 L_1 L_3 L_4 R_1 g_m + 2 C_3 C_4 L_1 L_3 L_4) + s^4 (2 C_1 C_3 L_1 L_3 R_1 + C_1 C_3 L_1 L_4 R_1 + 2 C_1 C_4 L_1 L_4 R_1 + 2 C_3 C_4 L_1 L_4 R_1 R_3 g_m + 2 C_3 C_4 L_1 L_4 R_3 + 2 C_3 C_4 L_3 L_4 R_1) + s^3 (2 C_1 C_3 L_1 R_1 R_3 + 2 C_3 C_4 L_4 R_1 R_3 + 2 C_3 L_1 L_3) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + 2 C_3 C_4 L_1 R_1 R_3 g_m + C_3 C_4 L_1 R_1 R_4 g_m + 2 C_3 C_4 L_1 R_3 + C_3 C_4 L_1 R_4 + 2 C_3 C_4 L_3 R_1) + s (2 C_3 C_4 R_1 R_3 + C_3 C_4 R_1 R_4 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1) + R_1 R_4}$$

$$\mathbf{9.799 \quad X-INVALID-ORDER-799} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_1 g_m s^4 + L_1 R_1 g_m + s^3 (C_3 C_4 L_1 L_3 R_1 R_4 g_m + C_3 C_4 L_1 L_4 R_1 R_3 g_m) + s^2 (C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_3 L_1 L_3 R_1 g_m + C_4 L_1 L_4 R_1 g_m) + s (C_3 L_1 R_1 R_3 g_m + C_4 L_1 R_1 R_4 g_m)}{C_3 R_1 + 2 C_4 R_1 + s^4 (2 C_1 C_3 C_4 L_1 L_3 R_1 + C_1 C_3 C_4 L_1 L_4 R_1) + s^3 (2 C_1 C_3 C_4 L_1 R_1 R_3 + C_1 C_3 C_4 L_1 R_1 R_4 + 2 C_3 C_4 L_1 L_3 R_1 g_m + 2 C_3 C_4 L_1 L_3 + C_3 C_4 L_1 L_4 R_1 g_m + C_3 C_4 L_1 L_4) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + 2 C_3 C_4 L_1 R_1 R_3 g_m + C_3 C_4 L_1 R_1 R_4 g_m + 2 C_3 C_4 L_1 R_3 + C_3 C_4 L_1 R_4 + 2 C_3 C_4 L_3 R_1) + s (2 C_3 C_4 R_1 R_3 + C_3 C_4 R_1 R_4 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1) + R_1 R_4}$$

$$\mathbf{9.800 \quad X-INVALID-ORDER-800} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_1 R_4 s^6 + 2 R_1 R_4 + s^5 (2 C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 + 2 C_1 C_3 L_1 L_3 L_4 R_1 + 2 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2 C_3 C_4 L_1 L_3 L_4 R_4) + s^4 (2 C_1 C_3 L_1 L_3 R_1 R_4 + 2 C_1 C_3 L_1 L_4 R_1 R_3 + C_1 C_3 L_1 L_4 R_1 R_4 + 2 C_1 C_4 L_1 L_4 R_1 R_4 + 2 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + 2 C_3 C_4 L_1 L_4 R_3 R_4) + s^3 (2 C_1 C_3 L_1 R_1 R_3 + 2 C_3 C_4 L_4 R_1 R_3 + 2 C_3 L_1 L_3) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + 2 C_3 C_4 L_1 R_1 R_3 g_m + C_3 C_4 L_1 R_1 R_4 g_m + 2 C_3 C_4 L_1 R_3 + C_3 C_4 L_1 R_4 + 2 C_3 C_4 L_3 R_1) + s (2 C_3 C_4 R_1 R_3 + C_3 C_4 R_1 R_4 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1) + R_1 R_4}$$

$$\mathbf{9.801 \quad X-INVALID-ORDER-801} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m s^5 + L_1 R_1 R_4 g_m s + s^4 (C_3 C_4 L_1 L_4 R_1 R_3 L_4 g_m + C_3 C_4 L_1 L_4 R_1 R_4 g_m)}{2 C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + 2 R_1 + s^5 (2 C_1 C_3 C_4 L_1 L_4 R_1 R_3 + C_1 C_3 C_4 L_1 L_4 R_1 R_4 + 2 C_3 C_4 L_1 L_3 L_4 R_1 g_m + 2 C_3 C_4 L_1 L_3 L_4) + s^4 (2 C_1 C_3 L_1 L_3 R_1 + C_1 C_3 L_1 L_4 R_1 + 2 C_1 C_4 L_1 L_4 R_1 + 2 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_3 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_3 C_4 L_1 L_4 R_3 + C_3 C_4 L_1 L_4 R_4 + 2 C_3 C_4 L_3 R_1) + s^3 (2 C_1 C_3 L_1 R_1 R_3 + 2 C_3 C_4 L_4 R_1 R_3 + 2 C_3 L_1 L_3) + s^2 (C_1 C_3 L_1 R_1 + 2 C_1 C_4 L_1 R_1 + 2 C_3 C_4 L_1 R_1 R_3 g_m + C_3 C_4 L_1 R_1 R_4 g_m + 2 C_3 C_4 L_1 R_3 + C_3 C_4 L_1 R_4 + 2 C_3 C_4 L_3 R_1) + s (2 C_3 C_4 R_1 R_3 + C_3 C_4 R_1 R_4 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1) + R_1 R_4}$$

$$\mathbf{9.802 \quad X-INVALID-ORDER-802} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{2C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + 2R_1 + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_4 + 2C_1 C_3 C_4 L_1 L_4 R_1 R_3 + C_1 C_3 C_4 L_1 L_4 R_1 R_4 + 2C_3 C_4 L_1 L_3 L_4 R_1 g_m + 2C_3 C_4 L_1 L_3 L_4) + s^4 (2C_1 C_3 C_4 L_1 R_1 R_3 R_4 + 2C_1 C_3 L_1 L_3 R_1 + 2C_1 C_4 L_1 L_4 R_1 + 2C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2C_3 C_4 L_1 L_3 R_4 + 2C_3 C_4 L_1 L_4 R_1}{2C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + 2R_1 + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_4 + 2C_1 C_3 C_4 L_1 L_4 R_1 R_3 + C_1 C_3 C_4 L_1 L_4 R_1 R_4 + 2C_3 C_4 L_1 L_3 L_4 R_1 g_m + 2C_3 C_4 L_1 L_3 L_4) + s^4 (2C_1 C_3 C_4 L_1 R_1 R_3 R_4 + 2C_1 C_3 L_1 L_3 R_1 + 2C_1 C_4 L_1 L_4 R_1 + 2C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2C_3 C_4 L_1 L_3 R_4 + 2C_3 C_4 L_1 L_4 R_1)}$$

$$\mathbf{9.803 \quad X-INVALID-ORDER-803} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 R_1 R_3 R_4 g_m s^2}{C_1 C_3 L_1 L_3 R_1 R_3 R_4 s^4 + R_1 R_3 R_4 + s^3 (2C_1 L_1 L_3 R_1 R_3 + C_1 L_1 L_3 R_1 R_4 + C_3 L_1 L_3 R_1 R_3 R_4 g_m + C_3 L_1 L_3 R_3 R_4) + s^2 (C_1 L_1 R_1 R_3 R_4 + C_3 L_3 R_1 R_3 R_4 + 2L_1 L_3 R_1 R_3 g_m + L_1 L_3 R_1 R_4 g_m + 2L_1 L_3 R_3 + L_1 L_3 R_4) + s (L_1 R_1 R_3 R_4 g_m + L_1 R_3 R_4 + 2L_3 R_1 R_3 + L_3 R_1 R_4)}$$

$$\mathbf{9.804 \quad X-INVALID-ORDER-804} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 R_1 R_3 g_m s^2}{R_1 R_3 + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 + 2C_1 C_4 L_1 L_3 R_1 R_3) + s^3 (C_1 L_1 L_3 R_1 + C_3 L_1 L_3 R_1 R_3 g_m + C_3 L_1 L_3 R_3 + 2C_4 L_1 L_3 R_1 R_3 g_m + 2C_4 L_1 L_3 R_3) + s^2 (C_1 L_1 R_1 R_3 + C_3 L_3 R_1 R_3 + 2C_4 L_3 R_1 R_3 + L_1 L_3 R_1 g_m + L_1 L_3) + s (L_1 R_1 R_3 g_m + L_1 R_3 + L_3 R_1)}$$

$$\mathbf{9.805 \quad X-INVALID-ORDER-805} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 R_1 R_3 R_4 g_m s^2}{R_1 R_3 R_4 + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 R_4 + 2C_1 C_4 L_1 L_3 R_1 R_3 R_4) + s^3 (2C_1 L_1 L_3 R_1 R_3 + C_1 L_1 L_3 R_1 R_4 + C_3 L_1 L_3 R_1 R_3 R_4 g_m + C_3 L_1 L_3 R_3 R_4 + 2C_4 L_1 L_3 R_1 R_3 R_4 g_m + 2C_4 L_1 L_3 R_3 R_4) + s^2 (C_1 L_1 R_1 R_3 R_4 + C_3 L_3 R_1 R_3 R_4 + 2C_4 L_3 R_1 R_3 R_4 + 2L_1 L_3 R_1 R_3 g_m + L_1 L_3 R_1 R_4)}$$

$$\mathbf{9.806 \quad X-INVALID-ORDER-806} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 R_1 R_3 R_4 g_m s^3 + L_1 L_3 R_1 R_3 g_m s^2}{C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 s^5 + R_1 R_3 + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 + 2C_1 C_4 L_1 L_3 R_1 R_3 + C_1 C_4 L_1 L_3 R_1 R_4 + C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + C_3 C_4 L_1 L_3 R_3 R_4) + s^3 (C_1 C_4 L_1 R_1 R_3 R_4 + C_1 L_1 L_3 R_1 + C_3 C_4 L_3 R_1 R_3 R_4 + C_3 L_1 L_3 R_1 R_3 g_m + C_3 L_1 L_3 R_3 + 2C_4 L_1 L_3 R_1 R_3 g_m + C_4 L_1 L_3 R_3)}$$

$$\mathbf{9.807 \quad X-INVALID-ORDER-807} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_3 L_4 R_1 R_3 g_m s^4 + L_1 L_3 R_1 R_3 g_m s^2}{C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 s^6 + R_1 R_3 + s^5 (C_1 C_4 L_1 L_3 L_4 R_1 + C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_3) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 + 2C_1 C_4 L_1 L_3 R_1 R_3 + C_1 C_4 L_1 L_4 R_1 R_3 + C_3 C_4 L_3 L_4 R_1 R_3 + C_4 L_1 L_3 L_4 R_1 g_m + C_4 L_1 L_3 L_4) + s^3 (C_1 L_1 L_3 R_1 + C_3 L_1 L_3 R_1 R_3 g_m + C_3 L_1 L_3 R_3)}$$

$$\mathbf{9.808 \quad X-INVALID-ORDER-808} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 L_4 R_1 R_3 g_m s^2}{2L_3 R_1 R_3 + L_4 R_1 R_3 + s^4 (C_1 C_3 L_1 L_3 L_4 R_1 R_3 + 2C_1 C_4 L_1 L_3 L_4 R_1 R_3) + s^3 (C_1 L_1 L_3 L_4 R_1 + C_3 L_1 L_3 L_4 R_1 R_3 g_m + C_3 L_1 L_3 L_4 R_3 + 2C_4 L_1 L_3 L_4 R_1 R_3 g_m + 2C_4 L_1 L_3 L_4 R_3) + s^2 (2C_1 L_1 L_3 R_1 R_3 + C_1 L_1 L_4 R_1 R_3 + C_3 L_3 L_4 R_1 R_3 + 2C_4 L_3 L_4 R_1 R_3 + L_1 L_3 L_4 R_1 g_m + L_1 L_3 L_4 R_3)}$$

$$\mathbf{9.809 \quad X-INVALID-ORDER-809} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 L_4 R_1 R_3 g_m s^2}{C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 s^6 + R_1 R_3 + s^5 (C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 + C_1 C_4 L_1 L_3 L_4 R_1 + C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_3) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 + 2C_1 C_4 L_1 L_3 R_1 R_3 + C_1 C_4 L_1 L_3 R_1 R_4 + C_1 C_4 L_1 L_4 R_1 R_3 + C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + C_3 C_4 L_1 L_3 R_3 R_4 + C_3 C_4 L_3 L_4 R_1 R_3)}$$

$$\mathbf{9.810 \quad X-INVALID-ORDER-810} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 L_4 R_1 R_3 R_4 g_m s^2}{2L_3 R_1 R_3 R_4 + L_4 R_1 R_3 R_4 + s^4 (C_1 C_3 L_1 L_3 L_4 R_1 R_3 R_4 + 2C_1 C_4 L_1 L_3 L_4 R_1 R_3 R_4) + s^3 (2C_1 L_1 L_3 L_4 R_1 R_3 + C_1 L_1 L_3 L_4 R_1 R_4 + C_3 L_1 L_3 L_4 R_1 R_3 R_4 g_m + C_3 L_1 L_3 L_4 R_3 R_4 + 2C_4 L_1 L_3 L_4 R_1 R_3 R_4 g_m + 2C_4 L_1 L_3 L_4 R_3 R_4) + s^2 (2C_1 L_1 L_3 R_1 R_3 R_4 + C_1 L_1 L_4 R_1 R_3 R_4 + C_3 L_3 L_4 R_1 R_3 R_4)}$$

$$\mathbf{9.811 \quad X-INVALID-ORDER-811} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_3 L_4 R_1 R_3 R_4 s^2}{C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 s^6 + R_1 R_3 R_4 + s^5 (C_1 C_3 L_1 L_3 L_4 R_1 R_3 + 2C_1 C_4 L_1 L_3 L_4 R_1 R_3 + C_1 C_4 L_1 L_3 L_4 R_1 R_4 + C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 g_m + C_3 C_4 L_1 L_3 L_4 R_3 R_4) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 R_4 + C_1 C_4 L_1 L_4 R_1 R_3 R_4 + C_1 L_1 L_3 L_4 R_1 + C_3 C_4 L_3 L_4 R_1 R_3 R_4 + C_3 L_1 L_3 L_4 R_1 R_3)}$$

9.812 X-INVALID-ORDER-812 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 s^6 + R_1 R_3 R_4 + s^5 (2 C_1 C_4 L_1 L_3 L_4 R_1 R_3 + C_1 C_4 L_1 L_3 L_4 R_1 R_4 + C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 g_m + C_3 C_4 L_1 L_3 L_4 R_3 R_4) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 R_4 + 2 C_1 C_4 L_1 L_3 R_1 R_3 R_4 + C_1 C_4 L_1 L_4 R_1 R_3 R_4 + C_3 C_4 L_3 L_4 R_1 R_3 R_4 + 2 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_4 L_1 L_3 L_4 R_1 R_3 R_4)}{C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 s^6 + R_1 R_3 R_4 + s^5 (2 C_1 C_4 L_1 L_3 L_4 R_1 R_3 + C_1 C_4 L_1 L_3 L_4 R_1 R_4 + C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 g_m + C_3 C_4 L_1 L_3 L_4 R_3 R_4) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 R_4 + 2 C_1 C_4 L_1 L_3 R_1 R_3 R_4 + C_1 C_4 L_1 L_4 R_1 R_3 R_4 + C_3 C_4 L_3 L_4 R_1 R_3 R_4 + 2 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_4 L_1 L_3 L_4 R_1 R_3 R_4)}$$

9.813 X-INVALID-ORDER-813 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 R_1 R_3 R_4 g_m s^3 + L_1 L_3 R_1 R_4 g_m s^2 + L_1 R_1 R_3 R_4 g_m s}{2R_1 R_3 + R_1 R_4 + s^4 (2C_1 C_3 L_1 L_3 R_1 R_3 + C_1 C_3 L_1 L_3 R_1 R_4) + s^3 (2C_1 L_1 L_3 R_1 + 2C_3 L_1 L_3 R_1 R_3 g_m + C_3 L_1 L_3 R_1 R_4 g_m + 2C_3 L_1 L_3 R_3 + C_3 L_1 L_3 R_4) + s^2 (2C_1 L_1 R_1 R_3 + C_1 L_1 R_1 R_4 + 2C_3 L_3 R_1 R_3 + C_3 L_3 R_1 R_4 + 2L_1 L_3 R_1 g_m + 2L_1 L_3) + s (2L_1 R_1 R_3 g_m + L_1 R_1 R_4)}$$

9.814 X-INVALID-ORDER-814 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 R_1 R_3 g_m s^3 + L_1 L_3 R_1 g_m s^2 + L_1 R_1 R_3 g_m s}{2C_1 C_3 C_4 L_1 L_3 R_1 R_3 s^5 + R_1 + s^4 (C_1 C_3 L_1 L_3 R_1 + 2C_1 C_4 L_1 L_3 R_1 + 2C_3 C_4 L_1 L_3 R_1 R_3 g_m + 2C_3 C_4 L_1 L_3 R_3) + s^3 (2C_1 C_4 L_1 R_1 R_3 + 2C_3 C_4 L_3 R_1 R_3 + C_3 L_1 L_3 R_1 g_m + C_3 L_1 L_3 + 2C_4 L_1 L_3 R_1 g_m + 2C_4 L_1 L_3) + s^2 (C_1 L_1 R_1 + C_3 L_3 R_1 + 2C_4 L_1 R_1 R_3 g_m + 2C_4 L_1 R_3 +$$

9.815 X-INVALID-ORDER-815 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 R_1 R_3 R_4 g_m s^3 + L}{2C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 s^5 + 2R_1 R_3 + R_1 R_4 + s^4 (2C_1 C_3 L_1 L_3 R_1 R_3 + C_1 C_3 L_1 L_3 R_1 R_4 + 2C_1 C_4 L_1 L_3 R_1 R_4 + 2C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + 2C_3 C_4 L_1 L_3 R_3 R_4) + s^3 (2C_1 C_4 L_1 R_1 R_3 R_4 + 2C_1 L_1 L_3 R_1 + 2C_3 C_4 L_3 R_1 R_3 R_4 + 2C_3 L_1 L_3 R_1 R_3 g_m + C_3 L_1 L_3 R_1 R_4 g_m + 2C_3 L_1$$

9.816 X-INVALID-ORDER-816 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m s^4 + L_1 R_1 R_3 g_m s + s^3 (C_3 L_1 L_3 R_1 R_3 g_m + C_4 L_1 L_3 R_1 R_4 g_m) + s^2 (C_4 L_1 R_1 R_3 R_4}{R_1 + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_3 + C_1 C_3 C_4 L_1 L_3 R_1 R_4) + s^4 (C_1 C_3 L_1 L_3 R_1 + 2 C_1 C_4 L_1 L_3 R_1 + 2 C_3 C_4 L_1 L_3 R_1 R_3 g_m + C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_3 C_4 L_1 L_3 R_3 + C_3 C_4 L_1 L_3 R_4) + s^3 (2 C_1 C_4 L_1 R_1 R_3 + C_1 C_4 L_1 R_1 R_4 + 2 C_3 C_4 L_3 R_1 R_3 + C_3 C_4 L_3 R_1 R_4 + C_3 L_1 L_3 R_1 g_m + C_3 L_1 L_3 R_1 g_m}.$$

9.817 X-INVALID-ORDER-817 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m s^5 + C_4 L_1 L_3 L_4 R_1 g_m s^4 + L_1 L_3 R_1 g_m s^2 + L_1 R_1 R_3 g_m s + s^3 (C_3 L_1 L_3 R_1 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_1 R_3 + C_4 L_1 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4)}{C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + R_1 + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_3 + C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 L_1 L_3 R_1 + 2 C_1 C_4 L_1 L_3 R_1 + C_1 C_4 L_1 L_4 R_1 + 2 C_3 C_4 L_1 L_3 R_1 R_3 g_m + 2 C_3 C_4 L_1 L_3 R_3 + C_3 C_4 L_3 L_4 R_1) + s^3 (2 C_1 C_4 L_1 R_1 R_3 + 2 C_3 C_4 L_3 R_1 R_3 + C_3 L_1 L_3 R_1 g_m +$$

9.818 X-INVALID-ORDER-818 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_1 R_3 g_m s^4 + L}{2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 s^6 + 2R_1 R_3 + s^5 (C_1 C_3 L_1 L_3 L_4 R_1 + 2C_1 C_4 L_1 L_3 L_4 R_1 + 2C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + 2C_3 C_4 L_1 L_3 L_4 R_3) + s^4 (2C_1 C_3 L_1 L_3 R_1 R_3 + 2C_1 C_4 L_1 L_4 R_1 R_3 + 2C_3 C_4 L_3 L_4 R_1 R_3 + C_3 L_1 L_3 L_4 R_1 g_m + C_3 L_1 L_3 L_4 + 2C_4 L_1 L_3 L_4 R_1 g_m + 2C_4 L_1 L_3 L_4) + s^3}$$

9.819 X-INVALID-ORDER-819 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m s^5 + L_1 R_1 R_3 g_m s + s^4 (C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + R_1 + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_3 + C_1 C_3 C_4 L_1 L_3 R_1 R_4 + C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 L_1 L_3 R_1 + 2 C_1 C_4 L_1 L_3 R_1 + C_1 C_4 L_1 L_4 R_1 + 2 C_3 C_4 L_1 L_3 R_1 R_3 g_m + C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_3 C_4 L_1 L_3 R_3 + C_3 C_4 L_1 L_3 R_4 + C_3 C_4 L_3 L_4 R_1))}{C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + R_1 + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_3 + C_1 C_3 C_4 L_1 L_3 R_1 R_4 + C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 L_1 L_3 R_1 + 2 C_1 C_4 L_1 L_3 R_1 + C_1 C_4 L_1 L_4 R_1 + 2 C_3 C_4 L_1 L_3 R_1 R_3 g_m + C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_3 C_4 L_1 L_3 R_3 + C_3 C_4 L_1 L_3 R_4 + C_3 C_4 L_3 L_4 R_1)}$$

9.820 X-INVALID-ORDER-820 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{2C_1C_3C_4L_1L_3L_4R_1R_3R_4s^6 + 2R_1R_3R_4 + s^5(2C_1C_3L_1L_3L_4R_1R_3 + C_1C_3L_1L_3L_4R_1R_4 + 2C_1C_4L_1L_3L_4R_1R_4 + 2C_3C_4L_1L_3L_4R_1R_3R_4g_m + 2C_3C_4L_1L_3L_4R_3R_4) + s^4(2C_1C_3L_1L_3R_1R_3R_4 + 2C_1C_4L_1L_4R_1R_3R_4 + 2C_1L_1L_3L_4R_1 + 2C_3C_4L_3L_4R_1R_3R_4 + 2C_3L_1L_3L_4R_1R_3R_4 + 2C_3C_4L_3L_4R_1R_3R_4 + 2C_3C_4L_3L_4R_1R_3R_4)}{2C_1C_3C_4L_1L_3L_4R_1R_3R_4s^6 + 2R_1R_3R_4 + s^5(2C_1C_3L_1L_3L_4R_1R_3 + C_1C_3L_1L_3L_4R_1R_4 + 2C_1C_4L_1L_3L_4R_1R_4 + 2C_3C_4L_1L_3L_4R_1R_3R_4g_m + 2C_3C_4L_1L_3L_4R_3R_4) + s^4(2C_1C_3L_1L_3R_1R_3R_4 + 2C_1C_4L_1L_4R_1R_3R_4 + 2C_1L_1L_3L_4R_1 + 2C_3C_4L_3L_4R_1R_3R_4 + 2C_3L_1L_3L_4R_1R_3R_4 + 2C_3C_4L_3L_4R_1R_3R_4 + 2C_3C_4L_3L_4R_1R_3R_4)}$$

9.821 X-INVALID-ORDER-821 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{2R_1 R_3 + R_1 R_4 + s^6(2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 + C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4) + s^5(C_1 C_3 L_1 L_3 L_4 R_1 + 2C_1 C_4 L_1 L_3 L_4 R_1 + 2C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_3 C_4 L_1 L_3 L_4 R_3 + C_3 C_4 L_1 L_3 L_4 R_4) + s^4(2C_1 C_3 L_1 L_3 R_1 R_3 + C_1 C_3 L_1 L_3 R_1 R_4 + 2C_1 C_4 L_1 L_4 F$$

9.822 X-INVALID-ORDER-822 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{2R_1R_3 + R_1R_4 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3 + C_1C_3C_4L_1L_3L_4R_1R_4) + s^5(2C_1C_3C_4L_1L_3R_1R_3R_4 + 2C_1C_4L_1L_3L_4R_1 + 2C_3C_4L_1L_3L_4R_1R_3g_m + C_3C_4L_1L_3L_4R_1R_4g_m + 2C_3C_4L_1L_3L_4R_3 + C_3C_4L_1L_3L_4R_4) + s^4(2C_1C_3L_1L_3R_1R_3 + C_1C_3L_1L_3R_1R_4 + 2C_1C_4L_1L_3R_1R_3g_m + C_1C_4L_1L_3R_1R_4g_m + 2C_1C_4L_1L_3R_3 + C_1C_4L_1L_3R_4) + s^3(2C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4 + 2C_1C_4L_1L_3R_3g_m + C_1C_4L_1L_3R_4g_m + 2C_1C_4L_1L_3R_3 + C_1C_4L_1L_3R_4) + s^2(2C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4 + 2C_1C_4L_1L_3R_3g_m + C_1C_4L_1L_3R_4g_m + 2C_1C_4L_1L_3R_3 + C_1C_4L_1L_3R_4) + s(C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4 + C_1C_4L_1L_3R_3g_m + C_1C_4L_1L_3R_4g_m + C_1C_4L_1L_3R_3 + C_1C_4L_1L_3R_4) + C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4 + C_1C_4L_1L_3R_3g_m + C_1C_4L_1L_3R_4g_m + C_1C_4L_1L_3R_3 + C_1C_4L_1L_3R_4}{2R_1R_3 + R_1R_4 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3 + C_1C_3C_4L_1L_3L_4R_1R_4) + s^5(2C_1C_3C_4L_1L_3R_1R_3R_4 + 2C_1C_4L_1L_3L_4R_1 + 2C_3C_4L_1L_3L_4R_1R_3g_m + C_3C_4L_1L_3L_4R_1R_4g_m + 2C_3C_4L_1L_3L_4R_3 + C_3C_4L_1L_3L_4R_4) + s^4(2C_1C_3L_1L_3R_1R_3 + C_1C_3L_1L_3R_1R_4 + 2C_1C_4L_1L_3R_1R_3g_m + C_1C_4L_1L_3R_1R_4g_m + 2C_1C_4L_1L_3R_3 + C_1C_4L_1L_3R_4) + s^3(2C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4 + 2C_1C_4L_1L_3R_3g_m + C_1C_4L_1L_3R_4g_m + 2C_1C_4L_1L_3R_3 + C_1C_4L_1L_3R_4) + s^2(2C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4 + 2C_1C_4L_1L_3R_3g_m + C_1C_4L_1L_3R_4g_m + 2C_1C_4L_1L_3R_3 + C_1C_4L_1L_3R_4) + s(C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4 + C_1C_4L_1L_3R_3g_m + C_1C_4L_1L_3R_4g_m + C_1C_4L_1L_3R_3 + C_1C_4L_1L_3R_4) + C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4 + C_1C_4L_1L_3R_3g_m + C_1C_4L_1L_3R_4g_m + C_1C_4L_1L_3R_3 + C_1C_4L_1L_3R_4}$$

9.823 X-INVALID-ORDER-823 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 R_1 R_3 R_4 g_m s^3 + L_1 R_1 R_3 R_4 g_m s}{2 R_1 R_3 + R_1 R_4 + s^4 (2 C_1 C_3 L_1 L_3 R_1 R_3 + C_1 C_3 L_1 L_3 R_1 R_4) + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 + 2 C_3 L_1 L_3 R_1 R_3 g_m + C_3 L_1 L_3 R_1 R_4 g_m + 2 C_3 L_1 L_3 R_3 + C_3 L_1 L_3 R_4) + s^2 (2 C_1 L_1 R_1 R_3 + C_1 L_1 R_1 R_4 + C_3 L_1 R_1 R_3 R_4 g_m + C_3 L_1 R_3 R_4 + 2 C_3 L_3 R_1 R_3 + C_3 L_3 R_1 R_4) + s (C_3 R_1 R_3 R_4)}$$

9.824 X-INVALID-ORDER-824 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 R_1 R_3 g_m s^3 + L_1 R_1 R_3 g_m s}{2C_1 C_3 C_4 L_1 L_3 R_1 R_3 s^5 + R_1 + s^4 (C_1 C_3 L_1 L_3 R_1 + 2C_3 C_4 L_1 L_3 R_1 R_3 g_m + 2C_3 C_4 L_1 L_3 R_3) + s^3 (C_1 C_3 L_1 R_1 R_3 + 2C_1 C_4 L_1 R_1 R_3 + 2C_3 C_4 L_3 R_1 R_3 + C_3 L_1 L_3 R_1 g_m + C_3 L_1 L_3) + s^2 (C_1 L_1 R_1 + C_3 L_1 R_1 R_3 g_m + C_3 L_1 R_3 + C_3 L_3 R_1 + 2C_4 L_1 R_1 R_3 g_m + 2C_4 L_1 R_3) + s}$$

9.825 X-INVALID-ORDER-825 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 R_1 R_3 R_4 g_m s^3 + L_1 R_1 R_3 R_4 g_m s}{2C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 s^5 + 2R_1 R_3 + R_1 R_4 + s^4 (2C_1 C_3 L_1 L_3 R_1 R_3 + C_1 C_3 L_1 L_3 R_1 R_4 + 2C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + 2C_3 C_4 L_1 L_3 R_3 R_4) + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 + 2C_1 C_4 L_1 R_1 R_3 R_4 + 2C_3 C_4 L_3 R_1 R_3 R_4 + 2C_3 L_1 L_3 R_1 R_3 g_m + C_3 L_1 L_3 R_1 R_4 g_m + 2C_3 L_1 L_3 R_3 + C_3 L_1 L_3 R_4)}.$$

9.826 X-INVALID-ORDER-826 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m s^4 + C_3 L_1 L_3 R_1 R_3 g_m s^3}{R_1 + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 + C_1 C_3 C_4 L_1 L_3 R_1 R_4) + s^4 (C_1 C_3 C_4 L_1 R_1 R_3 R_4 + C_1 C_3 L_1 L_3 R_1 + 2C_3 C_4 L_1 L_3 R_1 R_3 g_m + C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2C_3 C_4 L_1 L_3 R_3 + C_3 C_4 L_1 L_3 R_4) + s^3 (C_1 C_3 L_1 R_1 R_3 + 2C_1 C_4 L_1 R_1 R_3 + C_1 C_4 L_1 R_1 R_4 + C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_3 C_4 L_1 R_1 R_3 R_4 g_m)}$$

9.827 X-INVALID-ORDER-827 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m s^5 + L_1 R_1 R_3 g_m s + s^3 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + R_1 + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_3 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 + C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 L_1 L_3 R_1 + C_1 C_4 L_1 L_4 R_1 + 2 C_3 C_4 L_1 L_3 R_1 R_3 g_m + 2 C_3 C_4 L_1 L_3 R_3 + C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_3 C_4 L_1 L_4 R_3 + C_3 C_4 L_3 L_4 R_1) + s^3 (C_1 C_3 L_1$$

9.828 X-INVALID-ORDER-828 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 L_4 R_1 R_3 g_m s^4 + L_1 L_4 R_1 R_3 g_m s^2}{2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 s^6 + 2R_1 R_3 + s^5 (C_1 C_3 L_1 L_3 L_4 R_1 + 2C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + 2C_3 C_4 L_1 L_3 L_4 R_3) + s^4 (2C_1 C_3 L_1 L_3 R_1 R_3 + C_1 C_3 L_1 L_4 R_1 R_3 + 2C_1 C_4 L_1 L_4 R_1 R_3 + 2C_3 C_4 L_3 L_4 R_1 R_3 + C_3 L_1 L_3 L_4 R_1 g_m + C_3 L_1 L_3 L_4) + s^3 (C_1 L_1 L_4 R_1 + 2C_3 L_1 L_3 R_1 R_3 g_m +$$

9.829 X-INVALID-ORDER-829 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + R_1 + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_3 + C_1 C_3 C_4 L_1 L_3 R_1 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 + C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 C_4 L_1 R_1 R_3 R_4 + C_1 C_3 L_1 L_3 R_1 + C_1 C_4 L_1 L_4 R_1 + 2 C_3 C_4 L_1 L_3 R_1 R_3 g_m + C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_3 C_4 L_1 L_3 R_3 + C_3 C_4 L_1 L_3 R_4)}{C_1 C_3 C_4 L_1 L_3 L_4 R_1 s^6 + R_1 + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_3 + C_1 C_3 C_4 L_1 L_3 R_1 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 + C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 C_4 L_1 R_1 R_3 R_4 + C_1 C_3 L_1 L_3 R_1 + C_1 C_4 L_1 L_4 R_1 + 2 C_3 C_4 L_1 L_3 R_1 R_3 g_m + C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_3 C_4 L_1 L_3 R_3 + C_3 C_4 L_1 L_3 R_4)}$$

9.830 X-INVALID-ORDER-830 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{2C_1C_3C_4L_1L_3L_4R_1R_3R_4s^6 + 2R_1R_3R_4 + s^5(2C_1C_3L_1L_3L_4R_1R_3 + C_1C_3L_1L_3L_4R_1R_4 + 2C_3C_4L_1L_3L_4R_1R_3R_4g_m + 2C_3C_4L_1L_3L_4R_3R_4) + s^4(2C_1C_3L_1L_3R_1R_3R_4 + C_1C_3L_1L_4R_1R_3R_4 + 2C_1C_4L_1L_4R_1R_3R_4 + 2C_3C_4L_3L_4R_1R_3R_4 + 2C_3L_1L_3L_4R_1R_3g_m +$$

9.831 X-INVALID-ORDER-831 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{1}{2R_1 R_3 + R_1 R_4 + s^6(2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 + C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4) + s^5(C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 + C_1 C_3 L_1 L_3 L_4 R_1 + 2C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_3 C_4 L_1 L_3 L_4 R_3 + C_3 C_4 L_1 L_3 L_4 R_4) + s^4(2C_1 C_3 L_1 L_3 R_1 R_3 + C_1 C_3 L_1 L_3 R_1 R_4 + C_1 C_3 L_1$$

9.832 X-INVALID-ORDER-832 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{2R_1R_3 + R_1R_4 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3 + C_1C_3C_4L_1L_3L_4R_1R_4) + s^5(2C_1C_3C_4L_1L_3R_1R_3R_4 + C_1C_3C_4L_1L_4R_1R_3R_4 + 2C_3C_4L_1L_3L_4R_1R_3g_m + C_3C_4L_1L_3L_4R_1R_4g_m + 2C_3C_4L_1L_3L_4R_3 + C_3C_4L_1L_3L_4R_4) + s^4(2C_1C_3L_1L_3R_1R_3 + C_1C_3L_1L_3R_1R_4 + 2C_1C_3L_1L_3R_3R_4 + C_1C_3L_1L_3R_3R_4 + 2C_1C_3L_1L_3R_4R_3 + C_1C_3L_1L_3R_4R_4) + s^3(2C_1C_3L_1L_3R_3g_m + C_1C_3L_1L_3R_4g_m + 2C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4) + s^2(2C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4) + s(C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4) + C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4}{2R_1R_3 + R_1R_4 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3 + C_1C_3C_4L_1L_3L_4R_1R_4) + s^5(2C_1C_3C_4L_1L_3R_1R_3R_4 + C_1C_3C_4L_1L_4R_1R_3R_4 + 2C_3C_4L_1L_3L_4R_1R_3g_m + C_3C_4L_1L_3L_4R_1R_4g_m + 2C_3C_4L_1L_3L_4R_3 + C_3C_4L_1L_3L_4R_4) + s^4(2C_1C_3L_1L_3R_1R_3 + C_1C_3L_1L_3R_1R_4 + 2C_1C_3L_1L_3R_3R_4 + C_1C_3L_1L_3R_3R_4 + 2C_1C_3L_1L_3R_4R_3 + C_1C_3L_1L_3R_4R_4) + s^3(2C_1C_3L_1L_3R_3g_m + C_1C_3L_1L_3R_4g_m + 2C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4) + s^2(2C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4) + s(C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4) + C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4}.$$

9.833 X-INVALID-ORDER-833 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 R_1 R_3 g_m s^2 + L_1 R_3 g_m s + R_1 R_3 g_m}{R_1 g_m + s^3 (2C_1 C_4 L_1 R_1 R_3 g_m + 2C_1 C_4 L_1 R_3) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 + 2C_4 L_1 R_3 g_m) + s (2C_4 R_1 R_3 g_m + 2C_4 R_3 + L_1 g_m) + 1}$$

9.834 X-INVALID-ORDER-834 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 R_1 R_3 R_4 g_m s^2 + L_1 R_3 R_4 g_m s + R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^3 (2C_1 C_4 L_1 R_1 R_3 R_4 g_m + 2C_1 C_4 L_1 R_3 R_4) + s^2 (2C_1 L_1 R_1 R_3 g_m + C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_3 + C_1 L_1 R_4 + 2C_4 L_1 R_3 R_4 g_m) + s (2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4 + 2L_1 R_3 g_m + L_1 R_4 g_m)}$$

9.835 X-INVALID-ORDER-835 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 R_1 R_3 R_4 g_m s^3 + R_1 R_3 g_m + s^2 (C_1 L_1 R_1 R_3 g_m + C_4 L_1 R_3 R_4 g_m) + s (C_4 R_1 R_3 R_4 g_m + L_1 R_3 g_m)}{R_1 g_m + s^3 (2 C_1 C_4 L_1 R_1 R_3 g_m + C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 + 2 C_4 L_1 R_3 g_m + C_4 L_1 R_4 g_m) + s (2 C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2 C_4 R_3 + C_4 R_4 + L_1 g_m) + 1}$$

9.836 X-INVALID-ORDER-836 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 R_3 g_m s^4 + C_4 L_1 L_4 R_3 g_m s^3 + L_1 R_3 g_m s + R_1 R_3 g_m + s^2 (C_1 L_1 R_1 R_3 g_m + C_4 L_4 R_1 R_3 g_m)}{R_1 g_m + s^4 (C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4) + s^3 (2C_1 C_4 L_1 R_1 R_3 g_m + 2C_1 C_4 L_1 R_3 + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 + 2C_4 L_1 R_3 g_m + C_4 L_4 R_1 g_m + C_4 L_4) + s (2C_4 R_1 R_3 g_m + 2C_4 R_3 + L_1 g_m) + 1}$$

9.837 X-INVALID-ORDER-837 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_4 R_1 R_3 g_m s^3 + L_1 L_4 R_3 g_m s^2 + L_4 R_1 R_3 g_m s}{2R_1 R_3 g_m + 2R_3 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_3 g_m + 2C_1 C_4 L_1 L_4 R_3) + s^3 (C_1 L_1 L_4 R_1 g_m + C_1 L_1 L_4 + 2C_4 L_1 L_4 R_3 g_m) + s^2 (2C_1 L_1 R_1 R_3 g_m + 2C_1 L_1 R_3 + 2C_4 L_4 R_1 R_3 g_m + 2C_4 L_4 R_3 + L_1 L_4 g_m) + s (2L_1 R_3 g_m + L_4 R_1 g_m + L_4)}$$

9.838 X-INVALID-ORDER-838 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 R_3 g_m s^4 + R_1 R_3 g_m + s^3 (C_1 C_4 L_1 R_1 R_3 R_4 g_m + C_4 L_1 L_4 R_3 g_m) + s^2 (C_1 L_1 R_1 R_3 g_m + C_4 L_1 R_3 R_4 g_m + C_4 L_4 R_1 R_3 g_m) + s (C_4 R_1 R_3 R_4 g_m + L_1 R_3 g_m)}{R_1 g_m + s^4 (C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4) + s^3 (2C_1 C_4 L_1 R_1 R_3 g_m + C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4 + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 + 2C_4 L_1 R_3 g_m + C_4 L_1 R_4 g_m + C_4 L_4 R_1 g_m + C_4 L_4) + s (2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4 + L_1)}$$

9.839 X-INVALID-ORDER-839 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_4 R_1 R_3 R_4 g_m s^3 + L_1 L_4 R_3 R_4 g_m s^2 + L_4 R_1 R_3 R_4 g_m s}{2 R_1 R_3 R_4 g_m + 2 R_3 R_4 + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_3 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_3 R_4) + s^3 (2 C_1 L_1 L_4 R_1 R_3 g_m + C_1 L_1 L_4 R_1 R_4 g_m + 2 C_1 L_1 L_4 R_3 + C_1 L_1 L_4 R_4 + 2 C_4 L_1 L_4 R_3 R_4 g_m) + s^2 (2 C_1 L_1 R_1 R_3 R_4 g_m + 2 C_1 L_1 R_3 R_4 + 2 C_4 L_4 R_1 R_3 R_4 g_m + 2 C_4 L_4 R_3 R_4 + 2 L_1 L_4 R_3 g_m + L_1 L_4 R_4) + s (2 C_1 L_1 R_1 R_3 R_4 + 2 C_4 L_4 R_1 R_3 R_4 + 2 L_1 L_4 R_3) + 2 L_1 L_4 R_4}.$$

$$\mathbf{9.840 \quad X-INVALID-ORDER-840} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad R_3, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 R_3 R_4 g_m s^4 + R_1 R_3 R_4 g_m + s^3 (C_1 L_1 L_4 R_1 R_3 g_m + C_4 L_1 L_4 R_3 R_4 g_m) + s^2 (C_1 L_1 R_1 R_3 R_4 g_m + C_4 L_4 R_1 R_3 R_4 g_m + L_1 L_4 R_3 g_m) + s (L_1 R_3 R_4 g_m + L_4 R_1 R_3 g_m)}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4) + s^3 (C_1 L_1 L_4 R_1 g_m + C_1 L_1 L_4 + 2 C_4 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (2 C_1 L_1 R_1 R_3 g_m + C_1 L_1 R_1 R_4 g_m + 2 C_1 L_1 R_3 + C_1 L_1 R_4 + 2 C_4 L_4 R_1 R_3 g_m)}$$

$$\mathbf{9.841 \quad X-INVALID-ORDER-841} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad R_3, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 R_3 R_4 g_m s^4 + C_4 L_1 L_4 R_3 R_4 g_m s^3 + L_1 R_3 R_4 g_m s + R_1 R_3 R_4 g_m + s^2 (C_1 L_1 R_1 R_3 R_4 g_m + C_4 L_4 R_1 R_3 R_4 g_m)}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4) + s^3 (2 C_1 C_4 L_1 R_1 R_3 R_4 g_m + 2 C_1 C_4 L_1 R_3 R_4 + 2 C_4 L_1 L_4 R_3 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (2 C_1 L_1 R_1 R_3 g_m + C_1 L_1 R_1 R_4 g_m + 2 C_1 L_1 R_3 + C_1 L_1 R_4 + 2 C_4 L_4 R_1 R_3 g_m)}$$

$$\mathbf{9.842 \quad X-INVALID-ORDER-842} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_1 R_4 g_m s^2 + L_1 R_4 g_m s + R_1 R_4 g_m}{2 R_1 g_m + s^3 (C_1 C_3 L_1 R_1 R_4 g_m + C_1 C_3 L_1 R_4) + s^2 (2 C_1 L_1 R_1 g_m + 2 C_1 L_1 + C_3 L_1 R_4 g_m) + s (C_3 R_1 R_4 g_m + C_3 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.843 \quad X-INVALID-ORDER-843} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_1 g_m s^2 + L_1 g_m s + R_1 g_m}{s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1) + s^2 (C_3 L_1 g_m + 2 C_4 L_1 g_m) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.844 \quad X-INVALID-ORDER-844} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_1 R_4 g_m s^2 + L_1 R_4 g_m s + R_1 R_4 g_m}{2 R_1 g_m + s^3 (C_1 C_3 L_1 R_1 R_4 g_m + C_1 C_3 L_1 R_4 + 2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_4) + s^2 (2 C_1 L_1 R_1 g_m + 2 C_1 L_1 + C_3 L_1 R_4 g_m + 2 C_4 L_1 R_4 g_m) + s (C_3 R_1 R_4 g_m + C_3 R_4 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.845 \quad X-INVALID-ORDER-845} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 R_1 R_4 g_m s^3 + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_4 L_1 R_4 g_m) + s (C_4 R_1 R_4 g_m + L_1 g_m)}{s^4 (C_1 C_3 C_4 L_1 R_1 R_4 g_m + C_1 C_3 C_4 L_1 R_4) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1 + C_3 C_4 L_1 R_4 g_m) + s^2 (C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4 + C_3 L_1 g_m + 2 C_4 L_1 g_m) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.846 \quad X-INVALID-ORDER-846} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 g_m s^4 + C_4 L_1 L_4 g_m s^3 + L_1 g_m s + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_4 L_4 R_1 g_m)}{C_3 C_4 L_1 L_4 g_m s^4 + s^5 (C_1 C_3 C_4 L_1 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_4) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1 + C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (C_3 L_1 g_m + 2 C_4 L_1 g_m) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.847 \quad X-INVALID-ORDER-847} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 R_1 g_m s^3 + L_1 L_4 g_m s^2 + L_4 R_1 g_m s}{2 L_1 g_m s + 2 R_1 g_m + s^4 (C_1 C_3 L_1 L_4 R_1 g_m + C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4) + s^3 (C_3 L_1 L_4 g_m + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_1 L_1 R_1 g_m + 2 C_1 L_1 + C_3 L_4 R_1 g_m + C_3 L_4 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4) + 2}$$

$$\mathbf{9.848 \quad X-INVALID-ORDER-848} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 g_m s^4 + R_1 g_m + s^3 (C_1 C_4 L_1 R_1 R_4 g_m + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 R_1 g_m + C_4 L_1 R_4 g_m + C_4 L_4 R_1 g_m) + s (C_4 R_1 R_4 g_m + L_1 g_m)}{s^5 (C_1 C_3 C_4 L_1 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_4) + s^4 (C_1 C_3 C_4 L_1 R_1 R_4 g_m + C_1 C_3 C_4 L_1 R_4 + C_3 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1 + C_3 C_4 L_1 R_4 g_m + C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4 + C_3 L_1 g_m + 2 C_4 L_1 g_m) + s (C_3 R_1 g_m + C_3 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.849 \quad X-INVALID-ORDER-849} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 R_1 R_4 g_m s^3 + L_1 L_4 R_4 g_m s^2 + L_4 R_1 R_4 g_m s}{2 R_1 R_4 g_m + 2 R_4 + s^4 (C_1 C_3 L_1 L_4 R_1 R_4 g_m + C_1 C_3 L_1 L_4 R_4 + 2 C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_4) + s^3 (2 C_1 L_1 L_4 R_1 g_m + 2 C_1 L_1 L_4 + C_3 L_1 L_4 R_4 g_m + 2 C_4 L_1 L_4 R_4 g_m) + s^2 (2 C_1 L_1 R_1 R_4 g_m + 2 C_1 L_1 R_4 + C_3 L_4 R_1 R_4 g_m + C_3 L_4 R_4 + 2 C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_4 + L_1 L_4 R_1 R_4 g_m) + s (L_1 R_4 g_m + L_4 R_1 g_m) + R_1 R_4 g_m}$$

$$\mathbf{9.850 \quad X-INVALID-ORDER-850} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 R_4 g_m s^4 + R_1 R_4 g_m + s^3 (C_1 L_1 L_4 R_1 g_m + C_4 L_1 L_4 R_4 g_m) + s^2 (C_1 L_1 R_1 R_4 g_m + C_4 L_4 R_1 R_4 g_m + L_1 L_4 g_m) + s (L_1 R_4 g_m + L_4 R_1 g_m) + R_1 R_4 g_m}{2 R_1 g_m + s^5 (C_1 C_3 C_4 L_1 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_4) + s^4 (C_1 C_3 L_1 L_4 R_1 g_m + C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + C_3 C_4 L_1 L_4 R_4 g_m) + s^3 (C_1 C_3 L_1 R_1 R_4 g_m + C_1 C_3 L_1 R_4 + C_3 C_4 L_4 R_1 R_4 g_m + C_3 C_4 L_4 R_4 + C_3 L_1 L_4 g_m + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_1 L_1 R_1 g_m + 2 C_1 L_1 R_4 + C_3 L_4 R_1 g_m + C_3 L_4 R_4 + L_1 L_4 R_1 g_m) + s (L_1 R_4 g_m + L_4 R_1 g_m) + R_1 R_4 g_m}$$

$$\mathbf{9.851 \quad X-INVALID-ORDER-851} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 R_4 g_m s^4 + C_4 L_1 L_4 R_4 g_m s^3 + L_1 R_4 g_m s + R_1 R_4 g_m + s^2 (C_1 L_1 R_1 R_4 g_m + C_4 L_4 R_1 R_4 g_m) + s (L_1 R_4 g_m + L_4 R_1 g_m) + R_1 R_4 g_m}{2 R_1 g_m + s^5 (C_1 C_3 C_4 L_1 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_4) + s^4 (2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + C_3 C_4 L_1 L_4 R_4 g_m) + s^3 (C_1 C_3 L_1 R_1 R_4 g_m + C_1 C_3 L_1 R_4 + C_3 C_4 L_4 R_1 R_4 g_m + C_3 C_4 L_4 R_4 + C_3 L_1 L_4 g_m + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_1 L_1 R_1 g_m + 2 C_1 L_1 R_4 + C_3 L_4 R_1 g_m + C_3 L_4 R_4 + L_1 L_4 R_1 g_m) + s (L_1 R_4 g_m + L_4 R_1 g_m) + R_1 R_4 g_m}$$

$$\mathbf{9.852 \quad X-INVALID-ORDER-852} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_1 R_3 R_4 g_m s^2 + L_1 R_3 R_4 g_m s + R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 g_m + C_1 C_3 L_1 R_3 R_4) + s^2 (2 C_1 L_1 R_1 R_3 g_m + C_1 L_1 R_1 R_4 g_m + 2 C_1 L_1 R_3 + C_1 L_1 R_4 + C_3 L_1 R_3 R_4 g_m) + s (C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2 L_1 R_3 g_m + L_1 R_4 g_m) + R_1 R_3 R_4 g_m}$$

$$\mathbf{9.853 \quad X-INVALID-ORDER-853} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_1 R_3 g_m s^2 + L_1 R_3 g_m s + R_1 R_3 g_m}{R_1 g_m + s^3 (C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_3 + 2 C_1 C_4 L_1 R_1 R_3 g_m + 2 C_1 C_4 L_1 R_3) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 + C_3 L_1 R_3 g_m + 2 C_4 L_1 R_3 g_m) + s (C_3 R_1 R_3 g_m + C_3 R_3 + 2 C_4 R_1 R_3 g_m + 2 C_4 R_3 + L_1 g_m) + 1}$$

$$\mathbf{9.854 \quad X-INVALID-ORDER-854} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_1 R_3 R_4 g_m s^2 + L_1 R_3 R_4 g_m s + R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 g_m + C_1 C_3 L_1 R_3 R_4 + 2 C_1 C_4 L_1 R_1 R_3 R_4 g_m + 2 C_1 C_4 L_1 R_3 R_4) + s^2 (2 C_1 L_1 R_1 R_3 g_m + C_1 L_1 R_1 R_4 g_m + 2 C_1 L_1 R_3 + C_1 L_1 R_4 + C_3 L_1 R_3 R_4 g_m + 2 C_4 L_1 R_3 R_4 g_m) + s (C_3 R_1 R_3 R_4 g_m + C_3 R_3 R_4 + 2 C_4 R_1 R_3 R_4 g_m + 2 C_4 R_3 R_4 + L_1 R_3 g_m) + R_1 R_3 R_4 g_m}$$

$$\mathbf{9.855 \quad X-INVALID-ORDER-855} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 R_1 R_3 R_4 g_m s^3 + R_1 R_3 g_m + s^2 (C_1 L_1 R_1 R_3 g_m + C_4 L_1 R_3 R_4 g_m) + s (C_4 R_1 R_3 R_4 g_m + L_1 R_3 g_m) + R_1 R_3 g_m}{R_1 g_m + s^4 (C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 R_3 R_4) + s^3 (C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_3 + 2 C_1 C_4 L_1 R_1 R_3 g_m + C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4 + C_3 C_4 L_1 R_3 R_4 g_m) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 + C_3 C_4 R_1 R_3 R_4 g_m + C_3 C_4 R_3 R_4 + C_3 L_1 R_3 g_m + 2 C_4 L_1 R_3 g_m) + s (C_3 R_1 R_3 g_m + C_3 R_3 + 2 C_4 R_1 R_3 g_m + 2 C_4 R_3 + L_1 g_m) + R_1 R_3 g_m}$$

$$\mathbf{9.856 \quad X-INVALID-ORDER-856} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 R_3 g_m s^4 + C_4 L_1 L_4 R_3 g_m s^3 + L_1 R_3 g_m s + R_1 R_3 g_m + s^2 (C_1 L_1 R_1 R_3 g_m + C_4 L_4 R_1 R_3 g_m) + s (C_4 R_1 R_3 g_m + L_1 R_3 g_m) + R_1 R_3 g_m}{R_1 g_m + s^5 (C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_3) + s^4 (C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + C_3 C_4 L_1 L_4 R_3 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_3 + 2 C_1 C_4 L_1 R_1 R_3 g_m + 2 C_1 C_4 L_1 R_3 + C_3 C_4 L_4 R_1 R_3 g_m + C_3 C_4 L_4 R_3 + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 + C_3 L_4 R_1 R_3 g_m + C_3 L_4 R_3 + L_1 L_4 R_1 g_m) + s (L_1 R_4 g_m + L_4 R_1 g_m) + R_1 R_4 g_m}$$

$$\mathbf{9.857 \quad X-INVALID-ORDER-857} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 R_1 R_3 g_m s^3 + L_1 L_4 R_3 g_m s^2 + L_4 R_1 R_3 g_m s}{2 R_1 R_3 g_m + 2 R_3 + s^4 (C_1 C_3 L_1 L_4 R_1 R_3 g_m + C_1 C_3 L_1 L_4 R_3 + 2 C_1 C_4 L_1 L_4 R_1 R_3 g_m + 2 C_1 C_4 L_1 L_4 R_3) + s^3 (C_1 L_1 L_4 R_1 g_m + C_1 L_1 L_4 + C_3 L_1 L_4 R_3 g_m + 2 C_4 L_1 L_4 R_3 g_m) + s^2 (2 C_1 L_1 R_1 R_3 g_m + 2 C_1 L_1 R_3 + C_3 L_4 R_1 R_3 g_m + C_3 L_4 R_3 + 2 C_4 L_4 R_1 R_3 g_m + 2 C_4 L_4 R_3 + L_1 L_4 R_1 g_m) + s (L_1 R_4 g_m + L_4 R_1 g_m) + R_1 R_4 g_m}$$

9.858 X-INVALID-ORDER-858 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 R_3 g_m s^4 + R_1 R_3 g_m + s^3 (C_1 C_4 L_1 R_1 R_3 R_4 g_m + C_4 L_1 L_4 R_3 g_m)}{R_1 g_m + s^5 (C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_3) + s^4 (C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 R_3 R_4 + C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + C_3 C_4 L_1 L_4 R_3 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_3 + 2C_1 C_4 L_1 R_1 R_3 g_m + C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4 +$$

9.859 X-INVALID-ORDER-859 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_4 R_1 R_3 R_4 g_m s^3 + L_1 L_4 R_3 R_4 g_m s^2 + L_4 R_1 R_3 R_4 g_m s}{2R_1 R_3 R_4 g_m + 2R_3 R_4 + s^4 (C_1 C_3 L_1 L_4 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_4 R_3 R_4 + 2C_1 C_4 L_1 L_4 R_1 R_3 R_4 g_m + 2C_1 C_4 L_1 L_4 R_3 R_4) + s^3 (2C_1 L_1 L_4 R_1 R_3 g_m + C_1 L_1 L_4 R_1 R_4 g_m + 2C_1 L_1 L_4 R_3 + C_1 L_1 L_4 R_4 + C_3 L_1 L_4 R_3 R_4 g_m + 2C_4 L_1 L_4 R_3 R_4 g_m) + s^2 (2C_1 L_1 R_1 R_3 R_4 g_m + 2C_1 L_1 R_1 R_3 R_4 + 2C_1 L_1 R_1 R_4 + 2C_1 L_1 R_3 R_4 + 2C_1 L_1 R_4 + C_3 L_1 R_1 R_3 R_4 g_m + 2C_3 L_1 R_1 R_3 R_4 + 2C_3 L_1 R_1 R_4 + 2C_3 L_1 R_3 R_4 + 2C_3 L_1 R_4 + C_4 L_1 R_1 R_3 R_4 g_m + 2C_4 L_1 R_1 R_3 R_4 + 2C_4 L_1 R_1 R_4 + 2C_4 L_1 R_3 R_4 + 2C_4 L_1 R_4 + C_4 L_3 R_1 R_3 R_4 g_m + 2C_4 L_3 R_1 R_3 R_4 + 2C_4 L_3 R_1 R_4 + 2C_4 L_3 R_3 R_4 + 2C_4 L_3 R_4 + C_4 L_4 R_1 R_3 R_4 g_m + 2C_4 L_4 R_1 R_3 R_4 + 2C_4 L_4 R_1 R_4 + 2C_4 L_4 R_3 R_4 + 2C_4 L_4 R_4)}$$

9.860 X-INVALID-ORDER-860 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 R_3 R_4 g_m s^4 + (2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^5 (C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4) + s^4 (C_1 C_3 L_1 L_4 R_1 R_3 g_m + C_1 C_3 L_1 L_4 R_3 + 2C_1 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4 + C_3 C_4 L_1 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 g_m + C_1 C_3 L_1 R_1 R_3 R_4) + s^2 (C_1 C_3 L_1 R_1 R_3 R_4) + s (C_1 C_3 L_1 R_1 R_3 R_4) + C_1 C_3 L_1 R_1 R_3 R_4)}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^5 (C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4) + s^4 (C_1 C_3 L_1 L_4 R_1 R_3 g_m + C_1 C_3 L_1 L_4 R_3 + 2C_1 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4 + C_3 C_4 L_1 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 g_m + C_1 C_3 L_1 R_1 R_3 R_4) + s^2 (C_1 C_3 L_1 R_1 R_3 R_4) + s (C_1 C_3 L_1 R_1 R_3 R_4) + C_1 C_3 L_1 R_1 R_3 R_4}$$

9.861 X-INVALID-ORDER-861 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 R_3 R_4 g_m s^4 + 2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^5 (C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4) + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4 + C_3 C_4 L_1 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 g_m + C_1 C_3 L_1 R_3 R_4 + 2 C_1 C_4 L_1 R_1 R_3 R_4 g_m + 2 C_1 C_4 L_1 R_3 R_4 + 2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_3 g_m + 2 C_1 C_4 L_1 R_4 g_m + 2 C_3 C_4 L_1 R_1 R_3 g_m + 2 C_3 C_4 L_1 R_1 R_4 g_m + 2 C_3 C_4 L_1 R_3 g_m + 2 C_3 C_4 L_1 R_4 g_m + 2 C_3 C_4 L_1 R_3 R_4 + 2 C_3 C_4 L_1 R_4 R_4)}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4 + s^5 (C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4) + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4 + C_3 C_4 L_1 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 g_m + C_1 C_3 L_1 R_3 R_4 + 2 C_1 C_4 L_1 R_1 R_3 R_4 g_m + 2 C_1 C_4 L_1 R_3 R_4 + 2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_3 g_m + 2 C_1 C_4 L_1 R_4 g_m + 2 C_3 C_4 L_1 R_1 R_3 g_m + 2 C_3 C_4 L_1 R_1 R_4 g_m + 2 C_3 C_4 L_1 R_3 g_m + 2 C_3 C_4 L_1 R_4 g_m + 2 C_3 C_4 L_1 R_3 R_4 + 2 C_3 C_4 L_1 R_4 R_4)}$$

9.862 X-INVALID-ORDER-862 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 R_1 R_3 R_4 g_m s^3 + R_1 R_4 g_m + s^2 (C_1 L_1 R_1 R_4 g_m + C_3 L_1 R_3 R_4 g_m) + s (C_3 R_1 R_3 R_4 g_m + L_1 R_4 g_m)}{2 R_1 g_m + s^3 (2 C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_1 R_4 g_m + 2 C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4) + s^2 (2 C_1 L_1 R_1 g_m + 2 C_1 L_1 + 2 C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m) + s (2 C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2 C_3 R_3 + C_3 R_4 + 2 L_1 g_m) + 2}$$

9.863 X-INVALID-ORDER-863 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 R_1 R_3 g_m s^3 + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_3 L_1 R_3 g_m) + s (C_3 R_1 R_3 g_m + L_1 g_m)}{s^4 (2C_1 C_3 C_4 L_1 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 R_3) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_3 C_4 L_1 R_3 g_m) + s^2 (2C_3 C_4 R_1 R_3 g_m + 2C_3 C_4 R_3 + C_3 L_1 g_m + 2C_4 L_1 g_m) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

9.864 X-INVALID-ORDER-864 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 R_1 R_3 R_4 g_m s^3 + R_1 R_4 g_m + s^2 (C_1 L_1 R_1 R_4 g_m + C_3 L_1 R_3 R_4 g_m) + s (C_3 R_1 R_3 R_4 g_m + L_1 R_4 g_m)}{2 R_1 g_m + s^4 (2 C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_1 R_3 R_4) + s^3 (2 C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_1 R_4 g_m + 2 C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4 + 2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_4 + 2 C_3 C_4 L_1 R_3 R_4 g_m) + s^2 (2 C_1 L_1 R_1 g_m + 2 C_1 L_1 + 2 C_3 C_4 R_1 R_3 R_4 g_m + 2 C_3 C_4 R_3 R_4 + 2 C_3 L_1 R_1 g_m) + s (C_3 R_1 R_3 g_m + L_1 R_4 g_m) + R_1 R_4 g_m}.$$

9.865 X-INVALID-ORDER-865 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m s^4 + R_1 g_m + s^3 (C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_4 L_1 R_1 R_4 g_m + C_3 C_4 L_1 R_3 R_4 g_m) + s^2 (C_1 L_1 R_1 g_m + C_3 C_4 R_1 R_3 R_4 g_m + C_3 L_1 R_3 g_m + C_4 L_1 R_4 g_m) + s (C_3 R_1 R_3 g_m + C_4 R_1 R_4 g_m + L_1 g_m)}{s^4 (2C_1 C_3 C_4 L_1 R_1 R_3 g_m + C_1 C_3 C_4 L_1 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 R_3 + C_1 C_3 C_4 L_1 R_4) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_1 R_4 g_m) + s^2 (2C_3 C_4 R_1 R_3 g_m + C_3 C_4 R_1 R_4 g_m + 2C_3 C_4 R_3 + C_3 C_4 R_4 + C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

9.866 X-INVALID-ORDER-866 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m s^5 + R_1 g_m + s^4 (C_1 C_4 L_1 L_4 R_1 g_m + C_3 C_4 L_1 L_4 R_3 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 g_m + C_3 C_4 L_4 R_1 R_3 g_m + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 R_1 g_m + C_3 L_1 R_3 g_m + C_4 L_4 R_1 g_m) + s (C_3 R_1 R_3 g_m + L_1 g_m)}{s^5 (C_1 C_3 C_4 L_1 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_4) + s^4 (2 C_1 C_3 C_4 L_1 R_1 R_3 g_m + 2 C_1 C_3 C_4 L_1 R_3 + C_3 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1 + 2 C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (2 C_3 C_4 R_1 R_3 g_m + 2 C_3 C_4 R_3 + C_3 L_1 g_m + 2 C_4 L_1)}$$

9.867 X-INVALID-ORDER-867 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_4 R_1 R_3 g_m s^4 + L_4 R_1 g_m s + s^3 (C_1 L_1 L_4 R_1 g_m + C_3 L_1 L_4 R_3 g_m) + s^2 (C_3 L_4 R_1 R_3 g_m + L_1 L_4 g_m)}{2 R_1 g_m + s^5 (2 C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + 2 C_1 C_3 C_4 L_1 L_4 R_3) + s^4 (C_1 C_3 L_1 L_4 R_1 g_m + C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_1 L_4 R_3 g_m) + s^3 (2 C_1 C_3 L_1 R_1 R_3 g_m + 2 C_1 C_3 L_1 R_3 + 2 C_3 C_4 L_4 R_1 R_3 g_m + 2 C_3 C_4 L_4 R_3 + C_3 L_1 L_4 g_m + 2 C_4 L_1 L_4 g_m) + s^2 (2$$

9.868 X-INVALID-ORDER-868 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m s^5 + R_1 g_m + s^4 (C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_1 C_4 L_1 L_4 R_1 g_m + C_3 C_4 L_1 L_4 R_3 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_4 L_1 R_1 R_4 g_m + C_3 C_4 L_1 R_3 R_4 g_m + C_3 C_4 L_4 R_1 R_3 g_m + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 R_1 g_m + C_3 C_4 R_1 L_1 g_m)}{s^5 (C_1 C_3 C_4 L_1 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_4) + s^4 (2 C_1 C_3 C_4 L_1 R_1 R_3 g_m + C_1 C_3 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_1 R_3 + C_1 C_3 C_4 L_1 R_4 + C_3 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1 + 2 C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_1 R_4 g_m + C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4 g_m)}$$

9.869 X-INVALID-ORDER-869 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_4 R_1 R_3 R_4}{2R_1 R_4 g_m + 2R_4 + s^5 (2C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_1 L_4 R_3 R_4) + s^4 (2C_1 C_3 L_1 L_4 R_1 R_3 g_m + C_1 C_3 L_1 L_4 R_1 R_4 g_m + 2C_1 C_3 L_1 L_4 R_3 + C_1 C_3 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_4 R_4 + 2C_3 C_4 L_1 L_4 R_3 R_4 g_m) + s^3 (2C_1 C_3 L_1 R_1 R_3 R_4 g_m + 2C_1 C_3 L_1 R_3}$$

9.870 X-INVALID-ORDER-870 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m s^5 + R_1 R_4 g_m + s^4 (C_1 C_3 L_1 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_4 R_1 R_4 g_m + C_3 C_4 L_1 L_4 R_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 g_m + C_1 L_1 L_4 R_1 g_m}{2 R_1 g_m + s^5 (2 C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_1 L_4 R_3 + C_1 C_3 C_4 L_1 L_4 R_4) + s^4 (C_1 C_3 L_1 L_4 R_1 g_m + C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_1 L_4 R_4 g_m) + s^3 (2 C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_1 R_4 g_m + 2 C_1 C_3 L_1 R_1 R_3 + C_1 C_3 L_1 R_1 R_4) + s^2 (2 C_1 C_3 L_1 R_1 R_3 + C_1 C_3 L_1 R_1 R_4) + s (2 C_1 C_3 L_1 R_1 + C_1 C_3 L_1 R_1) + C_1 C_3 L_1 R_1}.$$

9.871 X-INVALID-ORDER-871 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m s^5 + R_1 R_4 g_m + s^4 (C_1 C_4 + 2 C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_1 L_4 R_3 + C_1 C_3 C_4 L_1 L_4 R_4) + s^4 (2 C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_1 R_3 R_4 + 2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_1 L_4 R_4 g_m) + s^3 (2 C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_1 R_3 + 2 C_1 C_3 L_1 R_1 R_4 + 2 C_1 C_3 L_1 R_3 R_4 + 2 C_1 C_3 L_1 R_4 R_3 + C_1 C_3 L_1 R_4 R_4) + s^2 (2 C_1 C_3 L_1 R_1 R_3 + C_1 C_3 L_1 R_1 R_4 + C_1 C_3 L_1 R_3 R_4 + C_1 C_3 L_1 R_4 R_3 + C_1 C_3 L_1 R_4 R_4) + s (C_1 C_3 L_1 R_1 R_3 + C_1 C_3 L_1 R_1 R_4 + C_1 C_3 L_1 R_3 R_4 + C_1 C_3 L_1 R_4 R_3 + C_1 C_3 L_1 R_4 R_4) + C_1 C_3 L_1 R_1 R_3 + C_1 C_3 L_1 R_1 R_4 + C_1 C_3 L_1 R_3 R_4 + C_1 C_3 L_1 R_4 R_3 + C_1 C_3 L_1 R_4 R_4}{2 R_1 g_m + s^5 (2 C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_1 L_4 R_3 + C_1 C_3 C_4 L_1 L_4 R_4) + s^4 (2 C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_1 R_3 R_4 + 2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_1 L_4 R_4 g_m) + s^3 (2 C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_1 R_3 + 2 C_1 C_3 L_1 R_1 R_4 + 2 C_1 C_3 L_1 R_3 R_4 + 2 C_1 C_3 L_1 R_4 R_3 + C_1 C_3 L_1 R_4 R_4) + s^2 (2 C_1 C_3 L_1 R_1 R_3 + C_1 C_3 L_1 R_1 R_4 + C_1 C_3 L_1 R_3 R_4 + C_1 C_3 L_1 R_4 R_3 + C_1 C_3 L_1 R_4 R_4) + s (C_1 C_3 L_1 R_1 R_3 + C_1 C_3 L_1 R_1 R_4 + C_1 C_3 L_1 R_3 R_4 + C_1 C_3 L_1 R_4 R_3 + C_1 C_3 L_1 R_4 R_4) + C_1 C_3 L_1 R_1 R_3 + C_1 C_3 L_1 R_1 R_4 + C_1 C_3 L_1 R_3 R_4 + C_1 C_3 L_1 R_4 R_3 + C_1 C_3 L_1 R_4 R_4}.$$

9.872 X-INVALID-ORDER-872 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, L_3 s + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 R_4 g_m s^4 + C_3 L_1 L_3 R_4 g_m s^3 + L_1 R_4 g_m s + R_1 R_4 g_m + s^2 (C_1 L_1 R_1 R_4 g_m + C_3 L_3 R_1 R_4 g_m)}{2 R_1 g_m + s^4 (2 C_1 C_3 L_1 L_3 R_1 g_m + 2 C_1 C_3 L_1 L_3) + s^3 (C_1 C_3 L_1 R_1 R_4 g_m + C_1 C_3 L_1 R_4 + 2 C_3 L_1 L_3 g_m) + s^2 (2 C_1 L_1 R_1 g_m + 2 C_1 L_1 + C_3 L_1 R_4 g_m + 2 C_3 L_3 R_1 g_m + 2 C_3 L_3) + s (C_3 R_1 R_4 g_m + C_3 R_4 + 2 L_1 g_m) + 2}$$

9.873 X-INVALID-ORDER-873 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 g_m s^4 + C_3 L_1 L_3 g_m s^3 + L_1 g_m s + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_3 L_3 R_1 g_m)}{2C_3 C_4 L_1 L_3 g_m s^4 + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 g_m + 2C_1 C_3 C_4 L_1 L_3) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_3 C_4 L_3 R_1 g_m + 2C_3 C_4 L_3) + s^2 (C_3 L_1 g_m + 2C_4 L_1 g_m) + s (C_3 R_1 g_m + C_3 + 2C_4 R_1 g_m + 2C_4)}$$

9.874 X-INVALID-ORDER-874 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, L_3 s + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 R_4 g_m s^4 + C_3 L_1 L_3 R_4 g_m s^3 + L_1 R_4 g_m s + R_1 R_4 g_m + s^2 (C_1 L_1 R_1 R_4 g_m + C_3 L_3 R_1 R_4 g_m)}{2 R_1 g_m + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_1 L_3 R_4) + s^4 (2 C_1 C_3 L_1 L_3 R_1 g_m + 2 C_1 C_3 L_1 L_3 + 2 C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_1 R_4 g_m + C_1 C_3 L_1 R_4 + 2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_4 + 2 C_3 C_4 L_3 R_1 R_4 g_m + 2 C_3 C_4 L_3 R_4 + 2 C_3 L_1 L_3 g_m) + s^2 (2 C_1 L_1 R_1 g_m +$$

9.875 X-INVALID-ORDER-875 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, L_3 s + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m s^5 + R_1 g_m + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (C_1 C_4 L_1 R_1 R_4 g_m + C_3 C_4 L_3 R_1 R_4 g_m + C_3 L_1 L_3 g_m) + s^2 (C_1 L_1 R_1 g_m + C_3 L_3 R_1 g_m + C_4 L_1 R_4 g_m) + s (C_4 R_1 R_4 g_m + L_1 g_m)}{s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 g_m + 2C_1 C_3 C_4 L_1 L_3) + s^4 (C_1 C_3 C_4 L_1 R_1 R_4 g_m + C_1 C_3 C_4 L_1 R_4 + 2C_3 C_4 L_1 L_3 g_m) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + C_3 C_4 L_1 R_4 g_m + 2C_3 C_4 L_3 R_1 g_m + 2C_3 C_4 L_3) + s^2 (C_3 C_4 R_1 R_4 g_m + C_3 C_4 R_4 + C_3 L_1 g_m + 2C_4 L_1)}$$

$$\mathbf{9.876 \quad X-INVALID-ORDER-876} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m s^6 + C_3 C_4 L_1 L_3 L_4 g_m s^5 + L_1 g_m s + R_1 g_m + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_4 L_1 L_4 R_1 g_m + C_3 C_4 L_3 L_4 R_1 g_m) + s^3 (C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 R_1 g_m + C_3 L_3 R_1 g_m + C_4 L_4 R_1 g_m)}{s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 g_m + 2C_1 C_3 C_4 L_1 L_3 + C_1 C_3 C_4 L_1 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_4) + s^4 (2C_3 C_4 L_1 L_3 g_m + C_3 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_3 C_4 L_3 R_1 g_m + 2C_3 C_4 L_3 + C_3 C_4 L_4 R_1 g_m + C_3 C_4 L_4) + s^2 (C_3 L_1 g_m + 2C_4 L_1 R_1 g_m + C_3 L_1 L_3 + C_4 L_1 L_4) + s (L_1 g_m + R_1 g_m) + R_1 g_m}$$

$$\mathbf{9.877 \quad X-INVALID-ORDER-877} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 L_4 R_1 g_m s^5 + C_3 L_1 L_3 L_4 g_m s^4 + L_1 L_4 g_m s^2 + L_4 R_1 g_m s + s^3 (C_1 L_1 L_4 R_1 g_m + C_3 L_3 L_4 R_1 g_m)}{2C_3 C_4 L_1 L_3 L_4 g_m s^5 + 2L_1 g_m s + 2R_1 g_m + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4) + s^4 (2C_1 C_3 L_1 L_3 R_1 g_m + 2C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 R_1 g_m + C_1 C_3 L_1 L_4 + 2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_3 C_4 L_3 L_4 R_1 g_m + 2C_3 C_4 L_3 L_4) + s^3 (2C_3 L_1 L_3 g_m + C_3 L_1 L_4 g_m) + s^2 (C_3 L_1 R_1 g_m + C_4 L_1 R_1 g_m) + s (L_1 g_m + R_1 g_m) + R_1 g_m}$$

$$\mathbf{9.878 \quad X-INVALID-ORDER-878} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m s^6 + R_1 g_m + s^5 (C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_4 L_1 L_4 R_1 g_m + C_3 C_4 L_1 L_3 R_4 g_m + C_3 C_4 L_3 L_4 R_1 g_m) + s^3 (C_1 C_4 L_1 R_1 R_4 g_m + C_3 C_4 L_3 R_1 R_4 g_m + C_3 L_1 L_3 g_m + C_4 L_1 L_4 g_m)}{s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 g_m + 2C_1 C_3 C_4 L_1 L_3 + C_1 C_3 C_4 L_1 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_4) + s^4 (C_1 C_3 C_4 L_1 R_1 R_4 g_m + C_1 C_3 C_4 L_1 R_4 + 2C_3 C_4 L_1 L_3 g_m + C_3 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + C_3 C_4 L_1 R_4 g_m + 2C_3 C_4 L_3 R_1 g_m + 2C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_4 R_1 g_m) + s^2 (C_3 L_1 R_1 g_m + C_4 L_1 R_1 g_m) + s (L_1 g_m + R_1 g_m) + R_1 g_m}$$

$$\mathbf{9.879 \quad X-INVALID-ORDER-879} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 L_4 R_1 R_4 g_m s^6 + R_1 R_4 g_m + s^5 (2C_1 C_3 L_1 L_3 L_4 R_1 g_m + 2C_1 C_3 L_1 L_3 L_4 + 2C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 L_1 L_3 R_4 + C_1 C_3 L_1 L_4 R_1 R_4 g_m + C_1 C_3 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_4 R_4) + s^3 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 R_1 g_m + C_1 C_3 L_1 L_4 + 2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4 R_4) + s^2 (C_3 L_1 R_1 g_m + C_4 L_1 R_1 g_m) + s (L_1 g_m + R_1 g_m) + R_1 g_m}{2R_1 R_4 g_m + 2R_4 + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (2C_1 C_3 L_1 L_3 L_4 R_1 g_m + 2C_1 C_3 L_1 L_3 L_4 + 2C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 L_1 L_3 R_4 + C_1 C_3 L_1 L_4 R_1 R_4 g_m + C_1 C_3 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_4 R_4) + s^3 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 R_1 g_m + C_1 C_3 L_1 L_4 + 2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4 R_4) + s^2 (C_3 L_1 R_1 g_m + C_4 L_1 R_1 g_m) + s (L_1 g_m + R_1 g_m) + R_1 g_m}$$

$$\mathbf{9.880 \quad X-INVALID-ORDER-880} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m s^6 + R_1 R_4 g_m + s^5 (C_1 C_3 L_1 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 R_4 g_m + C_1 C_4 L_1 L_4 R_1 R_4 g_m + C_3 C_4 L_3 L_4 R_1 R_4 g_m + C_3 C_4 L_3 L_4 R_4) + s^3 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 R_1 g_m + C_1 C_3 L_1 L_4 + 2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4 R_4) + s^2 (C_3 L_1 R_1 g_m + C_4 L_1 R_1 g_m) + s (L_1 g_m + R_1 g_m) + R_1 g_m}{2R_1 g_m + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4) + s^5 (C_1 C_3 C_4 L_1 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_4 + 2C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_1 g_m + 2C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 R_1 g_m + C_1 C_3 L_1 L_4 + 2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4 R_4) + s^3 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 R_1 g_m + C_1 C_3 L_1 L_4 + 2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4 R_4) + s^2 (C_3 L_1 R_1 g_m + C_4 L_1 R_1 g_m) + s (L_1 g_m + R_1 g_m) + R_1 g_m}$$

$$\mathbf{9.881 \quad X-INVALID-ORDER-881} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m s^6 + C_3 C_4 L_1 L_3 L_4 R_4 g_m s^5 + R_1 R_4 g_m + s^4 (2C_1 C_3 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 L_1 L_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_4 + 2C_3 C_4 L_1 L_3 L_4 g_m) + s^3 (2C_1 C_3 L_1 L_3 R_1 g_m + 2C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4 R_4) + s^2 (C_3 L_1 R_1 g_m + C_4 L_1 R_1 g_m) + s (L_1 g_m + R_1 g_m) + R_1 g_m}{2R_1 g_m + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4) + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_4 + 2C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 L_1 L_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_4 + 2C_3 C_4 L_1 L_3 L_4 g_m) + s^3 (2C_1 C_3 L_1 L_3 R_1 g_m + 2C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4 R_4) + s^2 (C_3 L_1 R_1 g_m + C_4 L_1 R_1 g_m) + s (L_1 g_m + R_1 g_m) + R_1 g_m}$$

$$\mathbf{9.882 \quad X-INVALID-ORDER-882} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_3 R_1 R_4 g_m s^3 + L_1 L_3 R_4 g_m s^2 + L_3 R_1 R_4 g_m s}{R_1 R_4 g_m + R_4 + s^4 (C_1 C_3 L_1 L_3 R_1 R_4 g_m + C_1 C_3 L_1 L_3 R_4) + s^3 (2C_1 L_1 L_3 R_1 g_m + 2C_1 L_1 L_3 + C_3 L_1 L_3 R_4 g_m) + s^2 (C_1 L_1 R_1 R_4 g_m + C_1 L_1 R_4 + C_3 L_3 R_1 R_4 g_m + C_3 L_3 R_4 + 2L_1 L_3 g_m) + s (L_1 R_4 g_m + 2L_3 R_1 g_m + 2L_3)}$$

$$\mathbf{9.883 \quad X-INVALID-ORDER-883} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_3 R_1 g_m s^3 + L_1 L_3 g_m s^2 + L_3 R_1 g_m s}{L_1 g_m s + R_1 g_m + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 R_1 g_m + 2C_1 C_4 L_1 L_3) + s^3 (C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 + C_3 L_3 R_1 g_m + C_3 L_3 + 2C_4 L_3 R_1 g_m + 2C_4 L_3) + 1}$$

$$\mathbf{9.884 \quad X-INVALID-ORDER-884} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_3 R_1 R_4 g_m s^3 + L_1 L_3 R_4 g_m s^2 + L_3 R_1 R_4 g_m s}{R_1 R_4 g_m + R_4 + s^4 (C_1 C_3 L_1 L_3 R_1 R_4 g_m + C_1 C_3 L_1 L_3 R_4 + 2C_1 C_4 L_1 L_3 R_1 R_4 g_m + 2C_1 C_4 L_1 L_3 R_4) + s^3 (2C_1 L_1 L_3 R_1 g_m + 2C_1 L_1 L_3 + C_3 L_1 L_3 R_4 g_m + 2C_4 L_1 L_3 R_4 g_m) + s^2 (C_1 L_1 R_1 R_4 g_m + C_1 L_1 R_4 + C_3 L_3 R_1 R_4 g_m + C_3 L_3 R_4 + 2C_4 L_3 R_1 R_4 g_m + 2C_4 L_3 R_4 + 2L_1 L_3 g_m) + s (L_1 R_4 g_m + 2L_3 R_1 g_m + 2L_3)}$$

$$\mathbf{9.885 \quad X-INVALID-ORDER-885} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_3 R_1 R_4 g_m s^4 + L_3 R_1 g_m s + s^3 (C_1 L_1 L_3 R_1 g_m + C_4 L_1 L_3 R_4 g_m) + s^2 (C_4 L_3 R_1 R_4 g_m + L_1 L_3 g_m)}{R_1 g_m + s^5 (C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + C_1 C_3 C_4 L_1 L_3 R_4) + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 R_1 g_m + 2C_1 C_4 L_1 L_3 + C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (C_1 C_4 L_1 R_1 R_4 g_m + C_1 C_4 L_1 R_4 + C_3 C_4 L_3 R_1 R_4 g_m + C_3 C_4 L_3 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s (C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + R_1 g_m + R_4}$$

$$\mathbf{9.886 \quad X-INVALID-ORDER-886} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_1 g_m s^5 + C_4 L_1 L_3 L_4 g_m s^4 + L_1 L_3 g_m s^2 + L_3 R_1 g_m s + s^3 (C_1 L_1 L_3 R_1 g_m + C_4 L_3 L_4 R_1 g_m)}{C_3 C_4 L_1 L_3 L_4 g_m s^5 + L_1 g_m s + R_1 g_m + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 R_1 g_m + 2C_1 C_4 L_1 L_3 + C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4) + s^3 (C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m + C_4 L_1 L_3) + s^2 (C_3 L_1 L_3 R_1 g_m + C_3 L_1 L_3 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s (C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + R_1 g_m + R_4}$$

$$\mathbf{9.887 \quad X-INVALID-ORDER-887} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_3 L_4 R_1 g_m s^3 + L_1 L_3 L_4 g_m s^2 + L_3 L_4 R_1 g_m s}{2L_3 R_1 g_m + 2L_3 + L_4 R_1 g_m + L_4 + s^4 (C_1 C_3 L_1 L_3 L_4 R_1 g_m + C_1 C_3 L_1 L_3 L_4 + 2C_1 C_4 L_1 L_3 L_4 R_1 g_m + 2C_1 C_4 L_1 L_3 L_4) + s^3 (C_3 L_1 L_3 L_4 g_m + 2C_4 L_1 L_3 L_4 g_m) + s^2 (2C_1 L_1 L_3 R_1 g_m + 2C_1 L_1 L_3 + C_1 L_1 L_4 R_1 g_m + C_1 L_1 L_4 + C_3 L_3 L_4 R_1 g_m + C_3 L_3 L_4 + 2C_4 L_3 L_4 R_1 g_m + 2C_4 L_3 L_4) + s (C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + R_1 g_m + R_4}$$

$$\mathbf{9.888 \quad X-INVALID-ORDER-888} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_1 g_m s^5 + L_3 R_1 g_m s + s^4 (C_1 C_4 L_1 L_3 R_1 R_4 g_m + C_4 L_1 L_3 L_4 g_m)}{R_1 g_m + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_3 L_4) + s^5 (C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + C_1 C_3 C_4 L_1 L_3 R_4 + C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 R_1 g_m + 2C_1 C_4 L_1 L_3 + C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + C_3 C_4 L_1 L_3 R_4 g_m + C_3 C_4 L_3 L_4 R_1 g_m) + s^3 (C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m + C_4 L_1 L_3) + s^2 (C_3 L_1 L_3 R_1 g_m + C_3 L_1 L_3 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s (C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + R_1 g_m + R_4}$$

$$\mathbf{9.889 \quad X-INVALID-ORDER-889} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_3 L_4 R_1 R_4 g_m s^3 + L_1 L_3 L_4 R_4 g_m s^2 + L_3 L_4 R_1 R_4 g_m s}{2L_3 R_1 R_4 g_m + 2L_3 R_4 + L_4 R_1 R_4 g_m + L_4 R_4 + s^4 (C_1 C_3 L_1 L_3 L_4 R_1 R_4 g_m + C_1 C_3 L_1 L_3 L_4 R_4 + 2C_1 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_3 L_4 R_4) + s^3 (2C_1 L_1 L_3 L_4 R_1 g_m + 2C_1 L_1 L_3 L_4 + C_3 L_1 L_3 L_4 R_4 g_m + 2C_4 L_1 L_3 L_4 R_4 g_m) + s^2 (2C_1 L_1 L_3 R_1 R_4 g_m + 2C_1 L_1 L_3 R_4 + C_1 L_1 L_4 R_1 R_4 g_m + C_1 L_1 L_4 R_4 + C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s (C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + R_1 R_4 g_m + R_4}$$

$$\mathbf{9.890 \quad X-INVALID-ORDER-890} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_1 R_4 g_m s^5 + L_3 R_1 R_4 g_m s + s^4 (C_1 C_4 L_1 L_3 R_1 R_4 g_m + C_4 L_1 L_3 L_4 g_m)}{R_1 R_4 g_m + R_4 + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (C_1 C_3 L_1 L_3 L_4 R_1 g_m + C_1 C_3 L_1 L_3 L_4 + 2C_1 C_4 L_1 L_3 L_4 R_1 g_m + 2C_1 C_4 L_1 L_3 L_4 + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 R_4 g_m + C_1 C_3 L_1 L_3 R_4 + C_1 C_4 L_1 L_4 R_1 R_4 g_m + C_1 C_4 L_1 L_4 R_4 + C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s^3 (2C_1 C_4 L_1 L_3 L_4 R_1 g_m + 2C_1 C_4 L_1 L_3 L_4 + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^2 (2C_1 L_1 L_3 R_1 R_4 g_m + 2C_1 L_1 L_3 R_4 + C_1 L_1 L_4 R_1 R_4 g_m + C_1 L_1 L_4 R_4 + C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s (C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + R_1 R_4 g_m + R_4}$$

$$\mathbf{9.891 \quad X-INVALID-ORDER-891} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_1 R_4 g_m s^5 + L_3 R_1 R_4 g_m s + s^4 (C_1 C_4 L_1 L_3 R_1 R_4 g_m + C_4 L_1 L_3 L_4 g_m)}{R_1 R_4 g_m + R_4 + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (2C_1 C_4 L_1 L_3 L_4 R_1 g_m + 2C_1 C_4 L_1 L_3 L_4 + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 R_4 g_m + C_1 C_3 L_1 L_3 R_4 + 2C_1 C_4 L_1 L_3 R_1 R_4 g_m + 2C_1 C_4 L_1 L_3 R_4 + C_1 C_4 L_1 L_4 R_1 R_4 g_m + C_1 C_4 L_1 L_4 R_4 + C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s^3 (2C_1 C_4 L_1 L_3 L_4 R_1 g_m + 2C_1 C_4 L_1 L_3 L_4 + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^2 (2C_1 L_1 L_3 R_1 R_4 g_m + 2C_1 L_1 L_3 R_4 + C_1 L_1 L_4 R_1 R_4 g_m + C_1 L_1 L_4 R_4 + C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s (C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + R_1 R_4 g_m + R_4}$$

$$\mathbf{9.892 \quad X-INVALID-ORDER-892} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 R_4 g_m s^4 + R_1 R_4 g_m + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 g_m + C_3 L_1 L_3 R_4 g_m) + s^2 (C_1 L_1 R_1 R_4 g_m + C_3 L_1 R_3 R_4 g_m + C_3 L_3 R_1 R_4 g_m) + s (C_3 R_1 R_3 R_4 g_m + L_1 R_4 g_m)}{2R_1 g_m + s^4 (2C_1 C_3 L_1 L_3 R_1 g_m + 2C_1 C_3 L_1 L_3) + s^3 (2C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_1 R_4 g_m + 2C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4 + 2C_3 L_1 L_3 g_m) + s^2 (2C_1 L_1 R_1 g_m + 2C_1 L_1 + 2C_3 L_1 R_3 g_m + C_3 L_1 R_4 g_m + 2C_3 L_3 R_1 g_m + 2C_3 L_3) + s (2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + R_1 g_m + R_4}$$

$$\mathbf{9.893 \quad X-INVALID-ORDER-893} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 g_m s^4 + R_1 g_m + s^3 (C_1 C_3 L_1 R_1 R_3 g_m + C_3 L_1 L_3 g_m) + s^2 (C_1 L_1 R_1 g_m + C_3 L_1 R_3 g_m + C_3 L_3 R_1 g_m) + s (C_3 R_1 R_3 g_m + L_1 g_m)}{s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 g_m + 2C_1 C_3 C_4 L_1 L_3) + s^4 (2C_1 C_3 C_4 L_1 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 R_3 + 2C_3 C_4 L_1 L_3 g_m) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_3 C_4 L_1 R_3 g_m + 2C_3 C_4 L_3 R_1 g_m + 2C_3 C_4 L_3) + s^2 (2C_3 C_4 R_1 R_3 g_m + 2C_3 C_4 R_3 + C_3 L_1 g_m + C_3 L_1) + s (C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2C_3 R_3 + C_3 R_4) + R_1 g_m + R_4}$$

9.894 X-INVALID-ORDER-894 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 R_4 g_m s^4 + R_1 R_4 g_m + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 g_m + 2 R_1 g_m + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_1 L_3 R_4) + s^4 (2 C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_1 R_3 R_4 + 2 C_1 C_3 L_1 L_3 R_1 g_m + 2 C_1 C_3 L_1 L_3 + 2 C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (2 C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_1 R_4 g_m + 2 C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4 + 2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_1 R_4) + s^2 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_1 L_3 R_4) + s (2 C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_1 R_3 R_4 + 2 C_1 C_3 L_1 L_3 R_1 g_m + 2 C_1 C_3 L_1 L_3 + 2 C_3 C_4 L_1 L_3 R_4 g_m) + C_1 C_3 L_1 L_3 R_1 R_4 g_m + R_1 R_4 g_m}{2 R_1 g_m + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_1 L_3 R_4) + s^4 (2 C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_1 R_3 R_4 + 2 C_1 C_3 L_1 L_3 R_1 g_m + 2 C_1 C_3 L_1 L_3 + 2 C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (2 C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_1 R_4 g_m + 2 C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4 + 2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_1 R_4) + s^2 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_1 L_3 R_4) + s (2 C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + 2 C_1 C_3 C_4 L_1 R_3 R_4 + 2 C_1 C_3 L_1 L_3 R_1 g_m + 2 C_1 C_3 L_1 L_3 + 2 C_3 C_4 L_1 L_3 R_4 g_m) + C_1 C_3 L_1 L_3 R_1 R_4 g_m + R_1 R_4 g_m}$$

9.895 X-INVALID-ORDER-895 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m s^5 + R_1 g_m + s^4 (C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 R_1 g_m + C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_4 L_1 R_1 R_4 g_m + C_3 C_4 L_1 R_3 R_4 g_m + C_3 C_4 L_3 R_1 R_4 g_m + C_3 L_1 L_3 g_m) + s^2 (C_1 L_1 R_1 g_m + C_3 C_4 L_1 L_3 g_m)}{s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 g_m + 2C_1 C_3 C_4 L_1 L_3) + s^4 (2C_1 C_3 C_4 L_1 R_1 R_3 g_m + C_1 C_3 C_4 L_1 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 R_3 + C_1 C_3 C_4 L_1 R_4 + 2C_3 C_4 L_1 L_3 g_m) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_1 R_4 g_m + 2C_3 C_4 L_3 R_1 g_m + 2C_3 C_4 L_3 g_m) + s^2 (C_1 L_1 R_1 g_m + C_3 C_4 L_1 L_3 g_m) + s (C_1 L_1 R_1 g_m + C_3 C_4 L_1 L_3 g_m) + R_1 g_m}.$$

9.896 X-INVALID-ORDER-896 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m s^6 + R_1 g_m + s^5 (C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_4 L_1 L_4 R_1 g_m + C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_3 L_4 R_1 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 g_m + C_3 C_4 L_4 R_1 R_3 g_m + C_3 L_1 L_3 g_m + C_4 L_1 L_3 g_m)}{s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 g_m + 2C_1 C_3 C_4 L_1 L_3 + C_1 C_3 C_4 L_1 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_4) + s^4 (2C_1 C_3 C_4 L_1 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 R_3 + 2C_3 C_4 L_1 L_3 g_m + C_3 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_3 C_4 L_1 R_3 g_m + 2C_3 C_4 L_3 R_1 g_m + 2C_3 C_4 L_3 g_m) + s^2 (C_1 C_3 L_1 + C_1 C_4 L_1 + C_3 C_4 L_1 + C_3 C_4 L_3) + s (C_1 C_3 + C_1 C_4 + C_3 C_4) + 1}.$$

9.897 X-INVALID-ORDER-897 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 L_4 R_1 g_m s^5 + L_4 R_1 g_m s + s^4 (C_1 C_3 L_1 L_4 R_1 R_3 g_m - 2 C_1 C_3 C_4 L_1 L_3 L_4) + s^5 (2 C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + 2 C_1 C_3 C_4 L_1 L_4 R_3 + 2 C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (2 C_1 C_3 L_1 L_3 R_1 g_m + 2 C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 R_1 g_m + C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_1 L_4 R_3 g_m + 2 C_3 L_1 L_3 L_4 R_1 g_m + 2 C_3 L_1 L_3 L_4 R_3 + 2 C_3 L_1 L_3 L_4 g_m)}{2 R_1 g_m + s^6 (2 C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m + 2 C_1 C_3 C_4 L_1 L_3 L_4) + s^5 (2 C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + 2 C_1 C_3 C_4 L_1 L_4 R_3 + 2 C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (2 C_1 C_3 L_1 L_3 R_1 g_m + 2 C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 R_1 g_m + C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_1 L_4 R_3 g_m + 2 C_3 L_1 L_3 L_4 R_1 g_m + 2 C_3 L_1 L_3 L_4 R_3 + 2 C_3 L_1 L_3 L_4 g_m) + s^3 (2 C_1 C_3 L_1 L_3 R_1 + 2 C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 R_1 + C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4 R_1 + 2 C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_1 L_4 R_3 + 2 C_3 L_1 L_3 L_4 R_1 + 2 C_3 L_1 L_3 L_4 R_3 + 2 C_3 L_1 L_3 L_4) + s^2 (2 C_1 C_3 L_1 L_3 R_1 + 2 C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 R_1 + C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4 R_1 + 2 C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_1 L_4 R_3 + 2 C_3 L_1 L_3 L_4 R_1 + 2 C_3 L_1 L_3 L_4 R_3 + 2 C_3 L_1 L_3 L_4) + s (2 C_1 C_3 L_1 L_3 R_1 + 2 C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 R_1 + C_1 C_3 L_1 L_4 + 2 C_1 C_4 L_1 L_4 R_1 + 2 C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_1 L_4 R_3 + 2 C_3 L_1 L_3 L_4 R_1 + 2 C_3 L_1 L_3 L_4 R_3 + 2 C_3 L_1 L_3 L_4) + 1}$$

9.898 X-INVALID-ORDER-898 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m s^6 + R_1 g_m + s^5 (C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_4 L_1 L_4 R_1 g_m + C_3 C_4 L_1 L_3 R_4 g_m + C_3 C_4 L_1 L_4 R_3 g_m + C_3 C_4 L_3 L_4 R_1 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_4 L_1 R_1 R_3 g_m)}{s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 g_m + 2C_1 C_3 C_4 L_1 L_3 + C_1 C_3 C_4 L_1 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_4) + s^4 (2C_1 C_3 C_4 L_1 R_1 R_3 g_m + C_1 C_3 C_4 L_1 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 R_3 + C_1 C_3 C_4 L_1 R_4 + 2C_3 C_4 L_1 L_3 g_m + C_3 C_4 L_1 L_4 g_m) + s^3 (C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2C_1 C_4 L_1 R_1 g_m + 2C_1$$

9.899 X-INVALID-ORDER-899 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{2R_1 R_4 g_m + 2R_4 + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (2C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2C_1 C_3 L_1 L_3 L_4 R_1 g_m + 2C_1 C_3 L_1 L_3 L_4 + 2C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 L_1 L_3 R_4 + 2C_1 C_3 L_1 L_4 R_1 R_3 g_m +$$

9.900 X-INVALID-ORDER-900 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m s^6 + R_1 R_4 g_m + s^5 (C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 R_4 g_m + C_1 C_3 L_1 L_4 R_1 R_3 g_m + C_1 C_3 L_1 L_4 R_4 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_4 R_3 + C_1 C_3 C_4 L_1 L_4 R_4 + 2C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_1 g_m + 2C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 R_1 g_m + C_1 C_3 L_1 L_4 + 2C_1 C_4 L_1 L_4 R_1 g_m)}{2R_1 g_m + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4) + s^5 (2C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_4 R_3 + C_1 C_3 C_4 L_1 L_4 R_4 + 2C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_1 g_m + 2C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_4 R_1 g_m + C_1 C_3 L_1 L_4 + 2C_1 C_4 L_1 L_4 R_1 g_m)}$$

9.901 X-INVALID-ORDER-901 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{2R_1g_m + s^6(2C_1C_3C_4L_1L_3L_4R_1g_m + 2C_1C_3C_4L_1L_3L_4) + s^5(2C_1C_3C_4L_1L_3R_1R_4g_m + 2C_1C_3C_4L_1L_3R_4 + 2C_1C_3C_4L_1L_4R_1R_3g_m + C_1C_3C_4L_1L_4R_1R_4g_m + 2C_1C_3C_4L_1L_4R_3 + C_1C_3C_4L_1L_4R_4 + 2C_3C_4L_1L_3L_4g_m) + s^4(2C_1C_3C_4L_1R_1R_3R_4g_m + 2C_1C_3C_4L_1R_1R_3R_4)}{2R_1g_m + s^6(2C_1C_3C_4L_1L_3L_4R_1g_m + 2C_1C_3C_4L_1L_3L_4) + s^5(2C_1C_3C_4L_1L_3R_1R_4g_m + 2C_1C_3C_4L_1L_3R_4 + 2C_1C_3C_4L_1L_4R_1R_3g_m + C_1C_3C_4L_1L_4R_1R_4g_m + 2C_1C_3C_4L_1L_4R_3 + C_1C_3C_4L_1L_4R_4 + 2C_3C_4L_1L_3L_4g_m) + s^4(2C_1C_3C_4L_1R_1R_3R_4g_m + 2C_1C_3C_4L_1R_1R_3R_4)}$$

9.902 X-INVALID-ORDER-902 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 R_1 R_3 R_4 g_m s^3 + L_1 L_3 R_3 R_4 g_m s^2 + L_3 R_1 R_3 R_4 g_m s}{R_1 R_3 R_4 g_m + R_3 R_4 + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 R_3 R_4) + s^3 (2C_1 L_1 L_3 R_1 R_3 g_m + C_1 L_1 L_3 R_1 R_4 g_m + 2C_1 L_1 L_3 R_3 + C_1 L_1 L_3 R_4 + C_3 L_1 L_3 R_3 R_4 g_m) + s^2 (C_1 L_1 R_1 R_3 R_4 g_m + C_1 L_1 R_3 R_4 + C_3 L_3 R_1 R_3 R_4 g_m + C_3 L_3 R_3 R_4 + 2L_1 L_3 R_3 g_m + L_1 L_3 R_4 g_m) + s (C_1 L_1 R_1 R_3 R_4 + C_3 L_3 R_1 R_3 R_4 + 2L_1 L_3 R_3 + L_1 L_3 R_4) + R_1 R_3 R_4}$$

$$\mathbf{9.903 \quad X-INVALID-ORDER-903} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_3 R_1 R_3 g_m s^3 + L_1 L_3 R_3 g_m s^2 + L_3 R_1 R_3 g_m s}{R_1 R_3 g_m + R_3 + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_3 L_1 L_3 R_3 + 2C_1 C_4 L_1 L_3 R_1 R_3 g_m + 2C_1 C_4 L_1 L_3 R_3) + s^3 (C_1 L_1 L_3 R_1 g_m + C_1 L_1 L_3 + C_3 L_1 L_3 R_3 g_m + 2C_4 L_1 L_3 R_3 g_m) + s^2 (C_1 L_1 R_1 R_3 g_m + C_1 L_1 R_3 + C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + 2C_4 L_3 R_3 + L_1 L_3 g_m)}$$

$$\mathbf{9.904 \quad X-INVALID-ORDER-904} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_3 R_1 R_3 R_4 g_m s^3 + L_1 L_3 R_3 R_4 g_m s^2 + L_3 R_1 R_3 R_4 g_m s}{R_1 R_3 R_4 g_m + R_3 R_4 + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 R_3 R_4 + 2C_1 C_4 L_1 L_3 R_1 R_3 R_4 g_m + 2C_1 C_4 L_1 L_3 R_3 R_4) + s^3 (2C_1 L_1 L_3 R_1 R_3 g_m + C_1 L_1 L_3 R_1 R_4 g_m + 2C_1 L_1 L_3 R_3 + C_1 L_1 L_3 R_4 + C_3 L_1 L_3 R_3 R_4 g_m + 2C_4 L_1 L_3 R_3 R_4 g_m) + s^2 (C_1 L_1 R_1 R_3 R_4 g_m + C_1 L_1 R_3 R_4 + C_3 L_3 R_1 R_3 R_4 g_m + C_3 L_3 R_3 + 2C_4 L_3 R_1 R_3 g_m + 2C_4 L_3 R_3 + L_1 L_3 g_m)}$$

$$\mathbf{9.905 \quad X-INVALID-ORDER-905} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_3 R_1 R_3 R_4 g_m s^4 + L_3 R_1 R_3 R_4 g_m s^3}{R_1 R_3 g_m + R_3 + s^5 (C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_3 R_3 R_4) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_3 L_1 L_3 R_3 + 2C_1 C_4 L_1 L_3 R_1 R_3 g_m + C_1 C_4 L_1 L_3 R_1 R_4 g_m + 2C_1 C_4 L_1 L_3 R_3 + C_1 C_4 L_1 L_3 R_4 + C_3 C_4 L_1 L_3 R_3 R_4 g_m) + s^3 (C_1 C_4 L_1 R_1 R_3 R_4 g_m + C_1 C_4 L_1 R_3 R_4 + C_1 L_1 R_3 R_4 g_m)}$$

$$\mathbf{9.906 \quad X-INVALID-ORDER-906} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_1 R_3 g_m s^5 + C_4 L_1 R_3 g_m s^4}{R_1 R_3 g_m + R_3 + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_3) + s^5 (C_1 C_4 L_1 L_3 L_4 R_1 g_m + C_1 C_4 L_1 L_3 L_4 + C_3 C_4 L_1 L_3 L_4 R_3 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_3 L_1 L_3 R_3 + 2C_1 C_4 L_1 L_3 R_1 R_3 g_m + 2C_1 C_4 L_1 L_3 R_3 + C_1 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_4 R_3 + C_3 C_4 L_1 L_3 L_4 R_3 g_m)}$$

$$\mathbf{9.907 \quad X-INVALID-ORDER-907} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_3 L_4 R_1 R_3 g_m s^3 + L_1 L_3 L_4 R_3 g_m s^2 + L_3 L_4 R_1 R_3 g_m s}{2L_3 R_1 R_3 g_m + 2L_3 R_3 + L_4 R_1 R_3 g_m + L_4 R_3 + s^4 (C_1 C_3 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 L_1 L_3 L_4 R_3 + 2C_1 C_4 L_1 L_3 L_4 R_1 R_3 g_m + 2C_1 C_4 L_1 L_3 L_4 R_3) + s^3 (C_1 L_1 L_3 L_4 R_1 g_m + C_1 L_1 L_3 L_4 + C_3 L_1 L_3 L_4 R_3 g_m + 2C_4 L_1 L_3 L_4 R_3 g_m) + s^2 (2C_1 L_1 L_3 R_1 R_3 g_m + 2C_1 L_1 L_3 R_3 + C_1 L_1 L_3 R_4)}$$

$$\mathbf{9.908 \quad X-INVALID-ORDER-908} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_1 R_3 g_m s^4 + L_3 L_4 R_1 R_3 g_m s^3}{R_1 R_3 g_m + R_3 + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_3) + s^5 (C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_4 L_1 L_3 L_4 R_1 g_m + C_1 C_4 L_1 L_3 L_4 + C_3 C_4 L_1 L_3 L_4 R_3 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_3 L_1 L_3 R_3 + 2C_1 C_4 L_1 L_3 R_1 R_3 g_m + C_1 C_4 L_1 L_3 R_3)}$$

$$\mathbf{9.909 \quad X-INVALID-ORDER-909} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_3 L_4 R_1 R_3 R_4 g_m s^4}{2L_3 R_1 R_3 R_4 g_m + 2L_3 R_3 R_4 + L_4 R_1 R_3 R_4 g_m + L_4 R_3 R_4 + s^4 (C_1 C_3 L_1 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 L_4 R_3 R_4 + 2C_1 C_4 L_1 L_3 L_4 R_1 R_3 R_4 g_m + 2C_1 C_4 L_1 L_3 L_4 R_3 R_4) + s^3 (2C_1 L_1 L_3 L_4 R_1 R_3 g_m + C_1 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 L_1 L_3 L_4 R_3 + C_1 L_1 L_3 L_4 R_4 + C_3 L_1 L_3 L_4 R_3 R_4 g_m)}$$

$$\mathbf{9.910 \quad X-INVALID-ORDER-910} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{R_1 R_3 R_4 g_m + R_3 R_4 + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4) + s^5 (C_1 C_3 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 L_1 L_3 L_4 R_3 + 2C_1 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_3 L_4 R_3 + C_1 C_4 L_1 L_3 L_4 R_4 + C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 R_3 R_4 + 2C_1 C_4 L_1 L_3 R_1 R_3 g_m + C_1 C_4 L_1 L_3 R_3 R_4)}$$

$$\mathbf{9.911 \quad X-INVALID-ORDER-911} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_4 (C_4 L_4 s^2 + 1)}{R_1 R_3 R_4 g_m + R_3 R_4 + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4) + s^5 (2C_1 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_3 L_4 R_3 + C_1 C_4 L_1 L_3 L_4 R_4 + C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 R_3 R_4 + 2C_1 C_4 L_1 L_3 R_1 R_3 g_m + C_1 C_4 L_1 L_3 R_3 R_4)}$$

9.912 X-INVALID-ORDER-912 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 R_3 R_4 g_m s^4 + R_1 R_3 R_4 g_m + s^3 (C_1 L_1 L_3 R_1 R_4 g_m + C_3 L_1 L_3 R_3 R_4 g_m) + s^2 (C_1 L_1 R_1 R_3 R_4 g_m + C_3 L_3 R_1 R_3 R_4 g_m + L_1 L_3 R_4 g_m) + s (L_1 R_3 R_4 g_m + L_3 R_1 R_4 g_m)}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^4 (2C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_3 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_3 R_4) + s^3 (2C_1 L_1 L_3 R_1 g_m + 2C_1 L_1 L_3 + 2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 g_m) + s^2 (2C_1 L_1 R_1 R_3 g_m + C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_3 + C_1 L_1 R_4 + 2C_3 L_3 R_1 R_3)}$$

9.913 X-INVALID-ORDER-913 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 R_3 g_m s^4 + R_1 R_3 g_m + s^3 (C_1 L_1 L_3 R_1 g_m + C_3 L_1 L_3 R_3 g_m) + s^2 (C_1 L_1 R_1 R_3 g_m + C_3 L_3 R_1 R_3 g_m + L_1 L_3 g_m) + s (L_1 R_3 g_m + L_3 R_1 g_m)}{R_1 g_m + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3) + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 R_1 g_m + 2C_1 C_4 L_1 L_3 + 2C_3 C_4 L_1 L_3 R_3 g_m) + s^3 (2C_1 C_4 L_1 R_1 R_3 g_m + 2C_1 C_4 L_1 R_3 + 2C_3 C_4 L_3 R_1 R_3 g_m + 2C_3 C_4 L_3 R_3 + C_3 L_1 L_3 g_m + 2C_4 L_1 L_3 g_m) + s^2 (C_1$$

9.914 X-INVALID-ORDER-914 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 R_3}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4) + s^4 (2C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_3 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_3 R_4 + 2C_1 C_4 L_1 L_3 R_1 R_4 g_m + 2C_1 C_4 L_1 L_3 R_4 + 2C_3 C_4 L_1 L_3 R_3 R_4 g_m) + s^3 (2C_1 C_4 L_1 R_1 R_3 R_4 g_m)}$$

9.915 X-INVALID-ORDER-915 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m s^5 + R_1 R_3 g_m + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_4 L_1 L_3 R_1 R_4 g_m + C_3 C_4 L_1 L_3 R_3 R_4 g_m) + s^3 (C_1 C_4 L_1 R_1 R_3 R_4 g_m + C_1 L_1 L_3 R_1 g_m + C_3 C_4 L_1 R_3 R_4 g_m) + s^2 (C_1 C_3 C_4 L_1 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 + C_1 C_3 C_4 L_1 L_3 R_4) + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 R_1 g_m + 2C_1 C_4 L_1 L_3 + 2C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (2C_1 C_4 L_1 R_1 R_3 g_m + C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_3 R_4 g_m) + s^2 (C_1 C_4 L_1 R_1 g_m + C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m) + s (C_1 C_4 L_1 + C_3 C_4 L_1 R_3 + C_3 C_4 L_1 R_4) + C_1 C_4 L_1}{R_1 g_m + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 + C_1 C_3 C_4 L_1 L_3 R_4) + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 R_1 g_m + 2C_1 C_4 L_1 L_3 + 2C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (2C_1 C_4 L_1 R_1 R_3 g_m + C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_3 R_4 g_m) + s^2 (C_1 C_4 L_1 R_1 g_m + C_1 C_4 L_1 R_3 g_m + C_1 C_4 L_1 R_4 g_m) + s (C_1 C_4 L_1 + C_3 C_4 L_1 R_3 + C_3 C_4 L_1 R_4) + C_1 C_4 L_1}.$$

9.916 X-INVALID-ORDER-916 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m s^6 + R_1 R_3 g_m + s^5 (C_1 C_4 L_1 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4 R_3 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_4 L_1 L_4 R_1 R_3 g_m + C_3 C_4 L_3 L_4 R_1 R_3 g_m + C_3 C_4 L_3 L_4 R_3 g_m)}{R_1 g_m + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_3 L_4) + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 + C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 R_1 g_m + 2C_1 C_4 L_1 L_3 + C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_3 L_4)}$$

9.917 X-INVALID-ORDER-917 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 L_4 R_1}{2R_1 R_3 g_m + 2R_3 + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3) + s^5 (C_1 C_3 L_1 L_3 L_4 R_1 g_m + C_1 C_3 L_1 L_3 L_4 + 2C_1 C_4 L_1 L_3 L_4 R_1 g_m + 2C_1 C_4 L_1 L_3 L_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_1 R_3 g_m + 2C_1 C_3 L_1 L_3 R_3 + 2C_1 C_4 L_1 L_4 R_1 R_3 g_m + 2C_1 C_4 L_1 L_4 R_3 g_m)}$$

9.918 X-INVALID-ORDER-918 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m s^6 + R_1 R_3 g_m + s^5 (C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + C_1 C_4 L_1 L_3 L_4 R_1 g_m + C_3 C_4 L_1 L_3 L_4 R_3 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_4 L_1 L_3 R_1 R_4 g_m + R_1 g_m + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_3 L_4) + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_1 L_3 R_3 + C_1 C_3 C_4 L_1 L_3 R_4 + C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + 2 C_1 C_4 L_1 L_3 R_1 g_m + 2 C_1 C_4 L_1 L_3 + C_1 C_4 L_1 L_4 R_1 g_m + C$$

9.919 X-INVALID-ORDER-919 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{2R_1 R_3 R_4 g_m + 2R_3 R_4 + s^6(2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4) + s^5(2C_1 C_3 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 L_1 L_3 L_4 R_3 + C_1 C_3 L_1 L_3 L_4 R_4 + 2C_1 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_3 L_4 R_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 R_4 g_m) + s^4(2C_1 C_3$$

9.920 X-INVALID-ORDER-920 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 g_m s^6 + R_1 R}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (C_1 C_3 L_1 L_3 L_4 R_1 g_m + C_1 C_3 L_1 L_3 L_4 + 2C_1 C_4 L_1 L_3 L_4 R_1 g_m + 2C_1 C_4 L_1 L_3 L_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_4)}$$

9.921 X-INVALID-ORDER-921 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + 2C_1 C_4 L_1 L_3 L_4 R_1 g_m + 2C_1 C_4 L_1 L_3 L_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 g_m + C_3 C_4 L_1 L_3 L_4 R_4)}{(s^2 + 2\alpha_1 s + \alpha_0)(s^2 + 2\beta_1 s + \beta_0)}$$

9.922 X-INVALID-ORDER-922 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 R_3 R_4 g_m s^4 + C_3 L_1 L_3 R_3 R_4 g_m s^3 + L_1 R_3 R_4 g_m s + R_1 R_3 R_4 g_m + s^2 (C_1 L_1 R_1 R_3 R_4 g_m + C_3 L_3 R_1 R_3 R_4 g_m)}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^4 (2C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_3 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_3 R_4) + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 g_m + C_1 C_3 L_1 R_3 R_4 + 2C_3 L_1 L_3 R_3 g_m + C_3 L_1 L_3 R_4 g_m) + s^2 (2C_1 L_1 R_1 R_3 g_m + C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_3 + C_1 L_1 R_4 + C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_1 R_4 g_m) + s (C_1 L_1 R_1 R_3 + C_1 L_1 R_3 + C_3 L_3 R_1 R_3 + C_3 L_3 R_1 R_4) + C_1 L_1 R_1 R_3 + C_1 L_1 R_3 + C_3 L_3 R_1 R_3 + C_3 L_3 R_1 R_4}.$$

9.923 X-INVALID-ORDER-923 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 R_3 g_m s^4 + C_3 L_1 L_3 R_3 g_m s^3 + L_1 R_3 g_m s + R_1 R_3 g_m + s^2 (C_1 L_1 R_1 R_3 g_m + C_3 L_3 R_1 R_3 g_m)}{R_1 g_m + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3) + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + 2C_3 C_4 L_1 L_3 R_3 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_3 + 2C_1 C_4 L_1 R_1 R_3 g_m + 2C_1 C_4 L_1 R_3 + 2C_3 C_4 L_3 R_1 R_3 g_m + 2C_3 C_4 L_3 R_3 + C_3 L_1 L_3 g_m) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1}$$

9.924 X-INVALID-ORDER-924 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 R_3 R_4 g_m s}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4) + s^4 (2C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_3 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_3 R_4 + 2C_3 C_4 L_1 L_3 R_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 R_4 g_m + C_1 C_3 L_1 R_3 R_4 + 2C_1 C_4 L_1 R_1 R_3 R_4 g_m)}$$

9.925 X-INVALID-ORDER-925 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m s^5 + R_1 R_3 g_m + s^4 (C_1 C_3 L_1 L_3 R_1 R_3)}{R_1 g_m + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 + C_1 C_3 C_4 L_1 L_3 R_4) + s^4 (C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 R_3 R_4 + C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + 2C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_3 R_4 g_m) + s^3 (C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_3 + 2$$

9.926 X-INVALID-ORDER-926 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m s^6 + C_3 C_4 L_1 L_3 L_4 R_3 g_m s^5 + L_1 R_3 g_m s^4}{R_1 g_m + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_3 L_4) + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_3 g_m + 2 C_1 C_3 C_4 L_1 L_3 R_3 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_3 + C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2 C_3 C_4 L_1 L_3 R_3 g_m + C_3 C_4 L_1 L_3 R_3)}$$

9.927 X-INVALID-ORDER-927 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 L_4 R_1 R_3 g_m s^5}{2R_1 R_3 g_m + 2R_3 + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3) + s^5 (C_1 C_3 L_1 L_3 L_4 R_1 g_m + C_1 C_3 L_1 L_3 L_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 g_m) + s^4 (2C_1 C_3 L_1 L_3 R_1 R_3 g_m + 2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_4 R_1 R_3 g_m + C_1 C_3 L_1 L_4 R_3 + 2C_1 C_4 L_1 L_4 R_1 R_3 g_m + 2C_1 C_4 L_1 L_4 R_3)}$$

9.928 X-INVALID-ORDER-928 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{R_1 g_m + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_3 L_4) + s^5 (2 C_1 C_3 C_4 L_1 L_3 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_1 C_3 C_4 L_1 L_3 R_3 + C_1 C_3 C_4 L_1 L_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_3 + C_3 C_4 L_1 L_3 L_4 g_m) + s^4 (C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 R_3 R_4 +$$

9.929 X-INVALID-ORDER-929 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{2R_1R_3R_4g_m + 2R_3R_4 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3R_4g_m + 2C_1C_3C_4L_1L_3L_4R_3R_4) + s^5(2C_1C_3L_1L_3L_4R_1R_3g_m + C_1C_3L_1L_3L_4R_1R_4g_m + 2C_1C_3L_1L_3L_4R_3 + C_1C_3L_1L_3L_4R_4 + 2C_3C_4L_1L_3L_4R_3R_4g_m) + s^4(2C_1C_3L_1L_3R_1R_3R_4g_m + 2C_1C_3L_1L_3R_3R_4 + C_1C_3L_1L_3R_4R_3 + C_1C_3L_1L_3R_4R_4)}{2R_1R_3R_4g_m + 2R_3R_4 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3R_4g_m + 2C_1C_3C_4L_1L_3L_4R_3R_4) + s^5(2C_1C_3L_1L_3L_4R_1R_3g_m + C_1C_3L_1L_3L_4R_1R_4g_m + 2C_1C_3L_1L_3L_4R_3 + C_1C_3L_1L_3L_4R_4 + 2C_3C_4L_1L_3L_4R_3R_4g_m) + s^4(2C_1C_3L_1L_3R_1R_3R_4g_m + 2C_1C_3L_1L_3R_3R_4 + C_1C_3L_1L_3R_4R_3 + C_1C_3L_1L_3R_4R_4)}$$

9.930 X-INVALID-ORDER-930 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3g_m + C_1C_3C_4L_1L_3L_4R_1R_4g_m + 2C_1C_3C_4L_1L_3L_4R_3 + C_1C_3C_4L_1L_3L_4R_4) + s^5(C_1C_3C_4L_1L_4R_1R_3R_4g_m + C_1C_3C_4L_1L_4R_3R_4 + C_1C_3L_1L_3L_4R_1g_m + C_1C_3L_1L_3L_4 + 2C_3C_4L_1L_3L_4R_3g_m + C_3C_4L_1L_3L_4R_4)}{1 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3g_m + C_1C_3C_4L_1L_3L_4R_1R_4g_m + 2C_1C_3C_4L_1L_3L_4R_3 + C_1C_3C_4L_1L_3L_4R_4) + s^5(C_1C_3C_4L_1L_4R_1R_3R_4g_m + C_1C_3C_4L_1L_4R_3R_4 + C_1C_3L_1L_3L_4R_1g_m + C_1C_3L_1L_3L_4 + 2C_3C_4L_1L_3L_4R_3g_m + C_3C_4L_1L_3L_4R_4) + s^4(C_1C_3C_4L_1L_3L_4R_1R_3R_4g_m + C_1C_3C_4L_1L_3L_4R_1R_4R_4g_m + C_1C_3C_4L_1L_3L_4R_3R_4 + C_1C_3C_4L_1L_3L_4R_4R_4) + s^3(C_1C_3C_4L_1L_3L_4R_1R_3R_4 + C_1C_3C_4L_1L_3L_4R_1R_4 + C_1C_3C_4L_1L_3L_4R_3 + C_1C_3C_4L_1L_3L_4R_4) + s^2(C_1C_3C_4L_1L_3L_4R_1 + C_1C_3C_4L_1L_3L_4R_3 + C_1C_3C_4L_1L_3L_4R_4) + s(C_1C_3C_4L_1L_3L_4) + 1}$$

$$\mathbf{9.931 \quad X-INVALID-ORDER-931} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 g_m + 2C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 R_4 + 2C_3 C_4 L_1 L_3 L_4 R_4)}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 g_m + 2C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 + 2C_3 C_4 L_1 L_3 L_4 R_3 R_4 + 2C_3 C_4 L_1 L_3 L_4 R_4)}$$

$$\mathbf{9.932 \quad X-INVALID-ORDER-932} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_1 R_3 g_m s^2 + R_1 R_3 g_m}{R_1 g_m + s^3 (2C_1 C_4 L_1 R_1 R_3 g_m + 2C_1 C_4 L_1 R_3) + s^2 (2C_1 C_4 R_1 R_3 + C_1 L_1 R_1 g_m + C_1 L_1) + s (C_1 R_1 + 2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

$$\mathbf{9.933 \quad X-INVALID-ORDER-933} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_1 R_3 R_4 g_m s^2 + R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^3 (2C_1 C_4 L_1 R_1 R_3 R_4 g_m + 2C_1 C_4 L_1 R_3 R_4) + s^2 (2C_1 C_4 R_1 R_3 R_4 + 2C_1 L_1 R_1 R_3 g_m + C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_3 + C_1 L_1 R_4) + s (2C_1 R_1 R_3 + C_1 R_1 R_4 + 2C_4 R_1 R_3 R_4 g_m + 2C_4 R_3 R_4)}$$

$$\mathbf{9.934 \quad X-INVALID-ORDER-934} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 R_1 R_3 R_4 g_m s^3 + C_1 L_1 R_1 R_3 g_m s^2 + C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m}{R_1 g_m + s^3 (2C_1 C_4 L_1 R_1 R_3 g_m + C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4) + s^2 (2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 + C_1 L_1 R_1 g_m + C_1 L_1) + s (C_1 R_1 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4) + 1}$$

$$\mathbf{9.935 \quad X-INVALID-ORDER-935} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 R_3 g_m s^4 + R_1 R_3 g_m + s^2 (C_1 L_1 R_1 R_3 g_m + C_4 L_4 R_1 R_3 g_m)}{R_1 g_m + s^4 (C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4) + s^3 (2C_1 C_4 L_1 R_1 R_3 g_m + 2C_1 C_4 L_1 R_3 + C_1 C_4 L_4 R_1) + s^2 (2C_1 C_4 R_1 R_3 + C_1 L_1 R_1 g_m + C_1 L_1 + C_4 L_4 R_1 g_m + C_4 L_4) + s (C_1 R_1 + 2C_4 R_1 R_3 g_m + 2C_4 R_3) + 1}$$

$$\mathbf{9.936 \quad X-INVALID-ORDER-936} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 R_1 R_3 g_m s^3 + L_4 R_1 R_3 g_m s}{2R_1 R_3 g_m + 2R_3 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_3 g_m + 2C_1 C_4 L_1 L_4 R_3) + s^3 (2C_1 C_4 L_4 R_1 R_3 + C_1 L_1 L_4 R_1 g_m + C_1 L_1 L_4) + s^2 (2C_1 L_1 R_1 R_3 g_m + 2C_1 L_1 R_3 + C_1 L_4 R_1 + 2C_4 L_4 R_1 R_3 g_m + 2C_4 L_4 R_3) + s (2C_1 R_1 R_3 + L_4 R_1 g_m + L_4)}$$

$$\mathbf{9.937 \quad X-INVALID-ORDER-937} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 R_3 g_m s^4 + C_1 C_4 L_1 R_1 R_3 R_4 g_m s^3 + C_4 R_1 R_3 R_4 g_m s + R_1 R_3 g_m + s^2 (C_1 L_1 R_1 R_3 g_m + C_4 L_4 R_1 R_3 g_m)}{R_1 g_m + s^4 (C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4) + s^3 (2C_1 C_4 L_1 R_1 R_3 g_m + C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_3 + C_1 C_4 L_1 R_4 + C_1 C_4 L_4 R_1) + s^2 (2C_1 C_4 R_1 R_3 + C_1 C_4 R_1 R_4 + C_1 L_1 R_1 g_m + C_1 L_1 + C_4 L_4 R_1 g_m + C_4 L_4) + s (C_1 R_1 + 2C_4 R_1 R_3 g_m + C_4 R_1 R_4 g_m + 2C_4 R_3 + C_4 R_4)}$$

$$\mathbf{9.938 \quad X-INVALID-ORDER-938} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_4 R_1 R_3 R_4 g_m s^3 + L_4 R_1 R_3 R_4 g_m s}{2R_1 R_3 R_4 g_m + 2R_3 R_4 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_3 R_4 g_m + 2C_1 C_4 L_1 L_4 R_3 R_4) + s^3 (2C_1 C_4 L_4 R_1 R_3 R_4 + 2C_1 L_1 L_4 R_1 R_3 g_m + C_1 L_1 L_4 R_1 R_4 g_m + 2C_1 L_1 L_4 R_3 + C_1 L_1 L_4 R_4) + s^2 (2C_1 L_1 R_1 R_3 R_4 g_m + 2C_1 L_1 R_3 R_4 + 2C_1 L_4 R_1 R_3 + C_1 L_4 R_1 R_4 + 2C_4 L_4 R_1 R_3 R_4 g_m + 2C_4 L_4 R_1 R_3 R_4 + 2C_4 L_4 R_1 R_3 R_4 + 2C_4 L_4 R_1 R_4)}{2R_1 R_3 R_4 g_m + 2R_3 R_4 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_3 R_4 g_m + 2C_1 C_4 L_1 L_4 R_3 R_4) + s^3 (2C_1 C_4 L_4 R_1 R_3 R_4 + 2C_1 L_1 L_4 R_1 R_3 g_m + C_1 L_1 L_4 R_1 R_4 g_m + 2C_1 L_1 L_4 R_3 + C_1 L_1 L_4 R_4) + s^2 (2C_1 L_1 R_1 R_3 R_4 g_m + 2C_1 L_1 R_3 R_4 + 2C_1 L_4 R_1 R_3 + C_1 L_4 R_1 R_4 + 2C_4 L_4 R_1 R_3 R_4 g_m + 2C_4 L_4 R_1 R_3 R_4 + 2C_4 L_4 R_1 R_3 R_4 + 2C_4 L_4 R_1 R_4)}$$

$$\mathbf{9.939 \quad X-INVALID-ORDER-939} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad R_3, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_1 R_3 R_4 g_m s^4 + C_1 L_1 L_4 R_1 R_3 g_m s^3 + L_4 R_1 R_3 g_m s + R_1 R_3 R_4 g_m + s^2 (C_1 L_1 R_1 R_3 R_4 g_m + C_4 L_4 R_1 R_3 R_4 g_m)}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4 R_4) + s^3 (2C_1 C_4 L_4 R_1 R_3 + C_1 C_4 L_4 R_1 R_4 + C_1 L_1 L_4 R_1 g_m + C_1 L_1 L_4) + s^2 (2C_1 L_1 R_1 R_3 g_m + C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_3 + C_1 L_1 R_4 + C_1 L_4 R_1 + 2C_4 L_4 R_1 R_3 R_4 g_m + 2C_4 L_4 R_1 R_3 R_4 + 2C_4 L_4 R_1 R_3 R_4 + 2C_4 L_4 R_1 R_4)}$$

$$\mathbf{9.940 \quad X-INVALID-ORDER-940} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad R_3, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1L_4R_1R_3R_4g_ms^4 + R_1R_3R_4g_m + s^2(C_1L_1R_1R_3R_4g_m + C_4L_4R_1R_3R_4g_m)}{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^4(2C_1C_4L_1L_4R_1R_3g_m + C_1C_4L_1L_4R_1R_4g_m + 2C_1C_4L_1L_4R_3 + C_1C_4L_1L_4R_4) + s^3(2C_1C_4L_1R_1R_3R_4g_m + 2C_1C_4L_1R_3R_4 + 2C_1C_4L_4R_1R_3 + C_1C_4L_4R_1R_4) + s^2(2C_1C_4R_1R_3R_4 + 2C_1L_1R_1R_3g_m + C_1L_1R_1R_4g_m + 2C_1L_1R_1R_3R_4) + s(2C_1R_1R_3R_4 + 2C_1R_1R_4g_m + 2C_1R_1R_3R_4) + 2}$$

$$\mathbf{9.941 \quad X-INVALID-ORDER-941} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{1}{C_3s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1R_1R_4g_ms^2 + R_1R_4g_m}{2R_1g_m + s^3(C_1C_3L_1R_1R_4g_m + C_1C_3L_1R_4) + s^2(C_1C_3R_1R_4 + 2C_1L_1R_1g_m + 2C_1L_1) + s(2C_1R_1 + C_3R_1R_4g_m + C_3R_4) + 2}$$

$$\mathbf{9.942 \quad X-INVALID-ORDER-942} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{1}{C_3s}, \quad \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1R_1g_ms^2 + R_1g_m}{s^3(C_1C_3L_1R_1g_m + C_1C_3L_1 + 2C_1C_4L_1R_1g_m + 2C_1C_4L_1) + s^2(C_1C_3R_1 + 2C_1C_4R_1) + s(C_3R_1g_m + C_3 + 2C_4R_1g_m + 2C_4) + 2}$$

$$\mathbf{9.943 \quad X-INVALID-ORDER-943} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{1}{C_3s}, \quad \frac{R_4}{C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1R_1R_4g_ms^2 + R_1R_4g_m}{2R_1g_m + s^3(C_1C_3L_1R_1R_4g_m + C_1C_3L_1R_4 + 2C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_4) + s^2(C_1C_3R_1R_4 + 2C_1C_4R_1R_4 + 2C_1L_1R_1g_m + 2C_1L_1) + s(2C_1R_1 + C_3R_1R_4g_m + C_3R_4 + 2C_4R_1R_4g_m + 2C_4R_4) + 2}$$

$$\mathbf{9.944 \quad X-INVALID-ORDER-944} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{1}{C_3s}, \quad R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1R_1R_4g_ms^3 + C_1L_1R_1g_ms^2 + C_4R_1R_4g_ms + R_1g_m}{s^4(C_1C_3C_4L_1R_1R_4g_m + C_1C_3C_4L_1R_4) + s^3(C_1C_3C_4R_1R_4 + C_1C_3L_1R_1g_m + C_1C_3L_1 + 2C_1C_4L_1R_1g_m + 2C_1C_4L_1) + s^2(C_1C_3R_1 + 2C_1C_4R_1 + C_3C_4R_1R_4g_m + C_3C_4R_4) + s(C_3R_1g_m + C_3 + 2C_4R_1g_m + 2C_4) + 2}$$

$$\mathbf{9.945 \quad X-INVALID-ORDER-945} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{1}{C_3s}, \quad L_4s + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1L_4R_1g_ms^4 + R_1g_m + s^2(C_1L_1R_1g_m + C_4L_4R_1g_m)}{C_1C_3C_4L_4R_1s^4 + s^5(C_1C_3C_4L_1L_4R_1g_m + C_1C_3C_4L_1L_4) + s^3(C_1C_3L_1R_1g_m + C_1C_3L_1 + 2C_1C_4L_1R_1g_m + 2C_1C_4L_1 + C_3C_4L_4R_1g_m + C_3C_4L_4) + s^2(C_1C_3R_1 + 2C_1C_4R_1) + s(C_3R_1g_m + C_3 + 2C_4R_1g_m + 2C_4) + 2}$$

$$\mathbf{9.946 \quad X-INVALID-ORDER-946} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{1}{C_3s}, \quad \frac{L_4s}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1L_4R_1g_ms^3 + L_4R_1g_ms}{2C_1R_1s + 2R_1g_m + s^4(C_1C_3L_1L_4R_1g_m + C_1C_3L_1L_4 + 2C_1C_4L_1L_4R_1g_m + 2C_1C_4L_1L_4) + s^3(C_1C_3L_4R_1 + 2C_1C_4L_4R_1) + s^2(2C_1L_1R_1g_m + 2C_1L_1 + C_3L_4R_1g_m + C_3L_4 + 2C_4L_4R_1g_m + 2C_4L_4) + 2}$$

$$\mathbf{9.947 \quad X-INVALID-ORDER-947} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{1}{C_3s}, \quad L_4s + R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1L_4R_1g_ms^4 + C_1C_4L_1R_1R_4g_ms^3 + C_4R_1R_4g_ms + R_1g_m + s^2(C_1L_1R_1g_m + C_4L_4R_1g_m)}{s^5(C_1C_3C_4L_1L_4R_1g_m + C_1C_3C_4L_1L_4) + s^4(C_1C_3C_4L_1R_1R_4g_m + C_1C_3C_4L_1R_4 + C_1C_3C_4L_4R_1) + s^3(C_1C_3C_4R_1R_4 + C_1C_3L_1R_1g_m + C_1C_3L_1 + 2C_1C_4L_1R_1g_m + 2C_1C_4L_1 + C_3C_4L_4R_1g_m + C_3C_4L_4) + s^2(C_1C_3R_1 + 2C_1C_4R_1 + C_3C_4R_1R_4g_m + C_3C_4R_4) + s(C_3R_1g_m + C_3 + 2C_4R_1g_m + 2C_4) + 2}$$

$$\mathbf{9.948 \quad X-INVALID-ORDER-948} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{1}{C_3s}, \quad \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1L_4R_1R_4g_ms^3 + L_4R_1R_4g_ms}{2R_1R_4g_m + 2R_4 + s^4(C_1C_3L_1L_4R_1R_4g_m + C_1C_3L_1L_4R_4 + 2C_1C_4L_1L_4R_1R_4g_m + 2C_1C_4L_1L_4R_4) + s^3(C_1C_3L_4R_1R_4 + 2C_1C_4L_4R_1R_4 + 2C_1L_1L_4R_1g_m + 2C_1L_1L_4) + s^2(2C_1L_1R_1R_4g_m + 2C_1L_1R_4 + 2C_1L_4R_1 + C_3L_4R_1R_4g_m + C_3L_4R_4 + 2C_4L_4R_1R_4g_m + 2C_4L_4R_4) + s(C_3R_1g_m + C_3 + 2C_4R_1g_m + 2C_4) + 2}$$

$$\mathbf{9.949 \quad X-INVALID-ORDER-949} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{1}{C_3s}, \quad \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1L_4R_1R_4g_ms^4 + C_1L_1L_4R_1g_ms^3 + L_4R_1g_ms + R_1R_4g_m + s^2(C_1L_1R_1R_4g_m + C_4L_4R_1R_4g_m)}{2R_1g_m + s^5(C_1C_3C_4L_1L_4R_1R_4g_m + C_1C_3C_4L_1L_4R_4) + s^4(C_1C_3C_4L_4R_1R_4 + C_1C_3L_1L_4R_1g_m + C_1C_3L_1L_4 + 2C_1C_4L_1L_4R_1g_m + 2C_1C_4L_1L_4) + s^3(C_1C_3L_1R_1R_4g_m + C_1C_3L_1R_4 + C_1C_3L_4R_1 + 2C_1C_4L_4R_1 + C_3C_4L_4R_1R_4g_m + C_3C_4L_4R_4) + s^2(C_1C_3R_1R_4 + 2C_1C_4R_1R_4g_m + C_3C_4R_1R_4g_m + C_3C_4R_4g_m) + s(C_1R_1R_4 + C_3R_1R_4g_m + C_3R_4g_m) + 1}$$

$$\mathbf{9.950 \quad X-INVALID-ORDER-950} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{1}{C_3s}, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1L_4R_1R_4g_ms^4 + R_1R_4g_m + s^2(C_1L_1R_1R_4g_m + C_4L_4R_1R_4g_m)}{2R_1g_m + s^5(C_1C_3C_4L_1L_4R_1R_4g_m + C_1C_3C_4L_1L_4R_4) + s^4(C_1C_3C_4L_4R_1R_4 + 2C_1C_4L_1L_4R_1g_m + 2C_1C_4L_1L_4) + s^3(C_1C_3L_1R_1R_4g_m + C_1C_3L_1R_4 + 2C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_4 + 2C_1C_4L_4R_1 + C_3C_4L_4R_1R_4g_m + C_3C_4L_4R_4) + s^2(C_1C_3R_1R_4 + 2C_1C_4R_1R_4g_m + C_3C_4R_1R_4g_m + C_3C_4R_4g_m) + s(C_1R_1R_4 + C_3R_1R_4g_m + C_3R_4g_m) + 1}$$

$$\mathbf{9.951 \quad X-INVALID-ORDER-951} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3}{C_3R_3s+1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1R_1R_3R_4g_ms^2 + R_1R_3R_4g_m}{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^3(C_1C_3L_1R_1R_3R_4g_m + C_1C_3L_1R_3R_4) + s^2(C_1C_3R_1R_3R_4 + 2C_1L_1R_1R_3g_m + C_1L_1R_1R_4g_m + 2C_1L_1R_3 + C_1L_1R_4) + s(2C_1R_1R_3 + C_1R_1R_4 + C_3R_1R_3R_4g_m + C_3R_3R_4) + 1}$$

$$\mathbf{9.952 \quad X-INVALID-ORDER-952} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3}{C_3R_3s+1}, \quad \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1R_1R_3g_ms^2 + R_1R_3g_m}{R_1g_m + s^3(C_1C_3L_1R_1R_3g_m + C_1C_3L_1R_3 + 2C_1C_4L_1R_1R_3g_m + 2C_1C_4L_1R_3) + s^2(C_1C_3R_1R_3 + 2C_1C_4R_1R_3 + C_1L_1R_1g_m + C_1L_1) + s(C_1R_1 + C_3R_1R_3g_m + C_3R_3 + 2C_4R_1R_3g_m + 2C_4R_3) + 1}$$

$$\mathbf{9.953 \quad X-INVALID-ORDER-953} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3}{C_3R_3s+1}, \quad \frac{R_4}{C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1R_1R_3R_4g_ms^2 + R_1R_3R_4g_m}{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^3(C_1C_3L_1R_1R_3R_4g_m + C_1C_3L_1R_3R_4 + 2C_1C_4L_1R_1R_3R_4g_m + 2C_1C_4L_1R_3R_4) + s^2(C_1C_3R_1R_3R_4 + 2C_1C_4R_1R_3R_4 + 2C_1L_1R_1R_3g_m + C_1L_1R_1R_4g_m + 2C_1L_1R_3 + C_1L_1R_4) + s(2C_1R_1R_3 + C_1R_1R_4 + C_3R_1R_3R_4g_m + C_3R_3R_4g_m) + 1}$$

$$\mathbf{9.954 \quad X-INVALID-ORDER-954} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3}{C_3R_3s+1}, \quad R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1R_1R_3R_4g_ms^3 + C_1L_1R_1R_3g_ms^2 + C_4R_1R_3R_4g_ms + R_1R_3g_m}{R_1g_m + s^4(C_1C_3C_4L_1R_1R_3R_4g_m + C_1C_3C_4L_1R_3R_4) + s^3(C_1C_3C_4R_1R_3R_4 + C_1C_3L_1R_1R_3g_m + C_1C_3L_1R_3 + 2C_1C_4L_1R_1R_3g_m + C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_3 + C_1C_4L_1R_4) + s^2(C_1C_3R_1R_3 + 2C_1C_4R_1R_3 + C_1C_4R_1R_4 + C_1L_1R_1g_m + C_1L_1 + C_3C_4R_1R_3g_m + C_3C_4R_1R_4g_m + C_3C_4R_4g_m) + s(C_1R_1R_3 + C_1R_1R_4 + C_3R_1R_3R_4g_m + C_3R_3R_4g_m) + 1}$$

$$\mathbf{9.955 \quad X-INVALID-ORDER-955} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3}{C_3R_3s+1}, \quad L_4s + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1L_4R_1R_3g_ms^4 + R_1R_3g_m + s^2(C_1L_1R_1R_3g_m + C_4L_4R_1R_3g_m)}{R_1g_m + s^5(C_1C_3C_4L_1L_4R_1R_3g_m + C_1C_3C_4L_1L_4R_3) + s^4(C_1C_3C_4L_4R_1R_3 + C_1C_4L_1L_4R_1g_m + C_1C_4L_1L_4) + s^3(C_1C_3L_1R_1R_3g_m + C_1C_3L_1R_3 + 2C_1C_4L_1R_1R_3g_m + 2C_1C_4L_1R_3 + C_1C_4L_4R_1 + C_3C_4L_4R_1R_3g_m + C_3C_4L_4R_3) + s^2(C_1C_3R_1R_3 + 2C_1C_4R_1R_3 + C_1C_4R_1R_4 + C_1L_1R_1g_m + C_1L_1 + C_3C_4R_1R_3g_m + C_3C_4R_1R_4g_m + C_3C_4R_4g_m) + s(C_1R_1R_3 + C_1R_1R_4 + C_3R_1R_3R_4g_m + C_3R_3R_4g_m) + 1}$$

$$\mathbf{9.956 \quad X-INVALID-ORDER-956} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3}{C_3R_3s+1}, \quad \frac{L_4s}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1L_4R_1R_3g_ms^3 + L_4R_1R_3g_ms}{2R_1R_3g_m + 2R_3 + s^4(C_1C_3L_1L_4R_1R_3g_m + C_1C_3L_1L_4R_3 + 2C_1C_4L_1L_4R_1R_3g_m + 2C_1C_4L_1L_4R_3) + s^3(C_1C_3L_4R_1R_3 + 2C_1C_4L_4R_1R_3 + C_1L_1L_4R_1g_m + C_1L_1L_4) + s^2(2C_1L_1R_1R_3g_m + 2C_1L_1R_3 + C_1L_4R_1 + C_3L_4R_1R_3g_m + C_3L_4R_3 + 2C_4L_4R_1R_3g_m + 2C_4L_4R_3) + s(C_1R_1R_3 + C_1R_1R_4 + C_3R_1R_3R_4g_m + C_3R_3R_4g_m) + 1}$$

$$\mathbf{9.957 \quad X-INVALID-ORDER-957} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3}{C_3R_3s+1}, \quad L_4s + R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1L_4R_1R_3g_ms^4 + C_1C_4L_1R_1R_3R_4g_ms^3 + C_4L_4R_1R_3g_ms + R_1R_3g_m}{R_1g_m + s^5(C_1C_3C_4L_1L_4R_1R_3g_m + C_1C_3C_4L_1L_4R_3) + s^4(C_1C_3C_4L_1R_1R_3R_4g_m + C_1C_3C_4L_1R_3R_4 + C_1C_3C_4L_4R_1R_3 + C_1C_4L_1L_4R_1g_m + C_1C_4L_1L_4) + s^3(C_1C_3C_4R_1R_3R_4 + C_1C_3L_1R_1R_3g_m + C_1C_3L_1R_3 + 2C_1C_4L_1R_1R_3g_m + C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_3 + C_1C_4L_4R_1 + C_3C_4L_4R_1R_3g_m + C_3C_4L_4R_3) + s^2(C_1C_3R_1R_3 + 2C_1C_4R_1R_3 + C_1C_4R_1R_4 + C_1L_1R_1g_m + C_1L_1 + C_3C_4R_1R_3g_m + C_3C_4R_1R_4g_m + C_3C_4R_4g_m) + s(C_1R_1R_3 + C_1R_1R_4 + C_3R_1R_3R_4g_m + C_3R_3R_4g_m) + 1}$$

$$\mathbf{9.958 \quad X-INVALID-ORDER-958} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3}{C_3R_3s+1}, \quad \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1L_4R_1R_3R_4g_ms^3 + L_4R_1R_3R_4g_ms}{2R_1R_3R_4g_m + 2R_3R_4 + s^4(C_1C_3L_1L_4R_1R_3R_4g_m + C_1C_3L_1L_4R_3R_4 + 2C_1C_4L_1L_4R_1R_3R_4g_m + 2C_1C_4L_1L_4R_3R_4) + s^3(C_1C_3L_4R_1R_3R_4 + 2C_1C_4L_4R_1R_3R_4 + 2C_1L_1L_4R_1R_3g_m + C_1L_1L_4R_1R_4g_m + 2C_1L_1L_4R_3 + C_1L_1L_4R_4) + s^2(2C_1L_1R_1R_3R_4g_m + 2C_1L_1R_1R_3R_4 + 2C_1L_1R_1R_4g_m + 2C_1L_1R_1R_4 + 2C_1L_1R_3 + 2C_1L_1R_4) + s(C_1R_1R_3R_4g_m + C_1R_1R_3R_4 + C_1R_1R_4g_m + C_1R_1R_4 + C_1R_3 + C_1R_4) + 2}$$

$$\mathbf{9.959 \quad X-INVALID-ORDER-959} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3}{C_3R_3s+1}, \quad \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1L_4R_1R_3R_4g_ms^3 + L_4R_1R_3R_4g_ms}{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^5(C_1C_3C_4L_1L_4R_1R_3R_4g_m + C_1C_3C_4L_1L_4R_3R_4) + s^4(C_1C_3C_4L_4R_1R_3R_4 + C_1C_3L_1L_4R_1R_3g_m + C_1C_3L_1L_4R_3 + 2C_1C_4L_1L_4R_1R_3g_m + C_1C_4L_1L_4R_1R_4g_m + 2C_1C_4L_1L_4R_3 + C_1C_4L_1L_4R_4) + s^3(C_1C_3L_1R_1R_3R_4g_m + C_1C_3L_1R_1R_3R_4 + 2C_1C_4L_1R_1R_3g_m + 2C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_3 + C_1C_4L_1R_4) + s^2(2C_1L_1R_1R_3R_4g_m + 2C_1L_1R_1R_3R_4 + 2C_1L_1R_1R_4g_m + 2C_1L_1R_1R_4 + 2C_1L_1R_3 + 2C_1L_1R_4) + s(C_1R_1R_3R_4g_m + C_1R_1R_3R_4 + C_1R_1R_4g_m + C_1R_1R_4 + C_1R_3 + C_1R_4) + 2}$$

$$\mathbf{9.960 \quad X-INVALID-ORDER-960} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3}{C_3R_3s+1}, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1R_1R_3R_4g_ms^3 + C_4L_1R_1R_3R_4g_ms}{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^5(C_1C_3C_4L_1L_4R_1R_3R_4g_m + C_1C_3C_4L_1L_4R_3R_4) + s^4(C_1C_3C_4L_4R_1R_3R_4 + 2C_1C_4L_1L_4R_1R_3g_m + C_1C_4L_1L_4R_1R_4g_m + 2C_1C_4L_1L_4R_3 + C_1C_4L_1L_4R_4) + s^3(C_1C_3L_1R_1R_3R_4g_m + C_1C_3L_1R_1R_3R_4 + 2C_1C_4L_1R_1R_3g_m + 2C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_3 + C_1C_4L_1R_4) + s^2(2C_1L_1R_1R_3R_4g_m + 2C_1L_1R_1R_3R_4 + 2C_1L_1R_1R_4g_m + 2C_1L_1R_1R_4 + 2C_1L_1R_3 + 2C_1L_1R_4) + s(C_1R_1R_3R_4g_m + C_1R_1R_3R_4 + C_1R_1R_4g_m + C_1R_1R_4 + C_1R_3 + C_1R_4) + 2}$$

$$\mathbf{9.961 \quad X-INVALID-ORDER-961} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad R_3 + \frac{1}{C_3s}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3L_1R_1R_3R_4g_ms^3 + C_1L_1R_1R_4g_ms^2 + C_3R_1R_3R_4g_ms + R_1R_4g_m}{2R_1g_m + s^3(2C_1C_3L_1R_1R_3g_m + C_1C_3L_1R_1R_4g_m + 2C_1C_3L_1R_3 + C_1C_3L_1R_4) + s^2(2C_1C_3R_1R_3 + C_1C_3R_1R_4 + 2C_1L_1R_1g_m + 2C_1L_1) + s(2C_1R_1 + 2C_3R_1R_3g_m + C_3R_1R_4g_m + 2C_3R_3 + C_3R_4) + 2}$$

$$\mathbf{9.962 \quad X-INVALID-ORDER-962} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad R_3 + \frac{1}{C_3s}, \quad \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3L_1R_1R_3g_ms^3 + C_1L_1R_1g_ms^2 + C_3R_1R_3g_ms + R_1g_m}{s^4(2C_1C_3C_4L_1R_1R_3g_m + 2C_1C_3C_4L_1R_3) + s^3(2C_1C_3C_4R_1R_3 + C_1C_3L_1R_1g_m + C_1C_3L_1 + 2C_1C_4L_1R_1g_m + 2C_1C_4L_1) + s^2(C_1C_3R_1 + 2C_1C_4R_1 + 2C_3C_4R_1R_3g_m + 2C_3C_4R_3) + s(C_3R_1g_m + C_3 + 2C_4R_1g_m + 2C_4) + 2}$$

$$\mathbf{9.963 \quad X-INVALID-ORDER-963} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad R_3 + \frac{1}{C_3s}, \quad \frac{R_4}{C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3L_1R_1R_3R_4g_ms^3 + C_1L_1R_1R_4g_ms^2 + C_3R_1R_3R_4g_ms + R_1R_4g_m}{2R_1g_m + s^4(2C_1C_3C_4L_1R_1R_3R_4g_m + 2C_1C_3C_4L_1R_3R_4) + s^3(2C_1C_3C_4R_1R_3R_4 + 2C_1C_3L_1R_1R_3g_m + C_1C_3L_1R_1R_4g_m + 2C_1C_3L_1R_3 + C_1C_3L_1R_4 + 2C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_4) + s^2(2C_1C_3R_1R_3 + C_1C_3R_1R_4 + 2C_1C_4R_1R_4 + 2C_1L_1R_1g_m + 2C_1L_1 + 2C_3C_4R_1R_3g_m + 2C_3C_4R_3) + s(C_3R_1g_m + C_3 + 2C_4R_1g_m + 2C_4) + 2}$$

$$\mathbf{9.964 \quad X-INVALID-ORDER-964} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad R_3 + \frac{1}{C_3s}, \quad R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1R_1R_3R_4g_ms^4 + R_1g_m + s^3(C_1C_3L_1R_1R_3g_m + C_1C_4L_1R_1R_4g_m) + s^2(C_1L_1R_1g_m + C_3C_4R_1R_3R_4g_m) + s(C_3R_1R_3g_m + C_4R_1R_4g_m)}{s^4(2C_1C_3C_4L_1R_1R_3g_m + C_1C_3C_4L_1R_1R_4g_m + 2C_1C_3C_4L_1R_3 + C_1C_3C_4L_1R_4) + s^3(2C_1C_3C_4R_1R_3 + C_1C_3C_4R_1R_4 + C_1C_3L_1R_1g_m + C_1C_3L_1 + 2C_1C_4L_1R_1g_m + 2C_1C_4L_1) + s^2(C_1C_3R_1 + 2C_1C_4R_1 + 2C_3C_4R_1R_3g_m + C_3C_4R_1R_4g_m + 2C_3C_4R_3 + C_3C_4R_4) + s(C_3R_1g_m + C_3 + 2C_4R_1g_m + 2C_4) + 2}$$

$$\mathbf{9.965 \quad X-INVALID-ORDER-965} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad R_3 + \frac{1}{C_3s}, \quad L_4s + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_4R_1R_3g_ms^5 + C_1C_4L_1L_4R_1g_ms^4 + C_3R_1R_3g_ms + R_1g_m + s^3(C_1C_3L_1R_1R_3g_m + C_3C_4L_4R_1R_3g_m) + s^2(C_1L_1R_1g_m + C_4L_4R_1g_m)}{s^5(C_1C_3C_4L_1L_4R_1g_m + C_1C_3C_4L_1L_4) + s^4(2C_1C_3C_4L_1R_1R_3g_m + 2C_1C_3C_4L_1R_3 + C_1C_3C_4L_4R_1) + s^3(2C_1C_3C_4R_1R_3 + C_1C_3L_1R_1g_m + C_1C_3L_1 + 2C_1C_4L_1R_1g_m + 2C_1C_4L_1 + C_3C_4L_4R_1g_m + C_3C_4L_4) + s^2(C_1C_3R_1 + 2C_1C_4R_1 + 2C_3C_4R_1R_3g_m + 2C_3C_4R_3 + C_3C_4R_4) + s(C_3R_1g_m + C_3 + 2C_4R_1g_m + 2C_4) + 2}$$

$$\mathbf{9.966 \quad X-INVALID-ORDER-966} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad R_3 + \frac{1}{C_3s}, \quad \frac{L_4s}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3L_1L_4R_1R_3g_ms^4 + C_1L_1L_4R_1g_ms^3 + C_3L_4R_1R_3g_ms^2 + L_4R_1g_ms}{2R_1g_m + s^5(2C_1C_3C_4L_1L_4R_1R_3g_m + 2C_1C_3C_4L_1L_4R_3) + s^4(2C_1C_3C_4L_4R_1R_3 + C_1C_3L_1L_4R_1g_m + C_1C_3L_1L_4 + 2C_1C_4L_1L_4R_1g_m + 2C_1C_4L_1L_4) + s^3(2C_1C_3L_1R_1R_3g_m + 2C_1C_3L_1R_3 + C_1C_3L_4R_1 + 2C_1C_4L_4R_1 + 2C_3C_4L_4R_1R_3g_m + 2C_3C_4L_4R_3) + s^2(2C_1L_1R_1R_3R_4g_m + 2C_1L_1R_1R_3R_4 + 2C_1L_1R_1R_4g_m + 2C_1L_1R_1R_4 + 2C_1L_1R_3 + 2C_1L_1R_4) + s(C_1R_1R_3R_4g_m + C_1R_1R_3R_4 + C_1R_1R_4g_m + C_1R_1R_4 + C_1R_3 + C_1R_4) + 2}$$

$$\mathbf{9.976 \quad X-INVALID-ORDER-976} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad L_3s + \frac{1}{C_3s}, \quad \frac{L_4s}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3L_1L_3L_4R_1g_ms^5 + L_4R_1g_ms + s^3(C_1L_1L_4R_1g_m + C_3L_3L_4R_1g_m)}{2C_1C_3C_4L_3L_4R_1s^5 + 2C_1R_1s + 2R_1g_m + s^6(2C_1C_3C_4L_1L_3L_4R_1g_m + 2C_1C_3C_4L_1L_3L_4) + s^4(2C_1C_3L_1L_3R_1g_m + 2C_1C_3L_1L_3 + C_1C_3L_1L_4R_1g_m + C_1C_3L_1L_4 + 2C_1C_4L_1L_4R_1g_m + 2C_1C_4L_1L_4 + 2C_3C_4L_3L_4R_1g_m + 2C_3C_4L_3L_4) + s^3(2C_1C_3L_3R_1 + C_1C_3L_4R_1g_m)}$$

$$\mathbf{9.977 \quad X-INVALID-ORDER-977} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad L_3s + \frac{1}{C_3s}, \quad L_4s + R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4R_1g_ms^6 + C_1C_3C_4L_1L_3R_1R_4g_ms^5 + C_4R_1R_4g_ms + R_1g_m + s^4(C_1C_3L_1L_3R_1g_m + C_1C_4L_1L_4R_1g_m + C_3C_4L_3L_4R_1g_m) + s^3(C_1C_4L_1R_1R_4g_m + C_3C_4L_3R_1R_4g_m)}{s^5(2C_1C_3C_4L_1L_3R_1g_m + 2C_1C_3C_4L_1L_3 + C_1C_3C_4L_1L_4R_1g_m + C_1C_3C_4L_1L_4) + s^4(C_1C_3C_4L_1R_1R_4g_m + C_1C_3C_4L_1R_4 + 2C_1C_3C_4L_3R_1 + C_1C_3C_4L_4R_1) + s^3(C_1C_3C_4R_1R_4 + C_1C_3L_1R_1g_m + C_1C_3L_1 + 2C_1C_4L_1R_1g_m + 2C_1C_4L_1 + 2C_3C_4L_3R_1g_m + 2C_3C_4L_3L_4R_1g_m)}$$

$$\mathbf{9.978 \quad X-INVALID-ORDER-978} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad L_3s + \frac{1}{C_3s}, \quad \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4R_1R_4g_ms^5 + C_1C_3C_4L_1L_3L_4R_4)}{2R_1R_4g_m + 2R_4 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_4g_m + 2C_1C_3C_4L_1L_3L_4R_4) + s^5(2C_1C_3C_4L_3L_4R_1R_4 + 2C_1C_3L_1L_3L_4R_1g_m + 2C_1C_3L_1L_3L_4) + s^4(2C_1C_3L_1L_3R_1R_4g_m + 2C_1C_3L_1L_3R_4 + C_1C_3L_1L_4R_1R_4g_m + C_1C_3L_1L_4R_4 + 2C_1C_3L_3L_4R_1 + 2C_1C_4L_1L_4R_1R_4g_m)}$$

$$\mathbf{9.979 \quad X-INVALID-ORDER-979} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad L_3s + \frac{1}{C_3s}, \quad \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4R_1R_4g_ms^6 + C_1C_3L_1L_3L_4R_1g_ms^5 + L_4R_1g_ms + R_1R_4g_m + s^4(C_1C_3L_1L_3R_1R_4g_m + C_1C_3L_1L_3R_4 + C_1C_3L_1L_4R_1R_4g_m + C_1C_3L_1L_4R_4 + 2C_1C_4L_1L_4R_1g_m + 2C_1C_4L_1L_4 + 2C_3C_4L_3R_1g_m + 2C_3C_4L_3L_4R_1g_m)}{2R_1g_m + s^6(2C_1C_3C_4L_1L_3L_4R_1g_m + 2C_1C_3C_4L_1L_3L_4) + s^5(C_1C_3C_4L_1L_4R_1R_4g_m + C_1C_3C_4L_1L_4R_4 + 2C_1C_3C_4L_3L_4R_1) + s^4(C_1C_3C_4L_4R_1R_4 + 2C_1C_3L_1L_3R_1g_m + 2C_1C_3L_1L_3 + C_1C_3L_1L_4R_1g_m + C_1C_3L_1L_4 + 2C_1C_4L_1L_4R_1g_m + 2C_1C_4L_1L_4 + 2C_3C_4L_3R_1g_m + 2C_3C_4L_3L_4R_1g_m)}$$

$$\mathbf{9.980 \quad X-INVALID-ORDER-980} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad L_3s + \frac{1}{C_3s}, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4R_1R_4g_ms^3 + L_3R_1R_4g_ms}{2R_1g_m + s^6(2C_1C_3C_4L_1L_3L_4R_1g_m + 2C_1C_3C_4L_1L_3L_4) + s^5(2C_1C_3C_4L_1L_3R_1R_4g_m + 2C_1C_3C_4L_1L_3R_4 + C_1C_3C_4L_1L_4R_1R_4g_m + C_1C_3C_4L_1L_4R_4 + 2C_1C_3C_4L_3L_4R_1) + s^4(2C_1C_3C_4L_3R_1R_4 + C_1C_3C_4L_4R_1R_4 + 2C_1C_3L_1L_3R_1g_m + 2C_1C_3L_1L_3 + 2C_1C_4L_1L_4R_1g_m + 2C_1C_4L_1L_4 + 2C_3C_4L_3R_1g_m + 2C_3C_4L_3L_4R_1g_m)}$$

$$\mathbf{9.981 \quad X-INVALID-ORDER-981} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{L_3s}{C_3L_3s^2+1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1L_3R_1R_4g_ms^3 + L_3R_1R_4g_ms}{R_1R_4g_m + R_4 + s^4(C_1C_3L_1L_3R_1R_4g_m + C_1C_3L_1L_3R_4) + s^3(C_1C_3L_3R_1R_4 + 2C_1L_1L_3R_1g_m + 2C_1L_1L_3) + s^2(C_1L_1R_1R_4g_m + C_1L_1R_4 + 2C_1L_3R_1 + C_3L_3R_1R_4g_m + C_3L_3R_4) + s(C_1R_1R_4 + 2L_3R_1g_m + 2L_3)}$$

$$\mathbf{9.982 \quad X-INVALID-ORDER-982} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{L_3s}{C_3L_3s^2+1}, \quad \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1L_3R_1g_ms^3 + L_3R_1g_ms}{C_1R_1s + R_1g_m + s^4(C_1C_3L_1L_3R_1g_m + C_1C_3L_1L_3 + 2C_1C_4L_1L_3R_1g_m + 2C_1C_4L_1L_3) + s^3(C_1C_3L_3R_1 + 2C_1C_4L_3R_1) + s^2(C_1L_1R_1g_m + C_1L_1 + C_3L_3R_1g_m + C_3L_3 + 2C_4L_3R_1g_m + 2C_4L_3) + 1}$$

$$\mathbf{9.983 \quad X-INVALID-ORDER-983} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{L_3s}{C_3L_3s^2+1}, \quad \frac{R_4}{C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1L_3R_1R_4g_ms^3 + L_3R_1R_4g_ms}{R_1R_4g_m + R_4 + s^4(C_1C_3L_1L_3R_1R_4g_m + C_1C_3L_1L_3R_4 + 2C_1C_4L_1L_3R_1R_4g_m + 2C_1C_4L_1L_3R_4) + s^3(C_1C_3L_3R_1R_4 + 2C_1C_4L_3R_1R_4 + 2C_1L_1L_3R_1g_m + 2C_1L_1L_3) + s^2(C_1L_1R_1R_4g_m + C_1L_1R_4 + 2C_1L_3R_1 + C_3L_3R_1R_4g_m + C_3L_3R_4 + 2C_4L_3R_1R_4g_m + 2C_4L_3R_1R_4)}$$

$$\mathbf{9.984 \quad X-INVALID-ORDER-984} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{L_3s}{C_3L_3s^2+1}, \quad R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1L_3R_1R_4g_ms^4 + C_1L_1L_3R_1g_ms^3 + C_4L_3R_1R_4g_ms^2 + L_3R_1g_ms}{R_1g_m + s^5(C_1C_3C_4L_1L_3R_1R_4g_m + C_1C_3C_4L_1L_3R_4) + s^4(C_1C_3C_4L_3R_1R_4 + C_1C_3L_1L_3R_1g_m + C_1C_3L_1L_3 + 2C_1C_4L_1L_3R_1g_m + 2C_1C_4L_1L_3) + s^3(C_1C_3L_3R_1 + C_1C_4L_1R_1R_4g_m + C_1C_4L_1R_4 + 2C_1C_4L_3R_1 + C_3C_4L_3R_1R_4g_m + C_3C_4L_3R_4) + s^2(C_1C_4R_1R_4g_m + C_1C_4R_1R_4)}$$

9.985 X-INVALID-ORDER-985 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_1 g_m s^5 + L_3 R_1 g_m s + s^3 (C_1 L_1 L_3 R_1 g_m + C_4 L_3 L_4 R_1 g_m)}{C_1 C_3 C_4 L_3 L_4 R_1 s^5 + C_1 R_1 s + R_1 g_m + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 R_1 g_m + 2C_1 C_4 L_1 L_3 + C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + C_3 C_4 L_3 L_4 R_1 g_m + C_3 C_4 L_3 L_4) + s^3 (C_1 C_3 L_3 R_1 + 2C_1 C_4 L_3 R_1 + C_1 C_4 L_3 L_4 R_1 g_m + C_1 C_4 L_3 L_4)}$$

9.986 X-INVALID-ORDER-986 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 L_4 R_1 g_m s^3 + L_3 L_4 R_1 g_m s}{2L_3 R_1 g_m + 2L_3 + L_4 R_1 g_m + L_4 + s^4 (C_1 C_3 L_1 L_3 L_4 R_1 g_m + C_1 C_3 L_1 L_3 L_4 + 2C_1 C_4 L_1 L_3 L_4 R_1 g_m + 2C_1 C_4 L_1 L_3 L_4) + s^3 (C_1 C_3 L_3 L_4 R_1 + 2C_1 C_4 L_3 L_4 R_1) + s^2 (2C_1 L_1 L_3 R_1 g_m + 2C_1 L_1 L_3 + C_1 L_1 L_4 R_1 g_m + C_1 L_1 L_4 + C_3 L_3 L_4 R_1 g_m + C_3 L_3 L_4 + 2C_4 L_3 L_4 R_1 g_m +$$

9.987 X-INVALID-ORDER-987 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4 R_1 g_m s^5 + C_1 C_4 L_1 L_3 R_1 R_4 g_m s^4 + C}{R_1 g_m + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 g_m + C_1 C_3 C_4 L_1 L_3 L_4) + s^5 (C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + C_1 C_3 C_4 L_1 L_3 R_4 + C_1 C_3 C_4 L_3 L_4 R_1) + s^4 (C_1 C_3 C_4 L_3 R_1 R_4 + C_1 C_3 L_1 L_3 R_1 g_m + C_1 C_3 L_1 L_3 + 2C_1 C_4 L_1 L_3 R_1 g_m + 2C_1 C_4 L_1 L_3 + C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + C_3 C_4 L_3 L_4 R_1 g_m)}$$

9.988 X-INVALID-ORDER-988 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 L_4 R_1 R_4 g_m s^3 + L_3 L_4 R_1 R_4 g_m s}{2L_3 R_1 R_4 g_m + 2L_3 R_4 + L_4 R_1 R_4 g_m + L_4 R_4 + s^4 (C_1 C_3 L_1 L_3 L_4 R_1 R_4 g_m + C_1 C_3 L_1 L_3 L_4 R_4 + 2C_1 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_3 L_4 R_4) + s^3 (C_1 C_3 L_3 L_4 R_1 R_4 + 2C_1 C_4 L_3 L_4 R_1 R_4 + 2C_1 L_1 L_3 L_4 R_1 g_m + 2C_1 L_1 L_3 L_4) + s^2 (2C_1 L_1 L_3 R_1 R_4 g_m + 2C_1 L_1 L_3 R_4 + C_1)}$$

9.989 X-INVALID-ORDER-989 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \frac{L_3s}{C_3L_3s^2+1}, \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1}, \infty, \infty \right)$

$$H(s) = \frac{C_1}{R_1 R_4 g_m + R_4 + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_4 + C_1 C_3 L_1 L_3 L_4 R_1 g_m + C_1 C_3 L_1 L_3 L_4 + 2 C_1 C_4 L_1 L_3 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 L_1 L_3 R_1 R_4 g_m + C_1 C_3 L_1 L_3 R_4 + C_1 C_3 L_3 L_4 R_1 + C_1 C_4 L_1 L_4 R_1 R_4 g_m + C_1 C_4 L_1 L_4 R_4 g_m + C_1 C_4 L_1 L_4 R_4 + C_1 C_4 L_3 L_4 R_1 g_m + C_1 C_4 L_3 L_4 R_4) + s^3 (C_1 C_3 L_1 L_3 R_1 R_4 + C_1 C_3 L_1 L_3 R_4 + C_1 C_3 L_3 L_4 R_1 + C_1 C_4 L_1 L_4 R_1 R_4 + C_1 C_4 L_1 L_4 R_4 + C_1 C_4 L_3 L_4 R_1 + C_1 C_4 L_3 L_4 R_4) + s^2 (C_1 C_3 L_1 L_3 R_1 + C_1 C_3 L_1 L_3 R_4 + C_1 C_3 L_3 L_4 R_1 + C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_4 R_4 + C_1 C_4 L_3 L_4 R_1 + C_1 C_4 L_3 L_4 R_4) + s (C_1 C_3 L_1 L_3 R_1 + C_1 C_3 L_1 L_3 R_4 + C_1 C_3 L_3 L_4 R_1 + C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_4 R_4 + C_1 C_4 L_3 L_4 R_1 + C_1 C_4 L_3 L_4 R_4) + C_1}$$

9.990 X-INVALID-ORDER-990 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1}{R_1 R_4 g_m + R_4 + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_4 + 2 C_1 C_4 L_1 L_3 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 L_1 L_3 R_1 R_4 g_m + C_1 C_3 L_1 L_3 R_4 + 2 C_1 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_1 C_4 L_1 L_3 R_4 + C_1 C_4 L_1 L_4 R_1 R_4 g_m + C_1 C_4 L_1 L_4 R_4 + 2 C_1 L_1 L_3 L_4 R_1 R_4 g_m + 2 C_1 L_1 L_3 L_4 R_4) + s^3 (C_1 C_3 L_1 L_3 R_1 R_4 g_m + C_1 C_3 L_1 L_3 R_4 + 2 C_1 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_1 C_4 L_1 L_3 R_4 + C_1 C_4 L_1 L_4 R_1 R_4 g_m + C_1 C_4 L_1 L_4 R_4) + s^2 (C_1 C_3 L_1 L_3 R_1 R_4 g_m + C_1 C_3 L_1 L_3 R_4 + 2 C_1 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_1 C_4 L_1 L_3 R_4 + C_1 C_4 L_1 L_4 R_1 R_4 g_m + C_1 C_4 L_1 L_4 R_4) + s (C_1 C_3 L_1 L_3 R_1 R_4 g_m + C_1 C_3 L_1 L_3 R_4 + 2 C_1 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_1 C_4 L_1 L_3 R_4 + C_1 C_4 L_1 L_4 R_1 R_4 g_m + C_1 C_4 L_1 L_4 R_4) + C_1 C_4 L_1 R_1 R_4 g_m + C_1 C_4 L_1 R_1 R_4}.$$

9.991 X-INVALID-ORDER-991 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 R_4 g_m s^4 + C_1 C_3 L_1 R_1 R_3 R_4 g_m s^3 + C_3 R_1 R_3 R_4 g_m s + R_1 R_4 g_m + s^2 (C_1 L_1 R_1 R_4 g_m + C_3 L_3 R_1 R_4 g_m)}{2R_1 g_m + s^4 (2C_1 C_3 L_1 L_3 R_1 g_m + 2C_1 C_3 L_1 L_3) + s^3 (2C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_1 R_4 g_m + 2C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4 + 2C_1 C_3 L_3 R_1) + s^2 (2C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_4 + 2C_1 L_1 R_1 g_m + 2C_1 L_1 + 2C_3 L_3 R_1 g_m + 2C_3 L_3) + s (2C_1 R_1 + 2C_3 R_1 R_3 g_m + C_3 R_1 R_4 g_m + 2}$$

9.992 X-INVALID-ORDER-992 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 g_m s^4 + C_1 C_3 L_1 R_1 R_3 g_m s^3 + C_3 R_1 R_3 g_m s + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_3 L_3 R_1 g_m)}{s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 g_m + 2C_1 C_3 C_4 L_1 L_3) + s^4 (2C_1 C_3 C_4 L_1 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 R_3 + 2C_1 C_3 C_4 L_3 R_1) + s^3 (2C_1 C_3 C_4 R_1 R_3 + C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_3 C_4 L_3 R_1 g_m + 2C_3 C_4 L_3) + s^2 (C_1 C_3 R_1 + 2C_1 C_4 R_1 + 2C_3 C_4 R_1 R_3 g_m + 2C_3 C_4 R_1)}$$

9.993 X-INVALID-ORDER-993 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 R_4 g_m s^4 + C_1 C_3 L_1 R_1 R_4 g_m s^5 + 2C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_4 + s^4 (2C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_1 R_3 R_4 + 2C_1 C_3 C_4 L_3 R_1 R_4 + 2C_1 C_3 L_1 L_3 R_1 g_m + 2C_1 C_3 L_1 L_3) + s^3 (2C_1 C_3 C_4 R_1 R_3 R_4 + 2C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_1 R_4 g_m + 2C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4 + 2C_1 C_3 L_1 R_4 g_m)}{2R_1 g_m + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_4) + s^4 (2C_1 C_3 C_4 L_1 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_1 R_3 R_4 + 2C_1 C_3 C_4 L_3 R_1 R_4 + 2C_1 C_3 L_1 L_3 R_1 g_m + 2C_1 C_3 L_1 L_3) + s^3 (2C_1 C_3 C_4 R_1 R_3 R_4 + 2C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_1 R_4 g_m + 2C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_4 + 2C_1 C_3 L_1 R_4 g_m)}$$

9.1003 X-INVALID-ORDER-1003 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 R_1 R_3 R_4 g_m s^3 + L_3 R_1 R_3 R_4 g_m s}{R_1 R_3 R_4 g_m + R_3 R_4 + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 R_3 R_4 + 2 C_1 C_4 L_1 L_3 R_1 R_3 R_4 g_m + 2 C_1 C_4 L_1 L_3 R_3 R_4) + s^3 (C_1 C_3 L_3 R_1 R_3 R_4 + 2 C_1 C_4 L_3 R_1 R_3 R_4 + 2 C_1 L_1 L_3 R_1 R_3 g_m + C_1 L_1 L_3 R_1 R_4 g_m + 2 C_1 L_1 L_3 R_3 + C_1 L_1 L_3 R_4) + s^2 (C_1 L_1 R_1 R_3 R_4 g_m + C_1 L_1 R_3 R_4 g_m)}$$

9.1004 X-INVALID-ORDER-1004 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_3 R_1 R_4}{R_1 R_3 g_m + R_3 + s^5 (C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_3 R_3 R_4) + s^4 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 + C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_3 L_1 L_3 R_3 + 2 C_1 C_4 L_1 L_3 R_1 R_3 g_m + C_1 C_4 L_1 L_3 R_1 R_4 g_m + 2 C_1 C_4 L_1 L_3 R_3 + C_1 C_4 L_1 L_3 R_4) + s^3 (C_1 C_3 L_3 R_1 R_3 + C_1 C_4 L_1 R_1 R_3 R_4 g_m + C_1 C_4 L_1 R_1 R_3 R_4)}$$

9.1005 X-INVALID-ORDER-1005 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_3 L_4}{R_1 R_3 g_m + R_3 + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_3) + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 + C_1 C_4 L_1 L_3 L_4 R_1 g_m + C_1 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_3 L_1 L_3 R_3 + 2 C_1 C_4 L_1 L_3 R_1 R_3 g_m + 2 C_1 C_4 L_1 L_3 R_3 + C_1 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_4 R_3 + C_1 C_4 L_1 L_4)}.$$

9.1006 X-INVALID-ORDER-1006 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 L_4 R_1 R_3 g_m s^3 + L_3 L_4 R_1 R_3 g_m s}{2L_3 R_1 R_3 g_m + 2L_3 R_3 + L_4 R_1 R_3 g_m + L_4 R_3 + s^4 (C_1 C_3 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 L_1 L_3 L_4 R_3 + 2C_1 C_4 L_1 L_3 L_4 R_1 R_3 g_m + 2C_1 C_4 L_1 L_3 L_4 R_3) + s^3 (C_1 C_3 L_3 L_4 R_1 R_3 + 2C_1 C_4 L_3 L_4 R_1 R_3 + C_1 L_1 L_3 L_4 R_1 g_m + C_1 L_1 L_3 L_4) + s^2 (2C_1 L_1 L_3 R_1 R_3 g_m + 2C_1 L_1 L_3 R_3 + C_1 L_1)}$$

9.1007 X-INVALID-ORDER-1007 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_3 g_m + R_3 + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_3) + s^5 (C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_3 L_4 R_1 R_3 + C_1 C_4 L_1 L_3 L_4 R_1 g_m + C_1 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 + C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_3 L_1 L_3 R_3 + 2 C_1 C_4 L_1 L_3 R_1)}{R_1 R_3 g_m + R_3 + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_3) + s^5 (C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_3 L_4 R_1 R_3 + C_1 C_4 L_1 L_3 L_4 R_1 g_m + C_1 C_4 L_1 L_3 L_4) + s^4 (C_1 C_3 C_4 L_3 R_1 R_3 R_4 + C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_3 L_1 L_3 R_3 + 2 C_1 C_4 L_1 L_3 R_1)}$$

9.1008 X-INVALID-ORDER-1008 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{2L_3R_1R_3R_4g_m + 2L_3R_3R_4 + L_4R_1R_3R_4g_m + L_4R_3R_4 + s^4(C_1C_3L_1L_3L_4R_1R_3R_4g_m + C_1C_3L_1L_3L_4R_3R_4 + 2C_1C_4L_1L_3L_4R_1R_3R_4g_m + 2C_1C_4L_1L_3L_4R_3R_4) + s^3(C_1C_3L_3L_4R_1R_3R_4 + 2C_1C_4L_3L_4R_1R_3R_4 + 2C_1L_1L_3L_4R_1R_3g_m + C_1L_1L_3L_4R_1R_4g_m + 2C_1L_1L_3L_4R_3R_4) + s^2(C_1C_3L_1L_3L_4R_1R_3R_4g_m + C_1C_3L_1L_3L_4R_3R_4 + 2C_1C_4L_1L_3L_4R_1R_3R_4g_m + 2C_1C_4L_1L_3L_4R_3R_4) + s(C_1C_3L_3L_4R_1R_3R_4 + 2C_1C_4L_3L_4R_1R_3R_4 + 2C_1L_1L_3L_4R_1R_3g_m + C_1L_1L_3L_4R_1R_4g_m + 2C_1L_1L_3L_4R_3R_4) + C_1C_3L_1L_3L_4R_1R_3R_4g_m + C_1C_3L_1L_3L_4R_3R_4}{2L_3R_1R_3R_4g_m + 2L_3R_3R_4 + L_4R_1R_3R_4g_m + L_4R_3R_4 + s^4(C_1C_3L_1L_3L_4R_1R_3R_4g_m + C_1C_3L_1L_3L_4R_3R_4 + 2C_1C_4L_1L_3L_4R_1R_3R_4g_m + 2C_1C_4L_1L_3L_4R_3R_4) + s^3(C_1C_3L_3L_4R_1R_3R_4 + 2C_1C_4L_3L_4R_1R_3R_4 + 2C_1L_1L_3L_4R_1R_3g_m + C_1L_1L_3L_4R_1R_4g_m + 2C_1L_1L_3L_4R_3R_4) + s^2(C_1C_3L_1L_3L_4R_1R_3R_4g_m + C_1C_3L_1L_3L_4R_3R_4 + 2C_1C_4L_1L_3L_4R_1R_3R_4g_m + 2C_1C_4L_1L_3L_4R_3R_4) + s(C_1C_3L_3L_4R_1R_3R_4 + 2C_1C_4L_3L_4R_1R_3R_4 + 2C_1L_1L_3L_4R_1R_3g_m + C_1L_1L_3L_4R_1R_4g_m + 2C_1L_1L_3L_4R_3R_4) + C_1C_3L_1L_3L_4R_1R_3R_4g_m + C_1C_3L_1L_3L_4R_3R_4}$$

9.1009 X-INVALID-ORDER-1009 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_4 g_m + R_3 R_4 + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4) + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 + C_1 C_3 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 L_1 L_3 L_4 R_3 + 2C_1 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_3 L_4 R_3 + C_1 C_4 L_1 L_3 L_4 R_4) + s^4 (C_1 C_3 L_1 L_3 R_1)}{...}$$

9.1010 X-INVALID-ORDER-1010 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_3 R_4 g_m + R_3 R_4 + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4) + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 + 2 C_1 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_1 L_3 L_4 R_3 + C_1 C_4 L_1 L_3 L_4 R_4) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 R_3 R_4 + 2 C_1 C_4 L_1 L_3 R_1 R_3 R_4)}{R_1 R_3 R_4 g_m + R_3 R_4 + s^6 (C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_3 R_4) + s^5 (C_1 C_3 C_4 L_3 L_4 R_1 R_3 R_4 + 2 C_1 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_1 L_3 L_4 R_3 + C_1 C_4 L_1 L_3 L_4 R_4) + s^4 (C_1 C_3 L_1 L_3 R_1 R_3 R_4 g_m + C_1 C_3 L_1 L_3 R_3 R_4 + 2 C_1 C_4 L_1 L_3 R_1 R_3 R_4)}$$

9.1011 X-INVALID-ORDER-1011 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_1 R_3 R_4 g_m s^4 + C_1 L_1 L_3 R_1 R_4 g_m s^3 + L_3 R_1 R_4 g_m s + R_1 R_3 R_4 g_m + s^2 (C_1 L_1 R_1 R_3 R_4 g_m + C_3 L_3 R_1 R_3 R_4 g_m)}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^4 (2C_1 C_3 L_1 L_3 R_1 R_3 g_m + C_1 C_3 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 L_1 L_3 R_3 + C_1 C_3 L_1 L_3 R_4) + s^3 (2C_1 C_3 L_3 R_1 R_3 + C_1 C_3 L_3 R_1 R_4 + 2C_1 L_1 L_3 R_1 g_m + 2C_1 L_1 L_3) + s^2 (2C_1 L_1 R_1 R_3 g_m + C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_3 + C_1 L_1 R_4 + 2C_1 L_3 R_1 + 2C_3 L_3 R_1 R_3) + s (2C_1 L_3 R_1 R_3 + C_3 L_3 R_1 R_3) + R_1 R_3}$$

$$\mathbf{9.1012 \quad X-INVALID-ORDER-1012} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{C_3L_3R_3s^2+L_3s+R_3}{C_3L_3s^2+1}, \quad \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3L_1L_3R_1R_3g_ms^4 + C_1L_1L_3R_1g_ms^3 + L_3R_1g_ms + R_1R_3g_m + s^2(C_1L_1R_1R_3g_m + C_3L_3R_1R_3g_m)}{R_1g_m + s^5(2C_1C_3C_4L_1L_3R_1R_3g_m + 2C_1C_3C_4L_1L_3R_3) + s^4(2C_1C_3C_4L_3R_1R_3 + C_1C_3L_1L_3R_1g_m + C_1C_3L_1L_3 + 2C_1C_4L_1L_3R_1g_m + 2C_1C_4L_1L_3) + s^3(C_1C_3L_3R_1 + 2C_1C_4L_1R_1R_3g_m + 2C_1C_4L_1R_3 + 2C_1C_4L_3R_1 + 2C_3C_4L_3R_1R_3g_m + 2C_3C_4L_3R_3) + s^2(2C_1C_3C_4L_1L_3R_1R_3g_m + 2C_1C_3C_4L_1L_3R_3) + s(2C_1C_3C_4L_1L_3R_1R_3 + 2C_1C_3C_4L_1L_3R_3) + 2C_1C_3C_4L_1L_3R_1R_3 + 2C_1C_3C_4L_1L_3R_3}$$

$$\mathbf{9.1013 \quad X-INVALID-ORDER-1013} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{C_3L_3R_3s^2+L_3s+R_3}{C_3L_3s^2+1}, \quad \frac{R_4}{C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^5(2C_1C_3C_4L_1L_3R_1R_3R_4g_m + 2C_1C_3C_4L_1L_3R_3R_4) + s^4(2C_1C_3C_4L_3R_1R_3R_4 + 2C_1C_3L_1L_3R_1R_3g_m + C_1C_3L_1L_3R_1R_4g_m + 2C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4 + 2C_1C_4L_1L_3R_1R_4g_m + 2C_1C_4L_1L_3R_4) + s^3(2C_1C_3L_3R_1R_3 + C_1C_3L_3R_1R_4 + 2C_1C_4L_1R_1R_3g_m + 2C_1C_4L_1R_3 + 2C_1C_4L_3R_1 + 2C_3C_4L_3R_1R_3g_m + 2C_3C_4L_3R_3) + s^2(2C_1C_3C_4L_1L_3R_1R_3g_m + 2C_1C_3C_4L_1L_3R_3) + s(2C_1C_3C_4L_1L_3R_1R_3 + 2C_1C_3C_4L_1L_3R_3) + 2C_1C_3C_4L_1L_3R_1R_3 + 2C_1C_3C_4L_1L_3R_3}$$

$$\mathbf{9.1014 \quad X-INVALID-ORDER-1014} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{C_3L_3R_3s^2+L_3s+R_3}{C_3L_3s^2+1}, \quad R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_3R_1R_3R_4g_ms^5 + R_1R_3g_m + s^4(C_1C_3L_1L_3R_1R_3g_m + C_1C_4L_1L_3R_1R_4g_m) + s^3(C_1C_4L_1R_1R_3 + C_1C_4L_1R_3 + 2C_1C_4L_3R_1 + 2C_3C_4L_3R_1R_3g_m + 2C_3C_4L_3R_3) + s^2(2C_1C_3C_4L_1L_3R_1R_3g_m + 2C_1C_3C_4L_1L_3R_3) + s(2C_1C_3C_4L_1L_3R_1R_3 + 2C_1C_3C_4L_1L_3R_3) + 2C_1C_3C_4L_1L_3R_1R_3 + 2C_1C_3C_4L_1L_3R_3}$$

$$\mathbf{9.1015 \quad X-INVALID-ORDER-1015} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{C_3L_3R_3s^2+L_3s+R_3}{C_3L_3s^2+1}, \quad L_4s + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4R_1R_3g_ms^6 + C_1C_4L_1L_3L_4R_1g_ms^5 + L_3R_1g_ms + R_1R_3g_m + s^4(C_1C_3L_1L_3R_1R_3g_m + C_1C_4L_1L_3R_1R_4g_m) + s^3(C_1C_4L_1R_1R_3 + C_1C_4L_1R_3 + 2C_1C_4L_3R_1 + 2C_3C_4L_3R_1R_3g_m + 2C_3C_4L_3R_3) + s^2(2C_1C_3C_4L_1L_3R_1R_3g_m + 2C_1C_3C_4L_1L_3R_3) + s(2C_1C_3C_4L_1L_3R_1R_3 + 2C_1C_3C_4L_1L_3R_3) + 2C_1C_3C_4L_1L_3R_1R_3 + 2C_1C_3C_4L_1L_3R_3}$$

$$\mathbf{9.1016 \quad X-INVALID-ORDER-1016} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{C_3L_3R_3s^2+L_3s+R_3}{C_3L_3s^2+1}, \quad \frac{L_4s}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{2R_1R_3g_m + 2R_3 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3g_m + 2C_1C_3C_4L_1L_3L_4R_3) + s^5(2C_1C_3C_4L_3L_4R_1R_3 + C_1C_3L_1L_3L_4R_1g_m + C_1C_3L_1L_3L_4 + 2C_1C_4L_1L_3L_4R_1g_m + 2C_1C_4L_1L_3L_4) + s^4(2C_1C_3L_1L_3R_1R_3g_m + 2C_1C_3L_1L_3R_3 + C_1C_3L_3L_4R_1 + 2C_1C_4L_1L_4R_1R_3g_m + 2C_1C_4L_1L_4R_3 + 2C_3C_4L_3R_1R_3g_m + 2C_3C_4L_3R_3) + s^3(2C_1C_3C_4L_1L_3L_4R_1R_3g_m + 2C_1C_3C_4L_1L_3L_4R_3) + s^2(2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3) + s(2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3) + 2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3}$$

$$\mathbf{9.1017 \quad X-INVALID-ORDER-1017} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{C_3L_3R_3s^2+L_3s+R_3}{C_3L_3s^2+1}, \quad L_4s + R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4R_1R_3g_ms^6 + R_1R_3g_m + s^5(C_1C_3C_4L_1L_3R_1R_3R_4g_m + C_1C_4L_1L_3L_4R_1g_m) + s^4(2C_1C_3C_4L_3R_1R_3 + C_1C_3C_4L_3R_1R_4 + C_1C_3L_1L_3R_1g_m + C_1C_3L_1L_3 + 2C_1C_4L_1L_3R_1g_m + 2C_1C_4L_1L_3R_3 + 2C_1C_4L_3R_1 + 2C_3C_4L_3R_1R_3g_m + 2C_3C_4L_3R_3) + s^3(2C_1C_3C_4L_1L_3L_4R_1R_3g_m + 2C_1C_3C_4L_1L_3L_4R_3) + s^2(2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3) + s(2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3) + 2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3}$$

$$\mathbf{9.1018 \quad X-INVALID-ORDER-1018} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{C_3L_3R_3s^2+L_3s+R_3}{C_3L_3s^2+1}, \quad \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{2R_1R_3R_4g_m + 2R_3R_4 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3R_4g_m + 2C_1C_3C_4L_1L_3L_4R_3R_4) + s^5(2C_1C_3C_4L_3L_4R_1R_3R_4 + 2C_1C_3L_1L_3L_4R_1R_3g_m + C_1C_3L_1L_3L_4R_1R_4g_m + 2C_1C_3L_1L_3L_4R_3 + C_1C_3L_1L_3L_4R_4 + 2C_1C_4L_1L_3L_4R_1R_4g_m + 2C_1C_4L_1L_3L_4R_4) + s^4(2C_1C_3L_1L_3R_1R_3g_m + 2C_1C_3L_1L_3R_3 + C_1C_3L_3L_4R_1 + 2C_1C_4L_1L_4R_1R_3g_m + 2C_1C_4L_1L_4R_3 + 2C_3C_4L_3R_1R_3g_m + 2C_3C_4L_3R_3) + s^3(2C_1C_3C_4L_1L_3L_4R_1R_3g_m + 2C_1C_3C_4L_1L_3L_4R_3) + s^2(2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3) + s(2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3) + 2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3}$$

$$\mathbf{9.1019 \quad X-INVALID-ORDER-1019} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{C_3L_3R_3s^2+L_3s+R_3}{C_3L_3s^2+1}, \quad \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3g_m + C_1C_3C_4L_1L_3L_4R_1R_4g_m + 2C_1C_3C_4L_1L_3L_4R_3 + C_1C_3C_4L_1L_3L_4R_4) + s^5(2C_1C_3C_4L_3L_4R_1R_3 + C_1C_3C_4L_3L_4R_1R_4 + C_1C_3L_1L_3L_4R_1g_m + C_1C_3L_1L_3L_4 + 2C_1C_4L_1L_3L_4R_1g_m + 2C_1C_4L_1L_3L_4R_3 + 2C_1C_4L_3R_1 + 2C_3C_4L_3R_1R_3g_m + 2C_3C_4L_3R_3) + s^4(2C_1C_3C_4L_1L_3L_4R_1R_3g_m + 2C_1C_3C_4L_1L_3L_4R_3) + s^3(2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3) + s^2(2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3) + s(2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3) + 2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3}$$

$$\mathbf{9.1020 \quad X-INVALID-ORDER-1020} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{C_3L_3R_3s^2+L_3s+R_3}{C_3L_3s^2+1}, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3g_m + C_1C_3C_4L_1L_3L_4R_1R_4g_m + 2C_1C_3C_4L_1L_3L_4R_3 + C_1C_3C_4L_1L_3L_4R_4) + s^5(2C_1C_3C_4L_1L_3R_1R_3R_4g_m + 2C_1C_3C_4L_1L_3R_3R_4 + 2C_1C_3C_4L_3L_4R_1R_3 + C_1C_3C_4L_3L_4R_1R_4 + 2C_1C_4L_1L_3L_4R_1g_m + 2C_1C_4L_1L_3L_4R_3 + 2C_1C_4L_3R_1 + 2C_3C_4L_3R_1R_3g_m + 2C_3C_4L_3R_3) + s^4(2C_1C_3C_4L_1L_3L_4R_1R_3g_m + 2C_1C_3C_4L_1L_3L_4R_3) + s^3(2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3) + s^2(2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3) + s(2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3) + 2C_1C_3C_4L_1L_3L_4R_1R_3 + 2C_1C_3C_4L_1L_3L_4R_3}$$

$$\mathbf{9.1021 \quad X-INVALID-ORDER-1021} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3L_1L_3R_1R_3R_4g_ms^4 + R_1R_3R_4g_m + s^2(C_1L_1R_1R_3R_4g_m + C_3L_3R_1R_3R_4g_m)}{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^4(2C_1C_3L_1L_3R_1R_3g_m + C_1C_3L_1L_3R_1R_4g_m + 2C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4) + s^3(C_1C_3L_1R_1R_3R_4g_m + C_1C_3L_1R_3R_4 + 2C_1C_3L_3R_1R_3 + C_1C_3L_3R_1R_4) + s^2(C_1C_3R_1R_3R_4 + 2C_1L_1R_1R_3g_m + C_1L_1R_1R_4g_m + 2C_1L_1R_3g_m)}$$

$$\mathbf{9.1022 \quad X-INVALID-ORDER-1022} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \quad \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3L_1L_3R_1R_3g_ms^4 + R_1R_3g_m + s^2(C_1L_1R_1R_3g_m + C_3L_3R_1R_3g_m)}{R_1g_m + s^5(2C_1C_3C_4L_1L_3R_1R_3g_m + 2C_1C_3C_4L_1L_3R_3) + s^4(2C_1C_3C_4L_3R_1R_3 + C_1C_3L_1L_3R_1g_m + C_1C_3L_1L_3) + s^3(C_1C_3L_1R_1R_3g_m + C_1C_3L_1R_3 + C_1C_3L_3R_1 + 2C_1C_4L_1R_1R_3g_m + 2C_1C_4L_1R_3 + 2C_3C_4L_3R_1R_3g_m + 2C_3C_4L_3R_3) + s^2(C_1C_3R_1R_3 + 2C_1C_3R_3g_m)}$$

$$\mathbf{9.1023 \quad X-INVALID-ORDER-1023} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \quad \frac{R_4}{C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3L_1L_3R_1R_3R_4g_ms^4 + R_1R_3R_4g_m + s^2(C_1L_1R_1R_3R_4g_m + C_3L_3R_1R_3R_4g_m)}{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^5(2C_1C_3C_4L_1L_3R_1R_3R_4g_m + 2C_1C_3C_4L_1L_3R_3R_4) + s^4(2C_1C_3C_4L_3R_1R_3R_4 + 2C_1C_3L_1L_3R_1R_3g_m + C_1C_3L_1L_3R_1R_4g_m + 2C_1C_3L_1L_3R_3 + C_1C_3L_1L_3R_4) + s^3(C_1C_3L_1R_1R_3R_4g_m + C_1C_3L_1R_3R_4 + 2C_1C_3L_3R_1R_3 + C_1C_3L_3R_1R_4) + s^2(C_1C_3R_1R_3R_4 + 2C_1C_3R_3R_4g_m)}$$

$$\mathbf{9.1024 \quad X-INVALID-ORDER-1024} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \quad R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_3R_1R_3R_4g_ms^4 + R_1R_3R_4g_m + s^2(C_1L_1R_1R_3R_4g_m + C_3L_3R_1R_3R_4g_m)}{R_1g_m + s^5(2C_1C_3C_4L_1L_3R_1R_3g_m + C_1C_3C_4L_1L_3R_1R_4g_m + 2C_1C_3C_4L_1L_3R_3 + C_1C_3C_4L_1L_3R_4) + s^4(C_1C_3C_4L_1R_1R_3R_4g_m + C_1C_3C_4L_1R_3R_4 + 2C_1C_3C_4L_3R_1R_3 + C_1C_3C_4L_3R_1R_4 + C_1C_3L_1L_3R_1g_m + C_1C_3L_1L_3) + s^3(C_1C_3C_4R_1R_3R_4 + C_1C_3C_4R_1R_3g_m)}$$

$$\mathbf{9.1025 \quad X-INVALID-ORDER-1025} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \quad L_4s + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4R_1R_3g_ms^4 + R_1R_3R_4g_m + s^2(C_1L_1R_1R_3R_4g_m + C_3L_3R_1R_3R_4g_m)}{R_1g_m + s^6(C_1C_3C_4L_1L_3L_4R_1g_m + C_1C_3C_4L_1L_3L_4) + s^5(2C_1C_3C_4L_1L_3R_1R_3g_m + 2C_1C_3C_4L_1L_3R_3 + C_1C_3C_4L_1L_4R_1R_3g_m + C_1C_3C_4L_1L_4R_3 + C_1C_3C_4L_3L_4R_1) + s^4(2C_1C_3C_4L_3R_1R_3 + C_1C_3C_4L_4R_1R_3 + C_1C_3L_1L_3R_1g_m + C_1C_3L_1L_3 + C_1C_4L_1L_4R_1g_m)}$$

$$\mathbf{9.1026 \quad X-INVALID-ORDER-1026} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \quad \frac{L_4s}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4R_1R_3g_ms^4 + R_1R_3R_4g_m + s^2(C_1L_1R_1R_3R_4g_m + C_3L_3R_1R_3R_4g_m)}{2R_1R_3g_m + 2R_3 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3g_m + 2C_1C_3C_4L_1L_3L_4R_3) + s^5(2C_1C_3C_4L_3L_4R_1R_3 + C_1C_3L_1L_3L_4R_1g_m + C_1C_3L_1L_3L_4) + s^4(2C_1C_3L_1L_3R_1R_3g_m + 2C_1C_3L_1L_3R_3 + C_1C_3L_1L_4R_1R_3g_m + C_1C_3L_1L_4R_3 + C_1C_3L_3L_4R_1 + 2C_1C_4L_1L_4R_1R_3g_m)}$$

$$\mathbf{9.1027 \quad X-INVALID-ORDER-1027} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \quad L_4s + R_4 + \frac{1}{C_4s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4R_1R_3g_ms^4 + R_1R_3R_4g_m + s^2(C_1L_1R_1R_3R_4g_m + C_3L_3R_1R_3R_4g_m)}{R_1g_m + s^6(C_1C_3C_4L_1L_3L_4R_1g_m + C_1C_3C_4L_1L_3L_4) + s^5(2C_1C_3C_4L_1L_3R_1R_3g_m + C_1C_3C_4L_1L_3R_1R_4g_m + 2C_1C_3C_4L_1L_3R_3 + C_1C_3C_4L_1L_3R_4 + C_1C_3C_4L_1L_4R_1R_3g_m + C_1C_3C_4L_1L_4R_3 + C_1C_3C_4L_3L_4R_1) + s^4(C_1C_3C_4L_1R_1R_3R_4g_m + C_1C_3C_4L_1R_3R_4 + 2C_1C_3C_4L_3R_1R_3g_m)}$$

$$\mathbf{9.1028 \quad X-INVALID-ORDER-1028} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \quad \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4R_1R_3g_ms^4 + R_1R_3R_4g_m + s^2(C_1L_1R_1R_3R_4g_m + C_3L_3R_1R_3R_4g_m)}{2R_1R_3R_4g_m + 2R_3R_4 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3R_4g_m + 2C_1C_3C_4L_1L_3L_4R_3R_4) + s^5(2C_1C_3C_4L_3L_4R_1R_3R_4 + 2C_1C_3L_1L_3L_4R_1R_3g_m + C_1C_3L_1L_3L_4R_1R_4g_m + 2C_1C_3L_1L_3L_4R_3 + C_1C_3L_1L_3L_4R_4) + s^4(2C_1C_3L_1L_3R_1R_3R_4g_m + 2C_1C_3L_1L_3R_3R_4 + C_1C_3L_1L_4R_1R_3g_m)}$$

$$\mathbf{9.1029 \quad X-INVALID-ORDER-1029} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \quad \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1C_3C_4L_1L_3L_4R_1R_3g_ms^4 + R_1R_3R_4g_m + s^2(C_1L_1R_1R_3R_4g_m + C_3L_3R_1R_3R_4g_m)}{2R_1R_3g_m + R_1R_4g_m + 2R_3 + R_4 + s^6(2C_1C_3C_4L_1L_3L_4R_1R_3g_m + C_1C_3C_4L_1L_3L_4R_1R_4g_m + 2C_1C_3C_4L_1L_3L_4R_3 + C_1C_3C_4L_1L_3L_4R_4) + s^5(C_1C_3C_4L_1L_4R_1R_3R_4g_m + C_1C_3C_4L_1L_4R_3R_4 + 2C_1C_3C_4L_3L_4R_1R_3 + C_1C_3C_4L_3L_4R_1R_4 + C_1C_3L_1L_3L_4R_1g_m)}$$

9.1030 X-INVALID-ORDER-1030 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 R_4 + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (2C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 R_4 + C_1 C_3 C_4 L_3 L_4 R_4) + s^3 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 R_4 + C_1 C_3 C_4 L_3 L_4 R_4) + s^2 (2C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 R_4 + C_1 C_3 C_4 L_3 L_4 R_4) + s (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 R_4 + C_1 C_3 C_4 L_3 L_4 R_4) + 2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + 2C_1 C_3 C_4 L_1 L_3 L_4 R_4}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4 + s^6 (2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + C_1 C_3 C_4 L_1 L_3 L_4 R_4) + s^5 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 R_4 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 R_4 + C_1 C_3 C_4 L_3 L_4 R_4) + s^4 (2C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 R_4 + C_1 C_3 C_4 L_3 L_4 R_4) + s^3 (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 R_4 + C_1 C_3 C_4 L_3 L_4 R_4) + s^2 (2C_1 C_3 C_4 L_1 L_3 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 R_4 + C_1 C_3 C_4 L_3 L_4 R_4) + s (2C_1 C_3 C_4 L_1 L_3 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 L_3 R_3 R_4 + C_1 C_3 C_4 L_1 L_4 R_1 R_3 g_m + C_1 C_3 C_4 L_1 L_4 R_3 R_4 + 2C_1 C_3 C_4 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_3 L_4 R_3 R_4 + C_1 C_3 C_4 L_3 L_4 R_4) + 2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_3 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_1 R_4 g_m + 2C_1 C_3 C_4 L_1 L_3 L_4 R_3 + 2C_1 C_3 C_4 L_1 L_3 L_4 R_4}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(L_1 s, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_3 C_4 L_1 R_3 R_4 g_m s^2 + L_1 g_m + s (C_3 L_1 R_3 g_m + C_4 L_1 R_4 g_m)}{C_3 + 2C_4 + s^2 (2C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_1 R_4 g_m) + s (2C_3 C_4 R_3 + C_3 C_4 R_4 + C_3 L_1 g_m + 2C_4 L_1 g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_3} \sqrt{C_4} \sqrt{L_1 g_m} \sqrt{C_3 + 2C_4} \sqrt{2R_3 + R_4} \sqrt{\frac{1}{g_m}}}{2C_3 C_4 R_3 + C_3 C_4 R_4 + C_3 L_1 g_m + 2C_4 L_1 g_m}$
wo: $\sqrt{C_3 + 2C_4} \sqrt{\frac{1}{2C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_1 R_4 g_m}}$
bandwidth: $\frac{(2C_3 C_4 R_3 + C_3 C_4 R_4 + C_3 L_1 g_m + 2C_4 L_1 g_m) \sqrt{\frac{1}{2C_3 C_4 L_1 R_3 g_m + C_3 C_4 L_1 R_4 g_m}}}{\sqrt{C_3} \sqrt{C_4} \sqrt{L_1 g_m} \sqrt{2R_3 + R_4} \sqrt{\frac{1}{g_m}}}$
K-LP: $\frac{L_1 g_m}{C_3 + 2C_4}$
K-HP: $\frac{R_3 R_4}{2R_3 + R_4}$
K-BP: $\frac{C_3 L_1 R_3 g_m + C_4 L_1 R_4 g_m}{2C_3 C_4 R_3 + C_3 C_4 R_4 + C_3 L_1 g_m + 2C_4 L_1 g_m}$
Qz: None
Wz: $\frac{1}{\sqrt{C_3} \sqrt{C_4} \sqrt{R_3} \sqrt{R_4}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_4 R_1 R_3 R_4 g_m s^2 + R_3 g_m + s (C_1 R_1 R_3 g_m + C_4 R_3 R_4 g_m)}{g_m + s^2 (2C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_3 + C_1 C_4 R_4) + s (C_1 R_1 g_m + C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m)}$$

Parameters:

Q: $\frac{2\sqrt{C_1} \sqrt{C_4} R_1 R_3 g_m \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}} + \sqrt{C_1} \sqrt{C_4} R_1 R_4 g_m \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}} + 2\sqrt{C_1} \sqrt{C_4} R_3 \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}} + \sqrt{C_1} \sqrt{C_4} R_4 \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}}{C_1 R_1 g_m + C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m}$
wo: $\sqrt{\frac{g_m}{2C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_3 + C_1 C_4 R_4}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{2C_1 C_4 R_1 R_3 g_m + C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_3 + C_1 C_4 R_4}} (C_1 R_1 g_m + C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m)}{2\sqrt{C_1} \sqrt{C_4} R_1 R_3 g_m \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}} + \sqrt{C_1} \sqrt{C_4} R_1 R_4 g_m \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}} + 2\sqrt{C_1} \sqrt{C_4} R_3 \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}} + \sqrt{C_1} \sqrt{C_4} R_4 \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}}$
K-LP: R_3
K-HP: $\frac{R_1 R_3 R_4 g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}$
K-BP: $\frac{C_1 R_1 R_3 g_m + C_4 R_3 R_4 g_m}{C_1 R_1 g_m + C_1 + 2C_4 R_3 g_m + C_4 R_4 g_m}$
Qz: None
Wz: $\frac{1}{\sqrt{C_1} \sqrt{C_4} \sqrt{R_1} \sqrt{R_4}}$

10.3 X-INVALID-WZ-3 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_3 R_1 R_3 R_4 g_m s^2 + R_4 g_m + s (C_1 R_1 R_4 g_m + C_3 R_3 R_4 g_m)}{2g_m + s^2 (2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4) + s (2C_1 R_1 g_m + 2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m)}$$

Parameters:

Q: $\frac{2\sqrt{2} \sqrt{C_1} \sqrt{C_3} R_1 R_3 g_m \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}} + \sqrt{2} \sqrt{C_1} \sqrt{C_3} R_1 R_4 g_m \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}} + 2\sqrt{2} \sqrt{C_1} \sqrt{C_3} R_3 \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}} + \sqrt{2} \sqrt{C_1} \sqrt{C_3} R_4 \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}}{2C_1 R_1 g_m + 2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m}$
wo: $\sqrt{2} \sqrt{\frac{g_m}{2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4}}$
bandwidth: $\frac{\sqrt{2} \sqrt{\frac{g_m}{2C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_1 R_4 g_m + 2C_1 C_3 R_3 + C_1 C_3 R_4}} (2C_1 R_1 g_m + 2C_1 + 2C_3 R_3 g_m + C_3 R_4 g_m)}{2\sqrt{2} \sqrt{C_1} \sqrt{C_3} R_1 R_3 g_m \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}} + \sqrt{2} \sqrt{C_1} \sqrt{C_3} R_1 R_4 g_m \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}} + 2\sqrt{2} \sqrt{C_1} \sqrt{C_3} R_3 \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}} + \sqrt{2} \sqrt{C_1} \sqrt{C_3} R_4 \sqrt{\frac{g_m}{2R_1 R_3 g_m + R_1 R_4 g_m + 2R_3 + R_4}}}$
K-LP: $\frac{R_4}{2}$

K-HP:

$$\frac{R_1 R_3 R_4 g_m}{2 R_1 R_3 g_m + R_1 R_4 g_m + 2 R_3 + R_4}$$

K-BP:

$$\frac{C_1 R_1 R_4 g_m + C_3 R_3 R_4 g_m}{2 C_1 R_1 g_m + 2 C_1 + 2 C_3 R_3 g_m + C_3 R_4 g_m}$$

Qz:

None

Wz:

$$\frac{1}{\sqrt{C_1} \sqrt{C_3} \sqrt{R_1} \sqrt{R_3}}$$

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X-PolynomialError